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CERTIFIED INFORMATION SYSTEMS AUDITOR

CISA

Complete Study Guide

Exam-focused explanations, revision and practice questions for all five domains

ABOUT THE AUTHOR

Muhammad Aamir

Seasoned Technology Professional – IT Risk Assurance & Technology Advisory | CISA | CDCP, CNCDP | ISO 27001 LA | Digital Transformation, Cyber Risk & IT Governance | Oracle, SAP, Cloud & AI Advisory | Technology Educator

18+ years in technology risk, IT audit, information security and digital transformation across public and private sectors.

Visiting Faculty at SKANS School of Accountancy (CAF 3 DSR); has taught computer science and information systems at the University of Lahore.

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Limit of liability. The author has made every effort to ensure the accuracy of this book, but it is provided for study purposes without warranty. Passing the examination and qualifying for the certification are separate steps, and this book cannot guarantee either result.

Dedicated to my mother,

my Jannah

Your prayers carried me through every exam and every walk of life.

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Preface

CISA preparation becomes easier when technical terms are connected to a business process and a decision. Knowing that **encryption** protects information is useful. Understanding what it protects, where **plaintext** remains and what evidence an auditor needs is what lets you work through a scenario.

Having experienced the rigour of the CISA examination first-hand, I wrote this guide with one objective: to help students and professionals pass on their first attempt, with enough understanding to apply the concepts long after the examination. Success requires more than memorization. It requires the IS auditor's mindset: knowing who holds **decision authority**, what **evidence** withstands scrutiny and how to eliminate tempting distractors under timed conditions.

Each topic connects explanations and examples with comparisons, revision aids and practice. Apply every study convention to the actor, objective, stage and **facts** in the question. Where a simplified reminder has limits, the explanation states the assumptions. Technical accuracy and **professional judgment** work together; a memory aid should never require learning an incorrect definition.

You should finish each topic able to answer four practical questions: What is the objective? What could go wrong? Who is responsible for the decision or control? What **evidence** would support a conclusion? These questions connect the technical material to the work of an information systems auditor.

Some IT audit experience will help, but unfamiliar terms are explained in the text and in the glossary at the end of each domain. Read at a pace that allows you to explain an idea in your own words. A **memorized phrase** is fragile when the scenario changes.

Muhammad Aamir
November 2026

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About the Author

Muhammad Aamir, MSCS (Network Security), CISA, **ISO/IEC 27001** Lead Auditor, serves as Senior Manager in IT **Risk** Assurance & Technology Advisory at A.F. Ferguson & Co., a member firm of the PwC network. Bringing more than 18 years of specialized assurance, enterprise risk and systems security experience across public and private sectors, he advises organizations on aligning technological architecture with resilient corporate **governance** and regulatory compliance. His industry background also includes hands-on technology, digital transformation and infrastructure advisory experience with etrends, bridging deep technical execution with enterprise risk assurance.

In academia and professional education, he has contributed extensively as a lecturer and faculty member. At the **University of Lahore** (UoL), he delivered undergraduate and postgraduate coursework in computer science and information systems, mentoring students in core domains including network security, database management systems and systems analysis. Alongside his university teaching at UoL, he serves as Visiting Faculty at the SKANS School of Accountancy, where he specializes exclusively in lecturing **CAF 3: Data, Systems and Risks** (DSR).

Across both UoL and SKANS, his instructional methodology centers on deconstructing complex technical engineering concepts, data architecture and security protocols into practical **risk**, internal control and **governance** frameworks tailored for aspiring accountants, IT auditors and cybersecurity professionals.

Professional profile: linkedin.com/in/aamiryaqoob

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Introduction: The CISA Examination

About the CISA certification

CISA stands for Certified Information Systems Auditor. It is an ISACA **certification** focused on auditing, controlling and protecting information systems and on evaluating how technology supports **business objectives**. It is relevant to audit, assurance, control, risk and information security work. [R1]

The perspective is broader than selecting a secure technical product. You may need to assess **governance**, **requirements**, evidence, service delivery, recovery or access. An appropriate technical control still needs an owner, suitable operation and evidence that it addresses the relevant risk.

The examination is one part of **certification**. ISACA also requires relevant professional experience, with permitted substitutions subject to its rules, and an application within the allowed period. Candidates should check their own eligibility and current **requirements** directly. Passing alone does not authorize someone to claim the designation. [R2]

The examination at a glance

Feature	What the student should understand
Format	150 multiple-choice questions in four hours; select one best answer from four options. [R3]
Passing score	450 on ISACA's 200–800 scale. A scaled score is not a simple percentage-correct conversion. [R3]
Question approach	Expect practical application of concepts. Read the role, facts, stage and requested objective.
Wrong answers	The current guide states there is no incorrect-answer penalty; answer every question. [R3]
Pacing	240 minutes divided by 150 is 1.6 minutes per question on average. Use timed practice and leave review time.
Administrative preparation	Read the current candidate guide, identity requirements and test-center or remote-proctor instructions. [R3]

A practice percentage is useful feedback, but it is not a guaranteed conversion to the official **scaled score**. Domain percentages describe exam content distribution. They are not multipliers that award extra points to an individual question. [R3]

Domains and weights

Use the **weights** to plan attention, then adjust for your actual **weaknesses**. Domains 4 and 5 together represent 52 percent of the published outline. Domains 1, 2 and 3 together represent 48 percent. None can safely be ignored. [R1]

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Domain and weight	Field of study	What to be ready to evaluate
1 Audit process - 18%	Planning and performing assurance work.	Risk-based scope, controls, evidence, sampling, analytics, reporting and audit quality.
2 Governance and management - 18%	Direction and oversight of enterprise IT.	Strategy, accountability, risk, policy, privacy, data, resources, vendors and performance.
3 Acquisition and implementation - 12%	Introducing or changing business systems.	Business cases, projects, SDLC, embedded controls, acceptance, migration and benefits.
4 Operations and resilience - 26%	Dependable service and recovery.	Assets, processing, interfaces, changes, incidents, databases, backups, continuity and recovery.
5 Protection of assets - 26%	Security controls and event management.	Access, network and endpoint security, cryptography, PKI, cloud, detection, response and evidence.

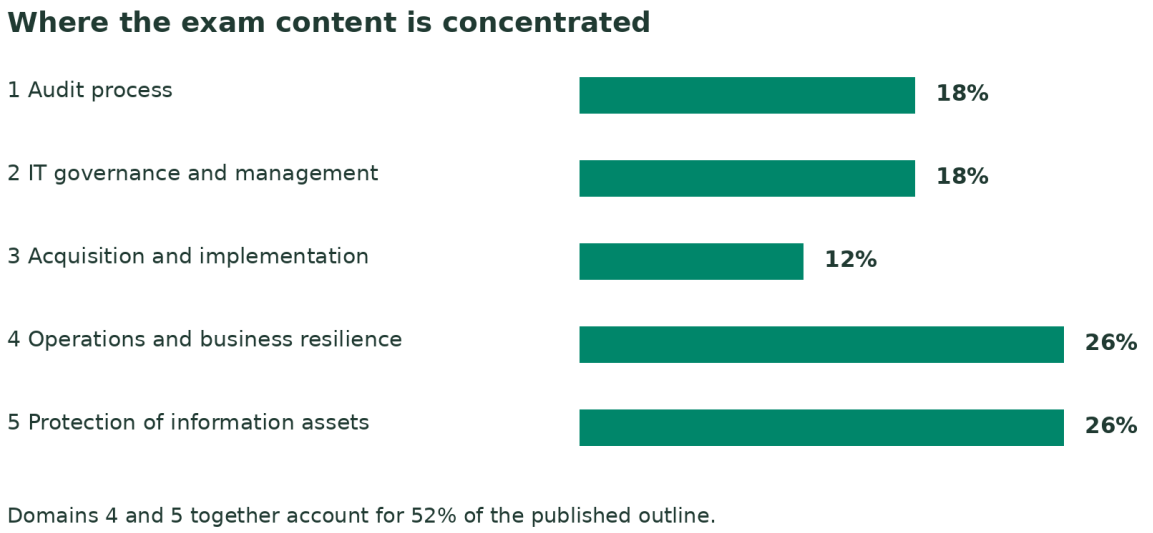


Figure I.1 Published domain weights are a planning guide; individual weaknesses also matter.

Why Domain 1 supports the other domains

Domain 1 provides the **audit method** used to evaluate technical and **governance** topics. A security scenario can require you to choose evidence, evaluate a control or decide how to communicate an exception. You need the Domain 5 concept and Domain 1 reasoning together. This is a study connection, not a claim that the official domain **weights** change.

Scenario	Subject knowledge	Audit reasoning
Access remains after an employee leaves.	Domains 2 and 5: ownership, identity lifecycle and authorization.	Domain 1: validate the leaver population, test timing, investigate and report supported findings.

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Scenario	Subject knowledge	Audit reasoning
A migrated customer file has matching totals but wrong balances.	Domain 3: conversion, mapping and reconciliation.	Domain 1: choose evidence that tests record identities and accuracy rather than relying on aggregates.
Recovery starts servers quickly but business payments still fail.	Domain 4: dependencies, RTO, RPO and service capability.	Domain 1: compare actual evidence with approved criteria and limit the conclusion to what was tested.

Study Domain 1 early. Then revisit its ideas while reading each later domain: objective, risk, **population**, **criteria**, evidence, conclusion and follow-up. Domain 2 supplies ownership and governance context; Domain 3 connects controls to delivery; Domains 4 and 5 show those decisions operating in live services.

How questions test judgment

Read the full stem before answering. Identify the actor, the **business objective**, what has already happened and the exact **qualifier**. Several choices can describe useful actions. The required answer is the one that best fits the stated objective and stage.

Wording	What it asks	Common error
FIRST	Which necessary action comes before the others in this situation?	Choosing a valuable later action or applying a fixed sequence despite the facts.
NEXT	Given what is already complete, what should follow?	Repeating the stage the question says is already done.
BEST / MOST appropriate	Which option most completely fits the objective and constraints?	Choosing a technically true statement that answers another question.
MOST effective	Which control or procedure most directly achieves the requested result?	Substituting a policy statement for evidence of operation, or a detective control for prevention.
PRIMARY	Which purpose or factor is central to this particular decision?	Choosing a secondary benefit that is real but less relevant.
GREATEST concern	Which supported exposure matters most in the stated circumstances?	Ranking by dramatic wording without assessing business impact and the facts.

A repeatable way to choose

First state the question in your own words. Identify the requested decision and the **evidence** already available. Eliminate choices that answer a different objective, use the wrong owner or belong to the wrong stage. Compare the remaining options against the actual **facts**. Select one and be able to explain why the closest alternative is less suitable.

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Do not add **facts** to rescue a distractor. Do not assume that the **most** senior escalation, most expensive tool or most restrictive control is always correct. Likewise, management ownership does not mean an auditor should remain silent about a serious unsupported risk.

Worked example of FIRST and BEST

An ordinary audit test identifies an apparent **approval** exception, but the auditor has not yet verified the record or understood the workflow. The question asks what to do FIRST. Establish the **facts** and relevant **criteria** and obtain appropriate clarification before writing a final conclusion. If the facts instead show evidence destruction or management interference, the audit mandate may require prompt escalation. The wording changes the judgment.

Now suppose the question asks for the BEST **evidence** that leaver access was removed on time during a year. A validated comparison of departure dates and **disablement** records is stronger for that objective than a **policy**, a manager's assurance or one current screenshot. The issue is the fit between evidence and conclusion, not whether those other items are ever useful.

Applying study conventions to the stated scenario

Use study conventions to organize your reasoning, then apply the complete scenario. A **qualifier** such as FIRST asks which necessary step comes next given what is already established. A BEST question asks which choice **most** directly satisfies the stated objective and constraints. Do not invent missing **facts** or apply a fixed sequence without checking its assumptions.

This is an independent study guide and does not claim access to examination keys or confidential questions. The practice rationales explain why one choice fits the written scenario and why its alternatives are weaker. Professional definitions should remain technically accurate: a **digital signature** is created with a **private signing key** and verified using the trusted corresponding **public key**; it is not generically private-key encryption.

> **Exam reasoning:** Identify the role, objective, established **facts** and **qualifier**. Choose the supported response that answers the question, and recognize the limits of evidence and control coverage.

How to use this book

Use four passes. In the first, read the explanation and example and define unfamiliar terms. In the second, reconstruct the process or comparison without looking. In the third, answer questions and explain every option. In the fourth, use The **Last Mile** and glossaries to revisit weak areas and connect domains.

The main topics follow ISACA's published five-domain structure and retain consistent section references for easy cross-checking. Each domain closes with The **Last Mile**, a glossary, practice questions and full answer explanations. Practical extensions connect selected modern technologies to their audit implications without claiming that every product acronym is a separately mandated examination topic.

Study activity	Observable result
Explain a concept	You can describe its purpose, owner and limitation without repeating the book word for word.
Work through an example	You can say what changes if one fact or objective changes.

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Study activity	Observable result
Answer a question	You can justify the chosen option and reject the strongest distractor for a specific reason.
Review a mistake	You can identify a knowledge gap, role confusion, qualifier error, missed fact or timing problem.
Retest later	You can apply the concept to a fresh scenario after a delay, rather than recognize a remembered answer.

A flexible eight-week study route

This is an example schedule, not a prescribed duration. Increase time for unfamiliar subjects and shorten only when you can demonstrate understanding. A learner with strong technical experience may still need more work on audit judgment and **governance**.

Period	Primary focus	Ongoing practice
Week 1	Domain 1 and an initial mixed diagnostic.	Record why answers were missed; establish an error log.
Week 2	Domain 2 and governance/ownership distinctions.	Revisit Domain 1 evidence and reporting decisions.
Week 3	Domain 3 and lifecycle decisions.	Trace requirements, acceptance and conversion evidence.
Weeks 4–5	Domain 4 and service/recovery scenarios.	Practice RTO/RPO, changes, reconciliation and dependencies.
Weeks 6–7	Domain 5 and security-control selection.	Practice IAM, key use, PKI, attacks, monitoring and response.
Week 8	Mixed questions, timed mocks and targeted repair.	Review fresh results, revisit weak topics and rehearse pacing.

Within a study session, combine focused reading, **closed-book recall** and questions. Revisit a difficult concept after a short delay and again later in the week. Allocate extra work using both the published content weight and **repeated weaknesses**, rather than reading every chapter for exactly the same time.

Use an error log

Record	Example
Question/topic	Recovery test: servers met RTO but data breached RPO.
My reasoning	I treated service startup as complete recovery.
Why it failed	The scenario asked whether all approved recovery objectives were met.
Correct principle	Check time, data currency, integrity, function and required service level.

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Record	Example
Next action	Revisit section 4.16 and answer a different recovery scenario later.

Practice questions and exam readiness

The book contains 100 **original domain practice questions** in the published domain proportions, plus 20 **original mixed scenarios**. Each question has one best answer for its stated facts and an explanation of every option. The mixed set supports transfer of understanding; it is not a full-length 150-question mock. None of these questions is an actual examination question.

Before exam day, check the current **official instructions**, identity details and test arrangements. Practice answering within the available time and reviewing uncertain questions without getting trapped on one item. Use the official tutorial and **answer all questions** under the current no-penalty rule. [R3]

You are better prepared when you can explain important distinctions, handle unfamiliar mixed scenarios, maintain pace and correct repeated errors. One high score on a familiar set is weaker **evidence** than consistent reasoning on fresh material.

Companion resources

mirakx.com is the publication home for this independent guide. Use the domain roadmaps, examples and **cross-domain reasoning** in the book to organize study.

certprepx.com is the companion practice tool, offering timed mock examinations, adaptive practice, domain analytics and spaced review. Use fresh questions, examine why each option fits or fails and return to weak topics. A platform **readiness** estimate is a study aid, not an official result or guaranteed conversion to ISACA's **scaled score**. Check the site for current features. [R4]

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Domain 1: Information System Auditing Process

1.0 Domain overview

An **IS audit** gives an independent, **evidence** based conclusion about information systems and their controls. The work begins with the business process and its risks. It ends with a supported conclusion, clear communication, and appropriate follow up. This domain accounts for 18 percent of the CISA exam.

Topics and reading route

Part	Section	Topic
A Planning	1.1	IS Audit Standards, Guidelines, Functions and Codes of Ethics
A Planning	1.2	Types of Audits, Assessments and Reviews
A Planning	1.3	Risk-Based Audit Planning
A Planning	1.4	Types of Controls and Considerations
B Execution	1.5	Audit Project Management
B Execution	1.6	Audit Testing and Sampling Methodology
B Execution	1.7	Audit Evidence Collection Techniques
B Execution	1.8	Audit Data Analytics
B Execution	1.9	Reporting and Communication Techniques
B Execution	1.10	Quality Assurance and Improvement of the Audit Process

Work through the explanations and examples, then use the comparisons and memory aids for revision. The final sections provide The **Last Mile**, a glossary, and questions with explanations. Keep a brief error log: record the concept you missed and why your chosen answer did not fit the situation.

Think through the audit

Suppose former employees can still access payroll. Understand how HR notifies IT of departures; assess the **exposure**; compare the leaver **population** with active accounts; investigate exceptions; and report the supported finding. Follow up to confirm the process now works.

Keep that sequence in mind: understand the process → assess risk → test → evaluate evidence → report and follow up. The charter provides authority for the audit function; it is not recreated for each engagement. The auditor evaluates controls. **Management owns** the controls and any **corrective action**.

Use the audit cycle in Figure 1.2 to connect the lessons. **Plan from objectives and risk** in sections 1.1–1.5. **Test controls and evaluate evidence** in sections 1.6–1.8. **Report supported conclusions**

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and follow up in sections 1.9–1.10. Apply the same sequence when the subject is governance, development, operations or security.

1.1 IS Audit Standards, Guidelines, Functions and Codes of Ethics

An **information systems (IS) audit** examines technology, the business processes it supports, and the controls that protect those processes. The auditor compares **evidence** with **criteria**, such as a policy, standard or agreed requirement, to reach a supported conclusion.

Consider an audit of payroll access. The auditor needs authority to obtain relevant records, a method for testing whether access was removed on time, and the freedom to report the results. The **audit charter** establishes the function's authority. **Professional standards** govern the work, and **independence and objectivity** protect the judgment behind the conclusion.

1.1.1 ISACA IS Audit and Assurance Standards

ISACA IS Audit and Assurance Standards establish mandatory professional requirements for applicable IS audit and assurance work. They set the minimum performance expected of the auditor and help management, the audit committee and other intended users understand what to expect from the work and its reporting.

Group	What it governs	Example
General standards	Conditions for credible audit work, including authority, independence, competence and professional care.	Confirm that the team has authority and skills for a cloud security review.
Performance standards	Planning, supervision, evidence collection, documentation and evaluation.	Test whether payroll access was removed on time and record the supporting evidence.
Reporting standards	Communication of results, conclusions and relevant limitations.	Explain the systems covered and how unavailable records affect the conclusion.

The auditor needs **sufficient appropriate evidence**. Sufficiency concerns how much evidence is needed. Appropriateness concerns its relevance and reliability. A signed payroll checklist records a claim about access removal. Compare authorized termination records with the system events showing when accounts were disabled to test that claim.

1.1.2 ISACA IS Audit and Assurance Guidelines

Guidelines help auditors apply standards to the circumstances of an engagement. Select a suitable approach and document significant departures when required. An **audit procedure** is a specific step taken to obtain or evaluate evidence.

Term	Purpose	Payroll access example
Standard	Sets the professional requirement.	Obtain sufficient appropriate evidence for the conclusion.
Guideline	Helps the auditor apply the requirement.	Consider the completeness and reliability of system records.

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Term	Purpose	Payroll access example
Procedure	Specifies an action to perform.	Compare required access removal times with actual account disablement times.

A **tool or technique** helps perform a procedure. A query can compare thousands of timestamps. Before relying on its output, check that the extract includes the relevant accounts and that the query uses the correct matching and timing rules.

1.1.3 ISACA Code of Professional Ethics

The **Code of Professional Ethics** guides the conduct of ISACA members and certification holders. Apply its duties when handling sensitive information, assessing evidence, accepting assignments and reporting findings.

- **Support sound governance and controls.** Encourage appropriate standards and practices for technology governance, audit, control, security and risk management.
- **Act objectively and with due care.** Apply professional standards and evaluate evidence impartially.
- **Serve stakeholders lawfully.** Maintain conduct that supports confidence in the profession.
- **Protect privacy and confidentiality.** Safeguard information, disclose it to appropriate parties under applicable requirements and protect it from personal use.
- **Maintain competence.** Keep skills current and undertake work that can be completed with the necessary knowledge, skills and support.
- **Communicate significant facts.** Inform appropriate parties of facts needed to understand the results fairly.
- **Support professional education.** Help stakeholders understand technology governance, audit, security, control and risk responsibilities.

A project sponsor asks audit to remove a supported finding because it could delay launch. Examine any new evidence and correct the finding if the facts warrant it. If the finding remains significant and supported, communicate it through the authorized reporting channel and raise continuing pressure with audit leadership.

Professional Competence, Due Care and Skepticism

Professional competence is the knowledge and skill needed for the work. **Due professional care** is the diligence and judgment reasonably expected of a competent auditor in the circumstances. Consider the significance of the risk, choose suitable procedures, evaluate the results and limit the conclusion to what the evidence supports.

Professional skepticism means questioning explanations fairly and checking them against evidence. If a system owner says a scheduled job disables every leaver's account, inspect the job results and actual account status. Investigate failed runs, excluded accounts and departures that do not match the system records.

1.1.4 ITAF™

ITAF™, ISACA's IT Audit Framework, brings professional standards and supporting guidance together for IT audit and assurance. Use it to connect a requirement to the planning, performance and reporting of the engagement. For example, when source records are incomplete, consider alternative procedures, evaluate the evidence obtained and communicate any limitation that affects the conclusion. [R8]

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1.1.5 IS Internal Audit Function

The **IS internal audit function** provides independent assurance and advice about technology risks and controls. It evaluates how controls are designed and how they operate. Management runs the controls, takes corrective action and makes risk decisions within its authority.

Audit Charter

The **audit charter** establishes the internal audit function's purpose, mandate, authority, responsibilities, reporting relationships and access. The **chief audit executive (CAE)** develops and maintains it with appropriate input. The **board**, or its authorized audit committee, approves it.

The charter gives the function standing authority. **Engagement scope** defines the systems, processes, locations and periods covered by one assignment. Access must support the objective and comply with applicable confidentiality and legal requirements.

A payroll manager declines to provide a log because it contains personal information. Explain why the records matter and discuss safeguards such as a restricted extract or supervised read-only access. If necessary records remain unavailable, consider alternative procedures and raise the issue with audit leadership. A restriction that prevents necessary work is a **scope limitation**. Assess and communicate its effect on the conclusion.

Management of the IS Audit Function

The CAE leads the audit function. **Functional reporting** gives the CAE direct access to the board on the charter, plan, significant results, resources and interference. **Administrative reporting** supports day-to-day matters such as staffing administration, facilities and expenses.



Diagram 1.1 Functional oversight protects the audit mandate. Administrative support keeps the function operating.

If the finance director handles audit expenses and proposes removing a high-risk finance system from the plan, assess the change against the risk and take it through the applicable governance approval process.

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Independence and Objectivity

Distinction	What it protects	Example of a threat
Organizational independence	The function's ability to set scope, obtain evidence and communicate results without improper interference.	A senior manager requires permission before audit may report a weakness in that manager's area.
Individual objectivity	An auditor's ability to evaluate evidence impartially.	An auditor is assigned to assess a control they recently designed and operated.

The second case is a **self-review threat**. Disclose the circumstances, evaluate the impairment and apply safeguards permitted by the applicable requirements. Assigning the work to a qualified person with no responsibility for the control may be appropriate.

Audit Charter, Audit Plan, Engagement Communication and Audit Program

Document	What it establishes	Responsibility
Audit charter	Standing authority and responsibilities of the function.	The CAE develops and maintains it. The board or authorized audit committee approves it.
Risk-based audit plan	Priorities, coverage and resources over a period.	The CAE develops it. The governing authority approves it and considers significant changes.
Engagement communication	Objectives, scope, timing and arrangements for one assignment.	Audit leadership communicates the arrangements. Relevant management provides information and coordinates access. Formal terms are agreed where applicable.
Audit program	What to test, how to test it and who performs the procedures.	The engagement team develops it. Audit leadership approves it under the function's methodology and reviews significant changes.

A company moves payroll to a cloud service. The **charter** supplies standing authority, the **plan** selects a payroll access review, and the **engagement communication** identifies the systems and period. The **program** directs the auditor to compare termination records with disablement events and investigate exceptions. Management can supply evidence to correct facts. Audit leadership remains responsible for the procedures and supported conclusion.

Internal and External Audit Roles

Area	Internal audit	External audit
Authority	Approved charter and applicable governance arrangements.	Engagement terms, applicable law or a regulatory mandate.
Scope	Risks and controls selected through internal audit planning.	The subject and scope of the engagement or mandate.
Reporting	Approved internal arrangements, with board oversight.	Intended recipients specified by the engagement or mandate.

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Area	Internal audit	External audit
Example	Review payroll access and procurement controls under the audit plan.	Examine technology controls relevant to financial reporting for a defined period.

An internal function can use employees, service providers or both. An external auditor may use relevant internal audit work when the applicable standards allow it, after assessing the work and how it was performed. Each auditor remains accountable for their conclusion.

IS Audit Resource Management

Match **skills, time, budget and tools** to the work. Before assigning a cloud security review, establish whether the team can interpret configuration evidence and supervise any specialist. Communicate material coverage gaps and their implications through the applicable governance channel.

Using the Services of Other Auditors and Experts

Assess the specialist's **competence, objectivity and experience**. Agree the scope, period, evidence access, confidentiality, supervision and reporting before work begins. Review the methods, assumptions, results and limitations before relying on the work.

In a cloud review, establish who designed, operates and assesses the relevant controls. A specialist who configured a control may face a **self-review threat**. A provider's assurance report can help when its system, controls and period match the need. Audit retains responsibility for its own conclusions.

Evidence and CISA Exam Perspective

Look for approved versions of the **charter and plan**, evidence of direct communication with the board, **reviewed programs**, and documented responses to access restrictions or conflicts. Compare the arrangements on paper with decisions made in practice.

If the question asks about	Focus on
The audit function's authority	The approved **audit charter**.
Which engagements receive priority and resources	The **risk-based audit plan**.
The boundaries and arrangements for one assignment	**Engagement scope and communication**.
Which tests will be performed and who approves them	The **audit program** and audit leadership.
A manager suppressing results or blocking access	**Organizational independence**, escalation and the effect on the work.
An auditor reviewing their own control	**Objectivity**, disclosure and appropriate safeguards or reassignment.

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1.2 Types of Audits, Assessments and Reviews

The **purpose of an engagement** determines what the auditor examines, which evidence is needed and what a reader can conclude. A financial audit of payroll may test reported expense. An operational audit may examine delays and rework. A compliance audit may test adherence to specified requirements.

An **audit** evaluates evidence against criteria to reach an independent conclusion. An **assessment** evaluates a condition or capability, often to identify gaps. A **review** may be a general examination or a formal engagement with a defined level of assurance. Read the engagement terms to understand what its results support.

Types of Audits and Assessments

Type	Primary objective	Practical example
IS audit	Evaluate whether systems and related resources are appropriately controlled, protected and reliable.	Examine access, change and recovery controls for online banking.
Compliance audit	Assess adherence to laws, policies, standards or contracts.	Verify retention against the applicable requirement.
Financial audit	Assess fair presentation of financial information under the applicable framework.	Test completeness and accuracy of payroll expense.
Operational audit	Assess how effectively and efficiently a process achieves objectives.	Examine delivery delays and manual order corrections.
Integrated audit	Combine relevant disciplines around a common process or objective.	Examine fulfillment, automated pricing and revenue reporting together.
Comprehensive audit	Examine a function broadly, including its financial records and controls.	Review purchasing records, supplier approvals and oversight.
Administrative audit	Assess whether administrative arrangements support effective operations.	Examine how staffing, approvals and roles affect service requests.
Specialized audit	Examine a focused subject, technology or risk.	Assess recovery capability for critical systems.
Third-party service audit	Evaluate outsourced services or relevant provider controls.	Examine a payroll processor's access to employee information.
Fraud audit	Investigate suspected intentional wrongdoing or its extent.	Trace payments to identify possible fictitious suppliers.
Forensic audit	Preserve and analyze evidence for investigation or potential proceedings.	Reconstruct changes to payment records and document evidence handling.
Computer forensic audit	Collect, preserve and analyze electronic evidence.	Examine a controlled device copy and system logs to reconstruct events.

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Type	Primary objective	Practical example
Functional audit	Assess whether delivered software performs required functions and meets relevant performance requirements.	Test billing calculations and response times against requirements.
Readiness assessment	Identify gaps before formal evaluation.	Review whether access reviews have supporting records before an examination.

CISA exam perspective. A readiness assessment may show that a procedure has been documented. A conclusion about operating effectiveness needs evidence of actual performance, exception handling and coverage of the relevant period.

Assurance and the Form of the Conclusion

Assurance is the confidence a user gains from independent evaluation against suitable criteria. Under applicable attestation standards, the type of engagement determines what work is performed and how results are communicated.

Engagement	Level or result	Example
Examination	**Reasonable assurance** within the stated scope and criteria.	An opinion supported by sufficient appropriate evidence.
Formal review	**Limited assurance** using procedures appropriate to that engagement.	A conclusion expressed in the form required by the framework.
Agreed-upon procedures (AUP)	Factual findings from specified procedures. Users draw their own conclusions.	Eight of 50 specified items lacked required approval.

Third-Party Service Audits and SOC Reports

Outsourcing assigns activities to a provider. The customer still manages its risks and operates its own controls. An independent service report can help assess the provider's part of the arrangement.

System and Organization Controls (SOC) is a suite of reporting services developed by the **American Institute of Certified Public Accountants (AICPA)**. The **service organization** provides the service, the **user entity** uses it and the **service auditor** performs the examination.

Report	Subject	Typical audit use
SOC 1	Provider controls relevant to user entities' **internal control over financial reporting (ICFR)**.	Assess payroll calculations and processing totals used in financial statements.
SOC 2	Controls for the described system against applicable **Trust Services Criteria**.	Assess detailed controls for a specified service, such as privileged access or recovery.
SOC 3	Assurance against applicable Trust Services Criteria in a shorter **general-use** report.	Obtain general assurance for initial evaluation. Obtain detailed evidence where needed.

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A payroll provider may have a **SOC 1** report for financial reporting controls and a **SOC 2** report for security and availability. Select the report that answers the audit question and read the service and criteria in scope.

Trust Services Criteria

Category	What it examines	Example
Security	Protection against unauthorized access, use or changes.	Privileged access is approved and monitored.
Availability	Service availability as committed.	Recovery testing checks whether agreed targets are achievable.
Processing integrity	Complete, valid, accurate, timely and authorized processing.	Orders are checked and processing exceptions resolved.
Confidentiality	Protection of information designated confidential.	Customer contracts and pricing records have restricted access.
Privacy	Handling of personal information over its life cycle under applicable criteria.	Employee data are retained and disposed of as required.

Check the **actual categories** covered. Confidentiality can concern commercial information such as pricing, while privacy focuses on personal information and its handling.

Type 1 and Type 2

For **SOC 1 and SOC 2**, the report family identifies the subject. The type identifies the time basis and whether operating effectiveness is addressed.

Type	Time basis	What the examination addresses
Type 1	A specified date.	The relevant system description and suitability of control design at that date.
Type 2	A specified period.	The relevant description, suitability of design and operating effectiveness over the period, with tests and results.

A **SOC 2 Type 1** dated 30 June can support a design assessment at that date. To assess operation from January through June, look for a relevant Type 2 report covering the period or obtain other evidence of operation.

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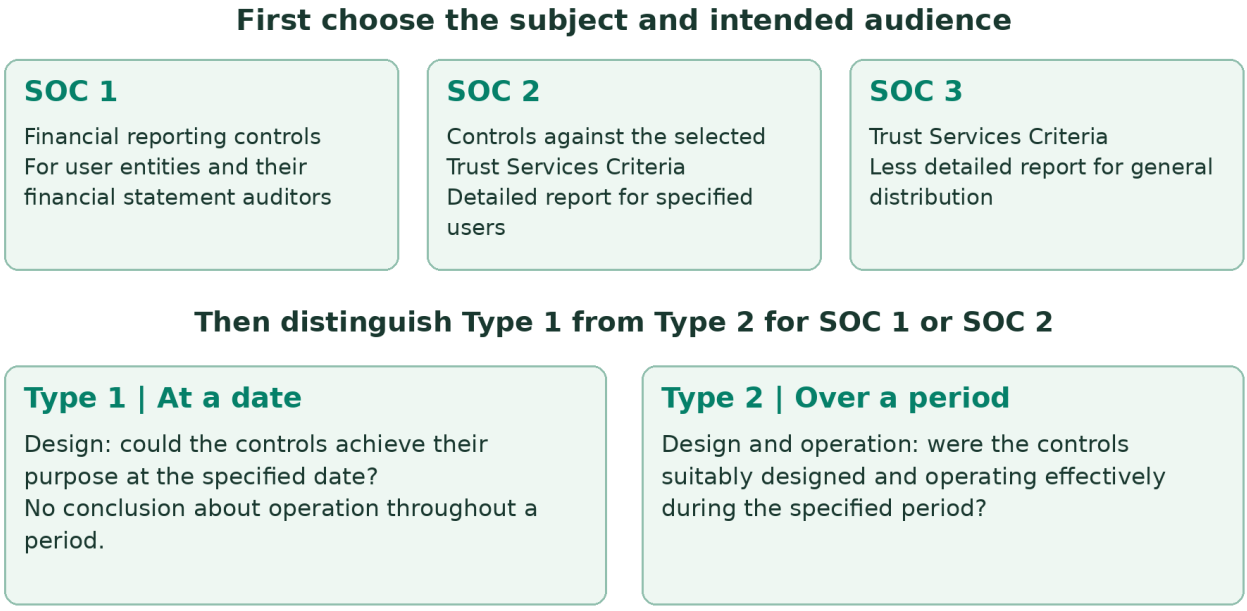


Figure 1.1 Select the report subject first, then evaluate its time and operating-effectiveness coverage.

Evaluating a SOC Report Before Relying on It

Start with your audit objective. Read the **system description**, **opinion**, and applicable **tests and results**. Compare the service, locations, criteria, type and dates with your need. Evaluate exceptions for their relevance to your risks and document which parts of the conclusion the report supports.

- **Complementary user entity controls (CUECs)** are controls the customer is expected to operate. If the provider assumes the customer approves pay changes, obtain evidence of the customer's approvals.
- A **subservice organization** is another provider used by the service organization. Under the **carve-out method**, specified subservice controls are excluded from the examination. Under the **inclusive method**, the specified subservice controls are included. Check the actual boundaries and obtain evidence for relevant gaps.
- A **bridge letter** is a management update about changes or events after a report period ends. Use it to understand the gap and decide what additional independent evidence is needed.

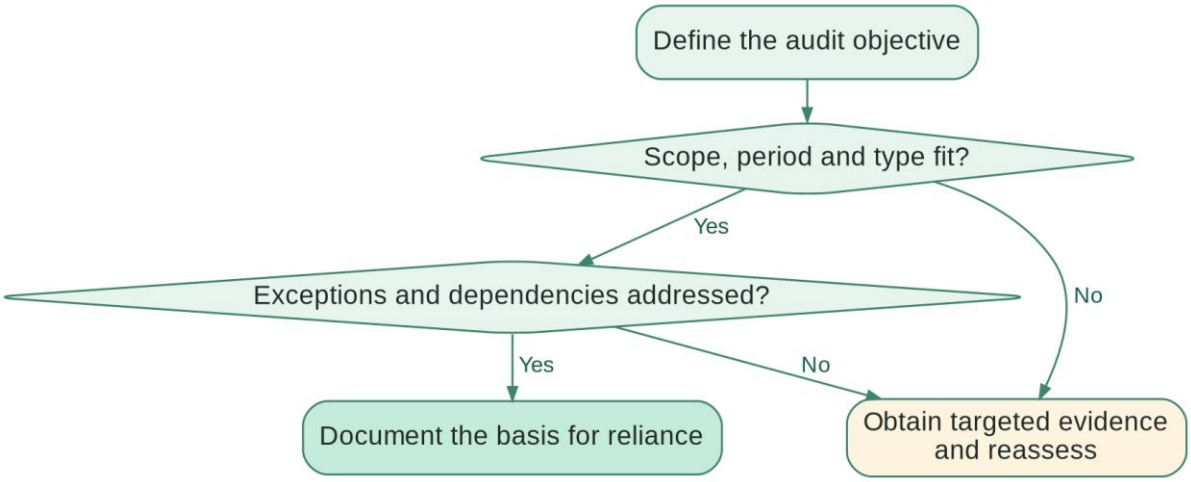


Diagram 1.2 Match third-party assurance to the objective, then address gaps before relying on it.

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Worked example. A payroll processor's SOC 1 Type 2 covers January through June. Your audit covers January through September. The report assumes customer approval of new hires and pay changes and carves out hosting controls. Verify service scope and relevant exceptions, test the customer's approvals, then evaluate evidence for hosting and the July-to-September period. A bridge letter may explain changes, while the evidence needed for control reliance depends on the risk and the remaining gap.

Apply the same reasoning when using **another auditor's work**. Assess competence, independence, scope, period, procedures and limitations. Document which conclusions are supported and what additional work remains.

1.2.1 Control Self-Assessment

Control self-assessment (CSA) is a structured exercise in which managers and staff identify risks and evaluate their own controls. They can use workshops, questionnaires or management-led reviews to identify improvements and assign owners.

A purchasing team examines supplier bank-detail changes. Staff explain that urgent requests sometimes bypass the independent callback. This reveals a practice absent from the written procedure. Management can assign corrective action, and audit can test actual changes to determine the extent and effect.

Objectives of CSA

Link **business objectives, risks and controls**. Identify how the control works in practice and assign an owner and due date to improvements.

Benefits of CSA

Staff involvement can uncover workarounds early, build control awareness and improve management's ownership of actions. Audit can use the results to identify areas for independent testing.

Disadvantages of CSA

Participants may overrate their work, lack assessment skills or be reluctant to challenge a manager. Ask clear questions, corroborate significant claims and track agreed actions.

The IS Auditor's Role in CSA

Audit may **facilitate**, explain control principles and challenge unsupported conclusions. Management owns design, operation and risk decisions. When assurance is needed, test actual records and evaluate the results independently.

1.2.2 Integrated Auditing

Integrated auditing combines relevant disciplines around a business process and its risks. In financial reporting, the term can also describe coordinated audits of the financial statements and internal control over financial reporting. Read the engagement scope to establish its meaning.

An online retailer accepts orders, calculates prices, dispatches goods, invoices customers and posts revenue. The team follows the same transactions through those stages.

Perspective	What to examine	Evidence
Operational	Whether orders are fulfilled accurately and on time.	Orders, dispatch records and unresolved exceptions.

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Perspective	What to examine	Evidence
Financial	Whether completed sales are billed accurately and recorded in the correct period.	Invoices, dispatch evidence and ledger entries.
IS	Whether automated pricing, interfaces and access controls support reliable processing.	Pricing configuration, change approvals, privileged access and reconciliations.

An automated price check may work in the items tested while administrators can change its rules without approval. Trace which conclusions depend on that check. Inspect change history or overrides and obtain additional transaction evidence where reliance is affected.

CISA Exam Perspective and Common Traps

If the question asks about	Focus on
The appropriate engagement	The **primary objective** and intended conclusion.
Reliance on a provider report	**Service, controls, criteria, type and period**, then exceptions and dependencies.
A relevant CUEC	Evidence that the **customer's control** is implemented and operates as required.
How to use CSA results	Use them to direct **independent validation**.
An IT control weakness in an integrated audit	Trace which operational and financial conclusions **depend on that control**.

1.3 Risk-Based Audit Planning

Risk is the effect uncertainty can have on a business objective. A supplier master file may occupy little storage, yet one unauthorized bank-detail change can divert a substantial payment. Consider the event, its likelihood and its consequences for finances, service, safety, compliance or reputation.

Risk-based audit planning uses business objectives, risks and obligations to decide what to audit, when to audit it and how much work to perform. Understand the process and its dependencies before selecting coverage.

1.3.1 Individual Audit Assignments

The **audit universe** is the inventory of processes, applications, infrastructure, projects, locations and providers that could be audited. Organize it around business objectives and information flows. Consider changes, dependencies, prior findings, control performance, mandatory work and existing assurance. A low-volume service that grants emergency administrator access may matter because many critical systems depend on it.

Planning level	Decision	Supplier payment example
Audit plan	Which areas receive coverage and when?	Prioritize payments after a supplier portal begins accepting bank-detail changes.

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Planning level	Decision	Supplier payment example
Engagement objective	What must the assignment determine?	Determine whether bank-detail changes are independently verified before use.
Scope	Which processes, systems and periods are covered?	Include portal and direct changes during the first six months after launch.
Criteria	What requirements apply?	Use the approved requirement for independent verification before release.
Audit program	Which procedures obtain evidence?	Trace actual changes to verification records and resulting payments.

The **annual plan** allocates coverage and resources. Longer-term planning anticipates changes in strategy, technology and skills. After a major incident, acquisition or migration, reassess coverage, obtain approval for significant plan changes under §1.1, and explain the consequences of deferring work.

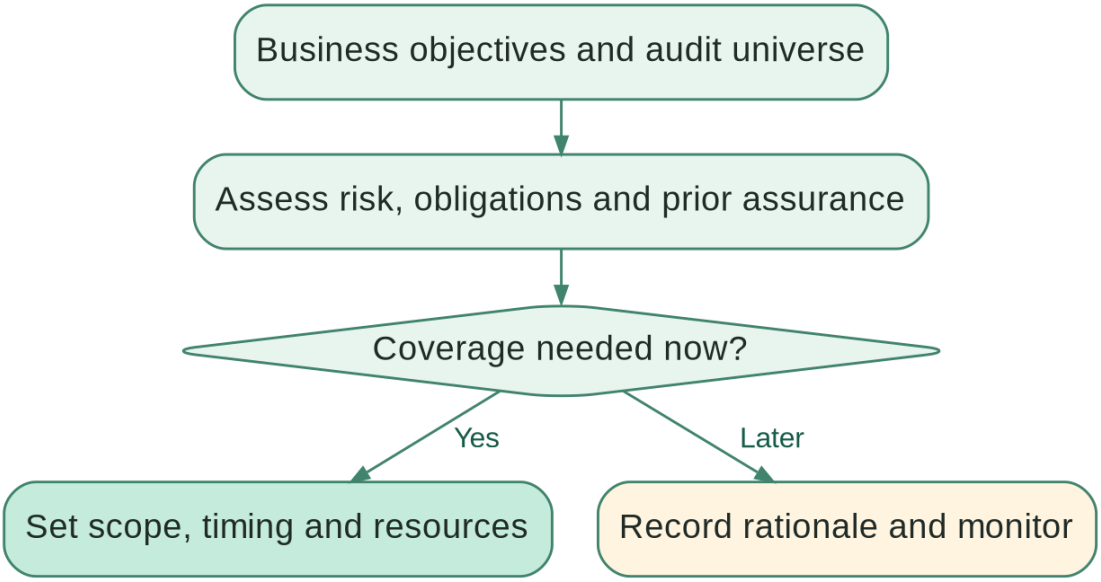


Diagram 1.3 A risk-based plan explains both current coverage and deferred work.

1.3.2 Effect of Laws and Regulations on IS Audit Planning

Identify applicable **legal, regulatory and contractual requirements** early. Confirm their scope, the period and activities covered, and who is responsible. Obtain specialist interpretation where needed.

Effect on planning	Audit decision	Example
Required coverage	Reserve time and skills for mandatory reviews and deadlines.	Include a required review of a regulated payment service.
Audit criteria	Translate requirements into testable conditions.	Determine which records must remain available and for how long.

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Effect on planning	Audit decision	Example
Evidence handling	Plan authorized access, transfer and retention.	Review sensitive records where held if transfer would conflict with a restriction.
Outsourcing	Identify duties and available evidence on both sides.	Verify that the provider can produce the required transaction logs.

A policy may state the correct retention period while configuration deletes records earlier. Test **actual configuration and retrievability**. Consider the significance of a breach and the authority for any risk decision within applicable obligations.

1.3.3 Audit Risk and Materiality

Business risk concerns the organization's objectives. **Audit risk** concerns reaching an inappropriate audit conclusion, such as supporting reliance when a material problem exists.

Term	Meaning	Supplier payment example
Inherent risk (IR)	Susceptibility to a material problem before considering relevant controls.	A fraudulent bank account could redirect a payment.
Control risk (CR)	Risk that controls fail to prevent, or detect and correct in time, a material problem.	Independent verification can be bypassed.
Detection risk (DR)	Risk that audit procedures miss an existing material problem.	Testing approved requests misses direct administrator changes.
Residual risk	Business exposure remaining after the controls and responses considered.	Direct changes or collusion can leave fraud exposure.

The Audit Risk Model

In the simplified audit risk model, audit risk relates to **inherent, control and detection risk**.

$$AR = IR \times CR \times DR$$

Assume **80% inherent risk**, **50% control risk** and **5% target audit risk**. Under the stated model, solve for acceptable detection risk.

$$DR = \frac{AR}{IR \times CR} = \frac{0.05}{0.80 \times 0.50} = 0.125 = 12.5\%$$

Higher assessed inherent and control risk requires **lower acceptable detection risk** for the same target audit risk. In practice, qualitative assessments can guide the **nature, timing and extent** of work. Use actual system changes as the starting population, cover the period after the portal launch and include privileged change routes.

Sampling risk can arise when selected items miss a material problem. **Nonsampling risk** can arise from an incorrect procedure, incomplete extract or misinterpretation even if all extracted records are examined. Identify the cause before deciding how to improve the work. Section 1.6 develops these distinctions.

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Materiality

Materiality concerns the significance of a matter to intended users. Consider **magnitude and nature**. One small unauthorized payment may reveal a privileged route that could affect every supplier. Evaluate the potential reach, obligations, service consequences and whether the weakness is systemic. Use that judgment when planning work and evaluating findings.

1.3.4 Risk Assessment

Risk assessment identifies risks, analyzes their significance and compares them with agreed criteria. **Analysis** develops understanding of an exposure. **Evaluation** compares it with criteria for priority or treatment. Establish the definitions adopted in the engagement. [R32]

Step	Question	Supplier payment example
Identify	What could interfere with the objective?	An unauthorized account change could redirect a payment.
Analyze	How could it happen, how likely is it and how serious could it be?	Examine change routes, verification and the amount that could be paid before detection.
Evaluate	How does the exposure compare with agreed criteria?	Determine whether the remaining exposure warrants earlier coverage or action.

Risk appetite describes the broad amount and type of risk the organization is willing to take or retain. **Risk tolerance** sets an acceptable boundary for a specific objective or exposure. A low appetite for interruption of critical payments may lead to a defined limit on payment-service downtime.

Response	Meaning	Example
Avoid	Stop the activity that creates the exposure.	Retire an obsolete change channel.
Mitigate	Reduce likelihood or impact.	Independently confirm changes using an established supplier contact.
Share or transfer	Allocate defined consequences to another party.	Insure specified financial losses under agreed terms.
Accept	Make an authorized decision to retain exposure.	Document the rationale, approval and conditions for reassessment.

Management owns risk assessment and treatment. Audit challenges management's assessment using other evidence and sets independent assurance priorities. Escalate acceptance outside authorized limits under the applicable governance arrangements.

1.3.5 IS Audit Risk Assessment Techniques

Choose the technique to fit the **decision and evidence**. Walkthroughs help explain how work occurs. Incidents, test results, system changes and prior findings help assess actual exposure. Document the basis, uncertainty and any override.

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Technique	How it works	Appropriate use
Qualitative	Uses defined descriptions such as low, medium and high, with reasons.	Compare service risks when credible probability and loss estimates are unavailable.
Semi-quantitative	Uses ordered ratings and an agreed scoring rule.	Rank candidates using defined criteria and thresholds.
Quantitative	Uses estimates with meaningful units, such as probability, event frequency or money.	Estimate loss for a defined scenario when inputs support the model.

Likelihood and Impact as a Score

On defined 1 to 5 scales, a **likelihood rating of 4** and an **impact rating of 5** yield the score below.

$$\text{Risk score} = 4 \times 5 = 20$$

Interpret the score using the scoring guide. Two equal scores may have different consequences or mandatory coverage. A weighted model can give factors different influence. For business impact rated 5 at 60% and recent change rated 3 at 40%, the weighted rating is **0.60 × 5 + 0.40 × 3 = 4.2** on the defined scale. Explain the weights and supporting evidence.

Probability and Impact as Expected Loss

For a simplified scenario with one possible event over a stated period, **P** is the estimated event probability and **I** is the estimated loss if it occurs. In this example, **R** denotes modeled expected loss.

$$R = P \times I = 0.20 \times \$500{,}000 = \$100{,}000$$

With a 20% probability over one year and loss of $500,000 if the event occurs, the modeled expected loss is **$100,000**. The loss conditional on the event remains $500,000. Use actual probability here, and use an appropriate frequency model for recurring events. Section 2.5 develops single and annual loss expectancy.

1.3.6 Risk Analysis

Risk analysis connects the objective to a plausible event, its causes, likelihood, consequences and existing controls. A **threat** is a potential cause of harm. A **vulnerability** is a weakness through which harm can occur.

Element	Supplier bank-detail example
Objective	Pay the correct supplier, in the correct amount, to the authorized account.
Threat and event	A fraudster impersonates a supplier and requests a bank-account change.
Vulnerability	Staff can enter the change without independent verification.
Impact	Funds may be diverted, legitimate suppliers remain unpaid and operations can be disrupted.
Control	A second employee confirms the request using an established supplier contact before release.

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Element	Supplier bank-detail example
Remaining exposure	Administrator changes may bypass the workflow, or the callback may be omitted.
Audit response	Identify change routes and compare actual changes with verification and payment evidence.

Assess how controls **actually operate** before estimating remaining exposure. Update the analysis when testing reveals a route or dependency missing from the original assessment.

CISA Exam Perspective and Common Traps

If the question asks about	Focus on
The first step for an unfamiliar process	The **business objective and relevant exposure** before selecting tests.
High assessed inherent and control risk	**Lower acceptable detection risk** and more persuasive procedures.
A small monetary issue	Its **nature, potential reach and significance** to users.
A major change after approval of the plan	Reassess risk and use the **applicable approval process** for coverage changes.
Reliance on management's rating	Assess the **evidence and assumptions**, then test controls where reliance matters.

1.4 Types of Controls and Considerations

A **control** is a measure that helps achieve an objective or manage risk. Determine what it should achieve, how it works, who performs it and what evidence shows that it worked when needed.

1.4.1 Internal Controls

Internal controls are policies, procedures, practices and organizational arrangements that provide **reasonable assurance** about operations, reporting, protection of assets and compliance. Management designs, implements and monitors controls. A **control owner** is accountable for a control and for responding to failures. A different person may perform it.

Controls work together. Access rules limit who changes a record, approval verifies that a change is legitimate, and reconciliation identifies changes that bypass the expected process. Assess the combination against the risk.

1.4.2 Control Objectives and Control Measures

A **control objective** states the desired result. A **control measure** is the chosen safeguard. The **control practice** is how the measure is carried out.

Element	Supplier payment example
Risk	An unauthorized bank-detail change diverts a payment.
Control objective	Only verified details become available for payment.
Control measure	Require independent verification before release.

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Element	Supplier payment example
Control practice	A second employee confirms the details using an established contact and records approval.
Evidence	The change record shows the verifier, result, approval time and when the details became available.

IS Control Objectives

Control area	Desired result
Access and identity	Authorized users receive appropriate rights and privileged actions are accountable.
Development and change	Requirements are approved and changes are tested and authorized before use.
Operations and infrastructure	Systems, networks and databases support dependable, protected processing.
Information processing	Authorized transactions are processed accurately, completely and in the correct period.
Continuity and incidents	Disruptions are detected, contained and recovered from within business requirements.
Outsourcing	Provider and customer duties support agreed security, processing and service needs.

General Control Methods

Administrative controls establish direction and duties. **Technical controls** enforce or record activity in systems. **Physical controls** protect premises and assets. A data center may use an access policy, electronic badge rules and locked cages together.

How performed	Example	Evidence to examine
Manual	A supervisor confirms a supplier request.	Who checked it, what they checked and when.
Automated	The system blocks release until approval is recorded.	Configuration, behavior and rights to change or bypass the rule.
IT-dependent manual	A reviewer examines a system-generated exception report.	Report completeness and accuracy, the review performed and action on exceptions.

IS-Specific Controls

IT general controls (ITGCs) support the operating environment, including access administration, change control, operations and backup. **Application controls** address specific transaction and data processing.

Stage	Objective	Example
Input	Accept complete, valid and authorized data.	Required fields, supplier validity, duplicate checks and authorization.

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Stage	Objective	Example
Processing	Process the right records accurately and completely.	Calculation rules, run-to-run reconciliation and review of rejects.
Output	Deliver accurate results to authorized recipients.	Reconcile payment outputs and restrict payroll report distribution.

An application rejects duplicate invoices. If administrators can alter that rule without approval, trace whether relevant access or change weaknesses affect reliance on the rule. Test the **application control** and the **general controls** on which it depends.

Input Validation and Processing Checks

Input validation checks data before acceptance or processing. Match the check to the error. A supplier ID can exist in the reference file and still belong to the wrong supplier for the transaction.

Check	What it evaluates	Example
Validity or existence	Value belongs to an approved set.	Supplier ID exists in the master file.
Completeness	Required fields or records are present.	Bank account number is supplied.
Range or limit	Value is within bounds or below a threshold.	Month is 1 through 12, or payment is within approval limit.
Reasonableness	Value is plausible in its context.	Overtime is plausible given hours and authorization.
Sequence	Expected order or continuity is present.	Investigate a missing invoice number.
Duplicate	Relevant identifiers repeat.	Flag a repeated supplier and invoice number before payment.
Check digit	A calculated extra digit detects specified input errors.	A mistyped identifier fails its check.
Run-to-run totals	Control figures reconcile between stages.	Account for posted, rejected and pending records.
Exception report	Items outside rules are identified for review.	Investigate and resolve rejected transactions.

Worked check-digit example. In a simple teaching rule, add a digit so all digits sum to a multiple of 10. Identifier **273** sums to **12**, so add **8**. Both **2738** and **7238** sum to **20**. This illustrates a transposition the rule misses. Establish which errors the actual algorithm detects.

Batch controls compare groups of records. For three invoices of 125, 250 and 375 monetary units, the **financial total** is **750** and the **record count** is **3**. For identifiers 101, 203 and 407, the **hash total** is **711**. This hash total is an arithmetic check over a nonfinancial field. If two records are lost and two duplicates are added, counts can still match. Use reconciliations and transaction-level checks suited to the likely error.

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Business Process Applications and Controls

Application or service	Exposure	Control focus
E-commerce	Malicious inputs or unauthorized orders affect processing.	Validate input, restrict access and examine handling of untrusted data.
Electronic data interchange (EDI)	Structured documents exchanged between organizations are lost, duplicated or altered.	Authenticate trading partners, protect messages, check acknowledgments and reconcile transactions.
Email	Impersonation leads to disclosure or fraudulent action.	Assess protection, user reporting and independent verification of sensitive requests.
Industrial control systems (ICS)	Failures or unauthorized actions affect physical processes.	Assess safety, availability, separation and remote access. Supervisory control and data acquisition (SCADA) provides supervisory monitoring and control. A programmable logic controller (PLC) executes process logic.
AI and expert systems	Poor data, rules or model behavior produce unreliable decisions.	Assess validation, change control, oversight and exception handling.

1.4.3 Control Classifications

Classify the **function described in the scenario**. A system may block an action, record it for review or help restore a service. Each serves a different purpose.

Class	Purpose	Example
Preventive	Stop or reduce the chance of an unwanted event.	Block payment until authorized approval exists.
Detective	Identify an event or deviation so someone can respond.	Investigate differences between released payments and approvals.
Corrective	Remedy consequences or restore the required condition.	Restore corrupted data and verify the recovered records.
Deterrent	Discourage misconduct.	A visible monitoring notice discourages unauthorized entry.
Compensating	Address the objective through an alternative when the primary control is impractical.	An independent supervisor verifies supplier changes and controls release when full separation of duties is infeasible.

A compensating control may be preventive or detective. Evaluate **independence, coverage, timing and action**. A monthly review after payments leave has a different effect from a reviewer who can block release before funds leave.

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1.4.4 Control Relationship to Risk

Preventive measures reduce the likelihood of an event. Detection can shorten the time to respond, and correction can limit consequences. Evaluate the **remaining exposure**, cost and effect on operations.

A temporary unauthorized production change may have been reversed before the audit. Current source files can match the approved version even though a change occurred. Protected deployment logs are more relevant to the historical question. When a safeguard causes frequent workarounds, examine the actual process and exception handling.

1.4.5 Prescriptive Controls and Frameworks

A **control framework** organizes objectives or practices. A **prescriptive control set** specifies safeguards for a defined scope. Start with business risks and applicable obligations, then select criteria and evidence.

Reference	Audit use
COBIT	Connect enterprise objectives with governance and management objectives for information and technology.
CIS Critical Security Controls	Structure coverage of prioritized security safeguards.
OWASP Software Assurance Maturity Model (SAMM)	Assess and improve software security practices.
Payment Card Industry Data Security Standard (PCI DSS)	Determine requirements for the applicable payment-card environment.
Cloud Security Alliance Cloud Controls Matrix (CCM)	Assess cloud controls and provider and customer responsibilities.

A **SOC report** supplies scoped assurance evidence as described in §1.2. A **control catalog** lists control descriptions, and an approved **baseline** identifies a starting set for a system or risk category. **Tailoring** adapts that set to the environment and obligations. For a legacy exception, verify applicable requirements, approval, rationale, alternative control, owner, expiry or review date, and actual operation.

1.4.6 Evaluation of the Control Environment

The **control environment** includes leadership conduct, oversight, accountability and staff competence. Examine how managers act when controls fail or a senior person requests an override.

Level	Audit question	Access termination evidence
Design effectiveness	Would the control meet the objective if performed as specified?	The workflow covers relevant accounts, defines timing and assigns responsibility for failures.
Implementation	Has the design been put into use?	A walkthrough and configuration show that the production workflow can remove an account.
Operating effectiveness	Did it work as intended over the relevant period?	Compare effective termination times with disablement and exception records.

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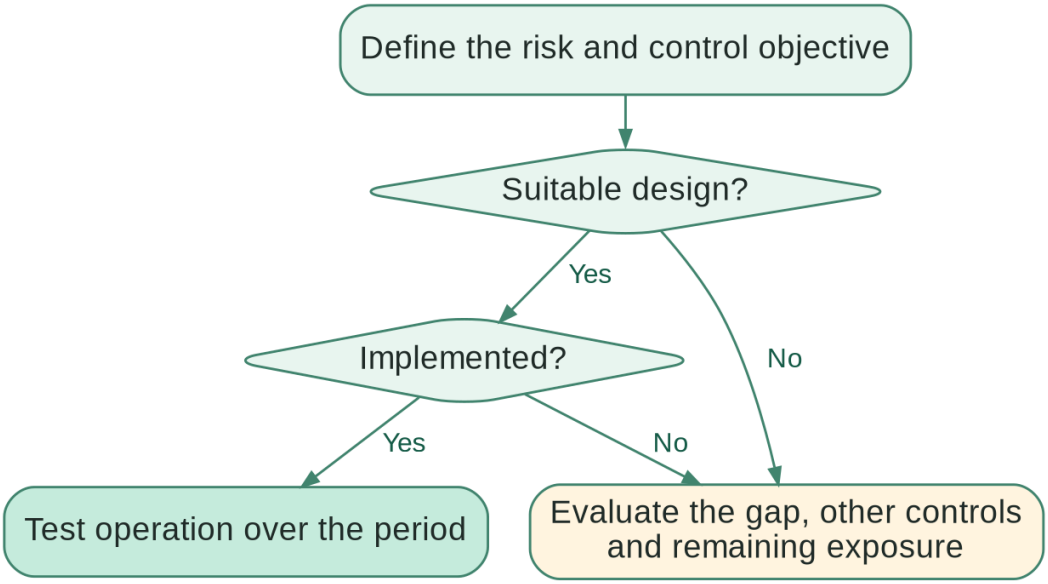


Diagram 1.4 Evaluate design, implementation and operation before concluding on the control.

Worked example. Policy requires access to end at the employee's **effective termination time**. Authorized HR records trigger removal, and support resolves failures. Check coverage of centrally managed and local accounts. Confirm the workflow is in use, then select from the complete **HR termination population** and compare actual disablement times.

An account stayed active for **two days** because a failed removal was left in the queue. If the design required review and timely correction, this is an **operating exception**. Examine privileges, use and other failures to evaluate significance. If local accounts were excluded without an alternative, assess the **design gap** separately.

Management Control Monitoring

Management monitors performance, investigates recurring failures and tracks corrective action. Check the completeness of the underlying population, the systems included, thresholds and evidence of follow-up. A dashboard reflects the scope and logic of its measurements.

Independent Evaluation of the Control Environment

Audit corroborates management's monitoring with suitable records, observation or reperformance. Examine overrides, overdue actions and exceptions closed without evidence. Error, collusion and management override can limit the assurance controls provide.

CISA Exam Perspective and Common Traps

If the scenario shows	Focus on
A procedure exists and period assurance is needed	Evidence of **actual operation** across the relevant period.
A system or change route is excluded	A possible **design gap** and any alternate controls.
A signed system-generated report	**Reliability of the report** and the review actually performed.
Administrators can alter automated rules	The **access and change dependencies** supporting reliance.

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If the scenario shows	Focus on
A proposed alternative to segregation of duties	Equivalent objective coverage, independence, timing and action.
Matching totals and suspected item errors	A **transaction-level test** suited to the error.

1.5 Audit Project Management

Audit project management organizes the people, time, procedures and evidence needed to answer an audit question. A bank's approved change tickets show which changes entered its approval process. To find changes that bypassed that process, the auditor also needs records of what actually reached production.

1.5.1 Audit Objectives

An **audit objective** states what the engagement must determine. **Scope** sets the boundaries. **Criteria** state what should have happened.

Element	Question	Production change example
Objective	What must the audit determine?	Whether production changes were authorized and tested as required.
Scope	Which activities, systems and periods are covered?	Application deployments and direct database changes to the payment platform from April through June.
Criteria	What requirements apply?	Normal changes need approval and testing before release. Emergency changes follow their approved procedure.
Audit program	What work obtains evidence?	Compare actual releases with approval, testing and version records.
Resources and schedule	Who performs and reviews the work, with what access and by when?	Assign someone who can interpret deployment records and arrange authorized access and review.

Discuss contacts, timing and access with management. If important evidence is unavailable, consider alternate sources and raise the issue with audit leadership. Assess how any remaining **scope limitation** affects the conclusion.

1.5.2 Audit Phases

Planning, **fieldwork and documentation**, and **reporting** connect the audit objective to the conclusion. **Follow-up** evaluates management's response. Communication, documentation and supervision continue throughout.

Planning Phase

Understand the process, risks and controls. Establish objectives, scope, criteria, evidence needs, skills and schedule. Audit leadership approves the program under the function's procedures. Use the **opening meeting** to confirm arrangements and identify access or scheduling issues early.

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Fieldwork and Documentation Phase

Perform procedures and validate the evidence. A **walkthrough** follows actual activity through a process, such as a request through testing, approval and deployment. Use it to confirm how controls operate and where activity can bypass them. Investigate exceptions and review work while evidence can still be obtained. Check time zones and emergency-change criteria before treating an apparent late approval as a confirmed deviation.

Reporting Phase

Validate the facts with responsible people, obtain management's response and complete audit review before issuing the supported conclusion. A **closing meeting** can resolve factual differences and identify unresolved disagreements. Communicate significant issues promptly if waiting would delay action. **Follow-up** verifies agreed actions and evaluates whether they address the risk.

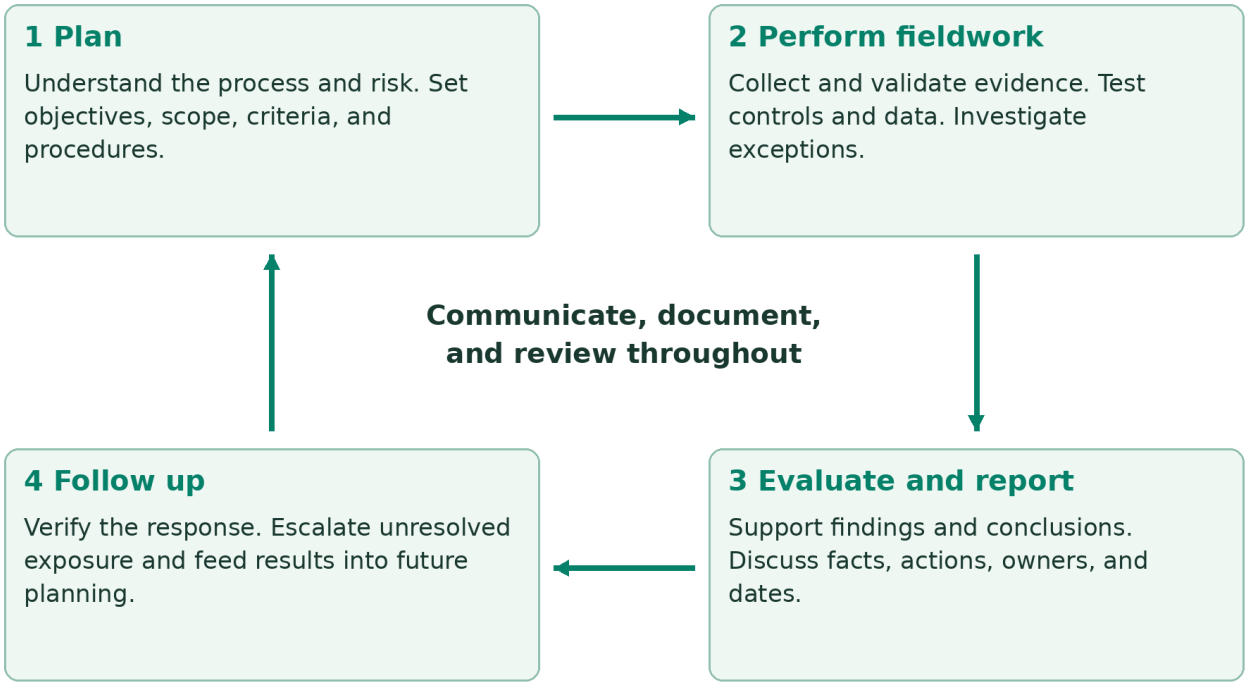


Figure 1.2 The audit cycle repeats as findings and follow-up inform future risk assessment. Communication, documentation, and review continue throughout.

Managing Progress and Scope Changes

Track completed procedures, evidence gaps, review points and time remaining. If 80 hours were planned, 60 were spent and 30 remain, the forecast is **90 hours**. Investigate the difference and raise significant resource or schedule changes with audit leadership.

Scope creep is unmanaged expansion beyond approved boundaries. If direct database changes are within scope but the program misses them, revise the procedure through the appropriate review. If the same account changes an unrelated system outside scope, assess significance and the link to this audit. Obtain a decision on expanded or separate work and record the effect on coverage, time and resources.

1.5.3 Audit Programs

An **audit program** translates objectives into procedures another competent auditor can perform. A **process narrative** describes work in words. A **flowchart** shows steps, decisions and handoffs. Confirm the documented process against actual practice before designing tests.

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Step	Procedure for the payment platform	Purpose
Establish the population	Obtain actual deployment and direct database change records for April through June.	Include activity outside the ticket system.
Validate the source	Check system coverage, filters, timestamps and identifiers.	Establish that the source can support the conclusion.
Select items	Apply the planned selection method and cover relevant change routes and categories.	Address significant risks, including emergency changes.
Test against criteria	Match actual releases and versions to approvals and test evidence. Apply the applicable normal or emergency procedure.	Assess the real change and timing.
Evaluate exceptions	Confirm facts, obtain supporting evidence, investigate causes and assess significance.	Distinguish a data issue, valid exception and control failure.

Send an **advance evidence request** specifying source, period, fields, responsible contacts, dates and a secure collection method. Include deployment time, system, account, version and ticket reference where available. Agree how to obtain evidence of changes with no ticket.

Minimum Skills to Develop an Audit Program

The team needs knowledge of the **business**, the **technology and controls**, and suitable **audit methods**. Arrange specialist support and review when the team needs expertise to interpret records or evaluate a control.

1.5.4 Audit Work Papers

Audit work papers document what the auditor did and how the evidence supports the conclusion. A deployment log is evidence. The work paper records its source, the items selected, comparisons performed and interpretation.

Field	Example entry
Objective and criteria	Determine whether normal releases had authorized approval before deployment.
Population and selection	April-through-June production changes. Record validated sources, filters and how 30 items were chosen.
Procedure and evidence	Match releases to approvals. Record source references and time zone adjustments.
Results and exceptions	Two normal releases had approvals recorded after deployment. Timestamp checks and supporting records established no valid prior approval for those releases.
Conclusion and next action	The control failed for two items tested. Assess significance and whether the cause warrants further work before concluding on wider operation.

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Record preparer, reviewer, dates, source references and how conflicting evidence was resolved. Retain query versions, filters and parameters needed to reproduce analysis. Protect access and retain records under applicable requirements.

1.5.5 Fraud, Irregularities and Illegal Acts

Fraud involves intentional deception. **Error** can occur without intent. An **irregularity** is a departure that may warrant investigation. An **illegal act** breaches applicable law. A **red flag** is an indicator requiring assessment.

Management override occurs when authority is used to bypass a control. **Collusion** occurs when people cooperate to defeat a control. Apparently independent approvals may be entered together after a release.

Several releases are labelled emergencies and were deployed by the same privileged account before approval. Verify the records and criteria, preserve evidence using authorized procedures and raise the matter through the applicable audit or investigation channel. Use an alternative authorized route if the normal contact may be involved. Management owns misconduct controls. Audit evaluates them, remains alert to indicators and communicates supported observations. Involve qualified investigative or legal specialists where appropriate.

1.5.6 Agile Auditing

Agile Auditing Overview

Agile auditing organizes work into short cycles called **sprints**. A **backlog** is a prioritized list of work. The team selects work, performs and reviews it, then adjusts priorities based on evidence. One auditor can validate deployment data while another completes the emergency-change walkthrough.

Practice	How it helps
Backlog prioritization	Keeps key risks and evidence needs visible.
Brief check-ins	Expose blockers such as missing database logs.
Sprint review	Reviews completed work and discusses supported observations and evidence gaps.
Retrospective	Improves how the team works in the next cycle.

Benefits of Agile Auditing

Short cycles can surface access issues earlier and give stakeholders timely feedback. The team can reprioritize when risks become clearer.

Agile Auditing Compared to Established Assurance Standards

Agile changes how the work is organized. Its **definition of done** includes evidence, documentation, review and a supported result. Apply the applicable approval process to significant scope or program changes.

CISA Exam Perspective and Common Traps

Situation	Audit decision
Identify unauthorized deployments	Start with **actual changes**, then trace to authorization.
Evidence unavailable	Seek suitable alternatives and assess the effect on the conclusion.

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Situation	Audit decision
Significant issue outside scope	Assess its link and significance, then obtain a scope or separate-coverage decision.
Management disputes a finding	Validate facts and document unresolved disagreement.
Possible fraud indicator	Preserve evidence, assess facts and use authorized escalation.
Sprint task marked complete	Confirm evidence, documentation and review support the result.

1.6 Audit Testing and Sampling Methodology

Audit testing turns an objective into evidence. Inspecting an invoice approval tests whether a **control operated**. Recalculating the amount tests whether the **information is correct**. The same invoice can serve both purposes when each procedure is designed and evaluated separately.

1.6.1 Compliance Versus Substantive Testing

A **compliance test**, or **test of controls**, evaluates whether a required control operated as intended. A **substantive test** directly examines transactions, balances or other information for validity, completeness or accuracy.

Test	Audit question	Invoice example
Compliance	Did the required control operate?	Inspect whether an authorized person approved the invoice before payment.
Substantive	Is the underlying transaction or amount correct?	Match the invoice to the order and receipt, then recalculate the amount.

A correct invoice amount and missing approval can occur together. Record the substantive result and the control deviation separately. Control evidence can affect planned reliance and substantive work. Where the objective is control effectiveness, obtain evidence of the control's operation.

Types of Substantive Procedures

Procedure	Purpose	Example
Test of details	Examine individual records or amounts.	Recalculate interest on selected loans using authorized rates and balances.
Substantive analytical procedure	Compare a reliable expectation with the reported result and investigate significant differences.	Estimate total interest from loan balances and rates, then investigate material differences.

The expectation and inputs must be suitable for the objective. A plausible trend can flag an issue. A precise expectation based on reliable data can provide stronger evidence about the amount.

Direction of Testing

Choose a starting population that contains the exception being sought.

Audit question	Start with	Compare with
Were all receipts recorded?	Goods receipt records.	Inventory or accounting records for **completeness**.

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Audit question	Start with	Compare with
Are recorded purchases genuine?	Recorded purchases.	Orders, invoices and receipts for **validity or occurrence**.
Were production changes authorized?	Actual deployments and direct changes.	Approval records, including changes that bypassed the ticket system.

1.6.2 Sampling

Audit sampling applies a procedure to less than the whole population to draw a conclusion about it. The **population** is the entire defined group. The **sampling unit** is an individual item selected. The **sampling frame** is the source list used to select units, and the **sample** is the subset examined.

To test required monthly access reviews, establish which systems and months **should have been reviewed**. The completed review register helps locate evidence. The expected schedule helps identify reviews never performed. Validate coverage before selecting items.

Statistical and Nonstatistical Sampling

Approach	Meaning	Interpretation
Statistical sampling	Uses **probability-based selection** and **statistical evaluation**.	Supports quantified sampling risk under the design and assumptions.
Nonstatistical sampling	Uses professional judgment to design and evaluate the sample without statistically quantifying sampling risk. Judgmental selection is one approach.	A reasoned projection, where appropriate, depends on the design and evidence.

Selecting the ten largest payments is useful for testing those high-value items. Define additional work before drawing conclusions about the rest of the population.

Attribute and Variable Sampling

Attribute sampling measures presence or absence of a characteristic, often the rate of control deviations. **Variable sampling** estimates a measurable quantity such as value or misstatement.

ACVS pair	Common association	Question
AC	**Attribute and Compliance**.	Did the required approval exist?
VS	**Variable and Substantive**.	By how much is the balance misstated?

Use **ACVS** as a cue. The objective determines the test, and the characteristic or amount determines the sampling method.

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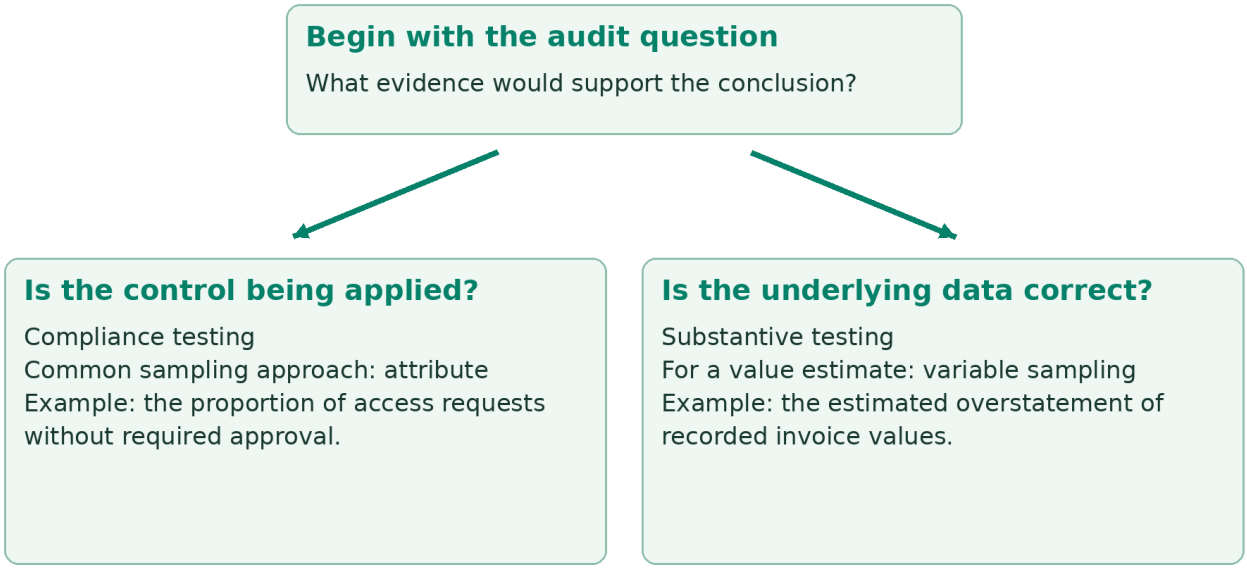


Figure 1.3 Start with the question. Attribute–compliance and variable–substantive are common associations, not mandatory pairings for every test.

Sampling Methods

Method	When to use it
Fixed-size attribute	Estimate a rate using a sample size set before testing.
Stop-or-go	Evaluate results in planned stages with defined rules for stopping or continuing. Useful when few deviations are expected.
Discovery	Seek a specified chance of detecting at least one occurrence at the design rate, under the method's assumptions. Use when even one occurrence matters.
Unstratified mean-per-unit	Estimate the total of one population using the sample mean.
Stratified mean-per-unit	Estimate totals within relevant groups and add them.
Difference estimation	Estimate total misstatement from recorded-to-audited differences.
Ratio projection	Apply the sample's error-to-recorded-value ratio to the population's recorded value when the design supports it.

For discovery sampling, interpret zero findings against the **design occurrence rate and detection probability**. Investigate a discovered event and assess whether the work needs to expand. Use predefined sequential rules for stop-or-go sampling.

Selecting Items

Selection method	How it works	Point to check
Simple random	Give every unit the same chance.	Validate the frame and document selection.
Systematic	Use a random start and an interval, such as every twentieth record.	Check for repeating patterns that could bias selection.

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Selection method	How it works	Point to check
Stratified	Group items by relevant characteristics and select within groups.	Combine results appropriately for the group sizes and methods.
Probability-proportional-to-size (PPS)	Give larger units more selection weight. Monetary-unit sampling can use positive recorded amounts.	Address zero, negative, understated or unrecorded items where relevant.
Targeted or judgmental	Choose specified items, such as unusual payments or privileged accounts.	Document the selection purpose and what the work supports.

A $10,000 item has **ten times the selection weight** of a $1,000 item under PPS using positive recorded amounts. Consider separate procedures for omitted items or exposures poorly represented by recorded values. [R40]

Key Sampling Terms

Set the parameters **before testing**. Document the objective, period, unit, what counts as a deviation, the selection method and the intended evaluation.

Term	Meaning and effect
Expected deviation rate	Anticipated control failure rate. Higher expected deviation generally increases the sample for planned reliance. Reconsider the approach when it approaches tolerance.
Tolerable deviation rate	Maximum control deviation compatible with intended reliance. Lower tolerance generally requires more evidence.
Tolerable misstatement	Monetary decision limit for the population being substantively tested.
Confidence level or confidence coefficient	The selected confidence for the statistical method. Higher confidence generally increases sample size, other factors equal.
Precision	Desired closeness of an estimate, often expressed as a margin or interval. Narrower precision generally needs more evidence.
Population variability	Dispersion of values. Greater variability commonly increases the effort needed for a value estimate.
Observed deviation rate	Deviations found divided by items tested.
Upper deviation limit	Upper bound produced by the statistical method at the selected confidence level, allowing for sampling uncertainty.

Under a design specifying **95% confidence**, the complement is **5%** for the stated inference under its assumptions. Testing 95 of 100 items gives a **95% sampling fraction**. Statistical confidence depends on the design and evaluation.

Mean, Standard Deviation and Monetary Projection

The **sample mean** is the average. The usual **sample standard deviation** measures dispersion around the sample mean. For sampled values x_i and sample size n:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{2 + 4 + 6}{3} = 4$$

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$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n - 1}} = \sqrt{\frac{8}{2}} = 2$$

For values **2, 4 and 6**, the mean is **4**. The squared differences are **4, 0 and 4**, so the sample standard deviation is **√(8 ÷ 2) = 2**. For the entire population, use population mean μ and divide by population size N.

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

Mean-per-unit example. An appropriate equal-probability sample has an audited mean of **$2,400** and the population contains **500 invoices**. The estimated total is **$2,400 × 500 = $1,200,000**. In a stratified design, estimate each stratum and add the totals.

Ratio projection example. A sample recorded at **$50,000** has an audited value of **$48,000**. Its overstatement is **$2,000**. If the method fits the design and the population is recorded at **$1,000,000**:

$$\text{Projected overstatement} = \frac{\$2{,}000}{\$50{,}000} \times \$1{,}000{,}000 = \$40{,}000$$

Compare the estimate and any allowance for sampling risk required by the method with tolerable misstatement. Consider the nature, cause and scope of errors. Use a projection only for the population it represents.

Sampling Risk

Sampling risk is the chance that a sample leads to a different conclusion from testing the full population with the same procedure. **Nonsampling risk** arises from other causes, such as a wrong procedure, incomplete extract or misinterpretation.

Test	Incorrect decision	Consequence
Controls	**Overreliance** on a control with too many deviations.	Insufficient further work.
Controls	**Underreliance** on a control that supports the planned reliance.	Unnecessary further work.
Substantive	**Incorrect acceptance** of a materially misstated population.	Inappropriate conclusion.
Substantive	**Incorrect rejection** of a population within acceptable limits.	Unnecessary investigation or testing.

A suitably designed larger sample can reduce sampling risk. Improve the source, procedure, supervision and review to address nonsampling risk. Even testing every extracted record depends on complete data and correct logic.

Worked Sampling Decision

A monthly access review has a **5% tolerable deviation rate**. The auditor examines **100** review instances and identifies **3 deviations**.

$$\text{Sample deviation rate} = \frac{3}{100} \times 100\% = 3\%$$

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The chosen statistical method produces an **upper deviation limit of 7.6%**. Compare that limit with **5% tolerance**. The result does not support planned reliance. If the exceptions followed a workflow change, investigate the affected period and assess whether revised reliance, further planned testing or more substantive work is appropriate.

CISA Exam Perspective and Common Traps

Situation	Focus on
Activity may bypass approval	Start with **actual activity**.
Rare condition where one occurrence matters	**Discovery sampling** under a defined design.
Few early exceptions	Apply planned **stop-or-go rules**.
Low observed rate	Compare the **upper limit with tolerance**, and investigate exceptions.
Wrong population or query	Correct the **nonsampling** cause.

1.7 Audit Evidence Collection Techniques

Audit evidence is information the auditor uses to support a finding or conclusion against criteria. Match it to the **claim**, **scope** and **period**. A successful backup job addresses job completion. Recoverability requires evidence that the required data can be restored and used, and a recovery time target requires reliable timing.

Sufficiency and Appropriateness

Sufficiency concerns how much evidence is needed. **Appropriateness** concerns its quality, including relevance and reliability. Consider risk, coverage and quality together.

Property	Question	Recovery example
Sufficiency	Is there enough evidence for the scope and conclusion?	Examine records for required systems and months, with further work where risk or gaps warrant it.
Relevance	Does it address the claim?	Validating a restored application addresses recoverability.
Reliability	Can the source, collection and accuracy be trusted?	Check timing definitions, log completeness and rights to modify records.

Evidence is **useful** when it answers the audit question and supports a decision. A folder of screenshots from the wrong environment contributes little to a conclusion about production.

Judging Reliability

Consider **independence, competence and objectivity** of the provider, and the **authenticity, completeness, accuracy and timing** of the evidence. Directly obtained evidence and independent corroboration can be persuasive when they match the objective and the method is sound.

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Source or form	Strength	What to check
Auditor-obtained evidence	Shows what the procedure tested.	Procedure suitability and input completeness.
Independent external evidence	Corroborates information from another source.	Respondent knowledge and independence, response authenticity, scope and date.
Internal records and reports	Can show operation over time.	Creation, processing, access and retention controls, including filters.
Oral evidence	Explains responsibilities, practice and possible causes.	Records, observation or tests that corroborate significant claims.
Copies and extracts	Preserve relevant evidence for analysis.	Traceability, completeness and integrity.

A validated restore addresses recoverability directly under the tested conditions. A job report supports an inference about an earlier step. The evidence's relevance depends on the specific claim.

Evidence Collection Techniques

Technique	Auditor action	Example and interpretation
Inquiry	Ask a knowledgeable person for information.	Ask why a job failed and corroborate important statements.
Inspection	Examine records, configurations, contracts or assets.	Inspect which jobs ran and when. Check source and coverage.
Observation	Watch an activity being performed.	Watch an authorized restore. Evidence relates to that observation.
Walkthrough	Follow actual activity through the process.	Trace a job from scheduling through failure handling to identify control points and implementation.
External confirmation	Obtain a response directly from a knowledgeable external party.	Ask a custodian to confirm copies held on a specified date. Verify respondent, channel and scope.
Recalculation	Independently check mathematical accuracy.	Recalculate elapsed recovery time.
Reperformance	Independently repeat a procedure or control.	Reconcile scheduled jobs to completed jobs and compare exceptions.
Analytical procedures	Examine relationships, trends and unusual differences.	Investigate an increase in failures after a configuration change.

Watching a reconciliation is **observation**. Checking its arithmetic is **recalculation**. Repeating the underlying check is **reperformance**. Control the delivery and receipt of external confirmations, and consider alternatives if the response is incomplete or unavailable.

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Reviewing IS Documents and Records

Review area	What to examine	What to verify
Organization structures	Organization charts, duties and approval authorities.	Actual responsibilities, reporting and rights.
Policies and procedures	Approved procedures and exception arrangements.	Version applicable in the audit period and actual practice.
Standards	Technical standards, configuration requirements and baselines.	Applicable systems, required safeguards and authorized deviations.
IS documentation	Architecture, configurations, test records, change history, contracts and recovery plans.	Scope, origin, approval history and connection to the actual version.

Electronic specifications, version histories and automated test outputs can be useful when their integrity and relevance are established.

Matching Evidence to the Claim

A critical application requires daily backups, monthly recovery tests and recovery within **four hours** during each test.

Claim	Evidence
Required backups completed	Reconcile the expected schedule with execution records. Investigate failures, retries and gaps.
A selected copy could be restored and used	Observe an authorized restore or inspect reliable test records, including validation of application use and data.
Recovery met the target	Verify the agreed start and finish events and recalculate elapsed time.
Monthly tests took place	Compare the expected schedule with dated records and follow-up on failures.

A test takes **four hours and twenty minutes** from the agreed start until the application is usable. It demonstrates recovery under those conditions and shows that the time target was exceeded. Evidence covering the relevant months is needed for a conclusion about monthly testing.

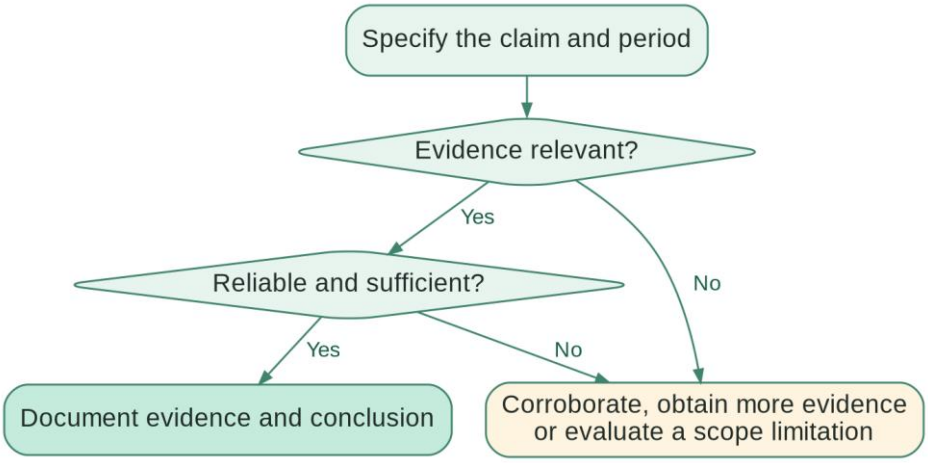


Diagram 1.5 Evidence must match the claim and support the conclusion for the stated period.

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Preserving Evidence and Chain of Custody

Record source system, extraction method and filters, audit period, collection date and time zone, record totals, collector and query version where needed. Collect short-lived records through authorized procedures. Protect copies from unauthorized access or change.

Chain of custody is the documented history of who collected, held, transferred and examined evidence. Use applicable preservation procedures when identity or integrity may be challenged, such as in a suspected fraud investigation.

- **Identify and collect.** Assign an identifier and record source, collector, date, method and condition.
- **Preserve and protect.** Secure the original or preserved copy. Analyze a controlled working copy where appropriate.
- **Record transfers.** Record who released and received the item, when, why and how integrity was checked.
- **Retain and reference.** Link the evidence to work papers and follow applicable retention and disposition requirements.

A **cryptographic hash** is a value calculated from digital content. Matching it with a trusted value recorded at collection helps verify that the content remained unchanged. Evidence of the original source's accuracy is a separate question.

1.7.1 Interviewing and Observing Personnel in Performance of Their Duties

Interviewing Personnel

Interview people who perform the work and those who supervise it. Ask open questions and request a demonstration or supporting record. For example, ask someone to walk through the last failed backup, explain how required systems were identified, and show who reviewed the result.

Record the person's role, date, facts and referenced evidence. Confirm important details. If a manager says all monthly tests occurred but a month is absent from the register, check whether the register is complete, whether other records support the test and whether the correct system was covered. Document what the evidence supports.

Observing Personnel in Performance of Their Duties

Observation shows **who performs the task**, the steps taken and the response to exceptions. Record when and where it occurred, people and systems covered, and whether it was a routine activity or demonstration. People can change their behavior while observed. Use records and other evidence for the relevant period.

If a person both performs and signs off a recovery test, compare actual duties and permissions with the approved independence requirement before deciding whether a control weakness exists.

CISA Exam Perspective and Common Traps

Situation	Focus on
Evidence of recoverability is required	A representative **restore and validation**, under conditions relevant to the objective.
A screenshot is offered for the whole year	Its **date and scope**, then historical evidence for the period.
Interview and records conflict	Investigate definitions, timing, scope and reliability. Corroborate.
Accuracy of a calculation	**Recalculation**. Repeat a control independently for **reperformance**.

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Situation	Focus on
Evidence identity or integrity may be challenged	Preserve and document **custody** under applicable procedures.
Evidence is incomplete	Obtain information that improves **relevance, reliability or coverage** for the claim.

1.8 Audit Data Analytics

Audit data analytics uses defined tests of data to answer an audit question. It can reveal unusual transactions, missing records or controls that may have failed. Define the objective, verify the data and test the analysis logic before interpreting the results.

Plan the analysis before choosing a tool

Choose a technique that suits the **population**, fields, criteria and follow-up the objective requires. Computer-assisted audit techniques, or **CAATs**, include audit software, scripts, test data and specialized testing utilities.

Analysis	Example
Matching and reconciliation	Compare HR departures with system accounts and investigate unmatched records.
Duplicate and gap detection	Find repeated invoice identifiers or missing sequence numbers; validate legitimate exceptions.
Recalculation	Recompute interest, tax, or payroll using approved rules.
Trend and outlier analysis	Find unusual changes, times, values, or activity; an outlier needs investigation.
Stratification and selection	Separate high value transactions or extract items meeting risk criteria.
Configuration and security analysis	Evaluate settings, logs, access rules, or authorized technical test results against criteria.

Validate data and the analysis

Reconcile extracted counts and relevant totals to independent source information. Check joins, filters, missing identifiers, duplicates, time periods, and field meanings. Retain the query or script and test its logic with known cases. A complete extract is useful only when the **procedure** tests the right question.

Example: A join on employee name misses accounts whose names have changed. Use a reliable identifier and investigate unmapped accounts before interpreting the result as a complete leaver test.

Whole **population** testing can remove **sampling risk** for the particular test and population. It does not remove incomplete data, faulty logic, or other nonsampling risks. Restrict access to sensitive extracts and avoid unnecessary collection.

The auditor should control or independently validate extraction **criteria**, selection, and interpretation. An auditee assisted extraction can be usable when the method and **completeness** are verified. Conversely, an auditor run extraction can be wrong. Preserve the query, parameters, totals, field definitions, and version of the script so that results can be reviewed.

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1.8.1 Computer-Assisted Audit Techniques

Generalized audit software and test techniques

Generalized audit software (GAS) supports selection, matching, recalculation, and reporting. Test data exercises the application with known valid and invalid transactions. Parallel simulation independently reprocesses real data and compares results. Obtain **authorization**, protect production, and ensure tests cannot distort actual business records.

CAATs include audit software, **database** utilities, scanners, debuggers, code tracing and mapping, **test data**, and expert systems. A database report generator retrieves information; a vulnerability scanner looks for specified weaknesses; a debugger or tracing tool helps examine execution; an audit expert system applies a knowledge base to guide assessment. GAS commonly supports reading files, sorting and joining records, filtering, sampling, statistical analysis, and recalculation.

Request read-only production access where practical. Perform transformations on controlled copies, obtain **approval** for installed tools, and protect sensitive extracts. Preserve field **definitions**, record layouts, query and script versions, instructions, source details, results, and review evidence. Link each analysis to its **audit objective**. Consider extraction effort, processing impact, software reliability, compatibility, maintainability, staff skills, and cost before choosing a tool.

Test data starts with inputs whose expected outcomes are known, including valid and invalid cases. The auditor submits them to the application and compares the actual results with the **expected results**. **Parallel simulation** starts with real business records and processes them through independently controlled audit logic. Comparing that result with the application output can reveal calculation or processing differences. In both cases, validate the logic and protect genuine business records.

A **test deck** is a prepared set of test transactions or inputs with known expected results. It should include relevant normal, boundary and invalid cases. Production data alone may lack the exceptions needed to evaluate a control.

Worked application test

Example: To test a credit **limit** rule, prepare cases below, at, and above the limit, including an invalid customer ID. Agree safe execution, **expected results**, evidence capture, and cleanup before running the cases. An above-limit transaction accepted without the required override is evidence about that rule; it does not prove every **application control** is ineffective.

Black box testing checks behavior against **expected results** without requiring knowledge of internal code. White box testing uses the implementation or code structure to select tests. Neither label guarantees sufficient coverage. Obtain the skills, permissions, and procedures needed for the objective.

Identify which system is being tested

Similar-sounding techniques differ in where processing occurs, who controls the logic, and whether the data are real. These distinctions matter because they determine both the **assurance** obtained and the **risk** of disturbing production.

Technique	Data and processing	What it can demonstrate
Generalized audit software	Audit tool analyzes selected or complete business records.	Patterns, reconciliations, calculations, and exceptions in available data.

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Technique	Data and processing	What it can demonstrate
Test data	Known test transactions run through the application under review.	Whether relevant application rules handle known cases correctly.
Parallel simulation	Independent audit logic reprocesses actual records.	Differences between expected processing and application results.
ITF	A dummy entity is included in controlled production processing.	How the live application processes the test cases, with effects kept out of genuine business results.
Embedded audit module	Embedded criteria collect selected processing information.	Recurring evidence about specified transactions or exceptions.
Snapshot	Selected processing information is captured along a transaction path.	What occurred at the captured stages.
Audit hook	A predefined condition triggers an alert or capture.	Timely identification of the specific condition being monitored.

Understand the supporting data and automation

The data infrastructure supports collection and analysis. A **database** organizes records for retrieval and processing. A **data warehouse** brings data together for analysis; a **data mart** focuses on a particular subject or business area. Data mining seeks patterns and relationships. An intelligent agent is software configured to observe conditions and take defined actions. These are supporting capabilities, not audit conclusions. XBRL tags reported business information in a structured form; the tags do not prove the underlying figures correct.

A **semantic network** represents concepts and their relationships as connected nodes and links. It is one way to structure knowledge, and differs from a decision tree or a diagram of data movement.

CAATs as a Continuous Online Audit Approach

Repeated automated procedures can collect evidence closer to the activity being audited. Identify the required records, audit trails and access early in system development so the necessary evidence will exist. Audit may advise on auditability while **management owns** implementation. Validate the tool, protect its **configuration** and reassess coverage when the system changes.

Tracing and tagging

Tracing and tagging is a CAAT technique in which selected transactions receive an identifier, or tag, so their processing path can be followed through the application. The auditor examines the relevant processing stages, decisions and outputs. The tag identifies what to follow; the trace records or reveals where it went and what processing occurred.

Example: Tag an authorized payment transaction and trace whether the required limit and **approval** checks execute before release. Confirm that the trace relates to the live version being assessed and that tagging does not bypass controls, expose **sensitive data** or disrupt production. This differs from an integrated test facility: **tracing and tagging** does not require a dummy entity or fictitious transaction.

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Source code comparison and production integrity

Source code comparison identifies differences between a source version under review and an approved source **baseline**. Compare like with like: source with approved source, and production executable or artifact with the approved executable or artifact. Source text cannot be directly compared with compiled binary bytes to establish equality. A trusted hash or signature can support artifact **integrity** only when the approved reference and verification process are trustworthy.

Investigate differences and their change approvals. To connect approved source to the deployed binary, examine **build provenance** and controlled inputs, toolchain, **dependencies** and release records; a reproducible-build comparison is meaningful only under the required controlled conditions. An unchanged executable does not prove that its **configuration**, data or business logic is appropriate. See section 3.3 for software supply-chain assurance. [R10, R13]

Object code is translated output from source code, commonly machine-level code that may still require linking to form an executable. Compare the appropriate approved object or executable artifacts and preserve source-to-build provenance. A filename alone does not establish artifact identity.

1.8.2 Continuous Auditing and Monitoring

Continuous auditing uses recurring or automated audit procedures to assess **risk** and controls more frequently. **Continuous monitoring** is management's ongoing **assessment** of its processes and controls. The two may use similar data, but their ownership and purpose differ. Management acts on its monitoring alerts; audit independently evaluates the evidence, coverage, and response without taking ownership of the control.

1.8.3 Continuous Auditing Techniques

Continuous auditing techniques

Select a continuous audit technique according to the evidence needed and its effect on production. For example, audit may independently compare a complete activity population with management's daily monitoring results to **test coverage** and response. **Continuous assurance** can draw on management monitoring and independent audit work when their responsibilities remain clear.

Continuous techniques capture evidence during or close to processing, which helps where a paper trail is limited. They need validated **source data**, carefully maintained rules, and control over operational impact. Automation can miss events outside its rules or produce **false** alarms.

An embedded audit module, or EAM, **is audit** functionality built into an application. It selects or records information according to defined **criteria**. A systems control audit review file, or SCARF, contains selected information captured for audit review. These concepts concern evidence capture; they do not mean that all application controls have been tested.

A snapshot records selected information at stages in a transaction's processing path. It helps reconstruct what happened. An **audit hook** triggers an **alert** or capture when a specified condition occurs, such as an unusual high-value payment. The hook's usefulness depends on the condition chosen.

An integrated test facility, or **ITF**, uses controlled dummy entities in the live application so test transactions can exercise production processing. Test effects must be isolated or reversed. Continuous and intermittent simulation, or CIS, selects transactions meeting defined conditions and simulates relevant processing for evaluation. These techniques differ in what is captured or tested and how they affect production.

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Technique	How it works
SCARF and embedded audit modules	Systems control audit review files and embedded modules capture selected transactions or exceptions using predefined criteria.
Snapshots	Record selected information along a transaction's processing path for later review.
Audit hooks	Triggers flag specified conditions and prompt review or response.
Integrated test facility or ITF	Dummy entities and controlled test transactions are processed within the live application. Isolate or reverse effects so real records remain correct.
Continuous and intermittent simulation	During processing, select transactions meeting criteria and simulate relevant processing for audit evaluation.

1.8.4 Artificial Intelligence in IS Audit

Distinguish algorithms AI ML and RPA

An **algorithm** is a specified method for solving a problem. A duplicate-invoice test can normalize **supplier** identifiers, compare supplier–invoice pairs, and flag matches. Changing the normalization rule may change the result. Audit should validate both the rule and its implementation against known cases.

Artificial intelligence, or AI, covers systems performing tasks associated with intelligent behavior. Machine learning, or ML, learns patterns from data. Robotic process automation, or **RPA**, executes configured workflow steps and need not use AI. Natural language processing, or NLP, analyzes or generates human language. None of these labels establishes audit reliability.

A classification model assigns an item to a category, such as a **policy** topic or a transaction-risk class. A **decision tree** uses branching tests, a **Bayesian classifier** uses probabilities, and a nearest-neighbor method compares the item with similar known examples. These are alternative ways of producing a result; the auditor still needs to validate whether the result is suitable for the audit question.

Technique	Audit use and limitation
Document classification	Assign policies or findings to categories. Validate labels and unusual cases; decision trees, Bayesian classifiers, and nearest-neighbor methods are possible approaches.
Text summarization	Condense long documents. Verify important conditions, exceptions, and numbers against the original.
Topic analysis	Find recurring themes across records. Review context before treating a theme as a supported finding.
Search and retrieval	Locate potentially relevant policies or earlier reports. Check currency, relevance, and completeness.
Statistical and predictive analysis	Explore trends or estimate expected outcomes. Understand assumptions, uncertainty, and whether the data represent the population.
Sentiment analysis	Estimate attitudes expressed in text. Sarcasm, language, and context can lead to misleading classifications.

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Technique	Audit use and limitation
RPA and rule engines	Automate repeatable collection, comparisons, and reporting steps. Validate the workflow and handle exceptions explicitly.

Evaluate alerts against **known outcomes** where possible and examine the cases that the model does not flag. A proprietary or complex model may require specialist help. Assess training and input data, unusual cases, **bias**, model changes, and changing performance over time. Human review must substantiate conclusions. A statistically detectable pattern is not automatically important to the **audit objective**.

Model or task	Meaning and distinction
Supervised learning	Learns from examples paired with known outcomes; classification and numerical regression are common tasks.
Unsupervised learning	Seeks structure in data without predefined outcome labels; a discovered grouping needs interpretation.
Classification versus regression	Classification predicts a category; regression estimates a numerical outcome. ML regression is different from regression testing after software changes.
Clustering	Groups similar observations without predefined outcome labels. It is different from server clustering for availability.
Naive Bayes	A probabilistic classifier using a simplifying conditional-independence assumption about features given the class. It is one Bayesian approach, not a synonym for all Bayesian methods.

AI algorithms and interpretation

For each proposed AI or machine-learning procedure, establish the task, expected output and evidence needed to evaluate it. A rules-based procedure can be checked against its explicit conditions; a learned model also needs evaluation against representative **known outcomes**. Document why the selected method is suitable for the **audit objective**.

A **false positive** flags a legitimate item as suspicious. A **false negative** misses an actual problem. Bias can make results systematically unsuitable for a particular group or situation. Model drift means performance changes as data or operating conditions change. Test these issues explicitly before using model outputs as evidence.

Concern	Audit response
Poor training or input data	Check provenance, completeness, accuracy, representativeness, and permitted use.
False positives and false negatives	Validate against known cases; investigate how many flagged items are legitimate and which problems are missed.
Bias or limited explainability	Evaluate relevant groups, understand decision limits, and retain a defensible explanation of the audit conclusion.

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Concern	Audit response
Drift or uncontrolled changes	Version data, model, thresholds, and scripts; monitor performance and approve changes.
Confidentiality and overreliance	Use approved tools, restrict sensitive data, verify outputs, and retain human accountability.

Example: A model flags payments made at night. Compare them with authorized **batch** schedules before alleging misconduct. Also test whether daytime fraud could be missed. A useful **alert** is the start of investigation, not its conclusion.

Fit problem	What it means	Audit implication
Overfitting	Learns training-specific detail or noise; performance deteriorates on new representative data.	Compare training, validation and independent test results.
Underfitting	Fails to capture important relationships and performs poorly even on training data.	Check model suitability, features and the learning setup.

Example: A transaction model identifies nearly every unusual item in its training file but misses many known cases in a separate period. Investigate overfitting, data leakage and changed conditions before treating its score as dependable audit evidence. [R37]

Interpretation of AI/ML Results

Statistical significance concerns whether a result is consistent with a statistical model and its **assumptions**; it does not establish business importance, causation or misconduct. Consider effect size, data coverage, uncertainty and representativeness. Compare model results with source records and suitable human-led tests before drawing a conclusion.

AI/ML Audit Risk and Considerations

Document the question, data, model or rules, version, test results and limitations. Proprietary logic can make explanation and **validation** harder, requiring specialist assistance or alternative procedures. Examine unusual cases and unflagged items as well as alerts. Maintain human **accountability** for the conclusion and protect confidential inputs and outputs.

Validating analytics and using AI responsibly

Data analytics can identify unusual transactions, missing approvals, duplicate payments and relationships across large populations. Establish **authorized access**, relevant fields, stable identifiers, time zones, data **definitions** and extraction completeness. Reconcile source and extract counts and totals where useful, but also inspect identities and significant fields; matching aggregates can conceal substitutions or offsetting errors.

Validate the **procedure** using known cases and independent checks. A join can multiply records when a supposed unique key is duplicated. A filter can discard nulls, late records or canceled transactions

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that matter to the objective. Preserve code, parameters and versions so the result can be reproduced. An exception is an investigative lead until its meaning is established.

Practical extension: Generative AI may summarize policies or suggest procedures, but can invent **requirements**, citations and explanations. Use approved tools and permitted data; verify source passages and calculations; protect confidential evidence; and retain human responsibility for the conclusion. For an AI system used by management, assess **data provenance**, model changes, **performance**, bias, drift and override arrangements. A fluent output or high aggregate accuracy is not sufficient assurance for every affected group or high-impact decision. [R11]

Failure	Audit implication	Useful validation
Incomplete extract	Missed records and false coverage	Reconcile source scope and boundary cases
Incorrect join or filter	False duplicates or missing exceptions	Test known cases and key uniqueness
Unverified generated output	Unsupported facts or conclusions	Verify against authoritative source evidence

Validate the Unit of Analysis

Example. An audit tests whether expense claims were paid twice. Joining each claim to every approver role increases 1,200 claim records to 1,500 rows. Establish the intended unit of analysis before counting duplicates. Check which keys are unique, examine unmatched and repeated keys, and reconcile claim identifiers and payment amounts to the original extracts. A one-to-many join can be legitimate for examining approval history and unsuitable for counting distinct claims. Preserve the original data and rerun the corrected logic against known cases before investigating duplicate payments.

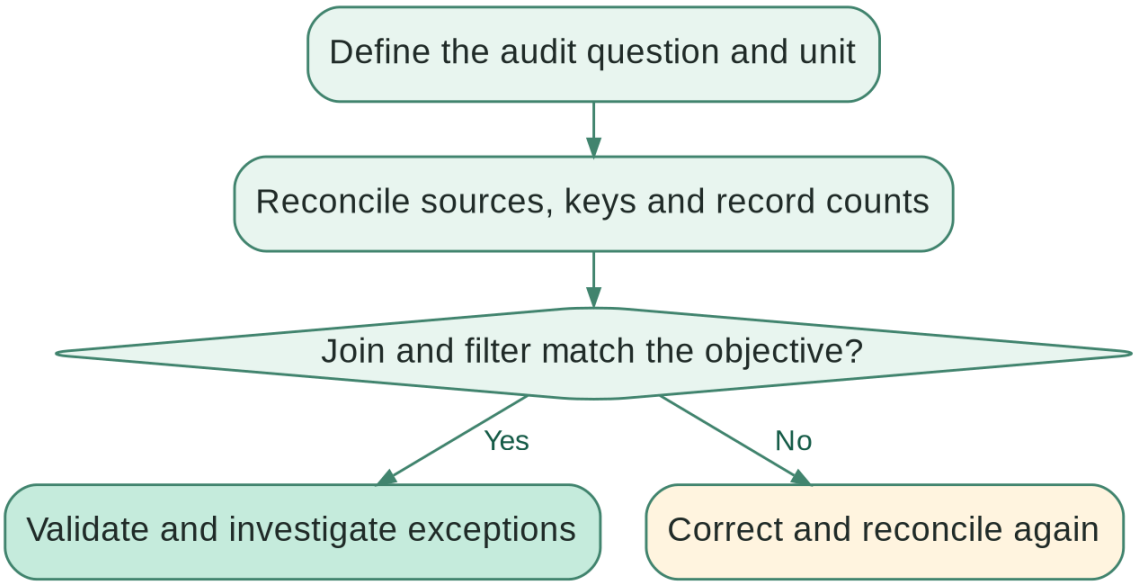


Diagram 1.6 Reconcile the source and test the analysis logic before interpreting exceptions.

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1.9 Reporting and Communication Techniques

Audit reporting communicates the supported results of the work to the people who need to act. A **finding** explains a significant difference between expected and observed conditions. Discussion with management can validate facts and clarify actions. Audit is responsible for its conclusion.

1.9.1 Communicating Audit Results

Separate investigation from timely communication

A suspected issue may need analysis before a defensible **finding** exists. An established serious condition may require prompt communication while further work continues. These are not competing rules: communicate what is known accurately, distinguish uncertainty, and continue the work needed to support the conclusion.

Situation	Appropriate approach
Suspicious access with unknown cause or extent	Corroborate the condition, assess exposure, and investigate before claiming a proven breach.
Confirmed serious control weakness	Communicate promptly through appropriate channels; establish extent, cause, and impact for the final conclusion.
Ongoing harmful activity	Use the defined escalation or incident procedures without waiting for the final audit report.
Management disputes verified facts	Review its evidence, correct inaccuracies, and document unresolved disagreement fairly.
Management accepts excessive residual risk	Confirm authority and rationale, document the decision, and escalate through audit leadership as needed.

Example: One access request appears to lack **approval**. Determine whether approval was recorded in a separate authorized workflow before alleging an unauthorized account. If a production administrator is confirmed to be changing and approving the same critical transactions, communicate the serious **segregation** issue promptly and assess the affected **activity**. Prompt communication does not eliminate the need for evidence.

1.9.2 Audit Report Objectives

An audit report communicates the supported results of an engagement to the people who need to act on them. It answers the **audit objective**, explains significant **exposure**, presents conclusions and recommendations, and records agreed actions. It also provides a reference for oversight and **follow-up**. Its usefulness depends on accurate evidence, a clear scope, and a conclusion the intended reader can understand.

1.9.3 Audit Report Structure and Contents

Finding elements and a worked example

A **finding** explains the gap between what is happening and what is required, why it matters, and what should change. The example below connects the **evidence** about former employees' access to a practical corrective response.

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The condition is what the **evidence** shows. **Criteria** describe what should happen. The **cause** explains the reason for the difference, if established. The effect or risk explains its actual or potential consequence. The recommendation addresses the weakness and its cause. Management then records the action it will take, who owns it, and the target date. Keep a suspected cause separate from a cause supported by evidence.

Element	Leaver access example
Condition	Three sampled payroll accounts remained active beyond the approved removal deadline.
Criteria	The access policy requires removal within the defined period after termination.
Cause	HR notifications did not reach the access team through a monitored workflow.
Effect or risk	Former staff could retain access to payroll information or functions.
Recommendation	Implement a monitored HR to IT notification process, reconcile leavers, and verify closure.
Management action	Record the responsible owner, agreed action, completion date, and any interim measure.

Verify **facts** and investigate the significance of **exceptions**. Discuss findings with responsible management, record disagreements fairly, and retain audit's independent conclusion. Communicate urgent significant issues through the appropriate channel without waiting for the final report.

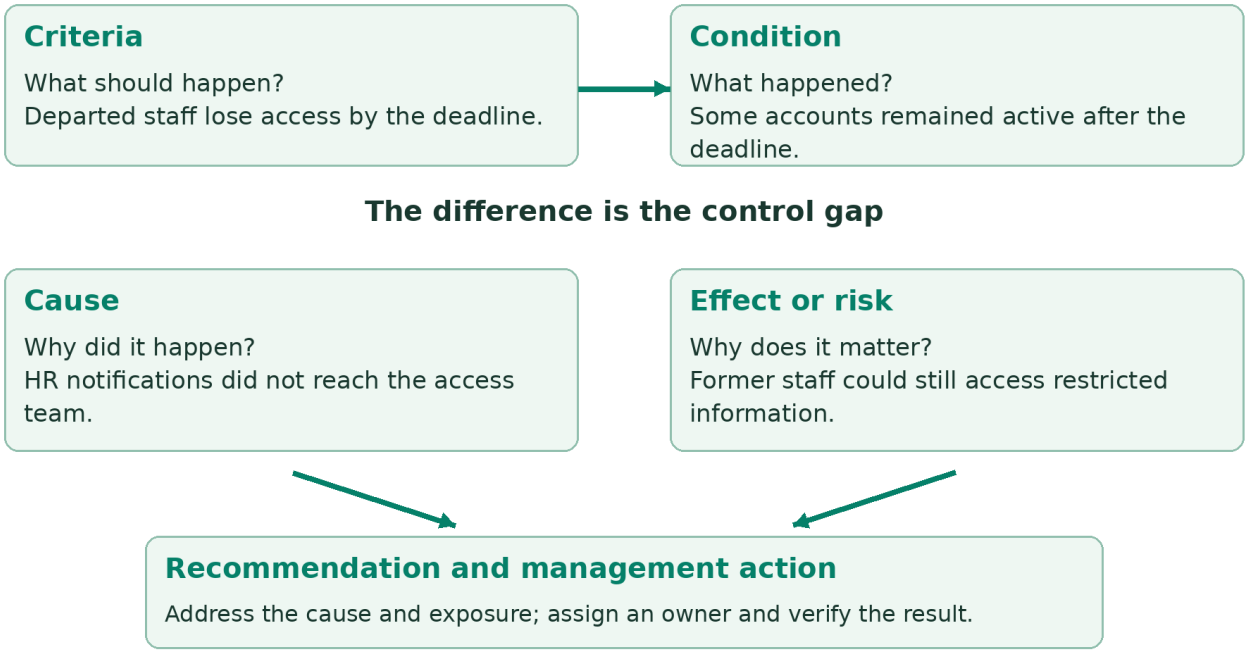


Figure 1.4 Compare the observed condition with criteria, then connect cause, consequence, and corrective action.

Write findings that lead to a clear response

Use condition, **criteria**, **cause**, effect, and recommendation to connect the evidence to the action needed. Avoid claiming a cause before investigating it. If the reason remains unknown, say so and recommend appropriate analysis rather than inventing a convenient explanation.

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The closing discussion checks **facts** and obtains responses, owners, and realistic dates. It may identify alternatives that address the same **risk** more effectively. Management's agreement is useful but does not determine whether a supported material finding is reported.

Describe severity within its framework

A control deficiency is a weakness in **control design** or operation. Significant deficiency and **material weakness** can have specific definitions in particular reporting frameworks. Evaluate severity using the applicable criteria, business effect, likelihood, scope, and compensating controls. Neither the number of exceptions nor a dramatic technical label determines severity by itself.

A final communication provides the overall conclusion and relevant limitations. Detailed security findings may require a restricted technical appendix. Retain the supporting work securely and keep an action record for **follow-up**. Closing a **finding** requires evidence proportionate to the risk, rather than merely a management promise.

Write for the report user and purpose

An audit report formally presents results, closes the engagement, provides the appropriate assurance and corrective recommendations, preserves a useful reference, supports **follow-up**, and demonstrates credible professional work. State the objective and answer it directly. Include **scope** and period, **criteria**, relevant methods, limitations, significant findings, overall conclusion, management actions, owners, and dates. A balanced report recognizes effective controls as well as weaknesses.

Use facilitation and negotiation to agree the **facts** and a workable response. At the exit meeting, confirm **accuracy**, realistic recommendations, and target dates. Explain unresolved differences and the **consequences** of inaction. Management's need to coordinate changes or obtain budget may affect timing; it does not remove a significant finding. The auditor decides the supported content within audit authority.

An interim communication can **alert** management to a serious issue before the final report. A restricted technical appendix can describe sensitive vulnerabilities for authorized recipients. Minor local matters may be communicated separately when appropriate, while the final report retains all significant conclusions and limitations. Identify report users and distribution **requirements** during planning.

Reporting depends on the engagement. An examination or audit may express an opinion; a review provides its specified lower assurance; **agreed upon procedures** report findings without an opinion. A separate technical report can supplement an executive report, but the accounts of material **facts** must remain consistent.

1.9.4 Audit Documentation

Protect retain and release the audit record

Maintain the **scope** and plan, process understanding, program, procedures, evidence, specialist contributions, findings, conclusions, report, and supervisory review. Use dates, authorship, identifiers, cross-references, and appropriate version control. Restrict access and transfer securely. **Retention** and destruction follow applicable law, contracts, professional requirements, and approved policy; suspend normal destruction when preservation is required.

Ownership and external access depend on the engagement and applicable arrangements. Do not assume an external requester is entitled to **working papers**. Obtain the required **authorization** and legal advice before release. **Follow-up** timing and evidence depend on the finding's risk and the agreed

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engagement: a low-risk status inquiry may suffice in one case, while a high-risk control failure calls for testing implementation and operation.

1.9.5 Follow-Up Activities

Follow up and accepted risk

Remediation is management's action to correct the weakness or address the **exposure**. **Follow-up** is audit's assessment of that response after reporting. **Management owns** the action; audit evaluates evidence of completion and effectiveness. The extent of follow-up depends on significance: a high-risk failure may need configuration inspection and retesting rather than an action owner's statement alone.

If management accepts **risk**, document the decision and authority. Where audit judges the accepted **exposure** excessive, escalate through audit leadership to the appropriate governing body. Do not suppress a supported finding or take over management's decision.

Memory aid: **Finding** logic: what is happening, what should happen, why the gap exists, why it matters, and what action addresses it.

1.9.6 Types of IS Audit Reports

An **IS audit** report may communicate results to internal management and the governing body, or provide independent assurance to an external user under an agreed reporting framework. Its purpose and terms determine its audience, **scope**, form, and distribution restrictions. Do not assume all reports provide the same assurance. For third-party service reports, distinguish the subject covered by **SOC 1**, SOC 2, and SOC 3 from the Type 1 or Type 2 time dimension explained in section 1.2.

Match the report to the engagement

The objective, work performed and agreed reporting framework determine the report and **assurance** expressed. A general management report may need a restricted technical appendix for sensitive findings. Different audiences can receive different detail, but the supported conclusions must remain consistent.

Engagement	Reporting distinction
Examination	A reasonable-assurance conclusion or opinion under the applicable framework, supported by sufficient appropriate work.
Review	A limited-assurance conclusion under the applicable framework; do not read it as equivalent to an examination.
Agreed-upon procedures	The procedures and factual findings are reported; no assurance opinion is expressed.
Internal IS audit	Reports the agreed scope, significant findings, conclusion and management actions under the audit function's requirements.

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Auditor Perspective

Confirm facts and communicate appropriately

For an ordinary apparent exception, establish the condition, **criteria** and significance and obtain relevant factual clarification. Do not confuse that step with asking management to approve the **audit conclusion**.

Suspected misconduct, interference or evidence destruction may require prompt escalation through the audit mandate. **Context** determines the next action.

Reporting supported findings and following through

A useful finding connects the condition with appropriate **criteria**, the supported **cause**, the business effect or **exposure** and a practical recommendation. Distinguish an observed failure from a possible consequence. Explain the population, period and limitations so readers understand how far the conclusion extends. Quantification should follow the evidence rather than a desire for a dramatic headline.

Discuss **facts** with the responsible people and consider contrary **evidence** fairly. Management disagreement does not require removing a supported finding. Record the response, action owner, target date and any authorized risk decision. Audit recommends and evaluates; management implements controls. Escalate significant unresolved **exposure** through the mandate when the accountable owner's authority or the enterprise's approved tolerance is exceeded.

Example: Four accounts lacked the **policy**-required written **approval**. An unsupported claim that managers approved them verbally does not establish compliance with that criterion. Seek corroboration, retain the supported condition and report the response. Prompt communication is appropriate for urgent risk or interference; the timing must fit severity and the reporting arrangements. Follow-up examines whether the **corrective action** addressed the risk in operation, rather than closing an item merely because a revised procedure was signed.

From a Recovery Test to a Supported Finding

Section 1.7's recovery test took **four hours and twenty minutes** against a four-hour target. The reporting decision depends on what the test covered and why the delay occurred.

Finding element	Supported wording or evidence
Criteria	The approved target is four hours from the agreed start until the application is usable.
Condition	The selected test exceeded the target by 20 minutes. Identify the application, test conditions and date.
Cause	Records show that the required recovery credential was unavailable to the assigned team for 25 minutes. Verify the cause with the team.
Effect	Recovery could exceed the approved interruption limit if the same dependency recurs.
Response and follow-up	Management agrees an owner and date for controlled alternate access. Audit verifies its design and the result of a representative retest.

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Report the supported scope. A late result from one exercise supports a finding about that exercise and its tested dependency. A conclusion about all services requires wider evidence. If the owner disputes the start time, apply the agreed measurement rule and confirm the source records before finalizing.

1.10 Quality Assurance and Improvement of the Audit Process

Audit quality is the degree to which the work follows applicable requirements and supports useful, defensible conclusions. **Quality assurance** evaluates whether the function applies its methodology effectively. **Quality improvement** addresses recurring weaknesses and checks whether changes help.

1.10.1 Audit Committee Oversight

The **audit committee** provides oversight of the audit function on behalf of the governing body. It considers the function's mandate, **independence**, risk coverage, resources, significant results, and quality arrangements. It challenges unresolved issues and receives escalation when management's actions or accepted risk warrant attention. Oversight supports **audit independence**; audit leadership remains responsible for managing and reviewing the work.

1.10.2 Audit Quality Assurance

Quality assurance oversight and review

Audit quality assurance evaluates whether the audit function and its work meet applicable professional and organizational **requirements**. Improvement uses the results to make the work better. This matters because an audit can miss its own objective through poor planning, weak evidence, or inadequate review even when its report looks professional. The governing body oversees the function; audit leadership owns the methods, supervision, review, and quality of its work.

Quality activity	What good evidence looks like
Engagement supervision	Approved objectives and program; timely review of test work, exceptions, and conclusions.
Working paper review	A reviewer can trace the report to sufficient relevant evidence and understand selection decisions and limitations.
Independence monitoring	Conflicts are declared, assessed, addressed, and reconsidered when circumstances change.
Training and development	Skills assessed against the audit plan; appropriate development and continuing education tracked.
Methodology monitoring	Recurring review errors lead to updated procedures, templates, training, or tools.
Function performance	Balanced measures of coverage, timeliness, quality, stakeholder usefulness, and follow up; speed alone is insufficient.

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Internal review external assessment and peer review

Internal quality activities include supervision, working paper review, self **assessment**, and **monitoring** of the function's performance. An external quality assessment provides an independent perspective on the function's conformance and effectiveness. A peer review uses qualified people outside the work under review; its scope and independence determine what assurance it provides.

The governing body oversees the quality arrangements. Audit leadership ensures that methods, resources, and development address the results. Internal **monitoring** can identify an independence problem, but it is not a substitute for a required independent external **assessment**.

Improvement in practice

Example: Several audits contain unvalidated spreadsheet populations. Audit leadership makes **reconciliation** a required planning step, trains staff, and checks later files for compliance. Merely recording the recurring weakness would not demonstrate improvement.

Quality **assurance** provides confidence that the audit work meets **requirements**. Improvement uses the results to change how work is performed. Track actions, responsibilities, and completion, then check whether the change corrected the recurring problem.

1.10.3 Audit Team Training and Development

Monitor competence and professional obligations

Audit leadership should maintain a documented method, a risk-appropriate review process, and a funded development plan for each role. Track training, continuing professional education, and applicable certifications or accreditations, assigning responsibility for deadlines. Reassess skills when technologies and the **audit plan** change. Monitor **independence** declarations throughout the year, not only on joining the team.

Summarize recurring review findings, **stakeholder** feedback, and quality trends for oversight. Record improvement actions, owners, target dates, and **evidence** that the changes worked. Completing a course, issuing reports faster, or closing review notes does not by itself demonstrate better assurance.

1.10.4 Monitoring

Measure useful performance

Compare delivery with the charter, **risk**-based plan, and quality objectives. Measures can include significant risk coverage, timeliness, **evidence** quality, stakeholder usefulness, overdue actions, and recurring review findings. No single count proves effectiveness. Issuing many reports is not a success if the reports avoid important risk or rest on weak evidence.

Example: A function meets every report deadline but repeatedly misses cloud services from its **audit universe**. A useful improvement updates the universe, reconciles it to procurement and asset information, reassesses coverage, and checks that future plans include the **exposure**. Faster report production would not address the cause.

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Domain case exercise

A payroll audit finds that management reconciles active accounts monthly, but the **reconciliation** omits a recently acquired subsidiary. The team's initial data extract also omitted it. The audit report is due tomorrow.

Assess the effect on **scope** and population, obtain and validate the missing data, and revise testing and timing as needed. Communicate the limitation and significance promptly. Do not issue an unqualified broad conclusion merely to meet the original deadline. Separately assess the incomplete management reconciliation as a **control design** or coverage weakness.

Exam focus A cleanly formatted report cannot compensate for incomplete **evidence**. Quality begins in planning and continues through testing, reporting, and follow up.

Bring the domain together

A dependable audit has a mandate, **risk** focused objectives, suitable procedures, persuasive evidence, independent evaluation, and clear communication. The sections in this chapter are connected: a flawed **population** weakens the tests, weak tests undermine the finding, and an unsupported finding cannot be repaired by confident report writing.

Assessing the quality of assurance itself

Audit quality includes competent staff, suitable methodology, independent judgment, supervision, sufficient appropriate **evidence** and clear supported reporting. Assess both individual engagements and the wider function. A report issued on schedule is an output; whether its conclusions are supported and its recommendations address important risk concerns quality and effectiveness.

Internal **monitoring** can include supervisory review, periodic self-**assessment**, stakeholder feedback and analysis of recurring work-paper or reporting defects. An external assessment brings independent evaluation under the applicable professional framework. Determine the requirements and intervals from the framework actually adopted; do not transfer an interval from another framework without establishing applicability.

Example: Reviews repeatedly identify unsupported **sample** projections. The response should improve competence, methodology and supervision and then verify improvement in later engagements. Adding another sign-off box without changing the work may not address the **cause**. Quality indicators can include rework, unresolved evidence limitations, coverage of significant risks and timely **follow-up**. Interpret them together: fewer findings can mean better controls, weak coverage or reluctance to challenge. The quality program must be able to identify deficiencies in audit's own work. [R8]

Check Whether the Quality Improvement Worked

Example. A supervisory review finds that several audit files contain query outputs without a record of source filters or completeness checks. Correct the affected files and determine whether the cause is unclear methodology, a competence gap or insufficient review. Audit leadership can update the procedure and train staff, then examine later files for reproducible results. An independent quality assessment can evaluate whether the function's improvement arrangements are effective. A record that staff attended training establishes participation. Subsequent files and review results show whether the weakness was addressed.

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1.11 The Last Mile - Memory Aids and Exam Traps

The decisions that organize the domain

Domain 1 questions often turn on the auditor's role, the stage of work, the purpose of a test, or the quality of **evidence**. Use the comparisons below to explain your choice, rather than selecting an answer merely because it contains a familiar phrase.

Ask yourself	What to distinguish
Who owns this decision?	Board authorization, audit leadership's work, and management's operational responsibility.
Where are we in the process?	Establishing the function, planning coverage, or performing a specific engagement.
What is being tested?	A control's operation or the correctness of the underlying data.
What supports the conclusion?	Relevant reliable evidence for the right population and period.
What is already established?	A concern requiring investigation, a supported finding, or an urgent condition requiring escalation.

Foundation check: Define risk before selecting coverage; distinguish the risk-assessment process from a score. PPS weights selection by the chosen size measure. ML regression, software regression testing and server clustering describe different concepts.

Essential distinctions

Pair	Recall
Charter and scope	Authority for the function versus boundaries of one engagement.
Plan and program	Coverage over time versus detailed test procedures.
Control objective and practice	Desired result versus how a measure is performed.
Design and operation	Could the control work? Did it actually work?
Detection and residual risk	Auditor may miss the problem versus business exposure remaining after controls.
Compliance and substantive	Test the control versus test the transaction or data.
Attribute and variable	Characteristic or rate versus quantity or value.
Sufficient and appropriate	Enough evidence versus relevant and reliable evidence.
Auditing and monitoring	Audit's assurance work versus management's control oversight.
Recommendation and risk acceptance	Audit advises; an authorized management role decides.

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Memory aids

Use each memory aid as a reminder after understanding the explanation. State the principle in your own words, identify its **assumptions** and **apply it** to the role, objective and facts in the question.

Memory aid	Meaning and use
CPP	Charter gives authority; Plan gives coverage; Program gives procedures.
ACVS	**Attribute sampling** → Compliance testing; Variable sampling → Substantive testing. Where control reliance is planned, test relevant operation to support it. The sequence and extent of substantive procedures depend on the objective, risk and evidence.
ICeD	Inherent × Control × Detection: the conceptual audit risk model. Audit procedures directly address detection risk.
Sam and non Sam	Sampling and nonsampling causes can affect detection. A larger sample does not correct a wrong procedure.
SRRU	Sufficient, Reliable, Relevant, Useful: evaluate evidence quantity and quality.
Independent, objective, qualified, timely	Assess competence, **independence**, objectivity, authenticity, completeness, relevance and timing. Direct collection may be persuasive for its purpose; source location alone does not decide reliability. Corroborate unsupported representations.
Prevent detect correct	Stop the event; identify it; recover from its effects. Backup and restore is corrective, not preventive.
IPO	Input, Processing, Output: follow the transaction and its controls across the application.
Reality → records	To find unauthorized or undocumented changes, start from the actual system changes and trace them to approvals. A change that was never logged cannot appear in a **sample** drawn from the log.
CCCER	Condition, Criteria, Cause, Effect, Recommendation: connect the facts to their business significance and response.
Observe without owning	Audit evaluates and advises while management owns controls and remediation.
Owners accept auditors advise	Risk acceptance belongs to the authorized risk owner. Audit challenges and escalates where appropriate.
Standards require guidelines guide	Distinguish mandatory professional requirements from supporting guidance and techniques.
Rate or value	Choose attribute sampling for a characteristic or rate; variable sampling for a quantitative estimate.
Real data own program	Parallel simulation: independently reprocess actual data.
Dummy entity live processing	ITF: controlled test cases in production, with genuine records protected.

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Exam traps

Each row contrasts a common error with a supported response. Apply the **stated facts**, **decision authority** and objective rather than treating any reminder as an unconditional answer rule.

Tempting wrong answer	Exam answer
Treating the engagement scope or a management request as audit's authority.	The audit charter, approved by the board or audit committee, is the source of authority; scope defines one engagement.
Letting the auditee approve the audit program.	Audit management approves the program. The auditee may clarify facts and practical arrangements; audit retains control of procedures and independent conclusions under its mandate.
Choosing a later step when asked what comes FIRST in establishing the audit function.	The charter comes first. If the stem says it already exists, move to the next stage: risk assessment, then the audit plan.
Treating residual risk as detection risk.	Residual risk is the business's exposure after controls; detection risk concerns audit procedures and is directly managed through their design and execution.
Reporting every observation at once, before any analysis.	Communicate urgent supported **exposure** or interference promptly through the mandate. For an ordinary unverified observation, establish relevant facts and assess implications before a final conclusion.
Choosing a detective control when a workable preventive control is offered.	A feasible **preventive control** can stop the relevant event; detection supports timely response. Select the control that best meets the objective, considering effectiveness, constraints, cost and residual risk.
Sampling from change documentation to find undocumented changes.	Sample from the system's actual changes and trace them to documentation.
Starting with substantive testing when the audit plans to rely on controls.	Where reliance on controls is planned, obtain **evidence** of their relevant operation to support reliance and the substantive work. If reliance is unavailable or inappropriate, design suitable substantive procedures for the objective and risk.
Classifying backup and restore as a preventive control.	Backup and restore is a corrective control: it recovers after a loss.
Treating ITAF guidelines as mandatory.	Standards are mandatory; guidelines explain how to apply them; tools and techniques are informational.
Accepting a manager's oral statement as sufficient evidence.	Unsupported oral representation alone ordinarily provides insufficient support for a material conclusion. Obtain appropriate corroboration and assess the quality of all sources.
Increasing the sample size to fix a flawed procedure.	A larger sample reduces sampling risk only; a wrong procedure is non-sampling risk.

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Tempting wrong answer	Exam answer
Assuming a full-population CAAT test cannot be wrong.	It removes sampling risk, but the completeness of the extract and the logic of the script must still be validated.
Having the auditor develop or maintain the DR plan.	The auditor observes DR tests and reports the results; management owns, develops and maintains the plan.
Removing a finding because management disagrees.	Retain a supported finding and include management's response in the report.
Overruling management's acceptance of a risk the auditor considers excessive.	The auditor neither accepts nor overrules risk; escalate to the audit committee or board.

Decision owners at a glance

Use the assigned authority and evidence to distinguish delivery responsibility from oversight and assurance.

Role	Responsibility	Boundary
Audit committee/board	Oversees the audit mandate and independence.	Does not perform engagement testing.
Chief audit executive	Directs the audit function and its quality arrangements.	Does not own management controls.
Engagement auditor	Plans, tests, evaluates and communicates supported conclusions.	Does not accept the auditee's residual business risk.
Process owner	Operates controls and implements agreed corrective action.	Cannot determine the independent audit opinion.

Threats and control purpose

The examples show complementary control purposes. Their classification depends on how the control is configured and used.

Risk	Preventive direction	Detective evidence	Corrective direction
Unauthorized access	Approved least-privilege access and timely removal.	Independent recertification and activity review.	Remove invalid rights and repair the lifecycle process.
Unsupported audit conclusion	Defined objectives, criteria and data-validation methods.	Supervision and independent file review.	Obtain missing evidence or qualify the conclusion appropriately.

1.12 Glossary

Look up terms alphabetically. Return to the lesson for the related example and audit decision.

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Term	Definition
Administrative audit	An audit examining administrative arrangements and their contribution to effective and efficient operations.
Administrative controls	The rules, procedures and practices dealing with operational effectiveness, efficiency and adherence to regulations and management policies.
Agile auditing	Iterative audit delivery that adapts priorities while maintaining evidence, review, and independence requirements.
Agreed upon procedures	Specified procedures reported as factual results without an assurance opinion.
Algorithm	A specified sequence of steps or rules for solving a problem.
Analytical procedures	Evaluation through plausible relationships, expectations, and investigation of differences.
Application controls	The policies, procedures and activities designed to provide reasonable assurance that objectives relevant to a given automated solution are achieved.
Artificial intelligence or AI	Systems performing tasks associated with intelligent behavior, using rules, models, or other techniques.
Assertion	A representation or characteristic to be tested, such as completeness, accuracy, validity, or correct period.
Assurance	Confidence supported by an independent evaluation against suitable criteria.
Attestation engagement	An engagement in which a practitioner reports on subject matter or an assertion about it against suitable criteria under the applicable framework.
Attribute sampling	Sampling used to estimate occurrence of a characteristic, commonly a deviation rate in control testing.
Audit backlog	A prioritized inventory of potential audit work.
Audit charter	The board or audit-committee-approved document granting the audit function its authority, responsibility and accountability. The source of the auditor's right to audit.
Audit committee	A governing committee overseeing audit under delegated authority, including mandate, plan, reporting, and significant unresolved issues.
Audit evidence	The information used to support the audit opinion.
Audit hook	A programmed trigger that flags a defined condition for attention.
Audit objective	The specific question or result the audit work is intended to address.
Audit plan	Selected audit coverage and work over a period; the term may also describe planning for an individual engagement, depending on context.
Audit program	A step-by-step set of audit procedures and instructions that should be performed to complete an audit. Approved within audit management.

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Term	Definition
Audit risk	The risk of reaching an incorrect conclusion based upon audit findings. Its three components are control risk, detection risk and inherent risk.
Audit scope	The boundaries of the work, including activities, systems, locations, and period.
Audit trail	A logical path linking a sequence of events, in the form of data, used to trace the transactions that have affected the contents of a record.
Audit universe	The inventory of entities, processes, systems, or activities potentially available for audit coverage.
Availability	Ensuring authorized users can access information and services when required.
Batch total	The sum of a meaningful amount field used to reconcile processing.
Bayesian classifier	A classification method using probabilities updated or evaluated using observed evidence.
Benchmarking	A systematic approach to comparing enterprise performance against peers and competitors in an effort to learn the best ways of conducting business.
Black box testing	Testing expected behavior without relying on knowledge of the internal implementation.
Bridge letter	A provider management update covering events or changes after a report period; it is not a new independent assurance report.
Business risk	Uncertainty affecting the organization’s ability to meet business objectives.
Carve-out approach	A report approach excluding specified subservice controls from the examination; evaluate the resulting assurance gap.
Chain of custody	The process of evidence handling from collection to presentation, necessary to maintain the validity and integrity of evidence, including documentation of who had access and when.
Check digit	A calculated digit added to an identifier to detect specified input errors.
Chief audit executive (CAE)	The head of the audit function; prepares the charter and plan for approval, and issues the final audit report.
Closing meeting	Discussion of findings, factual accuracy, management responses, and proposed actions before final communication.
Clustering in machine learning	ML clustering groups similar observations without predefined outcome labels. See 1.8.4.
Code of Professional Ethics	ISACA's binding conduct code: standards support, competence, confidentiality, objectivity, due care and disclosure of material facts.
Compensating control	An internal control that reduces the risk of an existing or potential control weakness resulting in errors and omissions.
Competent evidence	Evidence that is relevant and reliable for the audit purpose.

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Term	Definition
Complementary user entity controls	Customer controls assumed in the service organization's control design and assurance report.
Compliance audit	An audit evaluating adherence to specified laws, regulations, contracts, policies, or other requirements.
Compliance testing	Control tests designed to obtain evidence on both the effectiveness of the controls and their operation during the audit period.
Comprehensive audit	An audit designed to determine the accuracy of financial records and evaluate the internal controls of a function or department.
Computer-assisted audit technique (CAAT)	Any automated audit technique, such as generalized audit software, test data generators, computerized audit programs and specialized audit utilities.
Computer forensics	The application of the scientific method to digital media to establish factual information for judicial review.
Confabulation	Unsupported or false generated content; fluent output is not independently reliable evidence.
Confidence coefficient	The confidence level specified for the statistical sampling method, subject to its assumptions.
Confidentiality	Restricting information disclosure to authorized parties.
Continuing professional education or CPE	Ongoing learning used to maintain professional competence and meet relevant certification obligations.
Continuous and intermittent simulation or CIS	Simulation of selected transaction processing according to defined criteria for audit evaluation.
Continuous assurance	Assurance supported by management monitoring and independent continuous audit evaluation.
Continuous auditing approach	Recurring or ongoing audit evaluation and evidence gathering supported by suitable methods and technology.
Continuous monitoring	Management's ongoing oversight of controls, performance, and changing conditions.
Control baseline	Approved starting set of applicable controls for a defined system, service or risk category, subject to authorized tailoring and exception handling. See 1.4.
Control catalog	Organized collection of control descriptions from which applicable controls can be selected. Not every catalog control is mandatory for every system. See 1.4.
Control environment	The organizational culture, accountability, oversight, and practices supporting internal control.
Control measure	An activity or safeguard implemented to help achieve a control objective.

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Term	Definition
Control monitoring	Management's assessment of whether controls continue to operate as required.
Control objective	A statement of the desired result or purpose to be achieved by implementing control procedures in a particular process.
Control practice	The way a control measure is performed in day to day work.
Control risk	Risk that assets are lost or compromised, or that financial statements are materially misstated, due to lack of - or ineffective - design or implementation of internal controls.
Control self assessment or CSA	A structured process in which management and staff assess risks and controls in their own activities.
Corrective control	A control designed to correct errors, omissions, unauthorized uses and intrusions, once they are detected.
Corroboration	Additional evidence supporting an assertion or another item of evidence.
Countermeasure	A measure intended to reduce a threat, vulnerability, likelihood, or adverse consequence.
Criteria	The requirements or benchmarks used to judge the observed condition.
Cryptographic hash	A function producing a fixed-size digest of data; comparison can support integrity checking but does not prove the data were true at collection.
CSP independence	Assessment of audit independence and objectivity for the actual cloud provider, tenant, services and relationships; conflicting responsibilities require disclosure and appropriate resolution. See 1.1 and 2.9.
Data mart	An analytical data collection focused on a subject or business area.
Data mining	Methods for discovering patterns and relationships in data.
Data warehouse	An analytical repository combining data from multiple sources.
Decision tree	A model applying a sequence of branches or tests to classify an item or predict a result.
Design effectiveness	Whether a control, performed as intended, could address the relevant risk.
Detection risk	Risk that audit procedures fail to detect a material problem; affected by sampling and nonsampling causes.
Detective control	Controls designed to detect and report when errors, omissions and unauthorized uses or entries occur.
Deterrent control	A control that discourages an act without physically preventing it - warning banners, visible cameras, sanctions.
Difference estimation	Estimating total misstatement from sampled differences between recorded and audited values.
Discovery sampling	A form of attribute sampling used to determine a specified probability of finding at least one example of an occurrence in a population.

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Term	Definition
Due professional care	The diligence and competence a reasonably prudent auditor would apply, supported by supervision, review and adequate documentation.
Electronic data interchange or EDI	Structured electronic exchange of business documents between organizations.
Embedded audit module (EAM)	An integral part of an application system designed to identify and report specific transactions or other information based on pre-determined criteria, as part of real-time processing. Also known as a continuous auditing module.
Engagement letter	Communication recording the terms, objectives, boundaries, responsibilities, and arrangements of a specific engagement.
Evidence	Information that an auditor gathers in the course of performing an IS audit; relevant if it pertains to the audit objectives and has a logical relationship to the findings and conclusions it supports.
Examination engagement	An assurance engagement using sufficient work to support a reasonable-assurance opinion under its framework.
Exception report	A report generated by a program that identifies transactions or data that appear to be incorrect - outside a predetermined range or not conforming to specified criteria.
Expected error rate	The deviation rate the auditor anticipates before testing, based on prior results.
Expert system	A system applying a knowledge base and inference rules to provide specialized results or advice.
External confirmation	Evidence obtained directly from an independent third party about specified information.
False negative	A real problem incorrectly left unflagged.
False positive	A legitimate item incorrectly flagged as a problem.
Financial audit	Examination of financial reporting against the applicable reporting criteria.
Finding	A reportable condition, structured as condition, criteria, cause, effect and recommendation.
Fixed size attribute sampling	Attribute sampling with a planned sample size, commonly used to estimate a deviation rate.
Flowchart	A flowchart depicts process steps, decisions and their sequence. See 1.5.3.
Follow-up audit	Work performed after the report to verify that agreed remediation was implemented and is operating.
Forensic audit	An examination of suspected misconduct emphasizing evidence suitable for investigation and possible legal use.

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Term	Definition
Functional audit	Evaluation of whether a software product's actual functions and performance meet its approved requirements.
General controls (ITGC)	Controls applying across all systems - access, change, operations, physical and environmental. Weak ITGCs undermine reliance on application controls.
Generalized audit software (GAS)	Multipurpose audit software used for general processes such as record selection, matching, recalculation and reporting.
Guideline	Supporting advice on applying requirements using professional judgment.
Hash total	Sum of a numeric field used as a processing control rather than as a meaningful business amount; different from a cryptographic hash.
Homomorphic encryption	Encryption supporting specified computations on ciphertext, with results that can be decrypted to the corresponding plaintext computation.
Inclusive approach	A report approach including the stated subservice controls within the examination.
Incorrect acceptance	Concluding that a population or control is acceptable when it is not; threatens audit effectiveness.
Incorrect rejection	Concluding that a population or control is unacceptable when it is acceptable; can create unnecessary work.
Independence (in fact and in appearance)	Freedom from conditions that threaten objectivity - actual, and as perceived by a reasonable observer. Impairments must be disclosed.
Industrial control system or ICS	A system monitoring or controlling physical industrial processes.
Information system	People, processes, information, and technology working together to collect, process, store, distribute, and use information.
Information technology	Hardware, software, communications, and related technical facilities used to process and communicate information.
Inherent risk	The level of risk or exposure that does not account for the actions management has taken or might take, such as implementing controls.
Integrated audit	A coordinated examination of related financial, operational, and IT risks and controls.
Integrated test facility (ITF)	A testing methodology in which test data are processed in production systems, usually representing fictitious entities such as departments, customers or products; output reports are verified to confirm processing correctness.
Integrity	Protecting information from unauthorized or improper modification and preserving its accuracy and completeness.
Intelligent agent	Software configured to observe conditions and perform defined actions.

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Term	Definition
Interim report	Communication of significant results or urgent issues before the final report.
Internal control over financial reporting or ICFR	Controls intended to support reliable financial reporting under the applicable reporting framework.
Internal controls	The policies, procedures, practices and organizational structures designed to provide reasonable assurance that business objectives will be achieved and undesired events prevented or detected and corrected.
IS audit	Independent collection and evaluation of evidence about information systems, their safeguards, information, controls, and contribution to business objectives.
ITAF	ISACA's IT Audit Framework: professional standards and supporting guidance for IT audit and assurance.
Limited assurance	A lower level of assurance supported by a more limited engagement, commonly a formal review.
Machine learning or ML	Methods that learn patterns from data for tasks such as classification or prediction.
Management response	The auditee's stated action and owner for each finding. Disagreement is recorded; it does not remove the finding.
Materiality	The threshold above which an omission or misstatement would influence a user's decision. In IS audit it includes non-financial impact such as criticality, regulatory exposure and reputation.
Material weakness	A severe control deficiency defined by the applicable reporting framework; do not apply the label without its criteria.
Model drift	A change that reduces model performance as data patterns or operating conditions change.
Naive Bayes	Naive Bayes is a probabilistic classification approach that applies Bayes reasoning with a simplifying conditional-independence assumption about features given the class. See 1.8.4.
Natural language processing or NLP	Techniques for processing, analyzing, or generating human language.
Nearest neighbor method	Classification or prediction based on the most similar examples in a reference data set.
Neural network	A model using interconnected processing units with adjustable weights to learn relationships from data.
Non-sampling risk	The risk of a wrong conclusion for reasons unrelated to sample size - the wrong procedure, or a misinterpreted result.
Non-statistical (judgmental) sampling	Selection based on auditor judgment; results cannot be statistically projected to the population.

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Term	Definition
Object code / executable artifact	Object code is translated output from source code, commonly machine-level code that may still require linking to form an executable. See 1.8.1.
Objectivity	Impartial professional judgment free from inappropriate influence.
Opening meeting	Meeting confirming engagement objectives, scope, access, contacts, timing, and practical arrangements.
Operating effectiveness	Whether a control actually worked as intended during the relevant period.
Operational audit	An audit designed to evaluate the various internal controls, economy and efficiency of a function or department.
Overfitting	Overfitting occurs when a model fits training-specific detail or noise and performs poorly on new representative data. See 1.8.4.
Parallel simulation	The auditor reprocesses real data through independently written software and compares results with the system's output.
Parameterized query	A database operation separating command structure from supplied data to help prevent injection.
Peer review	Quality review by qualified people independent of the work being evaluated.
Physical control	A safeguard restricting or protecting physical access to facilities, equipment, or media.
Population	The complete group of items relevant to the audit test.
Population standard deviation	Dispersion of all population values around the population mean.
Precision and tolerable deviation	Precision is the width of the desired estimation range. Tolerable deviation is a decision limit; the two are distinct.
Prescriptive control framework	A structured set of specified safeguards or objectives used to guide implementation and assessment.
Preventive control	An internal control used to avoid undesirable events, errors and other occurrences that could have a negative material effect on a process or end product.
Probability-proportional-to-size sampling (PPS)	PPS selection gives larger units a greater selection probability according to the chosen size measure. See 1.6.2.
Professional skepticism	A questioning, evidence focused mindset that evaluates assertions and possible contradictory information fairly.
Programmable logic controller or PLC	An industrial controller executing logic for equipment or processes.
Quality assurance and improvement program	An organized approach to assessing and improving audit quality, conformance, and effectiveness.
Quality assurance (QA)	Systematic activities providing confidence that the audit function and engagements meet applicable requirements.

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Term	Definition
Readiness assessment	An assessment identifying gaps to address before a formal audit or assessment; design readiness does not establish operating effectiveness.
Reasonable assurance	A high but not absolute level of assurance supported by sufficient appropriate evidence.
Recalculation	Checking mathematical accuracy independently.
Record count	Number of records used as a completeness check; matching counts do not guarantee identical content.
Regression in machine learning	An ML regression model estimates a numerical outcome from input features. See 1.8.4.
Re-performance	Independent execution of a control or procedure by the auditor; its persuasiveness depends on the method and evidence.
Residual risk	The risk that remains after controls are applied. Distinct from detection risk.
Retrospective	A team review of completed work aimed at improving its methods.
Review engagement	A formal engagement providing limited assurance under the applicable framework; distinguish it from an informal document review.
Risk	Uncertainty about effects on objectives; audit assessment considers credible events, likelihood and consequences. See 1.3.
Risk acceptance	An authorized management decision to retain risk; audit documents, challenges, and escalates excessive unresolved exposure as appropriate.
Risk analysis	Analysis of the nature of identified risks, including their likelihood and consequences, to support evaluation against criteria and selection of treatment.
Risk appetite	The amount of risk, on a broad level, that an entity is willing to accept in pursuit of its mission.
Risk assessment / analysis / evaluation	Assessment supports risk decisions; analysis develops understanding of exposure; evaluation compares it with criteria. See 1.3.4.
Risk avoidance	Discontinuing or changing an activity to remove the associated exposure.
Risk evaluation	The process of comparing the estimated risk against given risk criteria to determine the significance of the risk.
Risk factor	A condition influencing a scenario’s likelihood or impact and used to assess risk or prioritize coverage.
Risk management	The coordinated activities to direct and control an enterprise with regard to risk.
Risk mitigation	The management of risk through the use of countermeasures and controls.
Risk tolerance	The acceptable level of variation that management is willing to allow for any particular risk as the enterprise pursues its objectives.

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Term	Definition
Risk transfer	Allocation of specified consequences or responsibilities to another party; retained exposure and accountability still need consideration.
Risk treatment	The process of selection and implementation of measures to modify risk.
Robotic process automation or RPA	Software automation executing configured workflow steps; it need not learn from data.
Sample	The subset of population items selected for examination.
Sample mean	The sum of the sampled values divided by the sample size.
Sample standard deviation	Dispersion of sample values around their mean; used to estimate variability.
Sampling risk	Risk of reaching a different conclusion from testing the full population with the same procedure.
Sampling risk level	The chosen risk level associated with the sampling method; in a simple confidence statement, one minus the confidence level.
Sampling unit	The individual item selected for examination, such as an invoice or account.
Sampling upper limit	A method-based upper estimate at a selected confidence level; compare it with the planned tolerance.
SCADA	Supervisory control and data acquisition, used to oversee and control industrial processes.
Scope limitation	A restriction or evidence gap affecting work or the supported conclusion; its significance and reporting effect must be assessed.
Segregation/separation of duties (SoD)	A basic internal control that prevents or detects errors and irregularities by assigning to separate individuals the responsibility for initiating and recording transactions and for the custody of assets.
Self-review threat	A threat to objectivity when an auditor evaluates work for which the auditor previously had responsibility.
Semantic network	A semantic network represents concepts and their relationships as connected nodes and links. See 1.8.1.
Service auditor	The independent practitioner examining and reporting on the service organization’s relevant controls.
Service level agreement or SLA	Documented service expectations, performance targets, measurement, and relevant responsibilities.
Service organization	The provider performing an outsourced service for user entities.
Snapshot	A technique capturing the state of a transaction at selected points during processing.
SOC 1 report	An independent report on service-organization controls relevant to customers’ internal control over financial reporting.

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Term	Definition
SOC 2 report	An independent report on the described service-organization system's controls against the applicable Trust Services Criteria.
SOC 3 report	A general-use report against Trust Services Criteria with less detail than a SOC 2 report.
Source code comparison	Compare reviewed source with an approved source baseline to detect differences. Compare executable artifacts separately with approved artifacts; source text and binary are not directly equivalent. See 1.8.1.
Specialized audit	An examination focused on a particular subject, service, technology, or risk area.
Sprint	A short planned period for completing a defined increment of work.
SQL injection	An attack in which untrusted input changes the intended structure or meaning of a database command.
Standard	An authoritative requirement; determine its applicability within the relevant professional or organizational framework.
Statistical sampling	A method of selecting a portion of a population, by means of mathematical calculations and probabilities, for making scientifically and mathematically sound inferences about the entire population.
Statistical significance	The relationship of an observed result to a statistical model and its assumptions; distinct from practical importance or causation.
Stop-or-go sampling	A sampling approach that permits stopping early when few errors are found, avoiding excessive sampling.
Stratified mean per unit	Estimating within more homogeneous groups and combining the estimates for the population.
Stratified sampling	Dividing the population into groups and selecting within them to improve efficiency or address different characteristics.
Structured Query Language or SQL	A language used to query and manage data in relational databases.
Subservice organization	Another provider used by the service organization in delivering the service.
Substantive testing	Testing that verifies the accuracy, completeness and validity of the data or balances themselves rather than the controls over them.
Sufficiency, reliability, relevance, usefulness	The four qualities evidence must have: enough of it; from a dependable source; related to the objective; and helpful in meeting the objective.
Systems control audit review file or SCARF	Selected processing information captured for audit review, commonly through embedded audit functionality.
Technical control	A safeguard implemented through technology; also called a logical control.
Test data	Auditor-prepared fictitious transactions run through a system to verify processing logic; must not contaminate production data.

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Term	Definition
Test deck	A test deck is a prepared set of test transactions or inputs with known expected results. See 1.8.1.
Tests of details	Substantive examination of individual transactions, balances, or data items.
Threat	A potential cause of harm to an asset, process, or business objective.
Tolerable error rate	The maximum deviation rate compatible with the intended control reliance. For value tests, tolerable error is expressed in the relevant quantity.
Tolerable misstatement	The maximum value error compatible with the purpose of a substantive test.
Tracing and tagging	CAAT technique that tags selected transactions and follows their processing path and results. Does not require an ITF dummy entity. See 1.8.1.
Trust Services Criteria or TSC	Criteria used to evaluate relevant controls over security, availability, processing integrity, confidentiality, or privacy.
Type 1 report	For SOC 1 or SOC 2, a report addressing the description and suitability of design at a specified date under the applicable requirements.
Type 2 report	For SOC 1 or SOC 2, a report also addressing operating effectiveness over a specified period, with relevant tests and results.
Underfitting	Underfitting occurs when a model fails to capture important relationships and performs poorly even on training data. See 1.8.4.
Unstratified mean per unit	Estimating a population total using a sample mean from the population without separate strata.
User entity	The customer organization using a service organization.
Variable sampling	Sampling used to estimate a quantitative characteristic such as a monetary amount or difference.
Vulnerability	A weakness that a threat could exploit.
Walkthrough	Tracing one transaction end-to-end through a process to confirm the auditor's understanding of the control design.
Weighted risk score	A rating combining risk factors with different assigned levels of influence.
White box testing	Testing informed by the application’s internal code, logic, or structure.
Working papers	Records of objectives, procedures, evidence, results, review, and conclusions that allow an experienced reviewer to understand the work.
XBRL	Extensible Business Reporting Language, used to tag and exchange structured business reporting information.

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1.13 Practice questions

These 18 original questions follow the CISA format: choose the one BEST answer from four options. Domain 1 carries 18 percent of the examination, so it has 18 of the book’s 100 domain questions. Allow about 1.6 minutes per question, then check section 1.14. Before you look, name the **qualifier** and say why your **second choice** is weaker.

Question 1.01 A newly appointed IS audit manager finds that several business units dispute the audit team's right to access production systems and records. Which document is the **PRIMARY** source of that right?

A. The engagement memo issued for each individual audit
B. The annual IS audit plan approved by the audit committee
C. The information security policy approved by the CISO
D. The IS audit charter endorsed by the audit committee

Question 1.02 An IS auditor is assigned to audit the change management process that she helped redesign as an internal consultant eight months ago. What should the auditor do **FIRST**?

A. Limit testing to controls she did not help to design
B. Disclose the potential impairment to audit management
C. Request that a peer review her completed work papers
D. Decline the assignment and ask to be reassigned

Question 1.03 An organization is introducing control self-assessment workshops for its IT operations teams. Which role is **MOST** appropriate for the IS auditor in this program?

A. Facilitator who guides and supports the workshops
B. Reviewer whose sign-off replaces the annual audit
C. Approver of the action plans that teams agree upon
D. Owner of the control ratings the workshops produce

Question 1.04 An IS audit manager is preparing next year's audit plan for a company that has just migrated its core finance applications to a cloud provider. What should the manager do **FIRST**?

A. Draft audit programs for the migrated applications
B. Estimate audit hours available for cloud-related work
C. Perform a risk assessment of the IT audit universe
D. Obtain the provider's latest SOC 2 Type 2 report

Question 1.05 During planning, an IS auditor concludes that inherent risk and control risk for a payments interface are both high. Which response is **MOST** appropriate?

A. Reduce control risk by recommending added controls
B. Lower detection risk by expanding substantive tests
C. Accept higher detection risk since controls are weak
D. Rely more on management's control self-assessment

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Question 1.06 An IS auditor finds a weakness in a system that processes low-value transactions but stores customer data subject to privacy law. In judging the materiality of the weakness, the auditor should give the **GREATEST** weight to:

A. the cost of implementing a compensating control
B. the monetary value of transactions the system processes
C. the number of users who have access to the system
D. the business impact if the weakness is exploited

Question 1.07 After a storage failure, operations staff restore a customer database from the previous night's backup. Which classification **BEST** describes the restoration?

A. Detective, because it reveals the corruption to operators
B. Compensating, because it offsets weak storage controls
C. Corrective, because it recovers from an event's effects
D. Preventive, because it limits the loss of customer data

Question 1.08 The IS audit team has drafted the audit program for a review of the treasury system. The treasury manager asks to approve the test steps before fieldwork starts. Which response is **MOST** appropriate?

A. Let the treasury manager approve the program as the system owner
B. Seek audit management approval and weigh the manager's comments
C. Submit the program to the audit committee for formal approval
D. Begin fieldwork and share the program at the closing meeting

Question 1.09 During fieldwork, an IS auditor notices vendor master file changes made by an accounts payable clerk outside business hours, followed by payments to those vendors. The auditor has preserved the relevant logs. What should the auditor do **NEXT**?

A. Expand sampling to quantify total losses before reporting
B. Report the indicators to audit management for escalation
C. Recommend the clerk's system access be suspended at once
D. Interview the clerk to obtain an explanation of the changes

Question 1.10 An IS auditor needs to estimate the total value of duplicate payments in a year of disbursements to support a recovery claim. Which sampling approach is **BEST** suited?

A. Stop-or-go sampling of disbursement records
B. Attribute sampling of payment approval evidence
C. Discovery sampling of invoices for duplicates
D. Variable sampling of the disbursement amounts

Question 1.11 An IS auditor tests a sample of registered change tickets to confirm that the changes represented by those tickets were approved before deployment. The **PRIMARY** objective of this test is to determine whether:

A. the value of unapproved changes falls within materiality
B. the changes deployed to production were free of coding errors
C. the change policy reflects current industry good practice
D. the approval control operated as intended during the period

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Question 1.12 An attribute sample of 60 user access reviews shows a deviation rate well below the tolerable rate. The auditor then learns the sample was drawn from reviews saved in a shared folder rather than from the system list of required reviews. What is the **GREATEST** concern?

A. Reviews that were missed could not enter the sample
B. The tolerable rate was set without management input
C. The sample size is too small for the confidence level
D. The deviation rate may understate the sampling risk

Question 1.13 An IS auditor is testing whether network accounts of terminated employees were disabled promptly during the year. Which evidence is **MOST** reliable?

A. A walkthrough given by the provisioning analyst
B. A disablement report extracted by the help desk
C. A signed confirmation from the IT security manager
D. Auditor's match of HR exit dates to account logs

Question 1.14 An IS auditor is testing whether firewall rule reviews were performed each quarter. Which of the following provides the **LEAST** reliable evidence?

A. Dated review sign-offs retained in the change tool
B. A statement by the network manager that reviews ran
C. Rule-base logs showing removals after each review
D. Tickets raised from review findings in each quarter

Question 1.15 A bank wants an audit technique that flags high-risk wire transfers for auditor review as they are processed, without interrupting production. Which technique **BEST** meets this need?

A. Integrated test facility using dummy bank accounts
B. Generalized audit software run against monthly extracts
C. Embedded audit module with transaction-selection criteria
D. Parallel simulation of the wire transfer program

Question 1.16 At the closing meeting, the application owner disputes a finding that 14 of 40 sampled privileged accounts lacked the documented approval required by policy, saying approvals were given verbally. No other evidence is offered. What is the **BEST** course of action?

A. Escalate the disagreement to the audit committee chair
B. Retain the finding and include management's response
C. Downgrade the rating to reflect the verbal approvals
D. Defer the report until the owner obtains written approval

Question 1.17 During follow-up, an IS auditor learns that management has accepted the risk of an unremediated high-risk finding. The auditor believes the residual risk exceeds the organization's risk appetite. Each of the following is an appropriate action **EXCEPT**:

A. Reopen the finding and require remediation by management
B. Discuss the basis for acceptance with senior management
C. Escalate the matter to the audit committee for resolution
D. Document management's acceptance in the follow-up file

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Question 1.18 While reviewing working papers before report issuance, an IS audit manager finds that a conclusion on backup controls cites a restore test that was never performed. What should the manager do **FIRST**?

A. Record the deficiency for the annual quality assessment
B. Ask the auditee to confirm in writing that restores work
C. Issue the report with a scope limitation on backup controls
D. Obtain the missing evidence before the report is issued

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1.14 Answers and explanations

Each explanation states the **deciding principle**, then shows why the key is best and why each other option, although often a sound practice, does not answer the question asked.

Question 1.01 Answer: D

Audit charter · Qualifier: PRIMARY · Review section 1.1

Principle: Audit's authority, including unrestricted access, derives from the charter approved by the board or **audit committee**.

A. The **engagement letter** communicates scope and arrangements under the authority already granted by the charter; it does not establish that organizational authority.

B. The **audit plan** sets coverage over time. It does not itself confer the organizational access rights needed to perform engagements.

C. A management **policy** may recognize audit access, but the board-approved charter is the source of audit's organizational authority in this scenario.

D. Correct. The charter, approved at board or **audit committee** level, establishes audit's purpose, authority and right of access across the organization.

Question 1.02 Answer: B

Independence · Qualifier: FIRST · Review section 1.1

Principle: Independence impairments are disclosed first so **audit management** can decide on reassignment or safeguards.

A. Changing scope may be a safeguard, but the auditor must first disclose the potential impairment so **audit management** can evaluate it.

B. Correct. Disclosure lets **audit management**, not the auditor alone, decide whether reassignment or safeguards are needed; it is the required first step.

C. **Independent review** may be a safeguard after assessment. Arranging peer review alone leaves the potential impairment undisclosed.

D. Reassignment may be appropriate, but **audit management** should decide after the auditor discloses the potential impairment.

Question 1.03 Answer: A

Control self-assessment · Qualifier: MOST · Review section 1.2

Principle: In CSA, line **management owns** the assessment and the IS auditor acts as facilitator.

A. Correct. The auditor facilitates and provides control expertise while **process owners** assess and own their controls, preserving independence.

B. CSA results can inform audit work. They supplement independent **assurance** rather than replacing the audit function's required coverage.

C. **Management owns** and approves its action plans. Audit can evaluate the proposed response without taking that decision from management.

D. Business teams own their self-**assessment** ratings. Audit taking ownership would blur responsibility and create an independence concern.

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Question 1.04 Answer: C

Risk-based audit planning · Qualifier: FIRST · Review section 1.3

Principle: The **audit plan** is built from a risk assessment; resourcing, evidence and programs follow from it.

A. An engagement **program** describes procedures for a selected engagement. The question concerns the earlier decision about risk-based annual coverage.

B. Resource constraints inform planning, but allocation should be guided by the risks and coverage needs established through the **risk assessment**.

C. Correct. A **risk assessment** identifies where the highest risks lie so the plan directs audit effort to them; every other step depends on it.

D. Detailed engagement **evidence** is collected after the **audit plan** has selected priorities. It is not the starting point for the annual coverage decision described.

Question 1.05 Answer: B

Audit risk · Qualifier: MOST · Review section 1.3

Principle: Detection risk is directly managed through audit procedure design and execution. High assessed inherent and **control risk** calls for procedures capable of reducing detection risk appropriately.

A. Recommending stronger controls may help future periods, but it does not establish that the controls operated effectively in the **period under review**.

B. Correct. The auditor changes the nature, timing and extent of appropriate procedures to obtain **sufficient evidence** at acceptable **audit risk**; this includes the necessary substantive work.

C. When inherent and **control risk** are high, the auditor needs an appropriately lower acceptable detection risk, supported by suitable procedures.

D. Management's self-assessment can inform the work. It does not by itself reduce **detection risk** or replace the independent procedures needed for sufficient evidence.

Question 1.06 Answer: D

Materiality · Qualifier: GREATEST · Review section 1.3

Principle: IS audit materiality is judged by overall impact, including regulatory and reputational consequences, not monetary value alone.

A. Control cost informs treatment decisions. The significance of this weakness depends on the **business consequences** of the exposure.

B. Transaction value is relevant in some audits, but here it would omit material privacy, regulatory and reputational **consequences**.

C. The number of users can influence exposure. It is an input to assessing **consequences**, not a complete measure of this weakness's significance.

D. Correct. IS audit materiality turns on the consequence to the business if exploited, here including privacy penalties and reputational harm.

Question 1.07 Answer: C

Control classification · Qualifier: BEST · Review section 1.4

Principle: Backup and **restoration** are corrective controls: they remedy the effects of an event after it occurs.

A. The incident has already been identified. Restoring the system addresses its **consequences** rather than detecting the original failure.

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B. A **compensating control** substitutes for another control to address its objective. Here restoration is described by its corrective recovery purpose.

C. Correct. Restoring from backup returns the system to a known good state after the incident, the defining purpose of a **corrective control**.

D. **Restoration** acts after the loss to recover service. It does not prevent the original failure from occurring.

Question 1.08 Answer: B

Audit program approval · Qualifier: MOST · Review section 1.5

Principle: Audit programs are approved within **audit management**; the auditee may give input but does not approve audit procedures.

A. **Auditee input** can improve understanding and logistics, but approval of the audit program belongs within the audit function in this scenario.

B. Correct. Approval stays within audit to protect independence, while **auditee input** on scope and logistics is still welcomed.

C. The committee provides oversight and approves the charter and **audit plan**. Routine approval of individual engagement procedures is an audit management responsibility here.

D. Audit authority permits appropriate procedures, but unnecessarily withholding **scope** until closing undermines engagement planning and communication.

Question 1.09 Answer: B

Fraud indicators · Qualifier: NEXT · Review section 1.5

Principle: On identifying fraud indicators, the auditor communicates them promptly to appropriate authority rather than investigating alone.

A. Quantification may be needed later. Delaying the required communication while the auditor completes a separate investigation can increase **exposure** and compromise evidence.

B. Correct. Prompt communication lets **audit management** involve the right officials, such as legal or investigators, while evidence is protected.

C. **Containment** may be necessary, but the auditor should promptly use the authorized escalation path rather than independently taking over management's access decision.

D. Questioning the suspected individual can **alert** them and compromise a formal investigation. Protect evidence and communicate through the established channel.

Question 1.10 Answer: D

Variable sampling · Qualifier: BEST · Review section 1.6

Principle: Estimating a monetary value is a substantive objective served by **variable sampling**.

A. This attribute approach addresses a control **deviation rate**. The question requires an estimate of total monetary value.

B. An attribute **compliance test** estimates a deviation rate, rather than the quantitative total required for the recovery claim.

C. Finding a duplicate can establish that a problem exists. It does not provide the **population**-value estimate requested here.

D. Correct. Variable sampling projects a quantitative value such as total misstatement, which is what the recovery claim requires.

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Question 1.11 Answer: D

Compliance testing · Qualifier: PRIMARY · Review section 1.6

Principle: Testing that approvals exist is a **compliance test** of control operation, not a substantive test of outcomes.

A. Quantifying the monetary effect is a different substantive objective. Testing whether changes had the required approvals evaluates **control operation**; the distinction does not impose a universal testing sequence.

B. **Approval** records do not establish that the code produces accurate results. That objective needs suitable code or processing procedures.

C. Reviewing policy against a benchmark concerns **control design**. Testing the change tickets asks whether the approval control actually operated.

D. Correct. A **compliance test** gathers evidence that a control operated as designed throughout the period under review.

Question 1.12 Answer: A

Sampling population · Qualifier: GREATEST · Review section 1.6

Principle: A sample is only as good as its **population**; to find missing items, sample from the source where omissions appear.

A. Correct. Drawing from completed reviews excludes the reviews that were never done, so the test cannot detect the deviation it targets.

B. The auditor should set a justified **tolerable deviation** rate, informed by relevant **context**. That does not correct selection from a folder that omits unperformed reviews.

C. Increasing the **sample size** does not fix the incomplete population: reviews never performed would still be excluded from selection.

D. **Sampling risk** matters when drawing a sample. The central defect here is the population's failure to include the deviations the test seeks.

Question 1.13 Answer: D

Evidence reliability · Qualifier: MOST · Review section 1.7

Principle: A validated comparison of complete HR departures with reliable account timing records directly addresses prompt **revocation**. Its quality depends on sources and **procedure**, not merely on the auditor doing it.

A. A walkthrough helps the auditor understand the process and identify controls, potentially using documents and **observation**. By itself it does not demonstrate timely **disablement** throughout the year.

B. An auditee-produced report can be useful evidence after appropriate **validation**. Here its completeness and integrity have not been established for the requested period.

C. A representation records management's claim. Without **corroboration**, it does not demonstrate when the relevant accounts were actually disabled.

D. Correct. Auditor-performed comparison of independent HR and system sources provides direct, reperformed **evidence** for the full period.

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Question 1.14 Answer: B

Evidence reliability · Qualifier: LEAST · Review section 1.7

Principle: An unsupported oral representation does not establish the disputed condition. Obtain suitable **corroboration** rather than ranking every source by its format alone.

A. Dated sign-offs provide documentary **evidence** of the claimed reviews. Their reliability and coverage still need evaluation, but they offer **corroboration** absent from the unsupported representation.

B. Correct. The unsupported representation does not demonstrate the claimed **control operation**. The other described evidence offers more direct corroboration for this objective.

C. The described logs offer **evidence** of actions following reviews. Validate their source, **integrity** and coverage; their evidential value is not guaranteed merely because they are system-generated.

D. The described tickets record review outcomes and provide **corroboration**. Their **completeness** and reliability still matter; the comparison does not create a universal evidence hierarchy.

Question 1.15 Answer: C

Continuous auditing techniques · Qualifier: BEST · Review section 1.8

Principle: An embedded audit module can select relevant live transactions for review. SCARF refers to selected audit-review information or the file; it is related to, rather than identical with, the module.

A. **Test data** exercises known cases to evaluate processing. It does not provide the requested recurring selection of actual live transfers.

B. Monthly extract analysis is **retrospective** and periodic. The scenario asks for flagging transactions during live processing.

C. Correct. An embedded module captures transactions meeting audit **criteria** during live processing, enabling continuous auditing.

D. **Parallel simulation** independently reprocesses data for comparison. It is not the embedded live-capture mechanism requested in the question.

Question 1.16 Answer: B

Management disagreement · Qualifier: BEST · Review section 1.9

Principle: A **finding** supported by evidence stands; management's disagreement is recorded with it in the report.

A. Escalation can be appropriate for unresolved interference or serious disagreement. Here the supported **finding** and management's response can be fairly presented in the report.

B. Correct. The **finding** is evidence-based, so it stays; including management's view gives readers a fair, balanced report.

C. New **evidence** should be considered, but an unsupported oral claim does not justify changing the finding supported by the records described.

D. Later approvals do not change the **historical condition** found. Record remediation where relevant without rewriting the audit-period facts or unnecessarily delaying reporting.

Question 1.17 Answer: A

Risk acceptance follow-up · Qualifier: EXCEPT · Review section 1.9

Principle: Management owns risk acceptance; the auditor challenges, documents and escalates but cannot compel remediation.

A. Correct. Not appropriate: requiring remediation would make audit the decision-maker over **risk**, a role that belongs to management.

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B. Appropriate: understanding **management's reasoning** is the first step before deciding whether escalation is warranted.

C. Appropriate: when accepted risk appears unacceptable, the auditor escalates to the **audit committee** for resolution.

D. Appropriate: the **follow-up** record must show the finding's status, including that management formally accepted the risk.

Question 1.18 Answer: D

Supervisory review · Qualifier: FIRST · Review section 1.10

Principle: Supervisory review must ensure every reported conclusion is supported by sufficient, appropriate **evidence** before issue.

A. Improving quality review may prevent **recurrence**. It does not supply the missing evidence for the conclusion about to be issued.

B. A written representation can contribute evidence, but it does not replace the missing restore **procedure** needed to support this specific conclusion.

C. A genuine inability to obtain **evidence** may require a limitation and its implications to be reported. Here the omitted **procedure** can still be performed; calling the omission a limitation does not support the conclusion.

D. Correct. Supervision exists to ensure conclusions rest on **evidence**; the test must be done or the conclusion cannot be reported.

1.15 Continue your preparation

Use your error log to identify the reason behind each mistake. Revisit the relevant explanation and **Last Mile** table, then test the concept with a fresh scenario. Register at certprepx.com for adaptive practice, timed full-length mocks and domain analytics. Available plans and features may change.

Scan to visit https://certprepx.com

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Domain 2: Governance and Management of IT

2.0 Domain overview

Technology creates value only when it supports a business purpose, has accountable owners, and is operated within acceptable **risk**. This domain examines how an enterprise directs IT, turns that direction into plans and services, protects its data, and checks the results. Domain 2 represents 18 percent of the CISA exam.

Use the explanations to build understanding, then revise with The **Last Mile**, the glossary, and the practice questions. The numbered core topics follow the source chapter structure. Additional teaching headings expand the subjects beneath them.

Topics and reading route

Part	Section	Topic
A IT Governance	2.1	Laws, Regulations and Industry Standards
A IT Governance	2.2	Organizational Structure, IT Governance and IT Strategy
A IT Governance	2.3	IT Policies, Standards, Procedures and Guidelines
A IT Governance	2.4	Enterprise Architecture and Considerations
A IT Governance	2.5	Enterprise Risk Management
A IT Governance	2.6	Data Privacy Program and Principles
A IT Governance	2.7	Data Governance and Classification
B IT Management	2.8	IT Resource Management
B IT Management	2.9	IT Vendor Management
B IT Management	2.10	IT Performance Monitoring and Reporting
B IT Management	2.11	Quality Assurance and Quality Management of IT

Follow the decision and its evidence

A retailer proposes a cloud customer platform. The board needs confidence that the investment supports growth without unacceptable **exposure**. Management needs a **business case**, data owners, **supplier** controls, funding, skilled staff, and measures of benefit. An auditor follows those connections: an approved purchase alone does not establish good governance, lawful data use, or successful delivery.

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2.1 Laws, Regulations and Industry Standards

Legal and regulatory **obligations** define constraints within which the business and IT must operate. Industry standards and contractual commitments may add **requirements**. An auditor first establishes which criteria apply, then tests the enterprise's arrangements against them.

2.1.1 Impact of Laws, Regulations and Industry Standards on IS Audit

An applicable requirement is one that governs the enterprise, **activity**, data, location, or agreement under review. Applicability may depend on where customers live, where processing takes place, the industry, and the service offered. A law in one country does not automatically apply everywhere, and a voluntary **standard** can become binding through a contract or regulatory requirement.

The auditor obtains a current understanding of the **obligations** with help from legal and compliance specialists where necessary. Translate each relevant obligation into an audit criterion, identify its owner and controls, and collect evidence of compliance. A **policy** that repeats the law is insufficient if the operational process does not meet it.

Requirement	Meaning	Audit implication
Law	Legislation enacted by a competent authority.	Determine applicability and examine the relevant controls and records.
Regulation	Rules issued under legal authority by a regulator.	Check sector requirements, reporting and supervisory obligations.
Industry standard	Agreed requirements or recognized practices for a subject.	Determine whether adoption is voluntary, contractual or mandated.
Contract	An enforceable agreement between parties.	Review service, data, licensing, assurance and termination duties.
Internal policy	Approved enterprise direction.	Evaluate conformance and whether the policy addresses external obligations.

Example: A payroll **supplier** hosts records outside the customer's home country. The audit considers the relevant employment, privacy, transfer, contractual and **retention** requirements. Local hosting rules cannot be assumed from the supplier's marketing statement; the actual data flow and applicable jurisdictions must be established.

Software licensing and intellectual property

Intellectual property includes legally protected creations such as software, designs, and proprietary content. A software license grants specified rights of use; buying access does not normally transfer ownership of the software. Restrictions can depend on users, devices, processors, **deployment** locations, or use by contractors.

Reconcile authorized software and subscription entitlements to deployment and actual use. Include virtual environments and cloud services, review renewal and **termination** terms, and investigate unauthorized installations. Protect the enterprise's own **source code** and confidential know-how through access controls and appropriate agreements.

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2.1.2 Governance, Risk and Compliance

Governance, risk and compliance (GRC) connects direction, uncertainty, and obligations. Governance establishes objectives and **decision rights**; risk management assesses uncertainty affecting those objectives; compliance determines and monitors the requirements the enterprise must meet. Coordinating them prevents conflicting policies, duplicated control testing, and important obligations falling between teams.

A shared **obligations** inventory or control library can map a single control to several **requirements**. That efficiency does not mean the requirements are identical. Owners still need to understand each obligation, its scope, and the evidence required.

Audit focus Review the **obligations** register, legal change process, responsible owners, control mappings, compliance reporting and unresolved breaches. Internal audit assesses this system independently; it does not become the legal adviser or accept management's **risk**.

Turning obligations into auditable requirements

Determine which **obligations** apply to the enterprise, processing activity, location, sector and contractual relationship. Laws and regulations create applicable external duties; industry standards and contractual commitments can add **requirements**. Internal policy should translate them into accountable action. A named standard in a policy is not proof that its full scope has been implemented.

Maintain a **requirements** register linking each applicable duty to a process owner, control, **evidence** and review date. Use qualified legal or regulatory interpretation where needed. The auditor evaluates whether management identifies and addresses requirements and whether the evidence supports compliance; the auditor should not invent a universal legal answer from a short technical scenario.

Example: A service provider moves backups to another jurisdiction. **Encryption** may reduce disclosure **exposure**, but does not by itself establish that the transfer, processing location and retention are permitted. Examine the approved arrangement, applicable duties, contract, access and deletion capability. Track changes in **obligations** and business activity, including new uses of data. A once-completed compliance checklist can become stale even if the technical environment has not changed.

Gap analysis compares the implemented state with the applicable **requirements** identified here. Establish the criteria and scope before interpreting a gap; section 2.2 explains its use in strategy and improvement planning.

Turn a Requirement Into Evidence

Example. A contract requires transaction records to be retrievable throughout a stated retention period. The provider produces a policy with the correct period, but the latest retrieval test covers only recent records. Confirm the requirement and which systems and providers it covers. Select older records near the retention boundary and verify that authorized users can retrieve and read them. If a legal restriction affects access, obtain specialist interpretation and agree a permitted evidence route. Evaluate the actual ability to meet the obligation using the approved criteria.

2.2 Organizational Structure, IT Governance and IT Strategy

IT governance establishes who decides, whose needs the decisions serve, and how results are overseen. Management makes those decisions operational. Organizational structure, strategy, committees and role **definitions** should reinforce that distinction.

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2.2.1 Enterprise Governance of Information and Technology

Enterprise **governance** of information and technology (**EGIT**) is the part of enterprise governance concerned with information and technology throughout the business. It extends beyond the IT department: business units using cloud services and information in automated decisions also need governance.

The governing body evaluates stakeholder needs, directs priorities, and monitors results. Executives plan and manage delivery within that direction. Both levels monitor, but for different purposes: the board considers value and **exposure**; an operations manager checks service **performance** and takes action.

Decision	Governance perspective	Management responsibility
Business alignment	Set and oversee enterprise direction.	Translate business objectives into the IT strategy and delivery plans.
Value	Oversee expected benefits and acceptable trade-offs.	Build cases, deliver capabilities and track realized benefits.
Risk	Oversee appetite and significant exposure.	Assess risk, operate controls and escalate exceptions.
Resources	Oversee adequacy and strategic allocation.	Manage people, budgets, suppliers and capacity.
Performance	Challenge outcomes and corrective action.	Measure delivery, investigate causes and improve services.



Figure 2.1 Governance sets direction; management delivers and reports; audit supplies independent assurance.

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2.2.2 Good Practices for EGIT

Good **governance** combines **strategic alignment**, value delivery, risk management, resource management, and performance measurement. These work together. A low-cost service that prevents a critical business transaction is not a successful investment simply because it meets its budget.

Establish clear **decision rights**, business participation, relevant policies, reliable information and escalation routes. Review the arrangements as the business, technology and risk change. The useful question is whether the chosen **governance** system produces sound decisions and accountable results.

Frameworks and standards in their proper roles

A framework organizes principles, objectives or practices; a **standard** sets **requirements** or guidance for a defined subject. Choose the reference that addresses the stated need and adapt its use to the enterprise. Framework adoption is evidence of an approach, not proof that the controls operate effectively.

Reference	Primary purpose	Distinction to remember
COBIT	Governance and management of enterprise information and technology.	Connects enterprise goals, alignment goals and governance or management objectives.
ISO/IEC 38500	Governance of IT for governing bodies.	Addresses governance principles rather than detailed service procedures.
ITIL / ISO/IEC 20000	Service management practices / service management system requirements.	Concern delivery and improvement of services.
ISO/IEC 27001	Information security management system requirements.	A management system for security, not a guarantee that every system is secure.
ISO 22301 / ISO 31000	Business continuity management / risk management guidance.	Continuity and enterprise risk have related but different purposes.
COSO	Internal control and enterprise risk management frameworks.	Enterprise-wide approaches extending beyond IT.
TOGAF / Zachman	Enterprise architecture method and framework / architecture classification structure.	A method for developing architecture differs from a taxonomy for organizing descriptions.
CMMI	Capability and process performance improvement.	An assessed maturity level is not a security certification.
NIST CSF 2.0	Cybersecurity risk management outcomes.	Six functions: Govern, Identify, Protect, Detect, Respond and Recover.
ISO 9001 / PMBOK / PRINCE2	Quality management / project management references.	Match the reference to quality or project management needs, respectively.

In **COBIT**, **governance** uses **EDM**: Evaluate, Direct and Monitor. Management objectives are organized into APO (Align, Plan and Organize), BAI (Build, Acquire and Implement), DSS (Deliver, Service and Support), and MEA (Monitor, Evaluate and Assess). “Plan, Build, Run and Monitor” is a useful broad teaching shorthand for management; it is not the formal set of COBIT domain names.

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Governance depends on more than processes. Organizational structures, policies, information, people and skills, culture and behavior, and services and infrastructure all affect whether it works. A committee with no authority or reliable reporting cannot compensate for these missing elements.

2.2.3 Audit's Role in EGIT

The auditor evaluates whether **governance** supports **business objectives**, assigns accountability, manages significant risk and enables useful oversight. Trace decisions from strategy through investment selection and operating results. Review what happens when a project fails, a risk exceeds tolerance, or reported benefits are not realized.

Three Lines Model

The **Three Lines Model** separates complementary responsibilities. First-line management delivers products and services and manages the associated **risk**. Second-line roles provide specialist support, **monitoring** and challenge on matters such as risk and compliance. Internal audit provides independent assurance to the governing body. First and second lines remain management roles; the second line is not independent in the same sense as internal audit.

Role	Typical work	Boundary
First line	Process owners operate controls and respond to service failures.	Owns the activity and its risk.
Second line	Risk, security oversight and compliance teams advise, monitor and challenge.	Supports and challenges management; does not replace audit independence.
Third line	Internal audit evaluates governance, risk management and controls.	Must avoid taking management responsibility for what it later audits.
Governing body	Sets direction and oversees accountability.	Receives management reporting and independent assurance.

2.2.4 Information Security Governance

Information security **governance** directs security so it supports the business and manages **exposure** appropriately. It establishes security priorities, accountability, resources, policy direction and performance oversight. A firewall configuration is an operating control; deciding the enterprise's security direction and holding management accountable are governance activities.

Effective Information Security Governance

Effective security governance links security objectives to **business objectives**, integrates risk decisions, assigns information ownership, funds necessary capability, and measures results. It also coordinates people, processes and technology across business units. Security performance should explain exposure and **business consequences** in terms decision-makers can use.

Example: A bank intends to expand digital lending. Security **governance** sets expectations for protecting customer information and controlling digital service risk. Management implements identity checks, secure development and **monitoring**. Audit checks whether the controls and reporting support the approved objectives.

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2.2.5 Information Systems Strategy

The **IS strategy** explains how information systems and technology will enable **business objectives**. It covers the capabilities required, the target operating approach, major investments, sourcing, architecture and resources. Business and IT should develop it together: technology may enable new business opportunities as well as support existing plans.

Alignment is demonstrated when IT initiatives map to **business outcomes**, with sponsors, benefits, resources and measures. A strategy written only as a list of products to purchase has not established why those products matter.

2.2.6 Strategic Planning

Strategic planning starts with business direction and the **current situation**. Assess existing systems, skills, costs, architecture, obligations and risks; define the future capabilities needed; identify gaps; and prioritize a roadmap. Strategic plans establish longer-term direction, tactical plans translate it into programs and resource decisions, and operational plans describe recurring execution.

Review strategy when material business or technology conditions change. Update **assumptions**, consult stakeholders and approve changes through established authority. A fixed three- or five-year cycle does not justify leaving an obsolete plan unchanged.

Planning element	What should be established
Business needs	The objective and the users or stakeholders who benefit.
Current and target states	Existing capability, desired capability and the gap between them.
Roadmap and dependencies	A feasible sequence, including retiring legacy systems and managing transition risk.
Ownership and resources	Business sponsors, accountable owners, funding, skills and capacity.
Benefits and review	Expected outcomes, measures, milestones and a process to reassess assumptions.

Audit focus Select major initiatives and trace them to strategy, approved cases, **benefit owners** and outcomes. Compare the documented process with committee decisions and **actual spending**. Investigate unsupported purchases, repeated overruns and priorities that change without business justification.

2.2.7 Business Intelligence

Business intelligence (BI) organizes data into information managers can use to understand **performance**, customers, costs and risk. A dashboard is the presentation of an analysis; its appearance does not establish that the underlying data are complete or meaningful.

Metadata describes data: its meaning, format, owner or rules. **Data lineage** shows where data came from, how it changed and where it is used. For a sales dashboard, metadata explains what “net sales” includes; lineage traces that figure through adjustments and interfaces to authorized source transactions.

An enterprise needs both a logical view of its data and controlled movement of that data between systems. **Definitions**, identifiers and relationships provide consistent meaning. Extraction,

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transformation, **validation** and loading provide a dependable data flow. Preserve sufficient detail and lineage to explain a reported result.

Component	Role in BI	Audit concern
Source data	Operational, external and other input records.	Coverage, authorized access and completeness.
Data access layer	Connects source stores to downstream processing.	Interfaces and selection rules do not omit or alter data improperly.
Staging and quality layer	Copies, validates, cleans and transforms data.	Reconciliation, rejected records and approved transformation logic.
Core data warehouse	An integrated analytical store of enterprise information.	Consistency, access, historical accuracy and recoverability.
Data preparation / data mart	Prepares analytical values / provides a subject or business-unit subset.	Derived measures reconcile and retain their intended meaning.
OLAP / data mining	Multidimensional analysis / discovery of patterns in data.	Valid dimensions, assumptions and interpretation.
Presentation layer	Reports, queries and dashboards used by decision-makers.	Correct labels, calculation rules and audience access.
Metadata repository	Definitions, formats, owners, business context and transformation rules.	Version control and traceability from the report back to source.
Warehouse control / messaging / network	Schedules processing, transports data and control messages, and enables communications.	Failed jobs, interface errors, delivery and security are monitored.

Example: Sales and finance use different **definitions** of “active customer.” A single dashboard cannot resolve the disagreement by combining the totals. **Business owners** first agree the definition, then implement the mapping and reconciliation. Audit traces the published metric through the transformation rules to source records.

Models that clarify business data

A context diagram shows the system or business boundary and its exchanges with external parties. A data flow diagram (**DFD**) traces movement and transformation of data between processes and stores. An entity relationship diagram describes entities, their attributes and relationships. These views help auditors understand completeness, ownership and dependencies; none proves that the implemented data flows match the model.

Activity and swim-lane diagrams

An **activity** diagram shows the steps and decision points in a process. A **swim-lane diagram** places those steps in lanes representing responsible roles or teams, making hand-offs visible. Use it to identify missing approvals, duplicated work and unclear **accountability**. Compare the model with actual records and access rather than assuming the documented flow is followed.

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2.2.8 Organizational Structure

The organizational structure assigns authority, reporting and responsibility. A **centralized IT** model can support consistency and shared control; a decentralized model can improve responsiveness to local needs. A federated model combines enterprise direction and shared services with selected local decisions. The appropriate arrangement depends on **business objectives** and risk.

IT Governing Committees

An **IT strategy committee** usually advises the board on direction, opportunities, significant investments and value. An **IT steering committee** usually brings business and IT management together to prioritize initiatives, allocate or recommend resources, resolve conflicts and monitor delivery. Authority to approve particular investments depends on the committee charter and delegated financial limits.

Body or role	Primary responsibility	Evidence to examine
Board of directors	Enterprise direction, governance accountability and significant risk oversight.	Approved strategy, appetite, oversight minutes and follow-up.
Senior management	Implement direction and remain accountable for results.	Plans, budgets, business ownership and escalation decisions.
IT strategy committee	Advice to the board on IT direction and value.	Charter, strategic recommendations and board reporting.
IT steering committee	Business and IT priorities, resource trade-offs and delivery oversight.	Business participation, investment decisions, minutes and actions.
Security standards committee	Coordinate and review security standards and related decisions.	Approved remit, stakeholder participation and recorded decisions.
CIO / CTO / CISO	Information and IT direction / technology direction / security leadership.	Role descriptions, authority and reporting; titles vary by enterprise.

Data Ownership

The business **data owner** is **accountable for the data** and determines its business value, classification and authorized access. A data custodian operates the technical safeguards and storage under that direction. A data steward helps maintain **definitions** and quality. A security administrator implements approved privileges; this is different from deciding whether a business user should have access.

Users must use information as authorized and report suspected misuse. New-user access should follow documented authorization, **identity verification**, **least privilege** and a record linking the request to the implemented privileges. The owner's approval is meaningful only if the actual access agrees with it.

IT Roles and Responsibilities

Job descriptions should identify responsibilities and incompatible access. Distinguish business ownership, technical administration and independent checking. **Outsourcing** a role does not remove the need to assign an internal owner for it.

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Function	Main responsibility
Systems development manager / project manager	Direct development work / plan and coordinate an assigned project.
Systems analyst	Translate business requirements into system specifications and analyze proposed solutions.
Application development and maintenance	Build and maintain application code through controlled development and change.
Infrastructure development and maintenance	Design and change the underlying platforms and technical infrastructure.
Infrastructure operations / system administration	Run scheduled services and operate system configurations, accounts and supporting resources.
Network management	Manage network connectivity, configuration, performance and availability.
Database administration	Administer database structure, performance, access mechanisms and recovery.
Security administration	Implement authorized security settings and privileges; maintain relevant records.
Security architect / security engineer	Design security patterns and requirements / build and integrate security mechanisms.
Data management / end-user support	Manage data architecture and assets / help users resolve service problems.
Media management / data entry	Control media custody and life cycle / capture input accurately with appropriate validation.
SCADA and industrial operations	Supervise or control physical processes; consider safety, availability and authorized access.
Quality assurance / vendor management	Evaluate process conformance / manage supplier obligations, performance and risks.

Executive abbreviation	Role and decision boundary
CEO	Chief executive officer; leads the enterprise under governing direction.
CFO	Chief financial officer; leads financial stewardship within the assigned mandate.
CLO	Chief legal officer; leads legal advice and legal-function responsibilities.
CSO	Often chief security officer in security contexts; some enterprises use the title differently. The actual mandate and delegated authority govern the role.

Separation of Duties Within IT

Separation, or segregation, of duties (SoD) divides incompatible activities so one person cannot both create an error or fraud and conceal it. **Authorization**, execution, recording and **reconciliation** should

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be separated where practical. A developer's unrestricted production deployment rights and a database administrator's ability to erase independent **audit evidence** are different examples of the same problem.

A RACI matrix states who is Responsible for doing the work, Accountable for its outcome, Consulted for input, and Informed of results. Assign a clear accountable owner for each activity and distinguish that **accountability** from the number of people who perform it. Compare the matrix with job descriptions, **permissions** and actual practice.

Control area	Suitable separation or safeguard
Transaction authorization	A suitably authorized person approves; a different role performs or independently reviews execution.
Ownership of assets	Ownership decisions are distinct from custody and routine processing.
Access to data	Business approval is distinct from technical provisioning and periodic independent review.
Authorization forms and user tables	Requests identify the person and privileges; approved records reconcile with actual access.
Development and production	Changes are independently approved and tested; production deployment and access are controlled.
Privileged activity	Logs reach a protected destination and are reviewed by someone independent of the activity.

Compensating Controls for Lack of Separation of Duties

A small team may be unable to separate every role. A **compensating control** must address the specific conflict: **independent review** of changes and transactions, protected audit trails, reconciliations, restricted emergency access, or external oversight. Calling a manager's occasional review "compensating" is insufficient without evidence that it could detect the relevant misuse.

Reporting arrangements also affect challenge and **objectivity**. Assess whether security or **risk** oversight can challenge operations and escalate concerns without suppression. The organizational chart is evidence of a potential conflict, not a universal rule that a particular title can never report to another. Duties, authority, safeguards and access to the governing body matter.

Example: One administrator maintains the payroll **database** and supports urgent corrections. A finance owner approves changes, a separate reviewer compares the database change log with approved requests, and protected logs prevent the administrator from hiding alterations. Review coverage and timeliness determine whether this arrangement adequately compensates for the conflict.

Matrix of Outcomes and Responsibilities

An outcomes-and-responsibilities matrix connects **governance** objectives with the roles expected to achieve or oversee them. It complements RACI by making the intended result explicit. The following examples show how **strategic alignment**, risk, value, resources and assurance connect to accountability.

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Role	Expected contribution
Board	Require evidence of alignment, oversee significant exposure and challenge results.
Executive management	Establish delivery, risk, resource and reporting processes that implement direction.
Steering committee	Coordinate priorities, resolve cross-functional conflicts and monitor initiatives.
Security leadership	Translate security direction into a managed program, metrics and coordinated assurance.
Audit leadership	Independently evaluate and report on alignment, risk management, performance and assurance coverage.

2.2.9 Auditing IT Governance Structure and Implementation

Evaluate **governance** as an operating system of decisions, reporting and **accountability**. Read the organizational chart, role descriptions, charters, strategy, policies, risk records, architecture, budgets and performance reports. Corroborate them through interviews, observed committee activity and samples of actual decisions.

Red flags include repeated unapproved purchases, unresolved ownership, obsolete plans, significant risks without owners, and projects monitored only for spending. Determine their business effect and **root cause**. Recommend improvements to **governance** without assuming management's decision-making role.

Auditor Perspective

Follow decision authority

A strategy body advises on direction; a steering body commonly coordinates priorities and delivery. Read the actual **charter and delegation** before assigning a decision.

An attractive technical solution does not establish **business alignment** without an approved objective, accountable owner and credible outcome measure.

Tracing governance decisions to business outcomes

Governance evaluates stakeholder needs, sets direction and monitors outcomes. Management plans, builds, runs and monitors activities to achieve that direction. In COBIT 2019, EDM is the governance domain; APO, BAI, DSS and MEA are management domains. A management **monitoring** objective remains management activity even though it informs governing oversight.

Assess **decision rights**, escalation routes, investment priorities, **risk appetite** and benefit accountability. A steering committee commonly coordinates priorities and delivery under its mandate; titles alone do not establish authority. The **Three Lines Model** describes complementary roles in delivery and risk management, expertise and challenge, and independent internal audit. Evaluate actual duties and conflicts rather than assuming every security job belongs in one fixed line.

Example: The board approves a digital strategy, but funding repeatedly favors projects with no link to its objectives. Inspect **portfolio** decisions, **benefit owners** and outcome reporting. The existence of the strategy is insufficient evidence of alignment. A chief information security officer's reporting line can create a potential conflict, but **objectivity** also depends on authority, independent challenge, escalation

and safeguards. Audit should evaluate those mechanisms and their actual operation rather than rank job titles mechanically.

Gap analysis and improvement priorities

Gap analysis compares the **current state** with an agreed target, requirement or capability and identifies the difference. Establish the **business objectives**, scope and comparison criteria before using it to prioritize improvements. A legal-compliance gap and a desired maturity improvement may require different decisions. Gap analysis supports strategy; it is not invariably the first action in every strategic-planning scenario.

Example: The target requires recovery of a payment service within two hours, but a representative exercise takes six. Validate both measurements, identify dependencies and causes, and develop owned, costed options. Management selects priorities and accepts **residual risk** within its authority; audit evaluates the evidence and **decision process**. The requirements mapping in section 2.1 supplies criteria for compliance-related **gap analysis**.

Follow a Governance Decision Into Operation

Example. The board approves a digital service strategy with explicit limits on disruption and retained supplier risk. Management selects a provider and configures controls within that authority. The auditor traces the decision from the approved objectives to the investment, contract, service measures and treatment of exceptions. If an operating manager accepts an outage exposure outside delegated limits, check the escalation record and governing decision. The board's direction and oversight correspond to governance. Selecting and operating the controls are management activities.

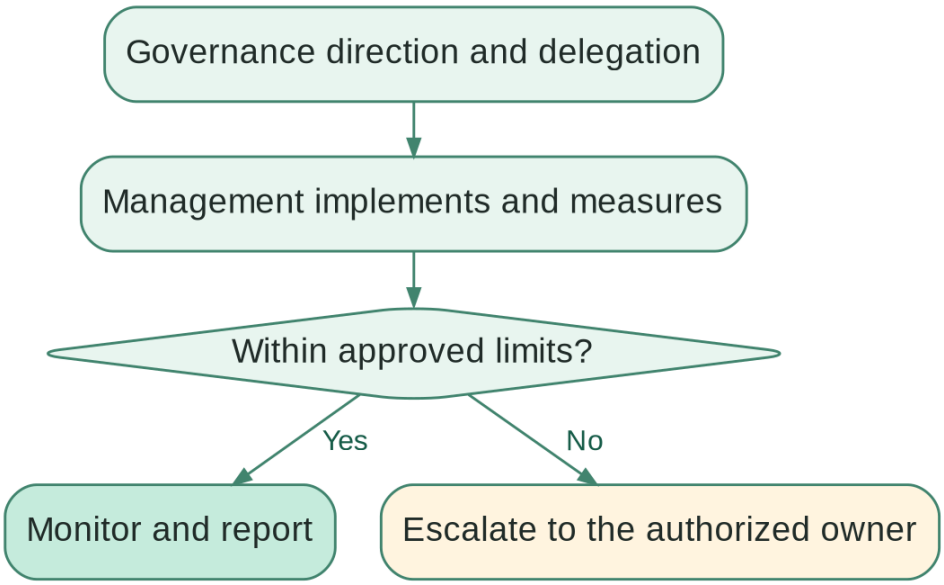


Diagram 2.1 Governance sets direction and delegation. Management implements and reports the results.

2.3 IT Policies, Standards, Procedures and Guidelines

Policies provide direction; **standards** and procedures make it operational; guidelines support judgment. The documents must reflect current **obligations** and risk, be approved at the appropriate level, and reach the people expected to follow them.

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2.3.1 Policies

A **policy** states approved intent, principles and expectations. It should identify its purpose, **scope**, accountable owner, responsibilities, compliance expectations and review arrangements. Approval comes from the authority established by enterprise governance, often the board or senior management depending on the policy.

A security **policy** establishes how the enterprise intends to protect information. Supporting topic policies can address access, acceptable use, data handling, remote work and **supplier** services. Communicate policies, train affected staff, track acknowledgments where appropriate, and manage exceptions through explicit authority and review dates.

Information Security Policy and Its Review

Review the information security **policy** for **business alignment**, legal obligations, ownership, scope, enforcement and exception handling. Confirm that significant changes in cloud use, data processing, remote access or threat exposure prompt reconsideration. A recently approved document can still be inadequate if important activities are outside its scope.

Document	Purpose	Example
Policy	States intent and expectations.	Sensitive customer information must be protected from unauthorized disclosure.
Standard	Specifies a mandatory requirement.	Approved strong authentication is required for privileged remote access.
Baseline	Defines an approved minimum configuration or control level.	The server build disables unnecessary services and applies the required logging settings.
Procedure	Describes the steps and responsibilities for a task.	Verify approval, provision the role, test access and record completion.
Guideline	Gives recommended ways to act where judgment is appropriate.	Suggested wording for clear data handling instructions.

2.3.2 Standards

An internal **standard** turns **policy** direction into a specific mandatory rule. It should be measurable and suitable for the systems it governs. A baseline is a minimum approved level, often expressed as a standard configuration for a platform or class of assets. Exceptions require **risk assessment**, approval, compensating measures where needed, and expiry or review.

2.3.3 Procedures

A **procedure** states how a task is performed, who performs it, what approvals are needed, and which records demonstrate completion. It should be usable by trained staff and match the actual process. Outdated instructions can create errors even where the **policy** and standard remain sound.

2.3.4 Guidelines

A **guideline** recommends practices and supports consistent judgment. It is generally advisory unless an enterprise explicitly makes a requirement mandatory through another instrument. Do not force

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policy, standard, baseline, procedure and guideline into a rigid five-step chronology: guidelines can support several levels, and a baseline commonly sits within standards.

Example: A new classification program first needs its objectives, approved scheme, ownership and handling **requirements** understood. A discovery tool can then support implementation. If the scheme already exists and the question asks how to locate unclassified files, choosing “write a **policy**” again may not address the actual gap.

Making direction enforceable and exceptions accountable

A **policy** states approved direction and expectations. A **standard** specifies mandatory requirements within its authority. A baseline establishes an approved minimum configuration or reference. A procedure describes how work is performed. A guideline generally advises unless the enterprise expressly makes its content mandatory. Terminology can vary, so inspect the organization’s definitions and approval arrangements.

Translate direction into rules people and systems can implement, with owners, approval, communication, monitoring and review. An exception process needs the reason, affected scope, **risk assessment**, compensating measures, **authorized decision**, expiry and reassessment. An exception is a managed departure, not permission to ignore external obligations or unlimited acceptance of risk.

Example: A hospital describes removable-media **encryption** only in an optional **guideline**. A mandatory **standard** establishes the internal requirement, while deployment, monitoring, training and exception handling make it effective. Renaming the document does not encrypt the drives. Sample actual devices and permitted exceptions and compare them with the approved rule. Verify that obsolete versions are withdrawn and that contractors and automated services receive the requirements relevant to their work.

Check Whether an Exception Still Has Authority

Example. A legacy application has an approved six-month exception to a mandatory access standard. Nine months later, the exception register still shows it as open and the alternative review has stopped. Examine the approved scope, expiry, decision authority and required alternate safeguard. Test whether the safeguard operated during the authorized period, then determine how the expiry was handled. The finding should distinguish the original authorized exception from the exposure after expiry and identify the responsible decision maker.

2.4 Enterprise Architecture and Considerations

Enterprise architecture connects business needs with information, applications and technology. It describes the current and desired states and guides the transition, helping the enterprise make coherent investment and design decisions.

Architecture layers and the transition roadmap

Enterprise architecture (EA) provides a coordinated view of how the enterprise works and the capabilities it needs. The business layer describes processes and capabilities; the data layer describes information and meaning; the application layer describes systems and interactions; and the technology layer describes supporting infrastructure. These layers influence one another.

A target architecture without a **transition roadmap** is an aspiration. Identify **dependencies**, standards, gaps, costs, risks and intermediate states. Establish architecture decision and exception processes so individual projects do not create incompatible solutions without a justified business reason.

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Layer	Question it answers	Retailer example
Business	Which capability and process are needed?	A single customer return process across stores and online sales.
Data / information	What information and definitions support it?	Common customer, order and product identifiers.
Application	Which systems provide the capability and exchange information?	The sales platform connects with stock, payment and returns services.
Technology	What infrastructure enables the systems?	Network, compute, identity and integration services with agreed resilience.

TOGAF provides an architecture approach, including the Architecture Development Method (ADM). **Zachman** organizes architecture descriptions by perspectives and questions such as what, how, where, who, when and why. They are not interchangeable with service management or information security management systems.

Audit focus Review approved architecture principles, current and target models, the roadmap, design decisions and **exceptions**. Trace selected investments to the architecture. Look for duplicated capabilities, inconsistent data, unsupported **interfaces**, technology obsolescence and concentration on a single provider.

Example: Two divisions buy customer platforms with incompatible identifiers. Each project meets its local **requirements**, but the enterprise cannot produce a reliable consolidated customer view. Architecture **governance** should expose the shared data requirement before investment and assign ownership of the integration decision.

Using enterprise architecture to expose dependencies

Enterprise architecture connects business capabilities, information, applications and technology. The **current state** shows how the enterprise operates; the **target state** and transition roadmap describe intended change. Assess whether architecture principles and decision processes support strategy, interoperability, resilience, security and manageable technical debt.

Useful evidence includes service and data-flow models, interface inventories, technology standards, approved deviations and transition decisions. Test a sample against reality. An attractive diagram that omits an unmanaged identity dependency may mislead project and recovery planning. Architecture **governance** should allow justified **exceptions** while making their risk and support consequences visible.

Example: Two business units propose separate customer platforms. A shared customer definition and controlled **interfaces** may improve consistency, while one central platform could create a concentration or migration risk. Evaluate the **business objective** and feasible options rather than assuming centralization is inherently best. Trace important **dependencies** into the BIA, change impact analysis and supplier oversight. Architecture is valuable when its information changes decisions and is maintained after implementation.

Use the Architecture to Find a Shared Dependency

Example. Three channels are scheduled to move to separate hosting environments, but each still uses one customer identity service and one data integration route. Compare the current and target architecture, data definitions, interfaces and owners. Follow a customer request through each channel

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to determine which dependencies remain shared. Ask whether the transition plan addresses service interruption and whether tests exercise the shared identity and data services. A hosting diagram can be accurate and still omit the dependency that determines the business outcome.

2.5 Enterprise Risk Management

Enterprise **risk management** (ERM) coordinates how uncertainty affecting **business objectives** is identified, assessed, treated and monitored. IT risk belongs within this enterprise view. A technology incident can affect financial results, customer safety, legal compliance and business strategy at the same time.

A **threat** is a potential **cause** of harm; a vulnerability is a weakness that can be exploited; an impact is a consequence for the enterprise. Risk analysis considers the likelihood and impact of a credible event in its business context. A vulnerability severity score alone does not establish business risk.

2.5.1 Developing a Risk Management Program

Begin with the enterprise's objectives, **context** and risk **criteria**. Establish sponsorship, policy, appetite, responsibilities, assessment methods, reporting and escalation. Assign accountable owners with the authority to make or escalate decisions. Embed the process in strategy, projects, operations, procurement and change rather than maintaining a disconnected annual exercise.

Term	Meaning	Decision use
Risk appetite	The amount and type of risk the enterprise is willing to pursue or retain in meeting objectives.	Guides strategy and significant trade-offs.
Risk tolerance	Acceptable variation or boundaries around a particular objective or risk.	Provides more specific operating and escalation limits.
Risk capacity	The amount of risk the enterprise can bear without threatening its viability or critical obligations.	Constrains how much exposure can reasonably be accepted.
Inherent risk	Exposure considered before the relevant controls or treatment.	Establishes the starting risk under the assessment method.
Residual risk	Exposure remaining after the controls or response considered.	Must be compared with the approved criteria and acceptance authority.

Appetite is a deliberate choice; **capacity** concerns the ability to absorb exposure. Neither permits management to disregard a legal obligation. The board oversees enterprise appetite, management implements the risk program, and authorized **risk owners** decide within delegated limits. Internal audit evaluates these arrangements without accepting risk on management's behalf.

Example: The retailer accepts some fluctuation in online demand but sets a specific **limit** for checkout disruption. A proposal that may save costs while exceeding that limit needs the established escalation and **decision process**. An IT manager cannot silently redefine enterprise tolerance to make the proposal appear acceptable.

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2.5.2 Risk Management Life Cycle

Risk management is iterative. Establish the objective and **context**, identify risk, assess it, select a response, and monitor the exposure and controls. Communication and consultation continue throughout. New information can require earlier assessments and decisions to be revisited.

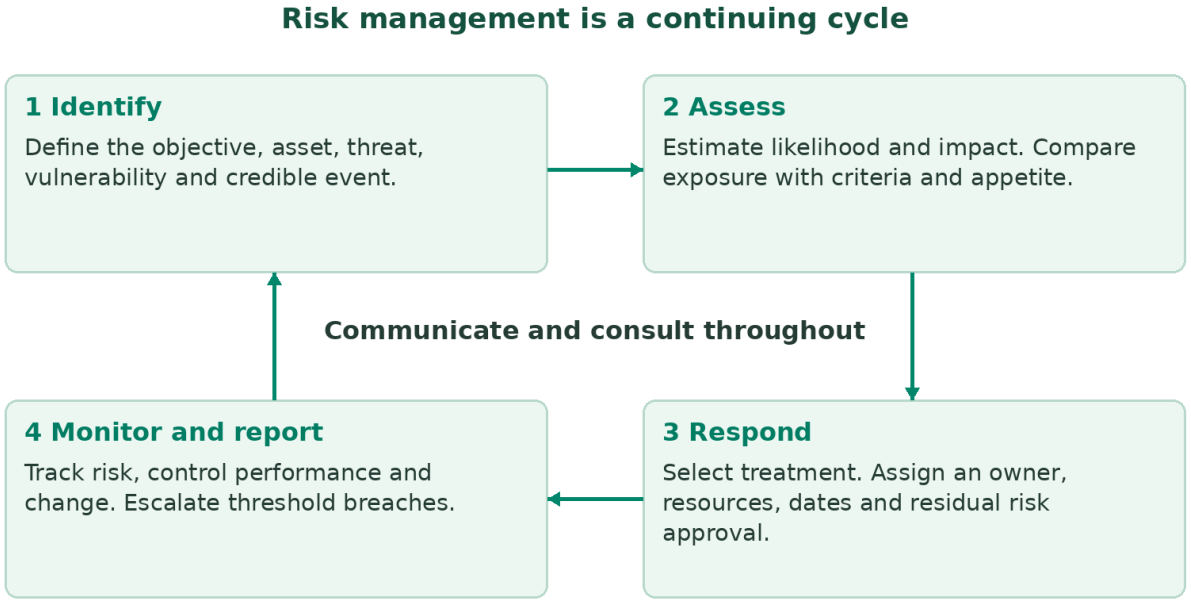


Figure 2.2 Identification, assessment, response and monitoring form a cycle, supported by continuing communication.

Step 1 IT Risk Identification

Identify assets, **dependencies**, business processes and credible events that could prevent the objective being achieved. Consider people, technology, facilities, suppliers, projects and emerging change. Describe the **cause**, event and consequence clearly enough for an owner to assess and manage it.

A **risk register** records the risk, **business objective**, owner, assessment, controls, response, action dates, residual exposure and review status. It should cover the enterprise's real activities, including services procured outside IT. A long register with no accountable owners is not an effective program.

Step 2 IT Risk Assessment

Analyze likelihood and impact over a stated period using reliable information and consistent definitions. Evaluate the resulting exposure against appetite, tolerance and other criteria to set priorities. Consider existing **control design** and operation, dependencies and aggregate exposure. Several individually small risks may combine into a significant **concentration risk**.

Use the assessment, analysis and evaluation distinctions introduced in section 1.3.4. A simple likelihood-times-impact score can support prioritization under a defined scale, but it does not replace the risk description, existing-control evidence or an authorized acceptance decision.

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Step 3 Risk Response and Mitigation

Select a response that addresses the identified exposure and respects **business objectives** and obligations. Define actions, owners, resources, deadlines and expected **residual risk**. Acceptance is an explicit authorized decision to retain exposure; inaction caused by an unassigned owner is not adequate evidence of acceptance.

Response	What changes	Example and limitation
Avoid	Stop or do not start the activity creating the exposure.	Cancel an unsupported service. Check that related obligations and transition risks are still handled.
Mitigate / reduce	Reduce likelihood, impact, or both through controls.	Add independent payment approval and exception review; some residual risk remains.
Share / transfer	Allocate specified consequences or responsibilities through another party.	Insurance or contract terms may shift defined losses; exclusions and retained accountability still matter.
Accept	Retain the exposure through an authorized informed decision.	Record rationale, authority, review triggers and monitoring within the relevant limits.

Cost is one consideration, not the only criterion. A control may be **required by law** or necessary to protect a critical service even where a simple expected-loss calculation appears unfavorable. A transfer agreement should be assessed for enforceability, coverage, **exclusions**, counterparty strength and the exposure the enterprise retains.

Step 4 Risk and Control Monitoring and Reporting

Monitor whether responses are implemented, controls work, **risk** changes, and thresholds are exceeded. Use incident trends, risk indicators, control indicators, testing and relevant business changes. Escalation should reach a person with authority to decide. Reassess after a new **supplier**, acquisition, major system change or significant incident.

Audit focus Sample risk register entries and trace them to assessments, **business owners**, implemented actions and monitoring. Challenge unjustified ratings, overdue actions and acceptance outside delegated authority. The **audit conclusion** should explain the business exposure rather than only the missing document.

2.5.3 Risk Analysis Methods

Qualitative analysis uses defined descriptive ratings such as low, medium and high. **Semiquantitative analysis** assigns scores or bands to those judgments. Quantitative analysis estimates numerical frequencies and consequences, often expressed in money. All depend on assumptions and evidence; a numerical answer is not automatically objective or precise.

Method	Useful when	Limitation
Qualitative	The enterprise needs a practical comparison and reliable financial data are limited.	Unclear scales or unsupported judgment can make ratings inconsistent.
Semiquantitative	A structured score helps compare factors or prioritize a portfolio.	Adding or multiplying arbitrary scores can suggest unjustified precision.

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Method	Useful when	Limitation
Quantitative	There is enough information to model event frequency and loss magnitude.	Input uncertainty, rare events and correlated losses can materially affect results.

Single loss and annualized loss expectancy

The simple asset-based model separates the loss from one event from the expected number of events in a year. Exposure factor (EF) is the proportion of asset value lost in one modeled event. Single loss expectancy (SLE) is that event's estimated loss. **Annualized rate of occurrence (ARO**) is the expected event frequency per year. **Annualized loss expectancy** (ALE) combines the two.

Measure	Formula or meaning
Asset value (AV)	The value used for the asset or exposure in the model.
Exposure factor (EF)	Proportion lost per event, expressed as a fraction from 0 to 1.
SLE	AV × EF.
ARO	Expected number of events per year; it may be fractional.
ALE	SLE × ARO.
Simple annual net benefit	Estimated annual loss reduction minus annual control cost.

Example: AV is $100,000 and EF is 25 percent. **SLE** = $100,000 × 0.25 = $25,000. If **ARO** is 2, ALE = $25,000 × 2 = $50,000 per year. A control reduces modeled ALE to $15,000 and costs $20,000 each year. The estimated annual loss reduction is $35,000 and the simple annual net benefit is $15,000. Document uncertainty and also consider compliance, operational and nonfinancial effects.

An **ARO** of 0.2 means an expected frequency of one event in five years under the model; it does not predict the actual year of occurrence. Do not multiply EF twice. When treatment changes both frequency and impact, recalculate the remaining loss using the revised **assumptions**.

Memory aid: One event: AV × EF gives **SLE**. One year: SLE × **ARO** gives ALE. Keep the units visible so that a one-time cost is not compared directly with an annual benefit without a common period.

Annual control net benefit = ALE before treatment − ALE after treatment − annual control cost. In the preceding example: $50,000 − $15,000 − $20,000 = $15,000 per year. The remaining $15,000 estimated annual loss is still exposure; reassess other consequences and control assumptions.

Interpreting simplified risk expressions

Some study material uses **asset × vulnerability × threat** as a risk mnemonic. These factors matter, but multiplying an asset's monetary value by undefined labels is not a universal quantitative method. A scoring model must define each scale and explain the result.

Residual risk is the exposure remaining after relevant controls or treatment. The shorthand "inherent risk minus control" is only conceptual unless both quantities have a defined, compatible measurement model. Estimate the remaining likelihood and consequences rather than subtracting a control name or an unsupported effectiveness percentage.

A probability applies to a defined event and period and lies between zero and one. **ARO** is an expected frequency per year and can exceed one. Probability × consequence and frequency × loss can both

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support expected-loss reasoning under their respective assumptions; neither formula removes uncertainty.

Auditor Perspective

Distinguish acceptance from inaction

Leaving a risk untreated is not necessarily authorized acceptance. Determine the **exposure**, obligations and delegated authority and obtain an **informed decision** by the proper owner.

Audit can assess or recommend treatment. It should not accept business risk on management's behalf.

Worked risk response and acceptance decision

Identify the objective, asset or process, threat event, vulnerability, consequences and existing controls. **Inherent risk** considers the exposure before the controls being assessed; **residual risk** is what remains after the selected controls or response. **Risk appetite** expresses broad willingness to pursue or retain risk; tolerance gives more specific acceptable boundaries. Definitions and measurement scales must be clear enough to support consistent decisions.

Example: A simplified annual-loss model estimates a PKR 4 million loss per event at 0.2 events per year, giving an expected annual loss of PKR 800,000. A proposed safeguard costs PKR 200,000 annually and reduces the modeled **exposure** to PKR 300,000. The modeled net annual benefit is PKR 300,000. This supports comparison, but uncertain **assumptions**, severe tail outcomes, **obligations** and operational effects also matter. Expected value alone does not decide acceptance.

Avoidance, reduction, sharing or transfer, and acceptance are response choices. Insurance can transfer defined financial **consequences** while leaving operational and reputational effects. An accountable owner accepts or escalates within delegated limits; a job title does not authorize unlimited out-of-tolerance acceptance. Audit challenges the **evidence**, response and authority and escalates as required, without becoming the **risk owner**.

Evaluate Who Can Retain the Remaining Exposure

Example. A business owner accepts the cost of a workaround after a supplier control failure. The modeled loss is within the owner's financial delegation, but the outage exposure exceeds a governing limit for the critical service. Compare each relevant dimension with the approved tolerance and authority. Review the controls, assumptions, proposed expiry and escalation record. Audit assesses whether the decision was informed and authorized. Management or the applicable governing authority decides how to treat the exposure.

2.6 Data Privacy Program and Principles

Privacy concerns how **personal information** is collected, used, shared, retained and made available to individuals. Security helps protect that information, but secure processing can still be unfair, excessive or outside an authorized purpose. A privacy program addresses those decisions across the information life cycle.

A **data subject** is an identifiable person to whom **personal information** relates. A controller determines the purposes and means of processing under the relevant framework; a processor processes information on a controller's behalf. These roles depend on actual decisions and activities, not merely the titles written in a contract.

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A privacy program needs sponsorship, accountable ownership, data visibility, applicable **requirements**, policies, training, individual-rights processes, supplier controls, incident arrangements and ongoing review. A data protection officer or **privacy officer** advises and monitors within the applicable mandate; appointing that person does not transfer all enterprise accountability to them.

Privacy principle	Practical meaning	Example
Lawfulness, fairness and transparency	Use an appropriate basis and explain the processing clearly.	Tell applicants why information is collected and how it will be used.
Purpose limitation	Use information for defined purposes and assess proposed changes.	Reassess use of employment records for a new profiling activity.
Data minimization	Collect and use what the purpose requires.	Do not request full identity documents where a less intrusive check is sufficient.
Accuracy	Maintain appropriate accuracy and correct errors.	Correct a verified customer address across relevant systems.
Storage limitation	Keep identifiable information only as justified by purpose and obligations.	Apply approved retention rules rather than keeping every record indefinitely.
Security	Protect information against inappropriate access, alteration and loss.	Restrict sensitive HR records and monitor privileged access.
Accountability	Be able to demonstrate that the required practices are implemented.	Maintain decisions, assessments, processing records and evidence of controls.

These are common **privacy** principles illustrated through GDPR-style terminology. The applicable jurisdiction determines precise duties, **exceptions** and time limits. Identify the actual requirement before treating a particular right, consent condition or notification deadline as universal.

Protected health information (PHI) is individually identifiable health information within the applicable scope of the US Health Insurance Portability and Accountability Act (HIPAA) framework. It can be electronic, paper or oral. A patient record held by a covered entity can qualify; not every health-related datum worldwide does. Check the applicable definitions and exclusions. [R35]

2.6.1 Privacy Documentation

Privacy documentation records both what the enterprise tells people and how it manages the processing internally. A clear notice supports transparency; an internal processing inventory supports control. Neither replaces the other, and a signed form does not make unnecessary or unlawful processing acceptable.

Types of Documentation

A **privacy notice** explains relevant processing to individuals in clear language. A consent record demonstrates an eligible person's choice where consent is the appropriate basis. The **personal information** inventory identifies what data exist, where they flow, their purposes, owners, recipients, safeguards and retention. Additional records support assessment, oversight and response.

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Record	Purpose	What an auditor checks
Privacy notice	Communicate processing, recipients, retention, rights and contact arrangements as required.	The notice is clear, accessible and consistent with actual processing.
Consent form and record	Record a specific informed choice where required.	Scope, time, affirmative action, relevant notice version and withdrawal handling.
Personal information inventory / processing records	Describe data, purposes, systems, owners, recipients and retention.	Coverage includes suppliers, copies, exports and unintentional collection.
Privacy impact assessment (PIA)	Identify effects on individuals and plan safeguards before or during change.	Decisions, risks, owners and remediation are acted on.
Data protection impact assessment (DPIA)	A legally defined impact assessment in frameworks that require it for relevant high-risk processing.	Applicability, required assessment and follow-up match the governing requirements.
Processing agreements	Define the parties' duties, instructions, safeguards and assistance.	Subprocessing, incidents, transfers, assurance and exit obligations.
Rights, incident and training records	Demonstrate responses and staff preparation.	Identity checks, timeliness, decisions, escalation and completed actions.
Privacy governance reports	Give leadership a view of compliance and unresolved exposure.	Reported measures are reliable and significant concerns reach decision-makers.

Example: A customer application collects location data although its notice describes only order delivery details. Encryption protects the data in storage, but it does not explain or authorize the extra collection. Review necessity, purpose, **lawful basis** and transparency before treating the **security controls** as a complete answer.

Privacy principles and frameworks

Privacy references help structure a program, but they have different purposes and do not all impose legal duties. Select applicable **requirements** separately, then use a suitable framework to make the controls complete and consistent.

Reference	Purpose and distinction
OECD Privacy Principles	Eight principles: collection limitation, data quality, purpose specification, use limitation, security safeguards, openness, individual participation and accountability.
AICPA Generally Accepted Privacy Principles	A privacy-management and assessment reference covering information handling across its life cycle.
ISACA Privacy Principles	A reference for organizing privacy responsibilities, practices and program evaluation.

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Reference	Purpose and distinction
ISO/TS 17975	Health-informatics guidance addressing consent in collecting, using or disclosing personal health information.
NIST Privacy Framework	A voluntary approach to privacy risk management; separate from the NIST Cybersecurity Framework.

2.6.2 Audit Process

Understand the privacy program's **scope**, applicable **obligations**, processing activities and responsible people. Assess the inventory and data flows before selecting samples. Trace representative information from collection through use, supplier handling, retention and disposal. Include new services and processing outside the core IT inventory.

Test whether notices reflect reality, processing has a suitable basis, access and sharing are controlled, rights requests are handled, and **retention** and **incident** processes operate. Examine both design and operation. Follow identified gaps through to management decisions and governing-body reporting where their significance requires it.

Audit focus Use a combination of processing records, contracts, impact assessments, configuration inspection, sample requests, deletion evidence and interviews. Avoid collecting unnecessary live **personal information** as **audit evidence**. A privacy finding should describe the affected individuals, processing and consequences as well as the missing control.

Auditing privacy throughout the information lifecycle

Privacy concerns **appropriate processing** and effects on people as well as confidentiality. Establish the applicable purpose, authority or legal basis, notices, collection limits, sharing arrangements, retention and individual rights. **Security controls** remain necessary, but encrypted processing can still be excessive, unfair or incompatible with the permitted purpose.

Use the chosen framework accurately. GDPR principles, OECD principles and the **NIST Privacy Framework** are related sources with different **scope** and status, rather than interchangeable vocabulary. NIST Privacy Framework 1.0 organizes outcomes using Identify-P, Govern-P, **Control-P**, Communicate-P and Protect-P. Check the version used before treating a draft or later document as an established requirement. [R16]

Example: Customer support recordings are proposed for model training. Review whether the new use is permitted, which people and data are affected, the provider's processing terms, access, **retention**, transfers and deletion arrangements. Removing names alone may leave identifiable content. Pseudonymized data can remain **personal information** when linkage is possible. Test controls with real processing flows and evidence of rights handling rather than relying solely on a published **privacy notice**.

Trace a Personal Information Request

Example. A customer asks for access to personal information. The service desk can find the account profile, but relevant recordings are held by a provider. Trace the request through identity verification, search, authorized release, documented exclusions and provider assistance. Check how the applicable requirements and deadlines are established. Test that a response contains only the requesting person's information and that unnecessary copies are removed under the approved retention arrangements.

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Encryption of the response supports confidentiality, while scope, purpose and authorized release address separate privacy questions.

2.7 Data Governance and Classification

Data governance establishes **decision rights** and accountability for data quality, meaning, use and protection. Classification turns differences in value and sensitivity into handling requirements. Both are business responsibilities supported by IT, security, privacy and data specialists.

2.7.1 Data Inventory and Classification

An **asset** is something the enterprise values and needs to manage. An information asset can be a customer dataset or business record; supporting IT assets include the applications, devices and services used to process it. Define the inventory scope and relationships so information ownership is not confused with technical custody.

A **data inventory** identifies information assets, systems, owners, purposes and flows. Classification assigns categories based on business value, **confidentiality**, integrity, availability and legal or contractual requirements. The classification scheme should be simple enough to use and detailed enough to distinguish materially different handling needs.

The owner determines or approves classification and access expectations; custodians implement the necessary safeguards. Labels help users apply handling rules, but a label alone does not enforce access, prevent copying or apply **retention**. Review classification when the information or its use changes.

Illustrative class	Meaning	Typical handling consideration
Public	Approved for unrestricted disclosure.	Maintain integrity and publish through approved channels.
Internal	Intended for authorized internal use.	Limit routine external sharing and apply the organization’s access rules.
Confidential	Unauthorized disclosure could materially harm the enterprise or people.	Use tighter access, approved sharing and appropriate protection.
Restricted	A particularly sensitive category under the enterprise scheme.	Apply the strongest relevant access, monitoring and handling controls.

These labels are examples, not universal legal categories. Personal data are not automatically a single sensitivity class: a public business contact and a health record have different **exposure**. An operational dataset may need strong **integrity** and availability even when its confidentiality requirement is low.

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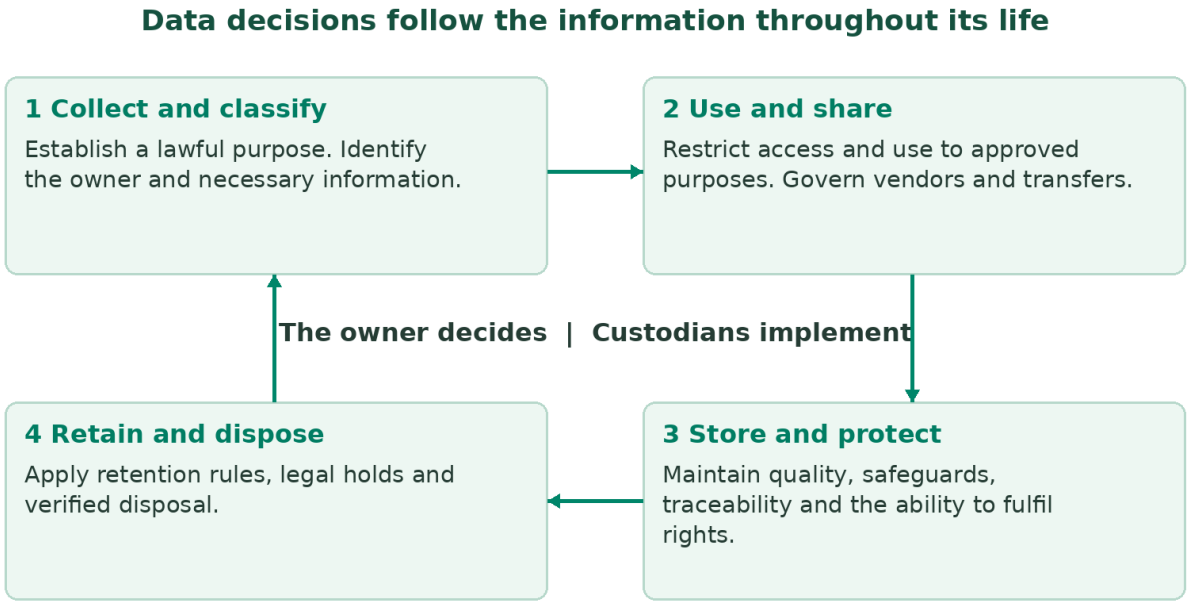


Figure 2.3 Classification and ownership guide handling throughout the data life cycle, not only at collection.

Data quality and traceability

Data quality means that information is fit for its intended use. Metadata explains the information; lineage traces its origin and transformations. Define those foundations before measuring completeness, accuracy, consistency or timeliness. Section 2.2.7 introduces the related BI example.

Data governance also addresses **completeness**, accuracy, consistency, timeliness and fitness for purpose. Define shared terms, identify authoritative sources, maintain metadata and lineage, and assign stewards to resolve quality issues. A complete data inventory supports privacy assessments, access reviews, architecture decisions and reliable BI.

Example: Finance and operations maintain different supplier bank details. A **data owner** defines the authoritative record and **change approval**; IT implements synchronization and access control; a steward monitors inconsistencies. Without these decisions, a new dashboard simply distributes conflicting data faster.

2.7.2 Legal Purpose, Consent and Legitimate Interest

A defined business purpose and an applicable **lawful basis** are related but different. “Useful for marketing” describes an intended purpose; it does not by itself establish legal **permission**. Identify the applicable law and basis before processing and maintain evidence of the decision. **Consent** is one possible basis, not a universal requirement or a cure for every processing problem.

Concept	Meaning	Important distinction
Legal purpose	A specified purpose compatible with the applicable legal requirements.	A legitimate business objective alone does not establish every legal permission needed.
Consent	An eligible individual’s informed and specific choice where the framework permits reliance on it.	Record the choice and support withdrawal; bundled or coerced agreement can be inadequate.

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Concept	Meaning	Important distinction
Legitimate interest	A basis available in some frameworks subject to defined conditions and balancing.	Document the interest, necessity and effects on the individual; do not assume it always applies.
Other lawful bases	Depending on the framework, grounds can include contract, legal obligation, vital interests or public task.	Determine the basis for each activity rather than assuming all processing needs consent.

Where **consent** is relied on, examine whether the choice is sufficiently informed, specific and freely made under the applicable rules. For legitimate interests, establish the interest pursued, necessity of the processing, and the balance with the person's rights and reasonable expectations. Special categories of information or particular jurisdictions may impose additional conditions.

2.7.3 Data Subject Rights

Privacy frameworks may give individuals rights to information, access, correction, deletion, restriction, portability, objection and protections concerning certain automated decisions. Their **availability**, limits and response periods depend on the applicable law and circumstances. Deletion is not an unconditional instruction to destroy records that must legally be retained.

Request or right	Practical control
Information and access	Provide the required explanation and eligible information through a secure channel.
Correction	Verify the issue and correct relevant records and downstream copies as required.
Deletion or restriction	Determine eligibility, retention conflicts and the systems or recipients affected.
Portability	Provide eligible information in the required usable form where the right applies.
Objection or withdrawal	Evaluate the request and stop or adjust relevant processing when required.
Automated decision safeguards	Identify applicable decisions, required explanations and routes for review or challenge.

A rights process needs intake channels, proportionate **identity verification**, an owner, deadline tracking, system and supplier coverage, and documented decisions. Identity checking prevents disclosing **personal information** to an impostor. Excessive verification can itself collect unnecessary information.

Transborder Data Flow

A cross-border data flow can occur through hosting, replication, support access, analytics or subcontracting. Map the actual locations, parties and access arrangements, then determine the relevant transfer **requirements** and safeguards. The location of a **supplier**'s head office does not establish where data are processed.

Review onward transfers, legal access risks, contractual commitments, technical safeguards and the ability to fulfil rights or recover data. A security **certification** does not automatically resolve transfer legality, residency restrictions or conflicting laws.

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Audit focus Trace a **sample** rights request across applications, archived records and suppliers. Check the legal basis for any refusal or exception and the **evidence** of completion. For transfers, corroborate contractual statements with architecture, subprocessor lists and operational access information.

NIST Privacy Framework functions supporting rights

The **NIST Privacy Framework** 1.0 uses Identify-P, Govern-P, **Control-P**, Communicate-P and Protect-P. Control-P supports granular processing decisions; Communicate-P supports understanding and dialogue. Use the relevant version when applying its category identifiers.

Category	Practical focus
CT.PO-P	Rules, roles and authorization for processing.
CT.DM-P	Manage data and enable individual participation.
CT.DP-P	Reduce unnecessary association of data with people.
CM.PO-P	Rules and responsibilities for transparent communication.
CM.AW-P	Make practices and risks understandable to individuals and organizations.

Auditor Perspective

Classify according to impact

The **business owner** decides classification and access expectations using the approved scheme and consequences of compromise. IT implements the assigned safeguards.

A **confidentiality** label alone may miss integrity and availability needs. Consider aggregated information and the complete data flow.

Testing ownership lineage and classification in use

Data governance assigns **accountability** for definitions, quality, permitted use, lifecycle and protection. Owners make business decisions; stewards coordinate definitions and quality; custodians implement handling under assigned responsibilities. One person can hold several roles, but accountability and conflicts must remain clear.

Lineage shows how data moves and changes from **origin** to use. A dashboard can be accurately calculated from the wrong source or from a field whose meaning changed. Trace important measures back through **interfaces** and transformations to authorized source records. Assess **completeness**, accuracy, consistency, validity, uniqueness and timeliness according to their business use, rather than assuming one quality percentage covers every purpose.

Example: A customer-risk report treats an empty risk field as low risk after an interface conversion. Matching record counts would not expose the semantic error. Inspect mapping rules, null handling, exception queues and owner **approval**. Classification should guide access, sharing, **retention** and disposal across copies, extracts and derived information. Discovery and **monitoring** can identify unclassified sensitive content; an enterprise need not wait for a manual label on every file before detecting exposure.

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Follow a Definition Across an Interface

Example. A credit report labels accounts with a missing closure date as active, while the source system uses the missing date for migrated historical accounts. Counts reconcile, but the resulting definition differs. Inspect the approved **metadata**, trace the transformation through **data lineage**, and compare selected source and target records. Ask the data owner to resolve the definition and approve the correction. Then test reports and downstream decisions that depend on the field. A record-count reconciliation checks quantity. The lineage test addresses meaning and accuracy.

2.8 IT Resource Management

IT resource management matches people, technology, information, finances and service **capacity** to business priorities. The objective is to sustain the capabilities the enterprise needs and obtain value from them, while maintaining appropriate control.

2.8.1 Value of IT

IT value is the contribution technology makes to **business outcomes**, considered alongside cost and risk. **Benefits** can include revenue growth, faster service, fewer errors, improved resilience or necessary compliance. A system delivered on time is an output; improved order fulfilment is an outcome. Assign a business benefit owner and measure whether the outcome occurs.

2.8.2 Implementing IT Portfolio Management

A portfolio groups investments, programs, projects or services so they can be selected and managed together. Compare strategic contribution, expected **benefits**, costs, risk, dependencies and resource constraints. **Portfolio management** can defer, stop or redirect an investment when assumptions change; it is more than maintaining a project list.

Consider the full life cycle, including ongoing service and retirement. A technically successful project may deliver little value if users do not adopt it or dependent process changes are not funded. **Business owners** remain responsible for realizing **benefits**.

Example: Three divisions request new analytics products while a common data-quality issue prevents reliable reporting. Portfolio review may prioritize the shared data capability before buying three tools. The decision should document expected benefit, **dependencies**, resource limits and the **exposure** of delay.

2.8.3 IT Management Practices

Management turns direction into operating plans, assigned responsibilities, service agreements, budgets and review routines. Maintain asset and service inventories, **capacity** forecasts, knowledge, skills and **supplier** visibility. Monitor deviations and make authorized changes rather than treating an approved annual budget as a permanent answer to changing demand.

2.8.4 Human Resource Management

People controls address competence, **integrity**, availability and **accountability** throughout employment. Apply them to employees and relevant contractors. HR and line management define the employment process; IT and security implement the associated access, training and technical safeguards.

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Practice	Purpose and audit evidence
Recruiting and hiring	Defined duties, verified qualifications, proportionate lawful screening and conflict checks.
Employee handbook	Communicated conduct, confidentiality, acceptable use, responsibilities and reporting routes.
Training	Role-based preparation, security awareness, skills development and evidence of completion.
Scheduling and time reporting	Coverage, reasonable workload, accurate project or service allocation and independent review where relevant.
Terms and conditions	Relevant confidentiality, intellectual property, acceptable use and continuing obligations.
Expectations during employment	Clear responsibilities, permitted access, competence and conduct expectations.
Performance management	Agreed goals, fair review and corrective development rather than only activity counts.
Disciplinary actions	Consistent authorized handling, appropriate investigation and documentation.
Promotion and transfers	Reassessed duties, authority and access; previous privileges removed when unnecessary.
Required vacations	An actual uninterrupted absence with work covered by another person can expose concealed irregularities.
Retention and succession	Retain needed skills and prepare qualified replacements for critical roles.
Termination	Timely removal of access, return of assets, notification and preservation of required records.

Key person risk and incompatible duties

Cross-training builds backup capability; **job rotation** moves people between duties; **succession planning** prepares replacements for critical roles. They address related but different dependencies. Granting every backup administrator permanent unrestricted access may reduce staffing dependence while creating a control conflict.

Mandatory vacation and rotation can help expose irregularities that depend on continuous concealment. They do not guarantee fraud **detection**, particularly where people collude or reviews are ineffective. Check that work is genuinely handed over and that the absent person does not keep operating the process remotely.

Termination and role changes

Use an authoritative HR event to trigger a coordinated process. Disable relevant network, cloud, application, remote, physical and **privileged access** at the appropriate time. Recover devices and credentials, transfer business ownership of records, address tokens and shared secrets, notify responsible teams, and preserve records subject to **retention** or investigation needs.

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For an involuntary departure with elevated **exposure**, coordinate HR, management, legal and security so access removal and communication occur at the appropriate time. Removing a directory account alone may leave independent **SaaS** sessions or vendor access active.

Audit focus Compare selected joiners, movers and leavers with actual **privileges** and completion times. Verify the **population** across HR and contractor records. Examine whether backups and succession arrangements can operate during an absence without uncontrolled access.

2.8.5 Enterprise Change Management

Enterprise **change management** prepares the organization to adopt changes in processes, technology, structure or behavior. It addresses sponsorship, affected people, communication, training, readiness, resistance and sustained adoption. **Technical change control** governs authorized alterations to systems; the two are complementary.

Involve users and **process owners** early, assess effects and **dependencies**, explain responsibilities, and check readiness. Training attendance is only one measure; assess whether people can perform the changed work and whether the intended business benefit is being realized.

Example: A new procurement system is technically stable, but employees continue ordering outside it because **approval** roles and training are unclear. More server **capacity** does not address the adoption failure. Management must resolve process ownership, prepare users and monitor actual use.

2.8.6 Financial Management Practices

Financial management plans, records, allocates and evaluates IT spending. Budgets should reflect business priorities, recurring **obligations**, projects and future **capacity**. Compare actual and committed spending with approved amounts and investigate variances. Low spending is not automatically favorable if critical maintenance or controls were deferred.

Concept	Meaning and audit consideration
Cost allocation	Assign costs using a transparent basis related to consumption, capacity or agreed service use.
Chargeback	Charge consuming business units for allocated IT costs; inspect the allocation method and approvals.
Showback	Report consumption and allocated costs without charging them; check whether the information influences decisions.
Capital expenditure / operating expense	Different accounting treatment for qualifying assets and recurring costs; evaluate against the applicable accounting policy.
Capitalization of software costs	Recognize eligible costs as an asset under the relevant criteria, rather than automatically capitalizing all development spending.
Total cost of ownership (TCO)	Consider acquisition, operation, support, security, change and exit costs over the relevant life.
Life cycle cost (LCC)	Estimate costs across the investment's stages, including transition and retirement.

A **business case** may use ROI to compare net benefit with investment, NPV to discount net future cash flows, IRR as the rate at which NPV is zero, and payback as the time to recover the investment. These

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measures address different questions. Payback alone omits later **benefits** and, in its simple form, the time value of money. Review **assumptions** and risk, not just the headline result.

2.8.7 Information Security Management

Information security management implements the security direction through a coordinated program of policies, **risk assessment**, controls, awareness, incident arrangements and **monitoring**. A management system establishes roles, repeatable processes, evidence and improvement. Security technology is part of that system, not its entirety.

Review whether resources match the information assets, **obligations** and risk. Ensure security is integrated into projects, procurement and operations. Security managers report exposure and control performance; business and **risk owners** make authorized decisions within enterprise governance.

Evaluating people capacity cost and knowledge

Resource management includes skills, staffing, knowledge, infrastructure, applications and financial **capacity**. Compare current and forecast needs with the enterprise's approved service and change commitments. Concentrated expertise, unsupported technology and unplanned license costs can threaten delivery even when current utilization appears low.

Assess cross-training, documented procedures, succession, contractor dependence and access removal. **Mandatory vacation** and rotation can expose concealment-dependent irregularities when work is actually handed over and independently checked; they do not guarantee fraud **detection**. Knowledge transfer should be demonstrated by another authorized person completing the task.

Example: Only one engineer can recover a business-critical **database**. A signed recovery plan and an unused training budget do not resolve the dependency. Verify trained alternate coverage, current instructions and an observed recovery exercise. Cost comparisons should include acquisition, operation, support, integration, **migration**, security and exit costs over the relevant period. A low initial cloud price can be outweighed by transfer charges or retained staffing. Audit evaluates how management measures these tradeoffs and whether decisions remain aligned with business **requirements**.

Verify a Resource Allocation Before Charging Users

Example. IT allocates shared cloud costs using the number of accounts, although a small batch processing team uses most of the compute time. Inspect the agreed allocation basis, source consumption data, shared costs, approvals and how disputes are handled. A **showback** report communicates use and allocated cost. A **chargeback** applies a charge under the approved method. Evaluate whether the measure supports the intended behavior and whether changing consumption or priority would require a revised allocation.

2.9 IT Vendor Management

Vendor management governs the relationship from selection through delivery, change and exit. It must address both the service purchased and the business **exposure** created by depending on another organization. **Accountability** for the enterprise's objectives and obligations remains with the enterprise.

2.9.1 Sourcing Practices

Sourcing determines how a capability will be provided. Insourcing uses internal resources; outsourcing obtains services from another organization; **selective or smart sourcing** uses a deliberate

combination. Location terms such as onshore, nearshore and offshore describe where delivery occurs, not the strength of the **supplier**'s controls.

Evaluate strategic importance, **capability**, cost, control, flexibility, resilience, legal **requirements** and exit feasibility. A low bid can conceal transition costs, retained staffing, integration work or lock-in. The best model depends on business need and risk, not a rule that internal or external delivery is always preferable.

Sourcing choice	Potential benefit	Exposure to assess
Insourcing	Direct control and retention of knowledge.	Skills, capacity, key-person dependence and ongoing investment.
Outsourcing	Access to expertise, scale or delivery capability.	Dependency, visibility, control gaps and contract or transition failure.
Selective sourcing	Choose delivery arrangements by service or function.	Integration, hand-offs and unclear accountability between suppliers.
Onshore / nearshore / offshore	Access to different locations, skills and delivery hours.	Jurisdiction, language, time zone, geopolitical exposure and data flows.
Cloud services	Rapid access to shared infrastructure, platforms or applications.	Shared responsibilities, configuration, concentration, consumption cost and exit.

Delivery location and staffing model

Onsite means work at the customer's premises; offsite means work elsewhere. These are different from onshore, nearshore and offshore, which describe geographic relationships. A hybrid staffing model mixes internal and **supplier** personnel. Evaluate access, supervision, data flows and **accountability** for the actual arrangement; location terminology alone does not establish its risk.

2.9.2 Outsourcing Practices and Strategies

Due diligence is a proportionate investigation of a proposed supplier, service or relationship before deciding to rely on it. Its purpose is to test the assumptions behind selection, including capability, financial viability and control responsibilities. A completed questionnaire is useful only when its answers are assessed and important claims corroborated.

Before selection, classify the service's criticality and data, assess alternatives and **retained responsibilities**, and perform proportionate due diligence. Review capability, control environment, security, privacy, **financial viability**, resilience, subcontractors, location and references. A vendor's large size or reputation does not replace this assessment.

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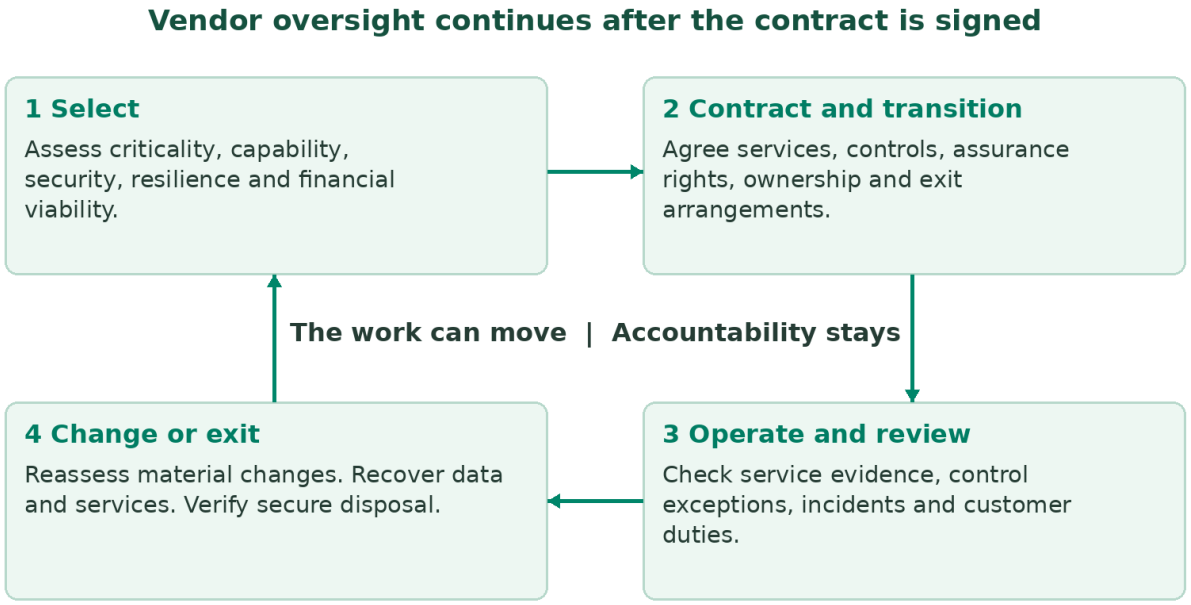


Figure 2.4 Due diligence, contracting, monitoring and exit form one continuous vendor management process.

Contracts and assurance rights

The contract defines enforceable duties and remedies. A **service level** agreement (SLA) specifies measurable **service commitments**, scope and measurement rules. An operating level agreement (OLA) coordinates supporting internal teams; other supporting agreements may govern subcontracted dependencies. Align the agreements so end-to-end commitments can actually be delivered.

Contract element	What should be clear
Scope and responsibility	Services, assets, interfaces, data ownership and the duties retained by each party.
Service levels	Availability, performance, response, recovery and support commitments; measurement windows and exclusions.
Security and privacy	Applicable requirements, access, confidentiality, handling, location and processing instructions.
Incidents and continuity	Notification, evidence preservation, cooperation, recovery capability and test participation.
Assurance and access	Right to audit or suitable independent evidence; access to information needed for oversight and regulators.
Subcontracting and change	Approval or notification, flow-down requirements, material changes and ongoing visibility.
Commercial and legal terms	Charges, remedies, liability, intellectual property, dispute handling and applicable law.
Termination and exit	Data return format, transition assistance, retention, verified deletion and continuity during migration.

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A right-to-audit clause creates an assurance route; it is useful only if it can be exercised and provides adequate coverage. Independent reports can support oversight when appropriate, but do not automatically replace all audit or regulatory **access rights**. The **most** significant contract gap depends on the service and exposure in the question.

A **nondisclosure agreement (NDA)** establishes agreed obligations over confidential information and permitted disclosure. Evaluate applicable parties, scope and handling requirements; it complements access and data-handling controls rather than replacing them.

Industry Standards and Benchmarking

Compare suppliers using relevant standards and comparable **performance** evidence. A certificate concerns a defined scope and management system; verify that the purchased service and locations are covered. Benchmark costs and **service levels** using consistent definitions and operating conditions. A cheaper or apparently faster peer service may involve materially different responsibilities.

Globalization Practices and Strategies

Global delivery can offer specialist skills and operating-hour coverage, but introduces additional jurisdictions, communication needs, cultural differences and dependency paths. Map where data and support access actually go, including **subcontractors**. Review applicable requirements, political and economic exposure, **concentration risk** and practical recovery or exit arrangements.

Outsourcing and Third-Party Audit Reports

A **SOC 1 report** addresses service-organization controls relevant to customers' internal control over financial reporting. A **SOC 2 report** addresses controls against applicable Trust Services **Criteria** for the described system. A SOC 3 report communicates assurance more generally with less detail. The subject and audience differ; SOC 2 is not simply a higher grade of SOC 1.

For SOC 1 or SOC 2, a Type 1 report addresses the description and suitability of control design at a specified date; a **Type 2 report** also addresses operating effectiveness over a specified period. Read the actual **scope**, criteria, opinion, tests, exceptions and dates. Consider subservice organizations, report-period gaps and **complementary user entity controls** (CUECs) the customer must implement.

Example: An outsourced payroll report covers January through June, but the customer needs **assurance** for the full year. The report may be useful for that period; it cannot establish July-to-December operation by itself. Obtain appropriate additional **evidence** and assess changes. A provider's **bridge letter** is a management statement, not an independent extension of the examination.

Software escrow, commonly implemented as **source-code escrow**, places agreed materials with an independent holder for release under defined conditions, such as specified **supplier failure**. Check release triggers, current complete deposits, build documentation, dependencies and rights to use the code. Escrow is of limited value if no one can build, maintain or operate the released software.

Service level management

Service level management (SLM) is the continuing process of agreeing, measuring, reviewing and improving **service commitments**. It includes defining business needs, negotiating achievable targets, assigning measurement responsibilities and addressing breaches. An SLA is a documented agreement produced and maintained through that process; signing one does not establish effective SLM.

For an outsourced service, trace one business requirement into its SLA target, measurement method, performance evidence, review and **corrective action**. Check dependencies and exclusions as well as

averages. A supplier can meet an infrastructure uptime target while the customer's business transaction still fails. Section 4.10 covers operational **service level** management.

2.9.3 Cloud Governance

Cloud services make shared computing capabilities available over a network with provisioning and consumption arrangements. Infrastructure as a service (**IaaS**) supplies infrastructure resources; platform as a service (**PaaS**) supplies a managed platform/runtime; software as a service (**SaaS**) supplies an application. The model helps locate responsibilities, while the actual service and contract determine the boundary. [R33]

Cloud governance establishes approved use, **decision rights**, policies, **risk assessment**, ownership, service oversight and exit arrangements. Shadow IT is technology acquired or used outside established oversight. Discover its business purpose and exposure, then apply proportionate approval and control rather than assuming blanket blocking resolves the need.

Model	Provider typically manages	Customer still needs to manage
IaaS	Underlying facilities, hardware and virtualization.	Guest operating systems, applications, data, identities and relevant configurations.
PaaS	Infrastructure and the offered platform or runtime.	Application code, data, identities and configurable service responsibilities.
SaaS	The supplied application and underlying service stack.	Authorized use, data, users, configuration, integration and retained obligations.

Exact responsibilities depend on the service and contract. The provider’s security does not automatically secure customer **permissions**, sharing settings or integrations. Review cloud inventories, identity federation, **privileged access**, logging, spend controls, data location, backup and export capability.

2.9.4 Governance in Outsourcing

Assign an internal service owner and relationship manager, establish a **governance** calendar, define reporting and escalation, and keep business stakeholders involved. Record decisions and unresolved issues. Ongoing governance should consider service value, risk, cost, satisfaction and **supplier** viability rather than only SLA credits.

Review material changes to service demand, processing location, **subcontractors** and technology. Ensure staff retained within the enterprise can challenge the **supplier**, interpret evidence and manage a transition. Moving the work must not eliminate the enterprise’s ability to understand the service it depends on.

2.9.5 Capacity and Growth Planning

Capacity planning forecasts the resources needed to meet expected demand and business growth. Include compute, storage, network, licenses, people, funding and supplier limits. Consider seasonal peaks, growth plans, resilience requirements and provisioning lead time. **Average utilization** can hide a bottleneck that affects critical transactions.

Use demand forecasts, historical trends and workload tests to establish thresholds and actions. Cloud scalability does not remove quota, cost, **database**, staffing or architectural constraints. Evaluate **capacity** within the approved budget and service expectations.

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Example: The retailer's average monthly server usage is low, yet checkout fails during a short annual sale. Review peak demand, the application and **database** bottlenecks, load-test **assumptions** and the time needed to scale. Purchasing general storage would not address a transaction-processing limit.

2.9.6 Third-Party Service Delivery Management

Monitor reports, incidents, changes, assurance and service outcomes against the agreement. Review recurring failures and verify **corrective action**. An SLA target met through broad **exclusions** may not reflect the business experience. Reconcile supplier reports to independent logs and customer records where appropriate.

Monitoring and Review of Third-Party Services

Hold scheduled reviews with the relevant owners, assess security and service evidence, follow unresolved **exceptions** and reassess significant risks. Monitor **financial viability** and concentration as well as technical delivery. Several important suppliers may depend on the same cloud platform or subcontractor.

Managing Changes to Third-Party Services

Changes to the **supplier**'s systems, staff, locations or **subcontractors** can change the risk and compliance position. Require the relevant notification or approval, reassess the effect, and update agreements and controls when needed. A relationship established with good due diligence can still become unacceptable after a material change.

Service Improvement and User Satisfaction

Use incidents, service trends, complaints and user feedback to agree improvements. Distinguish response time from resolution time and service availability from successful business transactions. Assign actions, owners and completion evidence. At exit, **test data** export and transition, revoke access, settle **retention** duties and obtain suitable evidence of deletion.

Audit focus Trace a critical service from **due diligence** through contract, transition, periodic review and exit planning. Check retained customer controls, not only the **supplier**'s report. Determine whether escalation and remediation worked when an agreed level or control failed.

Auditor Perspective

Supplier assurance has boundaries

A third-party report helps only if its **scope**, period, **criteria**, exceptions and customer responsibilities fit the service. Read those boundaries before relying on the opinion.

Contractual rights and independent reporting support oversight; customer control obligations still need evidence.

Following supplier risk through subcontracting and exit

Supplier oversight begins before selection and continues through contracting, delivery, change and exit. Assess **criticality**, financial and operational capability, security, privacy, concentration, subcontractors, location and legal obligations. Due diligence should match the actual service and risk rather than apply the same questionnaire to every vendor.

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Contracts should define measurable services, responsibilities, incident cooperation, access to assurance, data ownership and permitted use, **retention**, deletion, continuity, portability and **termination**. A right-to-audit clause is useful only if it can provide timely relevant evidence. **Service credits** compensate defined failures but do not restore a critical service. A provider certificate or report requires scope and exception analysis.

Practical extension: Supply-chain **integrity** includes software **dependencies**, build services, update channels and privileged remote support. Examine provenance, release verification, **vulnerability** response and the ability to revoke compromised access. An SBOM identifies components but does not certify a product secure. Example: Two contracted suppliers rely on the same cloud region and identity provider. Count the shared failure domain when assessing concentration. Test exit feasibility, including formats, keys, interfaces, retained evidence and independently verified deletion where required. [R13]

Independence in cloud assurance

When using or commissioning cloud assurance, assess the evaluator's **independence** for the actual subject, period and services involved. An auditor's work for a **CSP** does not by itself establish either independence or an unavoidable conflict in every tenant engagement. Examine overlapping responsibilities, **self-review** and other relationships, disclose relevant impairments and apply the requirements and safeguards explained in section 1.1. Keep the provider's controls and the tenant's retained controls distinct.

Check the Exit Route During the Contract Term

Example. A customer can download its records from a hosted service, but the export excludes audit history and attachments needed to continue processing. Read the contract and exit plan for format, completeness, timing, assistance, retention and deletion duties. Use an authorized test export and verify that a replacement process can interpret the data, including identifiers and relationships. Assess any subcontractor dependencies and restrictions. A successful download demonstrates delivery of a file. The validation shows whether the business can actually use the exported information.

2.10 IT Performance Monitoring and Reporting

Performance monitoring translates strategy and risk expectations into information that supports decisions. A good measure has a defined purpose, calculation, owner, source, period, target and action threshold. Reports should help the reader understand results and act on deviations.

2.10.1 Key Performance Indicators

A key **performance** indicator (**KPI**) measures performance relevant to an objective or process. A key goal indicator (**KGI**) describes whether a desired outcome has been achieved. The distinction depends on the level: an operating target for one team may be an input to a wider business outcome. Do not classify every KPI as automatically leading or every count as useful.

2.10.2 Key Risk Indicators

A key **risk** indicator (KRI) signals **exposure** or a change in the conditions associated with risk. It should connect to a risk statement and threshold, with an owner and response. An increase in overdue critical vulnerabilities can signal rising exposure; it does not by itself quantify the probability of a future breach.

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2.10.3 Key Control Indicators

A key control indicator (**KCI**) measures the operation or effectiveness of a control. For example, the proportion of privileged accounts reviewed by their due date measures review coverage. Whether those reviews actually identify and remove inappropriate access may require additional **evidence**. A completed checklist is not the same as effective challenge.

Measure	Question	Retailer example
KPI	How is the process or service performing?	Percentage of customer orders processed within the service target.
KGI	Was the desired outcome achieved?	The agreed reduction in failed checkouts was achieved this quarter.
KRI	What signals increased exposure?	Number of unresolved high-risk supplier control exceptions beyond their deadline.
KCI	Is a particular control operating as intended?	Percentage of required supplier reviews completed with exceptions followed up.
CSF	What must go well for the objective to succeed?	Reliable stock information and sufficient peak transaction capacity.

Example: A dashboard reports 100 percent **supplier** review completion. Audit finds that reviewers upload reports without reading **exceptions** or checking CUECs. Completion is a useful activity measure, but it does not establish that the control achieves its purpose. Refine the measure and test review quality.

2.10.4 Performance Optimization

Optimization improves **performance** in relation to business needs, resource constraints and risk. Understand the baseline and **root cause** before selecting an improvement. Leadership support, clear goals, reliable data, employee involvement and continuing review are important critical success factors (CSFs). A process can become faster while becoming less controlled.

Methodologies and Tools

Use suitable methods to understand the work, identify bottlenecks and evaluate improvements. Benchmarking compares performance with a relevant peer or reference. **Root cause analysis** investigates why a problem recurs. **Business process reengineering** (BPR) redesigns a process more fundamentally, while incremental improvement makes smaller repeated changes.

Capability concerns how well a particular process or practice performs. **Maturity** describes a broader progression in institutionalized, managed and improving practices under the chosen model. State which model is used; similar level names do not make different scales interchangeable.

CMMI maturity level	Recall meaning
0 Incomplete	Work may not be completed; the approach is not established.
1 Initial	Work occurs but is unpredictable and reactive.
2 Managed	Projects are planned and controlled.

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CMMI maturity level	Recall meaning
3 Defined	Organizational standards guide the work.
4 Quantitatively Managed	Quantitative understanding supports predictability and control.
5 Optimizing	Continuous improvement and adaptation are embedded.

The familiar **CMMI** progression is often taught as levels 1 through 5; the published model also identifies level 0 Incomplete. CMMI capability levels for individual practice areas use a different scale. A maturity **assessment** identifies current and target capability to prioritize improvement; the highest level is not automatically the right investment for every process.

Open Information Security Management Maturity Model

O-ISM3 is The Open Group's process-based approach to information security management and maturity. It connects defined security processes and performance measures with **business objectives** and improvement. Distinguish it from **CMMI**'s broader process capability focus and ISO/IEC 27001's management-system requirements. A maturity label alone does not prove that a particular control is effective.

2.10.5 Approaches and Techniques

Choose an approach suited to the objective and the nature of the problem. **Six Sigma** uses data and structured analysis to reduce defects and variation; DMAIC means Define, Measure, Analyze, Improve and Control. **Business agility** is the ability to adapt effectively to change. Agile practices use feedback, collaboration and incremental delivery; they do not remove the need for **governance**, authorization or evidence.

IT Balanced Scorecard

A **balanced scorecard** (BSC) links strategic objectives to measures across several perspectives. The traditional enterprise perspectives are financial, customer, internal process, and learning and growth. An IT adaptation commonly uses corporate contribution, user orientation, **operational excellence** and future orientation. These are related views of value, not two competing answers to the same wording.

Enterprise perspective	Typical IT perspective	Illustrative measure
Financial	Corporate contribution	Verified business benefits compared with the relevant life cycle cost.
Customer	User orientation	Business-user satisfaction and successful completion of the service needed.
Internal process	Operational excellence	Transaction reliability, incident recurrence and controlled delivery.
Learning and growth	Future orientation	Coverage of skills needed for the approved roadmap and reduction of obsolete dependencies.

The scorecard should tell a coherent strategy story: improve **capability**, improve the process, improve the user outcome, and demonstrate business contribution. Technical **availability** can support that story, but uptime alone cannot establish whether the investment creates the intended business benefit.

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IT Portfolio Management Versus Balanced Scorecard

Portfolio management selects, balances and steers investments and services. The **balanced scorecard** communicates objectives and measures progress. They complement each other: scorecard results may reveal a benefit shortfall that changes a portfolio decision, while portfolio changes may require revised objectives and measures.

Additional Approaches for Performance Measurement

Evaluate life cycle, **life cycle cost**, and benefit analysis together. A lower acquisition price may create higher support or exit costs; an apparent benefit may depend on adoption or changes outside IT. Compare alternatives over a consistent period and document **assumptions**. Use both financial and nonfinancial outcomes where they matter.

Build reporting from mission and strategy to objectives, measures and reliable sources. Define the **baseline**, targets, thresholds and responsibilities. Inspect trends and investigate causes. Communicate in language the audience can use; adding more technical measures does not necessarily improve oversight.

Audit focus Reperform selected **metric calculations**, reconcile source populations, inspect exclusions and test whether adverse results led to action. Consider **incentives**: teams can improve a reported closure rate by closing tickets prematurely without improving the service.

Interpreting performance and risk measures

A KPI measures progress or **performance** against an objective; a KRI indicates **exposure** or changing risk. The same measure may serve different purposes depending on its definition and use. Establish the owner, source, calculation, boundary, threshold, frequency and required response. An unexplained green indicator offers little assurance.

Example: Availability improves after maintenance outages are excluded from reporting. Recalculate using the agreed service window and understand both contractual treatment and **business impact**. Compare ticket closure speed with reopening rates and user outcomes; closing tickets quickly can hide poor resolution. Review trends and distribution as well as averages, especially for critical services or disadvantaged user groups.

NIST CSF 2.0 includes Govern, Identify, Protect, Detect, Respond and Recover. These functions organize concurrent and connected outcomes; they are not a mandatory linear incident sequence. A profile can identify current and target outcomes and support prioritized improvement. It is not by itself certification that controls operate effectively. Trace reported **performance** into governance decisions, funded **corrective action** and verification of outcomes. COBIT, balanced scorecards and service measures should support the organization's own objectives rather than replace them. [R14]

Recalculate a Measure After Its Definition Changes

Example. A security dashboard shows fewer overdue high-risk actions after the reporting query starts excluding actions reassigned to suppliers. Compare the metric definition and thresholds with the approved reporting objective. Reconcile the population before and after the query change, recalculate the measure using the agreed rule and trace significant exceptions to decisions. Explain the impact of the changed definition on trends. A **key risk indicator** should help the recipient see and respond to exposure, including actions awaiting a provider.

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2.11 Quality Assurance and Quality Management of IT

Quality arrangements help the enterprise deliver services and products that meet **requirements** consistently. Quality **assurance**, quality control and quality management work together, but answer different questions.

2.11.1 Quality Assurance

Quality assurance (**QA**) consists of planned, systematic activities that provide confidence **requirements** will be met. It emphasizes the processes used to produce and deliver the result. Examples include method reviews, process audits and checking whether required reviews occur. Quality control (QC) evaluates the output against requirements through inspection, testing or other checks.

The distinction concerns the purpose of the **activity**, not merely the team's job title. A QA department may coordinate product testing, but a test examining the delivered output is still serving a quality-control purpose.

Concept	Focus	Example
QA	Confidence in the process and its conformance.	Review whether development and release procedures require and retain independent testing.
QC	Whether the product or output meets requirements.	Test whether the released application calculates tax correctly.
QMS	The overall quality management system.	Quality policies, objectives, roles, methods, records and improvement arrangements.

Independence and release quality

A quality reviewer should not approve their own work or have an incompatible operational responsibility. QA arrangements can include standards, staff training, process reviews and checks that changes were authorized, tested and version-controlled before **release**. Small teams may combine roles, but still need suitably **independent review** and evidence that the review occurred.

2.11.2 Quality Management

Quality management directs and controls the enterprise's quality activities. A quality management system (QMS) brings together **policy**, objectives, responsibilities, resources, processes, evaluation and improvement. **ISO 9001** is a quality management system standard. Certification, where used, concerns its stated scope; it does not prove every output is defect-free.

Total quality management (**TQM**) emphasizes quality throughout the organization and continuing improvement with employee and customer involvement. Use feedback, defects, process reviews and **root cause analysis** to improve the system, rather than inspecting the same recurring defect at the end of every cycle.

Plan Do Check Act

Plan–Do–Check–Act (**PDCA**) is an improvement cycle. Plan an objective and change; implement it; evaluate the results against expectations; and standardize an effective change or revise the approach.

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Record owners and **evidence** of improvement. Closing an action because a document was issued does not show the recurring problem was corrected.

2.11.3 Operational Excellence

Operational excellence sustains reliable, effective and efficient service through clear ownership, sound processes, appropriate controls and continuing improvement. It considers customer outcomes, **resilience**, quality and cost together. It is not achieved simply by minimizing headcount or automating an unstable process.

Example: A **service desk** reduces average ticket duration by repeatedly closing unresolved cases. The indicator improves while user effort and repeat incidents rise. An effective improvement measures successful resolution and recurrence, investigates the **root cause**, and adjusts the workflow and incentives.

Audit focus Review the quality **policy**, standards, review independence, defect records, customer feedback, corrective actions and measured results. Trace a recurring issue through its cause, approved improvement and later evidence. Keep responsibility clear: **management owns** quality; internal audit independently evaluates it.

Separating quality activity from effective quality outcomes

A quality management system coordinates policies, roles, processes and improvement. Quality assurance supports confidence that processes can achieve **requirements**; quality control examines outputs or results. Testing is one quality activity. It does not replace governance of requirements, change, competence and **corrective action**.

Evaluate whether quality staff can raise findings and obtain resolution without inappropriate suppression. Reporting lines can reveal threats, but **actual authority**, **independent review** and escalation determine whether challenge is protected. A development manager's ability to delete adverse test findings without review is more direct evidence of a weakness than a job title alone.

Example: A team reports a 98 percent test pass rate while excluding all failed critical-security tests as optional. Inspect the **requirements** and defect population, severity rules, exclusions and acceptance authority. PDCA and DMAIC can organize improvement, but completed activities should lead to measured results and sustained control. Maturity assessments indicate capability under the chosen model; a higher maturity level is not certification that every application is secure. Audit should connect process quality to dependable **business outcomes**, including escaped defects, repeat incidents and the effectiveness of **corrective action**.

Separate a Product Check From a Process Check

Example. A team tests ten released transactions and finds the calculations correct. That **quality control** result concerns the tested outputs. To assess **quality assurance**, inspect how requirements, test cases, defects and release approvals were managed and whether failures could bypass the process. If a critical test was disabled to meet a date, evaluate the authorization and consequences. A quality management system links the process, responsibilities, performance evaluation and corrective action so that recurring weaknesses can be identified and addressed.

2.12 The Last Mile - Memory Aids and Exam Traps

Use these reminders after studying the chapter. Read the **facts** in each scenario before applying a familiar phrase: the correct role, artifact, control or measure depends on the decision being asked about.

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Memory aids

Memory aid	Meaning and limit
Directors direct; managers deliver.	Governance evaluates, directs and monitors. Management plans and performs the work. Both levels need evidence of results.
EDM; APO BAI DSS MEA.	EDM is the COBIT governance domain. The four management domains organize planning, implementation, service and assessment objectives.
Strategy looks ahead; steering sets priorities.	The strategy committee advises the board. The steering committee coordinates business and IT priorities within delegated authority.
Owner decides; custodian implements.	Business owners decide data classification and appropriate access. Custodians implement handling and protection.
Policy gives direction; procedure gives steps.	Standards set mandatory rules; baselines set minimum levels; guidelines advise. This is a distinction of purpose, not a rigid five-stage project plan.
Identify, assess, respond, monitor.	Establish context and criteria first; communicate throughout; repeat as conditions change.
TAMA: Transfer, Avoid, Mitigate, Accept.	Recall the four responses. Transfer does not erase the enterprise's retained obligations.
Appetite chooses; tolerance bounds; capacity constrains.	Appetite is the desired amount and type; tolerance provides specific limits; capacity is the ability to bear exposure.
Single loss, annual frequency, annual loss.	SLE = AV × EF. ALE = SLE × ARO. Keep the time period and units consistent.
The work moves; accountability stays.	A supplier can perform a service. The enterprise still needs an owner, evidence, monitoring and an exit route.
Performance, goal, risk, control.	KPI measures performance; KGI outcome; KRI exposure; KCI a control. Meaning depends on the stated objective.
QA checks the approach; QC checks the output.	Use the activity's purpose. A department name alone does not determine the classification.
Define, Measure, Analyze, Improve, Control.	DMAIC structures Six Sigma improvement. Measure the effect of the change and sustain it.
Plan, Do, Check, Act.	PDCA tests and embeds improvement. A completed action must still produce a useful result.

Calculation check: Distinguish a likelihood score, event probability and annual event frequency. Keep SLE, ALE and control cost in consistent units and periods. Metadata explains data; lineage traces it.

Two scorecards and a maturity progression

For the traditional enterprise BSC, recall Financial, Customer, Internal process, Learning and growth. For a common IT BSC adaptation, recall Corporate contribution, User orientation, **Operational excellence**, Future orientation. Read whether the question asks about the enterprise scorecard or its IT adaptation.

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CMMI recall	Meaning
0 Incomplete	Work is not reliably completed.
1 Initial	Reactive and unpredictable.
2 Managed	Planned and controlled at project level.
3 Defined	Organization-wide standards guide work.
4 Quantitatively Managed	Measured, analyzed and predictable.
5 Optimizing	Continuously improved and adaptable.

Exam traps

Tempting wrong answer	Exam answer
Having the board select or configure security controls.	The board directs and oversees (strategy, risk appetite, policy); management selects and implements controls.
Ranking IT investments by lowest cost.	Rank investments by alignment with business objectives and value over the life cycle.
Treating an approved IT strategy as proof of alignment.	The best evidence of alignment is that funded IT initiatives trace to stated business objectives.
Treating a quantitative risk estimate as objective fact.	ALE = SLE × ARO and SLE = AV × EF, but the result is only as good as its assumptions; review the source data and uncertainty.
Rejecting a legally required control because it costs more than the ALE it removes.	A control is cost-justified when its annual cost is below the ALE reduction it delivers, but legal and regulatory requirements apply regardless.
Treating unaddressed risk as accepted risk.	An authorized risk owner accepts or escalates within delegated limits, using an informed, documented decision.
Believing insurance or outsourcing transfers all risk.	Transfer shifts financial loss. Accountability, regulatory and reputational risk stay with the enterprise.
Every privacy activity requires consent.	Determine the applicable lawful basis and any additional conditions.
Encryption establishes privacy compliance.	Also assess purpose, minimization, transparency, rights and retention.
A deletion request overrides all retention.	Determine applicable rights, exceptions, legal holds and documented decisions.
A sensitive-data label protects the file.	The label guides handling; access and other safeguards must implement the rules.

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Tempting wrong answer	Exam answer
Assuming a reporting title alone establishes security objectivity.	Assess actual duties, authority, independent challenge and escalation. Reporting lines can create conflicts; evaluate safeguards and their operation.
Cross-training means granting all backup staff unrestricted access.	Build competence while controlling incompatible privileges.
One directory disablement completes a leaver process.	Cover applications, cloud sessions, privileged access, physical access and shared secrets.
Treating a right-to-audit clause as evidence that vendor controls work.	Right-to-audit is the key contract provision, but assurance comes from exercising it or from a relevant SOC 2 Type II report.
Any SOC 2 Type 2 report covers the customer's needs.	Read scope, period, criteria, exceptions, subservices and CUECs.
Using uptime statistics to show the board the value of IT.	Use an IT balanced scorecard: it links IT results to business objectives.
A high maturity rating means no weaknesses exist.	Check the model and scope; assess the relevant risk and actual operation.
Calling product inspection "quality assurance".	QA assures the process; QC inspects the product.
Buying a tool as the first step of a new program.	Policy, scope and ownership come first; tools follow the policy.

Work through an integrated case

The retailer launches a loyalty service through a cloud **supplier**. Customer **enrollment** rises, but management cannot identify all location-data exports. The vendor's assurance report ends four months before launch. No one owns the business benefit measure, and a former contractor retains access to the export tool.

Identify the separate issues: **data inventory** and purpose, supplier assurance coverage, **benefit ownership**, and leaver access. Establish the actual processing and access, assess business and privacy exposure, obtain appropriate evidence for the missing period, and communicate supported findings to the responsible owners. The fact that enrollment increased does not resolve the other risks. Internal audit recommends and evaluates; management decides and implements **corrective action**.

Decision owners at a glance

Use the assigned authority and evidence to distinguish delivery responsibility from oversight and assurance.

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Role	Responsibility	Boundary
Board/governing body	Sets direction, appetite and oversight.	Delegation needs clear boundaries.
Executive management	Translates direction into plans, resources and operation.	Retains assigned accountability when sourcing work.
Business/data owner	Defines value, classification, access expectations and outcomes.	Custodians implement rather than determine business meaning.
Supplier owner	Monitors service commitments, assurance and exit readiness.	A contract does not demonstrate actual performance.

Threats and control purpose

The examples show complementary control purposes. Their classification depends on how the control is configured and used.

Risk	Preventive direction	Detective evidence	Corrective direction
Misaligned investment	Business-case and architecture approval.	Benefit and portfolio performance review.	Reprioritize, redesign or terminate through authority.
Supplier control failure	Risk-based selection and contractual/control requirements.	Relevant assurance, incidents and service monitoring.	Remediation, tested contingency or controlled exit.

2.13 Glossary

Look up terms alphabetically. Return to the lesson for the related example and audit decision.

Term	Definition
Activity / swim-lane diagram	A view of process steps / a view placing steps in responsible roles or teams to show hand-offs.
Agile	An approach emphasizing collaboration, feedback and incremental adaptation in delivery.
Annualized loss expectancy (ALE)	Expected annual loss in the simple model; SLE × ARO.
Annualized rate of occurrence (ARO)	Expected event frequency per year.
APO	Align, Plan and Organize; a COBIT management domain.
Application architecture	A description of application capabilities and their interactions.
Asset value (AV)	The asset or exposure value used in a loss model.
BAI	Build, Acquire and Implement; a COBIT management domain.
Balanced scorecard	A set of linked strategic objectives and measures across complementary perspectives.

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Term	Definition
Baseline	An approved minimum level or configuration for a defined class of assets.
Benchmarking	Comparison with relevant peers or references using sufficiently comparable measures.
Benefits realization	Achieving and verifying the business outcomes expected from an investment.
Bridge letter	A provider management statement about the period after a report; not independent assurance for that gap.
Business agility	The ability to respond effectively to changing business needs and conditions.
Business architecture	A description of capabilities, processes and relationships supporting business objectives.
Business case	A justification of a proposed investment using objectives, alternatives, benefits, cost, risk and assumptions.
Business intelligence	Use of organized data and analysis to support management understanding and decisions.
Business process reengineering	Fundamental redesign of processes to achieve substantial improvement.
Capability	How well a particular process or practice performs under the chosen model.
Capacity management	Matching available service resources to present and forecast demand within cost and risk constraints.
Capital expenditure	Spending meeting the relevant criteria for asset recognition rather than immediate expense.
Centralized IT	A structure concentrating defined IT authority and services at enterprise level.
CEO / CFO / CLO / CSO	CEO: chief executive officer; CFO: chief financial officer; CLO: chief legal officer; CSO commonly means chief security officer in this context, but the actual enterprise mandate determines the role. See 2.2.8.
Chargeback	Charging consuming units for allocated IT costs.
CIO	Chief information officer; leads information and IT direction and delivery within the assigned mandate.
CISO	Chief information security officer; leads the information security program and reporting within the assigned mandate.
Cloud service provider (CSP) independence	Independence assessment for the actual cloud assurance subject and relationships, including self-review and management participation; not a blanket ban based only on work for a provider. See 1.1 and 2.9.
CMMI	Capability Maturity Model Integration; a process capability and performance improvement approach.

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Term	Definition
COBIT	ISACA framework for governance and management of enterprise information and technology.
Communicate-P	Privacy Framework function supporting understanding and dialogue about processing.
Compensating control	An alternative safeguard addressing a risk where the primary control or separation cannot be implemented.
Concentration risk	Exposure resulting from several important dependencies sharing a provider, location, technology or other common factor.
Consent	An eligible individual's qualifying informed choice for specified processing where the framework permits reliance on it.
Context diagram	A view of a system or process boundary and its external exchanges.
Controller	A party determining the purposes and means of personal-data processing under the relevant framework.
Control-P	Privacy Framework function supporting granular data-processing decisions.
COSO	Organization associated with enterprise internal control and ERM frameworks.
Cost allocation	Assigning costs to services, activities or consumers using an agreed method.
Cross-training	Developing backup competence in additional duties while controlling access conflicts.
CSF	Critical success factor; something that must go well for an objective or initiative to succeed.
CTO	Chief technology officer; leads technology direction or engineering capability within the assigned mandate.
CUECs	Complementary user entity controls assumed in the service report that customers need to implement.
Dashboard	A presentation of selected measures and results for an intended audience.
Data architecture	A description of information, structures, meaning, relationships and flows.
Data classification	Assignment of categories reflecting value, sensitivity and handling needs.
Data custodian	Role implementing technical handling and safeguards according to authorized direction.
Data flow diagram (DFD)	A model showing movement and transformation of data between processes and stores; See 2.2.7.
Data governance	Decision rights and accountability for the meaning, quality, use and protection of data.

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Term	Definition
Data inventory	An organized record of information assets, purposes, owners, systems, locations and flows.
Data lineage	Traceability from data origin through transformations to use. See 2.2.7.
Data mart	A subject-specific or business-unit subset organized for analysis.
Data minimization	Limiting collection and use to information needed for the purpose.
Data mining	Analysis of data to identify patterns, relationships or trends.
Data owner	Business role accountable for data and decisions about classification and appropriate use or access.
Data portability	Transfer or receipt of eligible data in a usable form where the applicable right exists.
Data quality	The extent to which data are suitable for their purpose, considering attributes such as accuracy and completeness.
Data steward	Role supporting data definitions, quality and day-to-day governance.
Data subject	The individual to whom personal information relates.
Data subject access request	A request to exercise applicable rights to obtain information or access personal data.
Data user	A person permitted to use information within assigned responsibilities and handling rules.
Data warehouse	An integrated analytical store of enterprise information, often including historical records.
Decentralized IT	A structure placing defined IT authority and delivery in business units or locations.
DMAIC	Define, Measure, Analyze, Improve and Control; an improvement cycle used in Six Sigma.
DPIA	Data protection impact assessment; a legally defined assessment required in specified circumstances under some frameworks.
DSS	Deliver, Service and Support; a COBIT management domain.
Due diligence	Proportionate investigation before deciding to rely on a supplier, service or relationship. See 2.9.2.
EDM	Evaluate, Direct and Monitor; the COBIT governance domain.
EGIT	Enterprise governance of information and technology; includes technology and information used throughout the business.
Enterprise architecture	A coordinated description of business, information, applications and technology, including current and target states.
Enterprise change management	Preparing people and the enterprise to adopt changes in processes, technology or behavior.

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Term	Definition
Entity relationship diagram	A model of entities, attributes and the relationships among them.
Erasure	Deletion of personal information where the applicable right or retention requirement calls for it.
ERM	Enterprise risk management; coordinated management of uncertainty affecting objectives.
ETL	Extract, Transform and Load; a common pattern for moving and preparing analytical data.
Exit plan	Arrangements for transitioning the service, retrieving data, removing access and disposing of retained information.
Exposure factor (EF)	The fraction of asset value estimated to be lost in a modeled event.
Federated IT	A structure combining shared enterprise direction or services with selected local authority.
Gap analysis	Comparison of current state with agreed target requirements or capabilities to identify differences and improvement needs. Objectives and criteria must first be established. See 2.1 and 2.2.
GRC	Coordinated governance, risk management and compliance arrangements.
Guideline	Recommended practices supporting judgment; generally advisory.
IaaS	Infrastructure as a Service; the customer retains significant operating-system and application responsibilities.
Impact	The consequence of an event for objectives, assets, people or obligations.
Information asset	Valuable information identified for ownership, use and protection within the adopted inventory/classification scope. See 2.7.1 and 5.1.
Inherent risk	Exposure considered before the relevant controls or treatment.
Insourcing	Provision of a capability using internal resources.
IRR	Internal rate of return; the discount rate at which the modeled NPV equals zero.
ISMS	Information security management system; coordinated processes for managing security within a defined scope.
ISO 22301	Requirements for a business continuity management system.
ISO 31000	Principles and guidance for risk management.
ISO 9001	Requirements for a quality management system.
ISO/IEC 20000	A family of service management standards, including management system requirements.
ISO/IEC 27001	Requirements for an information security management system.

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Term	Definition
ISO/IEC 38500	A standard addressing governance of IT for governing bodies.
IS strategy	The direction and capabilities through which information systems support enterprise objectives.
IT governance	Direction and oversight of information and technology so it supports enterprise objectives and accountable decisions.
ITIL	A body of service management practices for creating and delivering value through services.
IT management	Planning, implementation, operation and monitoring that deliver within enterprise direction.
IT steering committee	A cross-functional management body that prioritizes and monitors IT initiatives within its remit.
IT strategy committee	A committee that typically advises the governing body on IT direction and value.
Job rotation	Moving personnel between duties to build coverage and potentially expose concealed irregularities.
KCI	Key control indicator; a measure of control operation or effectiveness.
Key-person risk	Dependence on one person’s knowledge, access or availability.
KGI	Key goal indicator; a measure of whether a desired outcome was achieved.
KPI	Key performance indicator; a measure of performance relevant to an objective or process.
KRI	Key risk indicator; a measure signaling exposure or changing risk conditions.
Lawful basis	An applicable legal ground for a processing activity.
Legal hold	A requirement to preserve specified records rather than dispose of them routinely.
Legitimate interest	A processing basis available in some frameworks subject to specified conditions and balancing.
Life cycle cost	Estimated costs across an investment’s stages, including transition and retirement.
Likelihood	The assessed chance or frequency of an event within the stated context and period.
Mandatory vacation	A required absence with genuine work handover, potentially exposing concealed activity.
Maturity	A progression of institutionalized management and improvement practices under a defined model.
MEA	Monitor, Evaluate and Assess; a COBIT management domain.

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Term	Definition
Metadata	Information describing data, including meaning, format, ownership and rules. See 2.2.7.
NIST CSF 2.0	Cybersecurity outcomes organized into Govern, Identify, Protect, Detect, Respond and Recover.
NIST Privacy Framework	A voluntary privacy risk framework; version 1.0 uses five functions with the -P suffix.
Nondisclosure agreement (NDA)	An NDA establishes agreed obligations over confidential information and permitted disclosure. See 2.9.2.
NPV	Net present value; discounted net cash flows using the selected rate and period.
OECD Privacy Principles	Eight broad principles supporting appropriate personal-data practices.
O-ISM3	The Open Group's process-based information security management and maturity approach.
OLA	Operating level agreement coordinating supporting internal teams and responsibilities.
OLAP	Online analytical processing; multidimensional analysis of information.
Onshore / nearshore / offshore	Delivery in the same country / a nearby country / a more distant country, subject to the actual arrangement.
Onsite / offsite	Work performed at the customer's premises / elsewhere; distinct from country-based sourcing terms.
Operating expense	Cost treated as an expense under the applicable accounting policy.
Operational excellence	Sustained effective, efficient and reliable delivery supported by ownership, control and improvement.
Operational plan	A plan for recurring delivery and day-to-day execution.
Outsourcing	Obtaining a defined capability or service from another organization.
PaaS	Platform as a Service; the provider supplies an offered platform while customer application and data duties remain.
Payback period	The time required for modeled returns or savings to recover the investment.
PDCA	Plan, Do, Check and Act; a cycle for implementing, evaluating and sustaining improvement.
Personal information	Information relating to an identified or identifiable individual under the applicable definition.
PIA	Privacy impact assessment; examination of processing effects on individuals and suitable safeguards.
PMBOK / PRINCE2	Project management references; their role differs from governance, architecture or security standards.

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Term	Definition
Policy	Approved high-level direction and expectations.
Policy exception	An authorized documented deviation with risk assessment, conditions and review arrangements.
Portfolio management	Selection and oversight of investments or services as a whole to balance objectives, value, risk and resources.
Privacy	Appropriate handling of information about individuals throughout its collection, use, sharing and retention.
Privacy notice	Information communicated to individuals about relevant processing and rights.
Privacy officer / DPO	A role advising, monitoring and supporting privacy obligations under its assigned or legal mandate.
Procedure	The steps, roles and records for performing an activity.
Processor	A party processing personal data on behalf of a controller under the relevant framework.
Protected health information (PHI)	Protected health information (PHI) is individually identifiable health information within the applicable scope of the US Health Insurance Portability and Accountability Act (HIPAA) framework. See 2.6.
Purpose limitation	Defining appropriate purposes and assessing whether later use is compatible or otherwise permitted.
QA	Quality assurance; planned activities providing confidence that requirements will be met.
QC	Quality control; evaluation of an output against its requirements.
QMS	Quality management system; the coordinated policies, roles, processes and improvement arrangements for quality.
Qualitative analysis	Risk analysis using defined descriptive ratings and judgment.
Quantitative analysis	Risk analysis using numerical estimates of frequency and consequence.
RACI	Responsible, Accountable, Consulted and Informed; a matrix describing participation and decision ownership.
Rectification	Correction of personal information where the applicable conditions are met.
Residual risk	Exposure remaining after the controls or response considered.
Retention schedule	Defined retention and disposal requirements for categories of records.
Right-to-audit	An agreed assurance-access right; its scope and practical usability matter.
Risk	Uncertainty affecting objectives, assessed through credible events and their consequences.
Risk acceptance	An informed authorized decision to retain exposure with appropriate review and monitoring.

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Term	Definition
Risk appetite	The amount and type of risk the enterprise is willing to pursue or retain.
Risk assessment	Analysis and evaluation of exposure against defined criteria.
Risk avoidance	Not undertaking or discontinuing the activity that creates the exposure.
Risk capacity	The ability to bear risk without threatening viability or critical obligations.
Risk mitigation	Reducing likelihood, impact, or both through suitable action or controls.
Risk owner	The person with accountability and authority to manage or escalate a specified risk.
Risk register	A record of risks, owners, assessments, controls, responses, actions and review status.
Risk tolerance	Acceptable boundaries or variation for a particular objective or exposure.
Risk transfer or sharing	Allocating defined consequences or responsibilities to another party; retained exposure remains.
ROI	Return on investment; net return compared with the investment under the stated method.
Root cause analysis	Investigation of the underlying reasons for an issue rather than only its symptoms.
SaaS	Software as a Service; the provider supplies an application while customer use, access and data duties remain.
SCADA	Supervisory control and data acquisition; systems supervising or controlling physical processes.
Security administrator	Role implementing approved security configurations and privileges.
Segregation of duties	Separation of incompatible activities to reduce the ability to create and conceal errors or fraud.
Selective or smart sourcing	A deliberate combination of internal and external delivery by function or service.
Semiquantitative analysis	Risk analysis using numerical scores or bands derived from defined judgments.
Service level management (SLM)	Continuing process of agreeing, monitoring, reviewing and improving service commitments. An SLA is an agreement within that process. See 2.9 and 4.10.
Shadow IT	Technology obtained or used outside established enterprise oversight.
Shared responsibility	Division of service and security duties between provider and customer under the actual arrangement.
Showback	Reporting allocated consumption and cost without an actual internal charge.
Single loss expectancy (SLE)	Estimated loss per occurrence; in the simple model, AV × EF.

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Term	Definition
Six Sigma	A data-based approach to reducing defects and variation.
SLA	Service level agreement specifying measurable service commitments and related conditions.
SOC 1	Report on controls relevant to user entities' internal control over financial reporting.
SOC 2	Report on controls for the described service system against applicable Trust Services Criteria.
SOC 3	A general-use report on controls against Trust Services Criteria, providing less detail.
Software capitalization	Asset recognition of eligible software costs under the applicable accounting criteria.
Software escrow / source-code escrow	Independent holding of agreed source code and supporting materials for release under specified conditions. Verify current usable deposits, release rights and ability to build and maintain the software. See 2.9.
Staging area	An intermediate area for copying, checking and preparing data before downstream use.
Stakeholder	A person or group affected by, interested in, or able to influence the enterprise's decisions and outcomes.
Standard	A mandatory rule or specified requirement supporting a policy.
Strategic alignment	A demonstrable relationship between IT decisions and business objectives.
Subservice organization	Another service organization used in delivering the examined service.
Succession planning	Preparing qualified replacements for critical roles.
Tactical plan	A plan translating strategic direction into programs, resources and nearer-term action.
TCO	Total cost of ownership over the relevant period, including acquisition, operation, support and exit.
Technical change control	Authorization, testing, implementation and documentation of changes to systems.
Technology architecture	A description of infrastructure and technical services supporting applications and information.
Threat	A potential cause of an unwanted event.
Three Lines Model	A model connecting management's delivery, support and challenge roles with independent internal audit assurance.
TOGAF / ADM	An enterprise architecture approach / its Architecture Development Method.

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Term	Definition
TQM	Total quality management; an organization-wide commitment to quality and continuing improvement.
Transborder data flow	Movement of, or access to, information across national boundaries.
Transition roadmap	A planned sequence of changes from current to target capabilities.
Type 1 report	Report addressing the description and suitability of control design at a specified date under its framework.
Type 2 report	Report also addressing operating effectiveness over a specified period under its framework.
Vendor lock-in	Difficulty moving away from a supplier because of technical, commercial, data or operational dependencies.
Vulnerability	A weakness that may be exploited or contribute to an adverse event.
Zachman	A taxonomy organizing architecture descriptions by perspectives and fundamental questions.

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2.14 Practice questions

These 18 original questions follow the CISA format: choose the one BEST answer from four options. Domain 2 carries 18 percent of the examination, so it has 18 of the book’s 100 domain questions. Allow about 1.6 minutes per question, then check section 2.15. Before you look, name the **qualifier** and say why your **second choice** is weaker.

Question 2.01 An IS auditor is assessing whether a manufacturer's IT strategy supports its business strategy. Which evidence would **BEST** support a conclusion that the two are aligned?

A. IT spending as a share of revenue matches the industry average
B. The IT plan is structured around a recognized governance framework
C. Each funded IT initiative traces to a stated business objective
D. The board approved the IT strategic plan within the past year

Question 2.02 A retailer is chartering an IT steering committee composed of business unit heads and the CIO. Which should be the committee's **PRIMARY** function?

A. Evaluating the operating effectiveness of IT controls
B. Advising the board on IT's long-term strategic contribution
C. Approving the enterprise information security policy
D. Prioritising IT initiatives and overseeing their delivery

Question 2.03 At a small subsidiary, one administrator both provisions user access and applies changes to the payroll application. Management states that adding staff is not feasible. Which is the **BEST** control to recommend?

A. Independent supervisory review of the activity logs
B. Outsourcing payroll administration to a managed provider
C. Restricting the administrator's system access to business hours
D. An annual signed attestation of the administrator's duties

Question 2.04 A regional bank is revising its board charter to strengthen oversight of technology. Which activity is **LEAST** appropriate for the board to retain?

A. Selecting security controls for the core banking platform
B. Directing that IT investments be evaluated for business value
C. Monitoring IT performance against strategic objectives
D. Approving the enterprise's appetite for technology risk

Question 2.05 An IS auditor is planning a compliance audit of a newly outsourced customer contact centre that processes personal data in two countries. What should the auditor do **FIRST**?

A. Review the provider's most recent SOC 2 Type II report
B. Test a sample of call records for evidence of consent
C. Identify the applicable legal and contractual requirements
D. Interview the vendor manager about known compliance issues

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Question 2.06 A hospital's removable-media encryption requirement appears only in an advisory guideline. Departments explicitly treat it as optional and routinely copy patient data to unencrypted drives. Which change **BEST** makes the organizational requirement binding?

A. Deploying endpoint software that logs USB file transfers
B. Converting the guideline into a mandatory standard
C. Delivering awareness training on removable media risks
D. Reissuing the guideline with clearer wording and examples

Question 2.07 A steering committee approves a cloud CRM purchase for one division without an enterprise architecture review. Which is the **GREATEST** risk arising from this omission?

A. The vendor may not provide a current SOC 2 Type II report
B. The division may pay more than enterprise licensing rates
C. Division staff may need additional training on the new CRM
D. The CRM may duplicate capability and fail to integrate

Question 2.08 A risk assessment shows residual risk on a legacy trading platform exceeds the approved tolerance, and remediation will take a year. The IT manager offers to accept the risk in the interim. Who should **MOST** appropriately decide whether the risk is accepted?

A. The accountable senior risk owner through the applicable delegated approval and escalation limits
B. The CISO who performed and rated the risk assessment
C. The IT steering committee that funded the remediation
D. The IT manager responsible for operating the platform

Question 2.09 A distribution centre's warehouse system is valued at $500,000. A flood would destroy 40 percent of its value and is expected once every four years. Which figure **MOST** accurately represents the annualized loss expectancy?

A. $80,000
B. $200,000
C. $50,000
D. $125,000

Question 2.10 Following an acquisition, management proposes buying cyber insurance for the acquired company's e-commerce platform. What should be done **FIRST**?

A. Negotiate coverage limits and exclusions with the insurer
B. Assess the platform's risks against the risk appetite
C. Implement additional security controls on the platform
D. Record the residual risk as accepted in the risk register

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Question 2.11 A retailer plans to use loyalty-programme purchase histories, originally collected to administer rewards, to build marketing profiles shared with partners. What should the privacy officer evaluate **FIRST**?

A. Whether the data can be pseudonymised before transfer
B. Whether a lawful basis exists for the new purpose
C. Whether the profiles will be encrypted when shared
D. Whether partners have signed confidentiality agreements

Question 2.12 A marketing analyst asks the database team for read access to a customer table containing payment details. Whose approval is the **PRIMARY** requirement before access is granted?

A. The information security manager who sets access standards
B. The database administrator who manages the platform
C. The business owner accountable for the customer data
D. The analyst's line manager who confirms the business need

Question 2.13 A bank's treasury systems administrator has not taken leave in three years and is the only person who runs month-end reconciliation jobs. Which control would **BEST** help detect irregularities in this role?

A. Repeating background checks on the administrator periodically
B. Requiring the administrator to sign a non-disclosure agreement
C. Enforcing mandatory leave with duties covered by a peer
D. Documenting the reconciliation steps in a detailed runbook

Question 2.14 An IS auditor relies on a payroll provider's SOC 2 Type II report covering January to June, but the audit requires assurance for the full year. The report's scope and exceptions have been reviewed and found acceptable. What should the auditor do **NEXT**?

A. Accept the provider's bridge letter for the uncovered months
B. Report the coverage gap to management as an audit finding
C. Test the complementary user entity controls in the report
D. Obtain evidence of control operation for July to December

Question 2.15 A company is negotiating a contract to outsource hosting of its customer order system. The contract should provide for each of the following **EXCEPT**:

A. Notification and approval of the provider's subcontractors
B. Data return and verified deletion on contract termination
C. Transfer of regulatory compliance accountability to the vendor
D. The customer's right to audit or obtain independent assurance

Question 2.16 A logistics company licenses a route-optimisation application from a small software vendor and depends on it for daily dispatch. Source code escrow is proposed in the contract. Which risk does this **MOST** directly address?

A. The vendor going out of business and ceasing support
B. The vendor raising licence fees beyond agreed terms
C. Updates from the vendor introducing undetected defects
D. Vendor staff accessing dispatch data without authority

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Question 2.17 The board of an insurer wants a periodic view of whether IT is delivering value and supporting enterprise goals, not just operating efficiently. Which approach would **BEST** meet this need?

A. A key risk indicator dashboard for critical systems
B. An IT balanced scorecard covering four perspectives
C. A maturity assessment of IT processes against a framework
D. A monthly report of service level attainment by system

Question 2.18 An auditor reviews a software quality function whose findings can be changed or suppressed by the development manager without independent review or escalation. Which condition **MOST** directly explains this objectivity weakness?

A. The QA checklist was last updated eighteen months ago
B. QA reviews a sample of releases rather than each release
C. Defect metrics are reported to management each quarter
D. QA staff report to the development manager they review

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2.15 Answers and explanations

Each explanation states the **deciding principle**, then shows why the key is best and why each other option, although often a sound practice, does not answer the question asked.

Question 2.01 Answer: C

IT strategy alignment · Qualifier: BEST · Review section 2.2

Principle: Alignment is evidenced by tracing actual IT investments to the **business objectives** they serve, not by plan approval, spending ratios or framework use.

A. **Benchmarking** spend is a useful efficiency indicator; however, spending at the industry average says nothing about whether that money supports this enterprise's objectives.

B. A recognized framework improves the plan's structure and **completeness**; however, a well-structured plan does not prove its content reflects business priorities.

C. Correct. Mapping each funded initiative to a **business objective** directly demonstrates that IT investment decisions serve business strategy, which is the definition of alignment.

D. Board approval shows governance oversight of the plan; however, an approved plan can still fund initiatives that do not serve current **business objectives**.

Question 2.02 Answer: D

IT steering committee · Qualifier: PRIMARY · Review section 2.2

Principle: The stated steering committee coordinates investment priorities and delivery under its mandate. Governing bodies provide direction and oversight; titles and delegated authority can vary.

A. Evaluating control effectiveness is a valuable **assurance** activity; however, it is performed independently by internal audit, and a committee of business and **IT management** cannot independently assess itself.

B. Advising the board on strategic direction and value is a real **governance** role; however, it belongs to the **IT strategy committee**, not the management-level steering committee.

C. Security **policy** needs senior **approval**; however, that approval rests with the board or senior management, and policy approval is not the steering committee's defining purpose.

D. Correct. Business and IT leaders together prioritise competing initiatives, allocate resources and monitor delivery, which is the steering committee's core management function.

Question 2.03 Answer: A

Compensating controls for SoD · Qualifier: BEST · Review section 2.2

Principle: Where **segregation of duties** is infeasible, a compensating detective control such as independent review of protected activity logs addresses the specific conflict.

A. Correct. Independent review of protected logs can detect unauthorised access grants or payroll changes, directly compensating for the conflict that cannot be segregated.

B. **Outsourcing** can be a valid sourcing decision; however, it is a disproportionate response, **accountability** stays with the company, and the provider may have the same conflict.

C. **Time-based restrictions** narrow the window of opportunity; however, misuse during business hours remains possible and would still go undetected.

D. Attestations reinforce awareness of expected conduct; however, a signed statement cannot detect or prevent misuse of the **combined access**.

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Question 2.04 Answer: A

Governance versus management · Qualifier: LEAST · Review section 2.2

Principle: The board evaluates, directs and monitors; selecting specific controls is a management **activity** of planning, building and running.

A. Correct. Choosing specific platform controls is management's job; a board doing it blurs **accountability** and would weaken its ability to hold management to account.

B. Directing that investments be judged on business value is a **governance** 'direct' activity that sets expectations for management's portfolio decisions.

C. **Monitoring** performance against strategy is the board's 'monitor' role, giving it evidence that its direction is being achieved.

D. Setting **risk appetite** is a core board responsibility, because it frames how much technology risk management may pursue or retain.

Question 2.05 Answer: C

Compliance criteria · Qualifier: FIRST · Review section 2.1

Principle: The auditor must first establish which **obligations** apply, because those requirements become the audit criteria against which everything else is tested.

A. The **SOC 2 report** is useful evidence later; however, its relevance cannot be judged until the applicable requirements that it must address are known.

B. Consent testing may be needed; however, whether consent is the required **lawful basis** depends on which jurisdictions' requirements apply.

C. Correct. Determining applicable laws, regulations and contract terms defines the audit **criteria** and scope, so it must precede gathering or testing evidence.

D. Interviews provide helpful **context**; however, known issues can only be judged as non-compliance once the governing requirements are identified.

Question 2.06 Answer: B

Policies, standards and guidelines · Qualifier: MOST · Review section 2.3

Principle: The weakness is an advisory requirement treated as optional. A mandatory **standard** establishes the internal obligation alongside implementation, monitoring and applicable legal duties.

A. **Logging** gives detective visibility of transfers; however, it records unencrypted copying after it happens rather than correcting the advisory status of the rule.

B. Correct. Making **encryption** a mandatory standard removes the discretion that is causing the non-compliance and creates an enforceable, auditable requirement.

C. Training raises awareness of the **risk**; however, staff are not obliged to act on advice that the organisation has not made mandatory.

D. Clearer wording may improve understanding; however, a **guideline** remains advisory, so departments may still lawfully choose not to follow it.

Question 2.07 Answer: D

Enterprise architecture · Qualifier: GREATEST · Review section 2.4

Principle: Enterprise **architecture review** exists to keep investments coherent with the target state, so bypassing it chiefly risks duplicated capability and integration failure.

A. Assurance reporting matters for a cloud provider; however, it is addressed through vendor due diligence, not through an enterprise **architecture review**.

B. Higher licence cost is a real procurement concern; however, it is a financial inefficiency that procurement review addresses, not the core purpose of **architecture review**.

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C. Training needs arise with any new system; however, they are an implementation matter that **architecture review** is not designed to address.

D. Correct. Without **architecture review**, the CRM may duplicate existing systems and be unable to share customer data, undermining enterprise data **consistency** and wasting investment.

Question 2.08 Answer: A

Risk acceptance authority · Qualifier: MOST · Review section 2.5

Principle: An accountable **risk owner** decides or escalates within delegated limits. Exposure beyond approved tolerance does not grant an IT manager unilateral acceptance authority.

A. Correct. The accountable senior owner must obtain any governing **approval** required by the delegated limits and **exposure**, or require further treatment. A title alone confers no unlimited acceptance authority.

B. The CISO advises on and rates the risk; however, **independence** and accountability require that the assessor not also decide to accept it.

C. The steering committee prioritises and funds projects; however, its remit is investment oversight, not acceptance of out-of-tolerance **business risk**.

D. The IT manager understands the platform; however, an operator of the system lacks authority to accept **business risk** that exceeds approved tolerance.

Question 2.09 Answer: C

Annualized loss expectancy · Qualifier: MOST · Review section 2.5

Principle: SLE = AV x EF = $500,000 x 0.4 = $200,000, and ALE = SLE x ARO = $200,000 x 0.25 = $50,000.

A. This applies the **exposure factor** twice to the SLE; however, the correct annualising factor is the ARO of 0.25, not the 0.4 exposure factor.

B. This is the single loss expectancy for one flood; however, it has not been **annualised** by multiplying by the ARO of 0.25.

C. Correct. Multiplying the $200,000 single loss expectancy by the annual rate of 0.25 gives the expected annual loss of $50,000.

D. This multiplies asset value by the ARO; however, it omits the **exposure factor**, overstating the loss as if the whole asset were destroyed.

Question 2.10 Answer: B

Risk transfer · Qualifier: FIRST · Review section 2.5

Principle: A **risk response** such as transfer can be chosen only after the risk has been identified and analysed and evaluated against appetite.

A. Negotiating terms is part of implementing a transfer; however, coverage cannot be sized until the **exposure** being transferred has been assessed.

B. Correct. Assessing **likelihood** and impact against appetite determines whether insurance, mitigation or another response is appropriate and how much coverage is needed.

C. Mitigation may well be needed; however, choosing controls before **assessment** risks treating the wrong exposures or over-investing.

D. Acceptance is a valid response; however, it requires an **informed decision** by an authorised owner after assessment, not a default entry.

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Question 2.11 Answer: B

Purpose limitation · Qualifier: FIRST · Review section 2.6

Principle: Establish whether the new processing purpose is permitted under applicable **obligations** and basis. Security safeguards remain necessary but do not by themselves justify secondary use.

A. Pseudonymisation reduces **privacy risk**; however, it is a safeguard to consider once it is established that the new use is permitted.

B. Correct. Using data for a new purpose is permissible only with a compatible purpose or a valid **lawful basis**, so this determines whether the processing may proceed at all.

C. Encryption protects confidentiality in transit; however, encrypted data shared without a **lawful basis** is still unlawful processing.

D. **Confidentiality** agreements protect data once shared; however, they cannot make an impermissible secondary use lawful.

Question 2.12 Answer: C

Data ownership · Qualifier: PRIMARY · Review section 2.7

Principle: The business **data owner** classifies the data and authorises access to it; IT custodians implement that decision.

A. Security sets standards and may implement privileges; however, deciding whether a user should see the data is a **business owner** decision.

B. The **DBA** is the custodian who implements and protects access; however, custodians act on the owner's authorisation rather than deciding who may access data.

C. Correct. The **data owner** is accountable for the data, determines its classification and decides who may access it, so that **approval** is the primary requirement.

D. Line manager confirmation of need is a useful supporting step; however, the line manager is not **accountable for the data** and cannot authorise access to it.

Question 2.13 Answer: C

Mandatory leave · Qualifier: BEST · Review section 2.8

Principle: Mandatory leave with a substitute performing the duties is a **detective control** that can surface concealed irregularities.

A. Periodic screening can identify changed personal **risk** factors; however, it is preventive and would not uncover irregularities already concealed in the process.

B. An NDA protects confidential information; however, it does not reveal manipulation of **reconciliations** performed by the administrator.

C. Correct. When another trained person performs the duties, schemes that need continuous concealment are likely to surface, directly detecting irregularities.

D. A runbook reduces key-person dependency and supports continuity; however, **documentation** alone does not expose irregular activity.

Question 2.14 Answer: D

Third-party assurance period · Qualifier: NEXT · Review section 2.9

Principle: A Type II report covers only its stated period, so the auditor must obtain additional **evidence** for the uncovered months before concluding.

A. A bridge letter can provide context; however, it is a management assertion by the provider, not **independent evidence** of control operation.

B. Reporting may follow if **evidence** cannot be obtained; however, it is premature before the auditor has tried to close the gap.

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C. Testing **CUECs** is necessary for reliance; however, it addresses the customer's own controls, not the provider's unexamined months.

D. Correct. The report does not establish operation outside its stated period. Obtain and evaluate suitable additional **evidence** for the uncovered months and any material changes.

Question 2.15 Answer: C

Outsourcing contract terms · Qualifier: EXCEPT · Review section 2.9

Principle: Contracting out a service does not remove the customer's applicable **obligations** or accountable **risk** decisions. The provider can also have its own legal and contractual duties.

A. Subcontractor controls matter because fourth parties can change data location and **risk** without the customer's knowledge.

B. Exit terms ensure customer data is recoverable and that copies are destroyed, protecting continuity and **confidentiality** at termination.

C. Correct. The company retains its applicable regulatory **accountability**. Contractual allocation of duties and liability does not by itself remove those obligations.

D. Audit rights or independent **assurance** give the customer the visibility it needs to oversee controls it no longer operates directly.

Question 2.16 Answer: A

Source code escrow · Qualifier: MOST · Review section 2.9

Principle: Escrow releases **source code** to the customer on defined triggers such as vendor failure, so the software can still be maintained.

A. Correct. A suitable escrow arrangement can provide the source and supporting material under defined triggers. Current complete deposits, rights, tools and skills are also needed to maintain the service.

B. Price escalation is a commercial **risk**; however, it is managed through pricing clauses, and escrow release is not triggered by fee changes.

C. Defective updates are a real concern; however, they are addressed by testing and change control, not by depositing code with a **third party**.

D. Unauthorised vendor access is a genuine **risk**; however, it is controlled through access management and contract terms, not escrow.

Question 2.17 Answer: B

Balanced scorecard · Qualifier: BEST · Review section 2.10

Principle: An IT **balanced scorecard** links IT measures to business contribution, user, operational and future-orientation perspectives, measuring value and alignment.

A. KRIs signal changing **exposure**; however, they monitor risk rather than demonstrating the value IT delivers to the business.

B. Correct. A **balanced scorecard** combines business contribution with operational and future measures, giving the board a view of value and alignment beyond efficiency.

C. **Maturity** assessments show process capability; however, a mature process can still deliver little value to the enterprise.

D. **Service level** reports show operational performance; however, meeting SLAs indicates efficiency, not whether IT contributes to business goals.

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Question 2.18 Answer: D

Quality assurance independence · Qualifier: GREATEST · Review section 2.11

Principle: Suppression of adverse quality findings without **independent review** undermines objective challenge. Assess actual authority and safeguards rather than relying solely on reporting titles.

A. An outdated checklist may miss newer **requirements**; however, it can be updated readily and does not compromise reviewer objectivity.

B. Risk-based sampling is a common and acceptable QA practice; however, it limits coverage rather than undermining the **integrity** of the reviews performed.

C. Quarterly reporting may delay visibility of trends; however, it is a timeliness matter rather than a weakness in the **reliability of QA work**.

D. Correct. Reporting to the manager whose work is reviewed creates a conflict of interest that can suppress findings, undermining every QA conclusion.

2.16 Continue your preparation

Use your error log to identify the reason behind each mistake. Revisit the relevant explanation and **Last Mile** table, then test the concept with a fresh scenario. Register at certprepx.com for adaptive practice, timed full-length mocks and domain analytics. Available plans and features may change.

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Domain 3: Information Systems Acquisition, Development and Implementation

3.0 Domain overview

A system can be installed successfully and still fail the business. It may process the wrong information, allow unauthorized transactions, interrupt operations or deliver none of the promised **benefits**. Domain 3 follows the decisions that prevent those outcomes: justify the investment, define the **requirements**, design controls, build or acquire the solution, test it, introduce it safely and review the result.

This domain represents 12 percent of the current CISA exam outline. The eight main sections follow the established subject sequence. The explanations assume some IT auditing experience and concentrate on understanding the decision in a scenario, rather than memorizing an answer without its **context**.

Throughout the domain, distinguish the person who supplies evidence from the person who makes the decision. Developers build; users establish business acceptance; technical and operational owners accept their responsibilities; authorized management accepts **residual risk**. The auditor evaluates and advises while preserving **independence**.

Topics and reading route

Section	Subject
3.1	Project Governance and Management
3.2	Business Case and Feasibility Analysis
3.3	System Development Methodologies
3.4	Control Identification and Design
3.5	System Readiness and Implementation Testing
3.6	Implementation Configuration and Release Management
3.7	System Migration, Infrastructure Deployment and Data Conversion
3.8	Postimplementation Review
3.9–3.13	Last Mile, glossary, practice questions, answers and further practice

Read the explanations before using the tables. The **Last Mile** section is a revision aid; it does not replace the underlying reasoning. The practice questions are original learning exercises, not actual examination questions.

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3.1 Project Governance and Management

Project **governance** establishes authority, oversight and accountability. Project management organizes the work within those boundaries. A detailed schedule cannot compensate for an investment that has lost its **business justification**, and a strong sponsor cannot replace clear requirements and delivery controls.

3.1.1 Project Management Practices

A **project** is a temporary endeavor that delivers a defined result. Unlike continuing operations, it has a beginning, an intended end and agreed objectives. Its **scope**, time, cost, quality, resources and risks interact. Changing one can affect the others, so decisions must consider the whole project rather than a single target.

Initiating, planning, executing, **monitoring** and controlling, and closing are management activities. They are not interchangeable with technical SDLC phases. A project can use iterative development while still maintaining an approved **business case**, accountable owners and controlled changes.

Example: A payroll replacement is behind schedule. Removing recovery testing may protect the advertised date while creating unacceptable payroll disruption. Examine the **cause**, remaining risk and business priorities; management may need to change **scope**, resources, sequencing or the date through the approved process.

3.1.2 Project Management Structure

The structure determines how authority and resources are shared. In a functional structure, staff usually report to departmental managers and **project** authority may be limited. A projectized structure gives the project manager greater direct authority over a dedicated team. A matrix combines functional and project relationships, so priorities and escalation rights need particular clarity.

Assess the **actual authority** in use, not only the organization chart. Shared specialists, competing operational demands and unclear reporting can undermine an otherwise credible plan.

3.1.3 Project Management Roles and Responsibilities

Roles should make decisions and **accountability** visible. One person may hold several roles in a small enterprise, but incompatible responsibilities still require suitable safeguards.

Role	Primary responsibility
Sponsor	Owns the business justification, secures support and resolves escalated business issues.
Steering committee/project board	Provides oversight and makes decisions within delegated authority, including major changes and stage gates.
Project manager	Coordinates delivery, plans work, manages dependencies and reports status and issues.
Business/process owner	Defines business needs, owns business controls and accepts business suitability.
Development/configuration team	Builds or configures the approved solution and provides technical evidence.

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Role	Primary responsibility
Test and quality specialists	Plan and perform appropriate verification and report defects and results.
Operations/service owner	Establishes support, monitoring, recovery and operational readiness.
Security, risk and compliance specialists	Advise on applicable requirements and assess relevant risks.
IS auditor	Evaluates governance, risk, controls and evidence without becoming the project owner or approver.

3.1.4 Project Management Techniques

A **work breakdown structure (WBS)** decomposes the work required for a project into manageable elements. A work package can have an owner, estimate and completion criteria. This supports scope and accountability; a schedule adds timing and dependencies. Section 3.1.9 distinguishes the work view from a product or deliverable view.

Use techniques that support decisions: a work breakdown structure for **scope**, a dependency network for sequencing, a schedule for timing, a risk register for uncertainty and an issue log for problems already encountered. Status reports should compare the **approved baseline** with actual results and explain deviations.

Plan-driven methods emphasize defined phases and baselines. Adaptive methods deliver and learn in short cycles. Hybrid arrangements combine them. The audit question is whether the chosen method provides appropriate **accountability** and evidence for the **project**'s risk, not whether one label is universally superior.

3.1.5 Portfolio/Program Management

A portfolio is a collection of investments managed to achieve strategic objectives; its projects need not be directly related. A program coordinates related projects and activities to achieve **benefits** that require joint management. A project delivers its specified result. Assessing only individual project schedules can miss shared **dependencies** and competing demands across the portfolio.

Example: Identity modernization, customer onboarding and payment integration may form a **program** because the intended digital-service benefit depends on all three. The enterprise **portfolio** may also include an unrelated data-center replacement competing for the same funding.

3.1.6 Project Management Office

A **project** management office (PMO) supports **consistency** through methods, reporting, expertise and, where authorized, delivery oversight. A supportive PMO advises; a controlling PMO requires compliance with defined practices; a directive PMO can manage projects directly. Its actual mandate should be documented.

Project Portfolio Database

A portfolio **database** records investments, owners, status, costs, **benefits**, risks and dependencies. Its value depends on reliable updates and decision use. Reconcile material figures to financial and project records; a dashboard can look precise while being based on optimistic or outdated submissions.

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3.1.7 Project Benefits Realization

Benefits realization identifies, owns, measures and sustains the expected business value. Define a baseline, target, measure, timing and accountable owner. Delivery outputs such as installed servers are different from **business outcomes** such as reduced processing time. Benefits tracking starts during the business case and continues after project closure.

Example: A procurement system promises fewer duplicate payments. Measure the relevant **baseline** and later duplicate-payment rate, including **exceptions** processed outside the system. Completion of user training alone does not demonstrate the promised benefit.

3.1.8 Project Initiation

Initiation establishes why the project is needed, who sponsors it, what its high-level **scope** is and whether it should proceed. A project charter or equivalent authorizes the project and states objectives, ownership, authority, assumptions and constraints. It does not replace the detailed plan or the continuing **business case**.

3.1.9 Project Objectives

Objectives should state a result that can be assessed: what will change, by how much, by when and within what constraints. Distinguish mandatory business and control **requirements** from preferences. Vague objectives such as “modernize the system” make acceptance and **benefits** measurement difficult.

An object breakdown structure describes the product or deliverables. A work breakdown structure (WBS) decomposes the work needed to produce them into manageable elements. A **work package** is a unit of planned work that can have an owner, estimate and completion **criteria**. These terms describe different views of the same delivery effort.

3.1.10 Project Planning

Planning integrates **scope**, deliverables, resources, **dependencies**, schedule, budget, quality, communications, procurement and risk. Record assumptions so they can be challenged when conditions change. A baseline is an approved reference for comparing performance; revising it requires authority and a retained explanation.

Cost and software size estimation

Estimate the work using suitable historical data, expert judgment, analogous projects, parametric models or bottom-up work packages. Include testing, **migration**, training, **documentation**, controls, infrastructure and operating transition. A development-only estimate understates the investment.

Function Point Analysis (**FPA**) measures functional size from user-recognizable functions, commonly external inputs, external outputs, external inquiries, internal logical files and external **interface** files. Function points are not staff hours. Convert size into effort using a defensible productivity model and project **assumptions**. Lines of code depend heavily on language and implementation; COCOMO, the Constructive Cost Model, estimates effort using size and other cost drivers.

Scheduling and establishing time frames

A **Gantt chart** displays activities against time. A network model makes **dependencies** visible. The **critical path method (CPM**) uses the dependency network to identify the critical path. This is the

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longest-duration dependent path through the modeled work and determines the earliest completion under the model's assumptions. Total float is the delay an activity can absorb without delaying the modeled completion date. There can be multiple critical paths; resource constraints and changed estimates can alter the schedule.

Worked example: Requirements A takes 2 days. Build B takes 4 days after A; data preparation C takes 3 days after A. Testing D takes 2 days after both B and C, followed by **cutover** E of 1 day. A–B–D–E is 9 days and is critical. A–C–D–E is 8 days, so C has 1 day of total float under these **assumptions**.

PERT, the Program Evaluation and Review Technique, uses optimistic (O), **most** likely (M) and pessimistic (P) estimates. Its conventional weighted expected duration is (O + 4M + P) / 6. With 2, 5 and 8 days, the expected duration is 5 days. This estimate is not a guarantee.

Total float (total slack) = LS − ES = LF − EF. ES/EF are early start/finish; LS/LF are late start/finish. If ES = day 2, EF = day 5, LS = day 4 and LF = day 7, total float is 2 days. Use the same calendar and date convention.

Free float is the delay possible without delaying the early start of an immediate successor. In a simple finish-to-start network without lags, free float = earliest ES among immediate successors − activity EF. It can differ from total float. Subtracting an activity's duration from overall project duration does not work for every dependency network. [R41]

Timebox management

A **timebox** fixes a period for work and prioritizes what can be completed to the required **standard**. Reduce or reprioritize scope through the authorized process rather than quietly removing essential testing or controls. Cost may be constrained by a stable team, but timeboxing alone does not guarantee a fixed total cost.

3.1.11 Project Execution

Execution coordinates people and suppliers to produce the planned outputs. Maintain clear work ownership, issue escalation, quality reviews and controlled deliverables. Confirm that necessary specialists and users are genuinely available. Adding staff late can increase communication and training effort; **Brooks's law** describes that tendency in software projects, rather than a universal prohibition on adding resources.

3.1.12 Project Controlling and Monitoring

Compare actual progress, cost, quality and risk with the **approved baseline**. Distinguish completed work from effort spent. Manage scope changes through impact analysis and approval, update resource forecasts and reassess the **business case** when material assumptions change. A risk register concerns uncertain events; an issue log concerns problems that have occurred.

Earned value analysis and management

Earned value analysis (EVA), within earned value management (EVM), combines scope, schedule and cost evidence. **Planned value** (PV) is the budgeted value of work scheduled. Earned value (EV) is the budgeted value of work actually completed. **Actual cost** (AC) is what was spent. Schedule variance (SV) = EV − PV; cost variance (CV) = EV − AC. Schedule performance index (SPI) = EV/PV; cost performance index (CPI) = EV/AC. These measures depend on credible budgets and completion evidence.

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Worked example: PV is $120,000, EV is $90,000 and AC is $100,000. **Schedule variance** is −$30,000 and SPI is 0.75. **Cost variance** is −$10,000 and CPI is 0.90. The project has delivered less budgeted work than planned and spent more than the **earned value**. The schedule variance is expressed in value units, not a direct number of days.

Supporting forecast: budget at completion (BAC) is the approved total budget. If current cumulative cost efficiency is assumed to continue, **EAC = BAC/CPI**, where EAC is estimate at completion. Then **ETC = EAC − AC** estimates remaining cost, and **VAC = BAC − EAC** shows the forecast budget variance.

Example: BAC = $200,000 and CPI = 0.80 produce EAC = $250,000. With AC = $75,000, ETC = $175,000 and VAC = −$50,000. These are forecasts under the stated assumption; a revised estimate or a different future-efficiency assumption can require another EAC method. [R42]

3.1.13 Project Closing

Closure confirms agreed deliverables and acceptance, resolves or formally transfers remaining work, finalizes contracts and records, captures **lessons** and releases resources. Transfer support and benefit responsibilities to **named owners**. Administrative closure does not establish that benefits have already occurred.

3.1.14 IS Auditor's Role in Project Management

Assess the **business case**, roles, reporting, risk management, changes, quality gates and evidence behind major decisions. Report material concerns early enough to be useful. Advisory involvement can improve auditability, but audit should not manage the project, select its solution or accept its **residual risk**. Where prior involvement creates an objectivity concern, apply appropriate **independent review** arrangements.

Auditor Perspective

Reassess before choosing a rescue action

When cost, **scope** or expected benefits change materially, determine whether the **business case** and priorities remain valid. Adding staff or reducing testing is not an automatic recovery strategy.

The sponsor and **governance** bodies make authorized investment decisions. The project manager coordinates delivery; audit evaluates the evidence.

Reading project evidence and earned value together

Project governance establishes sponsorship, **decision rights**, **benefit ownership** and escalation. Project management coordinates scope, time, cost, quality, resources and risk. Audit can provide timely advice and independent assessment without authorizing the project's business decisions. A stage gate should use evidence of readiness, rather than merely a scheduled meeting.

Worked example: Apply the formulas in section 3.1.12 to a second case. **Planned value** is 100, **earned value** is 80 and **actual cost** is 90 in the same cost units. Schedule variance is -20, cost variance is -10, SPI is 0.80 and CPI is approximately 0.89. The project has earned less than planned and spent more than the value earned. These indicators need credible baselines and completion measurement; they do not directly express calendar delay.

Review **dependencies**, critical-path **assumptions**, resource conflicts and approved changes. PERT's conventional expected duration is (optimistic + 4 × most likely + pessimistic)/6; it is an estimate, not a

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guaranteed deadline. Example: A team moves unfinished security work outside scope to report on-time completion. Trace changed scope to the sponsor's approved objectives and **acceptance criteria**. A favorable dashboard cannot compensate for silently excluded requirements.

Read a Status Report Against the Approved Baseline

Example. A sponsor receives a green schedule status after the team removes delayed acceptance work from the reporting baseline. Review who authorized the scope and schedule change and what it does to benefits, risk and readiness. Trace earned value to completed deliverables that meet the agreed criteria. Inspect dependencies and forecast assumptions before accepting a new date. If work was deferred rather than completed, the board or steering body needs a clear account of the remaining commitment and its consequences.

3.2 Business Case and Feasibility Analysis

Cost-benefit analysis compares the relevant costs and benefits of alternatives over a defined period, using consistent assumptions. Include operating, transition and exit costs, uncertainty and material nonfinancial effects. A favorable financial estimate does not authorize ignoring a mandatory obligation.

The **business case** explains why an investment should proceed: the problem or opportunity, options, expected value, lifecycle costs, risks and alignment. **Feasibility** asks whether the option is practical within its constraints. A technically possible solution can still be unaffordable, unsuitable for users or impossible to deliver in time.

Assessing viability and value

Dimension	Question and example
Economic	Do credible benefits justify lifecycle costs and risk? Include implementation, licenses, support, training and exit.
Technical	Can available technology, skills and architecture meet the requirements?
Operational	Can the business operate and support the proposed process and controls?
Schedule	Can the necessary result be delivered within the real deadline and dependencies?
Legal/regulatory	Can the option meet applicable obligations, rights and contractual constraints?

Return on investment compares net benefit with investment under a stated formula and period. Payback measures the time to recover the initial outlay. Net present value (**NPV**) discounts future cash flows and includes the investment; a positive NPV indicates value under the **assumptions**. Internal rate of return (IRR) is the discount rate at which NPV is zero. Do not choose solely from the largest percentage: timing, risk, mandatory **obligations** and unreliable forecasts can change the decision.

Use sensitivity analysis to test how the result changes when **assumptions** move, such as user adoption, operating cost or delivery delay. Consider the do-nothing option where meaningful. For make-versus-buy, compare requirement fit, control, time, skills, ownership, maintenance and exit. A package can reduce development effort, but extensive **customization** can remove that advantage.

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Worked investment calculations

Use a consistent **cash-flow scope and period**. Consider a project with an initial outlay of $100,000 and four annual net operating cash inflows of $40,000, with no residual value or further outlay. Net inflows already reflect the operating costs included in this example.

Simple ROI = (total benefits − investment cost) / investment cost × 100%. Here ROI = ($160,000 − $100,000)/$100,000 × 100% = **60% over four years**. It is not an annualized return. State the convention: organizations can define benefits and investment bases differently.

Simple payback = initial outlay / constant net inflow per period. Here $100,000/$40,000 = **2.5 years**, assuming receipts accrue uniformly within each year. If cash is received only at year-end, recovery first occurs at the end of year 3.

For **uneven inflows**, accumulate them. An outlay of $100,000 followed by $30,000, $40,000 and $50,000 leaves $30,000 unrecovered after year 2. If year-3 receipts accrue uniformly, payback is 2 + 30,000/50,000 = **2.6 years**. Simple payback ignores discounting and receipts after recovery.

NPV = CF0 + Σ[CFt/(1 + r)^t], for t = 1 to n. CF0 is the time-zero cash flow, ordinarily negative for the initial outlay; CFt is net cash flow at period t; r is the per-period discount rate; n is the number of periods. Future outflows carry negative signs. Do not subtract the initial investment twice.

Using year-end inflows and a 10% annual discount rate for the four-year example: NPV = −100,000 + 40,000/1.10 + 40,000/1.10^2 + 40,000/1.10^3 + 40,000/1.10^4 ≈ **$26,794.62**. The positive result depends on the modeled cash flows and rate. Section 3.2 provides a separate two-year example.

IRR is a discount rate at which the modeled NPV equals zero: NPV(r = IRR) = 0. It need not have one useful solution for every cash-flow pattern. Assess scale, timing, risk, reliable benefit evidence and mandatory needs alongside the reported return.

3.2.1 IS Auditor's Role in Business Case Development

Evaluate whether the justification is complete, supported, approved and consistent with strategy and **risk appetite**. Challenge omitted costs, duplicated benefits, unrealistic schedules and unassigned **benefit ownership**. The auditor can recommend stronger analysis without becoming responsible for the investment decision.

Example: A proposed automation claims all staff time saved as cash savings while the staffing budget remains unchanged. Time may still be a valid **capacity** benefit, but the financial forecast should describe it accurately. Confirm the **baseline**, how the released capacity will be used and who will measure the outcome.

Testing a business case rather than accepting its arithmetic

A **business case** explains the problem, alternatives, expected **benefits**, lifecycle costs, risks, assumptions and accountable **benefit owners**. Assess technical, operational, financial, scheduling and other applicable feasibility. Include retaining or improving the current service where it is a credible alternative. Sunk costs should not dictate a forward decision, although exit obligations and future transition costs remain relevant.

Worked example: A **project** costs 100 now and produces 60 at each of the next two year-ends. At a 10 percent discount rate, NPV is −100 + 60/1.10 + 60/1.10^2, approximately 4.13 in the same units. The modeled financial result is positive under those **assumptions**. Payback gives a different view of timing and can ignore later flows; neither measure establishes that the **benefits** will actually occur.

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Example: The main saving assumes six posts disappear, but the approved staffing plan retains all six without a measurable redeployment benefit. Challenge the assumption and reassess the case. Sensitivity analysis should expose the **assumptions** capable of changing the decision. Include intangible outcomes where appropriate, describe how they will be assessed and avoid counting the same benefit twice. After **approval**, revisit the case when costs, **scope** or expected value materially change.

Test Whether a Benefit Can Actually Be Realized

Example. A business case counts a license saving from retiring a legacy application at launch. The migration plan retains the old system for reconciliation and regulatory access for another year. Recalculate the benefit using the actual retirement date, include dual-running and exit costs, and test the sensitivity of the result to further delay. Confirm who owns the benefit and which event triggers it. A favorable discounted result supports the stated assumptions. The audit tests whether the plans and evidence make those assumptions credible.

3.3 System Development Methodologies

A development methodology organizes how needs become an operational system. Whatever the method, the enterprise needs appropriate **requirements**, design, control, verification, acceptance and change evidence. Buying a package or using cloud services changes the delivery work; it does not remove **accountability** for the result.

3.3.1 Business Application Development

Business applications support activities such as ordering, payroll, banking and reporting. Their development should begin with the business process and information needs. Identify users, data owners, transactions, **dependencies**, applicable **obligations** and the consequences of incorrect or unavailable processing.

System and architecture types

These terms describe different purposes and structures. A decision-support system is not the same as a transaction system, and a service-based architecture is not automatically secure. The architecture determines where **interfaces**, shared **dependencies** and control responsibilities need attention.

System or concept	Purpose and audit focus
MIS	A management information system supplies routine structured reports. Check definitions, completeness, accuracy and timeliness.
DSS	A decision support system uses data, models and what-if analysis for semi-structured decisions. Validate assumptions, inputs and model limits; it supports the decision-maker.
EIS	An executive information system summarizes indicators and trends for senior management, often with drill-down. Verify provenance and indicator definitions.
Expert system	A knowledge base and inference engine apply expert rules. Review rule accuracy, updates, conflict handling and ownership.

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System or concept	Purpose and audit focus
AI, machine learning and neural networks	AI is the broader field; machine learning learns patterns from data. Neural networks are one model family. Assess **data quality**, bias, performance, explainability and oversight.
GIS	A geographic information system analyzes spatial information. Check location accuracy, coordinate systems and relevant source data.
EDI	Electronic data interchange exchanges structured business documents between organizations. Assess standards, mapping, authentication, acknowledgments and reconciliation.
E-commerce	Digital transactions require dependable authorization, order and payment integrity, privacy, availability and fulfillment interfaces.
Middleware and client-server	Middleware connects components; client-server separates client and server responsibilities. Assess trust boundaries, compatibility and shared dependencies.
SOA and microservices	Service-oriented architecture organizes capabilities as services. Microservices commonly use smaller independently deployable services; the terms are related, not identical. Control **interfaces**, identities, dependencies and observability.
API	An application programming interface defines program interactions. Assess authentication, authorization, validation, error handling, versioning and rate controls.
Data warehouse/data mart and ETL	A warehouse integrates data for analytics; a mart serves a subject or group. Extract-transform-load processes need controlled mapping, reconciliation, lineage and timely loads.
Decision table/decision tree	A table maps combinations of conditions to actions; a tree shows branching decisions. Both help identify missing, conflicting or untested rules.

Business system	Meaning and audit focus
Enterprise resource planning (ERP)	Integrates core business processes through coordinated application modules and shared or consistently integrated data. Check configuration, SoD, master data, interfaces, migration and complete transaction flows.
Human resource management system (HRMS)	Supports personnel records and HR processes. Check employee-data confidentiality, role permissions, interfaces and joiner/mover/leaver information.

3.3.2 SDLC Models

The systems development life cycle (**SDLC**) covers activities from identifying a need through development or acquisition, implementation and review. A waterfall model organizes work mainly in sequential phases. Iterative development refines the solution through repeated cycles; incremental development delivers portions of functionality. A **project** can be both iterative and incremental.

The V-model emphasizes the relationship between specification and verification. Business needs inform acceptance tests; system **requirements** inform system tests; **interfaces** inform integration tests;

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detailed module design informs unit tests. This is a traceability relationship, not a reason to defer all test thinking until coding ends.

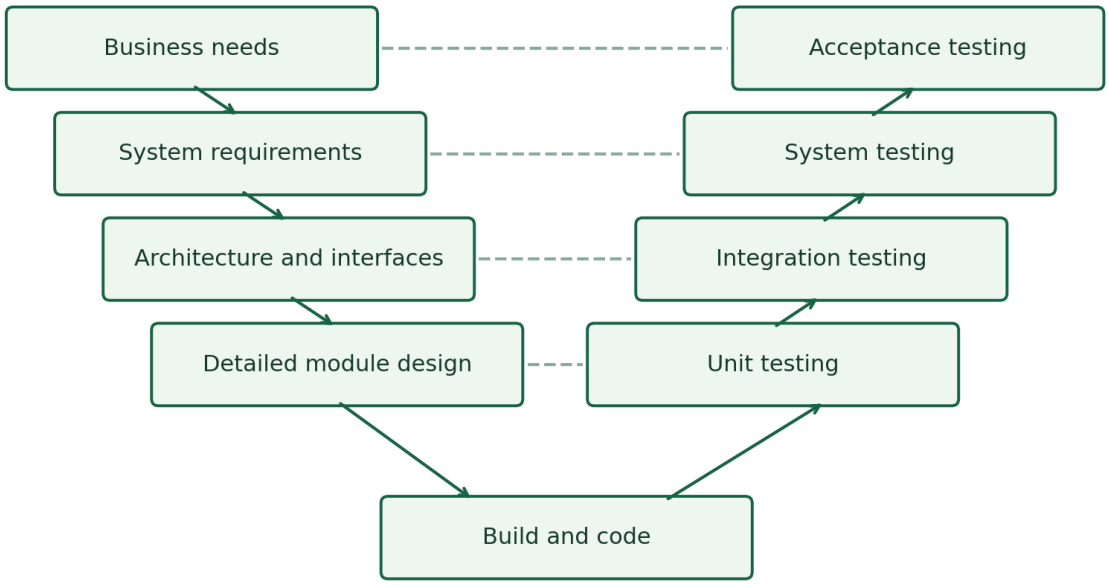


Figure 3.1 V-model traceability: plan corresponding tests while defining requirements and design.

3.3.3 SDLC Phases

The classic phases provide a useful decision sequence even when actual delivery overlaps or iterates. Establish the business need before choosing technology, define **acceptance criteria** as **requirements** develop, and retain approval and verification evidence through the lifecycle.

Phase	Purpose and expected evidence
Feasibility	Evaluate viability and options; establish a supported business case and decision to proceed.
Requirements definition	Specify business, functional, data, control and nonfunctional requirements, with measurable acceptance criteria.
Selection/acquisition or design	For a package, evaluate fit and terms. For a build, specify architecture, data, interfaces and embedded controls.
Configuration or development	Create the solution from approved specifications; control code, parameters, reviews and unit tests.
Final testing and implementation	Demonstrate readiness and obtain appropriate acceptance; convert data and deploy through an authorized plan.
Postimplementation review	Evaluate the live result, control effectiveness, costs, benefits and lessons.

Requirements definition and traceability

A **functional requirement** states what the system must do: calculate overtime under approved rules. A nonfunctional requirement states a quality or constraint: complete payroll within the agreed processing window, protect sensitive records or meet a defined **recovery objective**. Make requirements testable and identify the owner and priority.

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Gather needs through interviews, **observation**, workshops, existing records and prototypes. Include process **exceptions**, interfaces, audit trails, security, retention, reporting and recovery. Record assumptions and resolve conflicting requirements. User involvement matters because a perfectly coded specification can still describe the wrong business process.

A **requirements traceability** matrix connects requirements to design, build/**configuration**, tests and acceptance. Use it in both directions: important needs require evidence, and delivered features require an approved basis. Control changes after baselining and assess their effect on related controls and tests.

Design, data modeling and development

Design converts **requirements** into architecture, workflows, **interfaces**, data structures, screens and controls. An entity-relationship diagram (ERD) models entities, keys and relationships, including their cardinality. Review both normal operation and exceptions, such as canceled orders or failed payments.

Normalization organizes relational data to reduce redundancy and update anomalies. First normal form uses atomic values without repeating groups; second normal form removes partial dependency on a composite key; third normal form removes inappropriate transitive dependency of non-key attributes. Denormalization deliberately introduces redundancy, often for **performance**, and needs controls to preserve consistency.

Example: Storing a **supplier** address separately on every unpaid invoice can create conflicting versions. A supplier master can reduce repetition, while the design may intentionally preserve an invoice's historical address. The correct model depends on the business meaning, not a rule to eliminate every repeated value.

Development uses approved designs, coding standards, reviews, controlled repositories and testing. Integrated development environments and debugging tools improve productivity but do not replace **independent review** or formal defect handling. **Configuration** changes in purchased software deserve the same attention to authorization and testing as code changes.

Object-relational mapping (ORM) maps application objects to relational database structures and operations. Review the actual generated queries, authorization and transaction handling; using an ORM does not guarantee absence of injection or data-integrity defects.

3.3.4 IS Auditor's Role in SDLC Project Management

Assess whether the chosen lifecycle contains appropriate review points, user involvement, **risk analysis**, requirements traceability, **control design** and evidence. Review auditability early: who can reconstruct transactions, investigate exceptions and prove approval? Raise material concerns before design decisions become costly to change.

Maintain an advisory role. Audit can identify a missing **control objective** or evaluate a proposed control, but should not own the design, write the production code or sign business acceptance. Evaluate actual responsibilities and **objectivity** rather than relying only on a job title.

3.3.5 Software Development Methods

Choose a method suited to uncertainty, delivery constraints and **risk**. Each method needs disciplined ownership and **evidence**. Speed does not justify omission of critical controls or verification.

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Method	How it works	Main consideration
Waterfall	Progresses mainly through defined phases and baselines.	Useful with stable needs; late discoveries can be costly.
Iterative/incremental	Refines the solution and/or delivers functionality in portions.	Maintain integration, architecture, regression tests and an overall view of requirements.
Prototyping/evolutionary development	Uses models or early versions to clarify needs; some prototypes evolve into the final product.	Distinguish disposable demonstrations from production-ready code; control scope and expectations.
RAD	Rapid Application Development combines fast user feedback, prototyping and tools, often within timeboxes.	Requires available users, suitable architecture and retained quality controls.
Agile	Uses short feedback cycles, collaboration and adaptive planning to deliver value.	Keep risk, control and acceptance evidence proportionate and current.
Object-oriented development	Organizes software around objects that encapsulate data and behavior.	Review inheritance, interfaces, reuse and correct object interactions.
Component-based development	Assembles reusable components.	Assess provenance, licensing, vulnerabilities, compatibility and maintenance.
Web application development	Delivers functionality through web technologies and interfaces.	Address input validation, access control, sessions, APIs and deployment exposure.
DevOps/DevSecOps	Integrates development and operations; DevSecOps embeds security throughout delivery.	Control repositories, pipelines, build inputs, secrets, release approvals and production access.
Reengineering/reverse engineering	Reengineering changes an existing system; reverse engineering recovers its design or logic.	Recovered behavior may include defects; verify the intended business rules.

The **Common Gateway Interface (CGI)** is a legacy interface through which a web server invokes a program or script to generate a response. Review input handling, execution privileges, maintenance and testing. It is an interface mechanism, rather than a development methodology or an automatic control.

Object-oriented concept	Meaning and example
Encapsulation	Groups state and behavior behind a defined interface and controls access to implementation details. An account object can expose an approved update operation instead of unrestricted field changes.
Inheritance	A derived class reuses or specializes members and behavior of another class. This is different from inheriting a user's access permissions.
Polymorphism	A common interface or operation has different implementations for different object types. The same payment operation can execute differently for different payment types.

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These design mechanisms do not prove authorization or secure implementation. Test the relevant implementations and interfaces. Timeboxing is already explained in section 3.1.10. A data flow diagram (**DFD**) is introduced in section 2.2.7. [R38]

Scrum, delivery pipelines and process redesign

Scrum is a framework commonly used for adaptive development. The Product Owner orders the Product Backlog to maximize value; Developers create a usable increment; the Scrum Master supports effective Scrum. Sprints contain planning, daily coordination, review and **retrospective** activities. A Definition of Done makes required quality conditions explicit. Business acceptance arrangements and **release authorization** still need clarity.

Continuous integration regularly combines changes and runs checks. Continuous delivery keeps verified changes ready for **release**; continuous deployment automatically deploys changes that satisfy the pipeline's **criteria**. Automation makes reliable controls repeatable, but weak credentials, bypassed gates or incomplete tests can make errors repeatable too.

Business process reengineering redesigns how work is performed, rather than simply automating the existing steps. Benchmarking compares **performance** or practice with a relevant reference. Before copying another organization's approach, examine differences in objectives, obligations and operating constraints. Redesign roles, approvals and information flows alongside the technology.

3.3.6 System Development Tools and Productivity Aids

Computer-aided software engineering (CASE) tools support models, **documentation**, code or **database** generation and associated lifecycle activities. Upper CASE commonly supports analysis and requirements, middle CASE detailed design, and lower CASE implementation; usage can vary. A repository keeps related definitions and artifacts together, so access, consistency and version controls are important.

Code generators and fourth-generation languages (4GLs) can reduce detailed programming work. Generated software still needs appropriate testing. Protect the model, generation rules and **dependencies**, and confirm that generated artifacts match the approved inputs. User-built reports and applications also need **governance** when they support material business decisions.

3.3.7 Infrastructure Development/ Acquisition Practices

IT infrastructure is the supporting technology, facilities and operational resources on which information services run. Networks, servers, storage, system software and power or cooling dependencies are examples. Acquiring one device is therefore only part of establishing a usable service.

Infrastructure must support business service **requirements** for **performance**, security, capacity, availability and recoverability. Examine existing constraints and the target architecture before procurement. Include networks, facilities, hardware, system software, staffing and operating support.

Physical architecture analysis and implementation planning

Analyze the current environment, establish **requirements**, design and compare options, and validate uncertain **assumptions** through a proof of concept where appropriate. A proof of concept tests selected feasibility questions; it is not full acceptance. Define success criteria for compatibility, controls, performance and resilience.

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Coordinate procurement, delivery, site **readiness**, installation and testing. Power, cooling, space, network circuits or unavailable skills can become critical **dependencies**. Assign owners for hardening, installation checks, recovery, documentation and operational handover.

3.3.8 Hardware/Software Acquisition

Base procurement on **approved requirements** and objective evaluation. A request for information (RFI) explores market capabilities; a request for proposal (RFP) asks suppliers to propose solutions to stated needs; a request for quotation (RFQ) seeks price and terms for sufficiently specified items. Their use depends on the procurement **context**.

Set evaluation **criteria** and **weights** before final scoring. Assess mandatory requirements separately, then compare fit, controls, compatibility, supplier viability, support, references and lifecycle cost. A high average score should not hide failure of an essential control requirement.

Use scripted demonstrations or a conference-room **pilot** to test the enterprise's own critical scenarios. A **supplier**'s preferred demonstration may avoid the difficult interfaces or exceptions. Record gaps, proposed workarounds and evidence, and obtain specialist input for technical, security, legal and commercial terms.

3.3.9 System Software Acquisition

System software operates or supports the computing platform used by other programs. Operating systems and system utilities are examples; database and middleware platforms provide further shared services. **Application software** performs business tasks such as payroll or order processing. Acquisition must consider both the supporting platform and the business application's dependencies.

For operating systems, databases and middleware, evaluate **compatibility**, security, **performance**, capacity, vendor support, staffing and expected life. Include installation, training, licensing, maintenance, migration and exit costs. Both open-source and proprietary products require control of rights, versions, dependencies and support.

Contract terms should address deliverables, responsibilities, dates, price, **documentation**, acceptance, security, **confidentiality**, maintenance, support and appropriate assurance or continuity arrangements. Link payment to meaningful agreed milestones where suitable. Use appropriate legal review.

IS Auditor's Role in Software Acquisition

Evaluate whether the acquisition is justified, consistently assessed and approved. Review evidence of requirement fit, **supplier** claims, significant gaps, contract controls and acceptance arrangements. Audit evaluates selection **integrity** and risk; it does not choose the product for management.

Configuration changes supported parameters; **customization** changes or extends program logic. Prefer configuration when it meets approved needs, because customization can increase upgrade, testing and support complexity. Neither is automatically low risk.

Source-code escrow places agreed source and supporting materials with an independent holder for release under specified events. Check complete and current deposits, build tools and dependencies, documentation, release conditions and rights to use the material. Possession of **source code** alone does not guarantee that the enterprise can rebuild or maintain the service.

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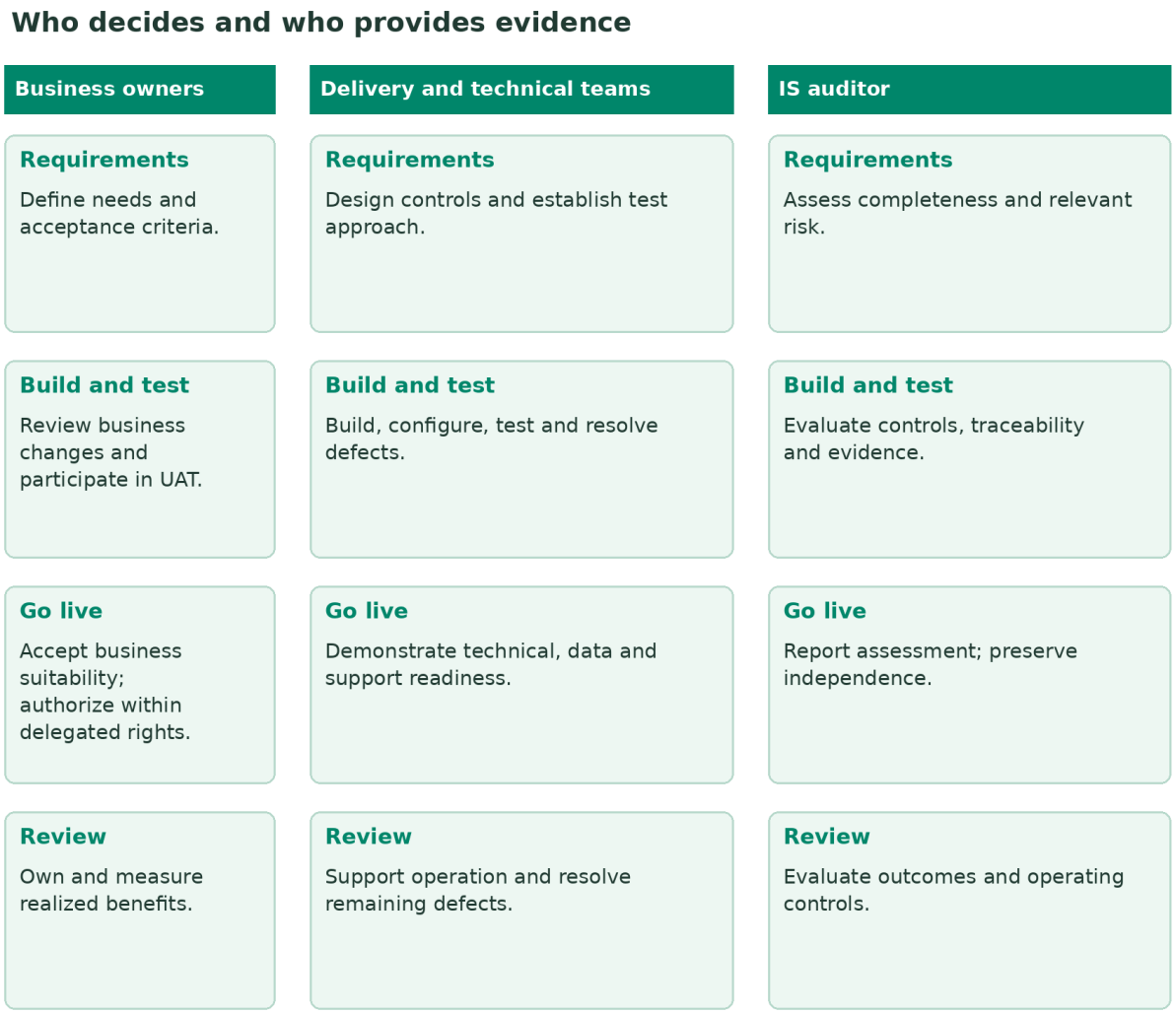


Figure 3.2 Decision ownership remains distinct through delivery.

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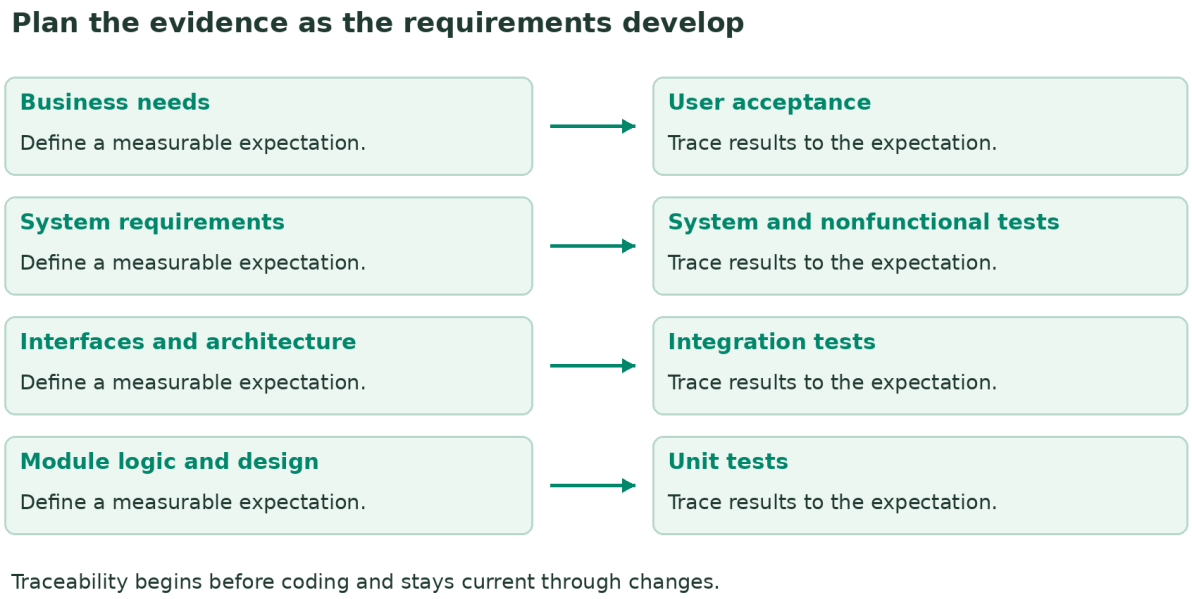


Figure 3.3 Requirements and tests remain traceable through change.

Assuring iterative delivery and the software supply chain

Lifecycle methods organize delivery differently, but each needs justified **requirements**, suitable design, controlled construction, testing, acceptance and support. Agile work can maintain these controls in small increments using backlog **criteria**, automated tests and reviewed changes. DevOps integrates development and operations; DevSecOps embeds security into the delivery process. None of these labels removes management **accountability** or the need for evidence.

Practical extension: **SAST** examines code or related representations without running the application. DAST exercises a running application. **Software composition analysis** examines third-party components and known issues. An SBOM identifies software components and versions, supporting vulnerability and licensing analysis; it is neither a vulnerability test nor a security certificate. **Test coverage**, tool limitations, false positives and exception handling matter more than the presence of a tool name. [R10, R13]

Protect repositories, build identities, **dependencies**, secrets, signing keys and deployment **authorization**. Use trusted inputs, controlled versions, artifact integrity and appropriate provenance. Example: An attacker compromises a permitted build action and signs an altered release with an accessible key. A **valid signature** alone does not establish a trustworthy build. Trace one production artifact back to its approved source, dependency set, build records, tests and deployment decision, and investigate the ability to bypass any gate.

Evidence	What it supports	Important limitation
SAST and DAST results	Specified code and running-behavior checks	Coverage and configuration can miss weaknesses
SBOM and composition analysis	Component visibility and relevant known issues	Unknown or omitted components remain possible
Signature and provenance	Artifact integrity and documented origin	Compromised signing or build trust undermines reliance

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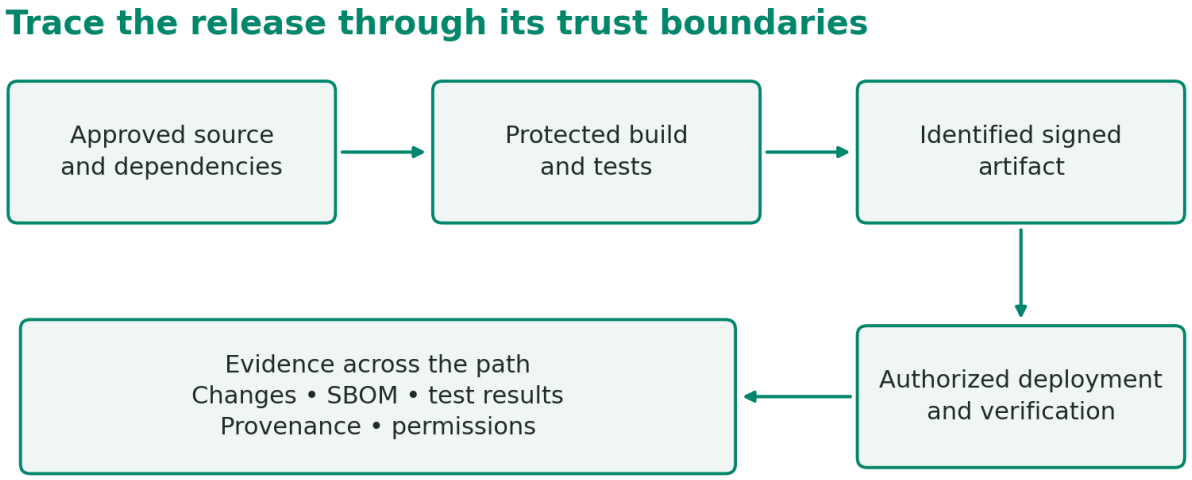


Figure 3.4 Assurance across source, dependencies, build artifacts and deployment.

Project metrics alongside lifecycle methods

Earned value analysis (EVA), used within earned value management (EVM), and the **critical path method (CPM**) support project **monitoring** under an appropriate plan; they are not alternative SDLC methodologies. Section 3.1.12 explains PV, EV, AC and their performance measures. Section 3.1.10 explains dependency paths and float. Compare these indicators with delivered quality, accepted scope and risk, rather than treating a favorable ratio or schedule as proof of control readiness.

Verify the Release Path for a Supplier Component

Example. A software bill of materials lists the approved components, but a maintenance pipeline can rebuild and publish an artifact with different dependencies. Compare the inventory and approved build provenance with the actual released artifact. Check dependency resolution, build identity, signing authority and whether security gates apply to every authorized release route. Examine an approved exception or bypass to determine what it permitted and how it was reviewed. An inventory helps identify exposure. The provenance and enforcement evidence show what was actually built and released.

3.4 Control Identification and Design

Design controls around the business risks while **requirements** and design are being developed. Specify the objective, owner, action, evidence and exception handling. Adding controls just before go-live is often costly and incomplete. Trace controls through input, processing and output, including interfaces and **privileged access**.

3.4.1 Application Controls

Application controls protect the **authorization**, completeness, accuracy and validity of transactions and information within a business application. They may be automated, manual or combined. General

IT controls support dependable operation: an automated approval rule is unreliable if unauthorized users can change it in production.

Input/Origination Controls and Input Authorization

Input controls apply at the source and every entry path: screens, files, **interfaces**, scanners and APIs. Confirm authorized **origin**, required approval and acceptable data. Separate transaction creation, approval and sensitive master-data maintenance where appropriate. A valid supplier and correctly formatted bank account do not prove that a bank-account change was authorized.

Input check	Meaning and example
Completeness	Required information is present: reject an invoice without supplier ID. Presence does not prove correctness.
Format/picture	Characters follow an expected pattern. A date also needs a valid-calendar check.
Validity/existence	Value belongs to an allowed set or exists in a reference file: supplier must exist and be active.
Range	Value lies between lower and upper bounds: month 1 through 12.
Limit	Value meets a single boundary: approval does not exceed the clerk's authority.
Reasonableness	Value makes sense in its business context: investigate implausible overtime.
Sequence	Expected numbering or order is maintained: investigate unexplained invoice gaps.
Duplicate	Repeated submissions or records are detected using an appropriate matching rule.
Check digit	A calculated digit helps detect identifier entry errors. It does not prove account ownership.
Key verification	Independent re-entry or comparison detects transcription differences.
Cross-field consistency	Related fields satisfy a rule: delivery date must not precede order date.

Batch Controls and Balancing

Establish reliable **batch** control information and compare it through processing. Include accepted, rejected and pending items, with explained adjustments. Do not force agreement by inserting unsupported balancing entries.

Control total	Meaning and limitation
Record count	Number of records. Equal counts can conceal an omission replaced by a duplicate or another record.
Financial/batch total	Sum of meaningful amounts or quantities. Offsetting differences can leave the total unchanged.

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Control total	Meaning and limitation
Hash total	Arithmetic sum of a normally nonfinancial numeric field, such as employee identifiers. This is not a cryptographic hash.
Run-to-run totals	Compare control information between processing stages, accounting for authorized differences.

Example: Of 100 invoices totaling $50,000, 98 totaling $49,500 post and two totaling $500 are rejected. **Reconciliation** must include both outcomes. The rejected items still need authorized correction, revalidation and controlled resubmission.

Error Reporting and Handling

Give rejected transactions visible owners, reasons, aging and resolution. Preserve the original data, link corrections to it, restrict suspense-file changes and prevent duplicate resubmission. An accurate posted **population** can still conceal incomplete business processing if **exceptions** remain unresolved.

Processing Procedures and Controls

Processing controls ensure that accepted transactions follow approved rules and produce complete, correct results once. Test calculations, decision branches, period boundaries, **interfaces**, retries and failures. Reconcile input to posted output, **exceptions** and pending items.

Control	Purpose
Reconciliation	Account for populations and values across stages and systems; investigate differences.
Duplicate prevention/idempotency	Ensure a repeated request does not repeat the business effect, such as charging a card twice.
Concurrency control	Protect simultaneous updates through suitable locking, version checks or equivalent mechanisms.
Reasonableness and limit checks	Detect implausible or out-of-bound calculated results.
Checkpoint/restart	Resume failed processing without losing or duplicating completed transactions.
Audit trail	Link origin, approval, processing, reversals and exception resolution to identifiable actors and events.

Data File Control Procedures

Protect files and databases from incorrect versions, unauthorized access and incomplete updates. Check identifiers, processing sequence, access, **retention**, backups, **reconciliation** and recovery. Referential integrity maintains valid relationships, such as an order referencing an existing customer. It does not establish the truth of every business value.

Parity detects an odd number of changed bits, including a single-bit change, and can miss an even number. A checksum or cyclic redundancy check (CRC) detects accidental corruption under its algorithm; neither establishes authenticity against deliberate alteration. A **cryptographic hash** supports **integrity** comparison when the expected digest is trusted. An unkeyed hash alone does not

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authenticate origin because an attacker may replace both message and digest. Digital signatures and **message authentication** codes provide integrity and source-authentication properties under their respective key and trust arrangements.

Even-parity example: data bits 1011 contain three 1s, so parity bit 1 makes the total number of 1s even. Simple parity can detect an odd number of bit changes but can miss an even number. It provides error detection, not cryptographic authenticity.

3.4.2 Output Controls

Output controls protect reports, messages, files, displays and printed material. Ensure that results are complete and accurate, reach authorized recipients and remain protected through **retention** and disposal. Temporary exports and print queues can expose **sensitive data** even when the main application is secure.

Area	Control example
Completeness and accuracy	Reconcile payroll totals, employee population and period before releasing reports.
Authorization and distribution	Restrict recipients and verify delivery destinations and channel security.
Exception visibility	Alert on failed reports, missing outputs and unusual values; assign resolution.
Confidentiality	Protect display, printing, transmission and stored exports.
Retention and disposal	Apply approved retention and secure disposal, including temporary extracts.
User review	Require timely review by a person who understands the business and can investigate differences.

Auditor Perspective

Validation is not authorization

A correctly formatted, existing account can still be used in a fraudulent request. Match the control to the **risk**: **authorization** and independent verification address legitimacy.

Counts and totals help reconcile processing, while record identities and significant values expose substitutions and offsetting errors.

Designing controls around business rules and misuse

Identify controls while **requirements** are being defined so they can be designed, built and tested. Trace the **business risk** to a specific rule, responsible role and evidence of operation. A general requirement to make the system secure is insufficient to test whether an unauthorized person can amend a supplier bank account.

Input controls address issues such as **completeness**, validity, ranges, duplicates and **authorization**. Processing controls address sequence, calculation, balancing and permitted changes. Output controls address completeness, authorized distribution, **reconciliation** and retention. An arithmetic batch total or check digit is different from a cryptographic hash. A check digit detects the error patterns supported by its algorithm; it does not establish that an identifier belongs to the correct person.

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Example: A payroll approver can approve a change, then edit its amount before **release**. A second-**approval** feature does not address the risk if post-approval changes remain permitted. Require invalidation or renewed approval for relevant changes and test bypass routes, including imports and APIs. Include **logging**, exception handling, recovery and safe failure. A control embedded in one screen may fail when an alternate interface reaches the same business function.

CMMI maturity and control design

Capability **Maturity** Model Integration (**CMMI**) characterizes process improvement and **performance**. Its familiar maturity progression is 1 Initial, 2 Managed, 3 Defined, 4 Quantitatively Managed and 5 Optimizing. Current CMMI also describes maturity level 0 Incomplete. Maturity levels concern a staged organizational progression; capability levels concern individual practice areas and use a different scale. See section 2.10.4 for the broader maturity discussion. [R19]

For development controls, ask whether practices are merely performed inconsistently, managed within projects, standardized across the organization, quantitatively controlled or continually improved. Verify the appraisal's model version, **scope** and supporting evidence. A **maturity** rating is not certification that every application is secure or that every control operated effectively during the audit period.

Capability Maturity Model (CMM) is the older model terminology; **Capability Maturity Model Integration (CMMI)** is its broader successor family. Recognize historical labels while using the correct model/version and keeping maturity levels distinct from practice-area capability.

Design the Approval Around the Risk

Example. A lending application records the approval of a payment instruction. A user can edit the recipient after approval without triggering another review. Define the control objective as approval of the **specific instruction released**, including the recipient and amount. Test whether any material change invalidates the approval and whether alternate channels enforce the same rule. Inspect the record that links an approver to the version used for release. Consider the risk, control action, owner, evidence and exception handling during design.

3.5 System Readiness and Implementation Testing

Software testing evaluates software and related work against defined requirements or expected behavior. It can examine individual components, their interactions and the complete business flow. A useful test states what is being checked, the expected result and what the actual evidence shows. Readiness also requires people, data, controls and support.

Readiness requires evidence about the system, controls, data, people and support arrangements. Testing supplies part of that evidence. A high pass rate is weak reassurance when critical **requirements** or failure scenarios were never included.

3.5.1 Testing Classifications

Test levels describe **scope**; test types describe objectives. Security and **performance** testing can occur at more than one level. Define expected outcomes before execution and retain actual results, defects, fixes and retest evidence.

Level	What is examined
Unit	An individual module or component, including logic and boundaries.

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Level	What is examined
Integration	Interactions between components or systems, including interface errors.
System	The complete integrated solution against its system requirements.
Acceptance	Suitability against agreed business and technical acceptance criteria.

Other Types of Testing

Type	Purpose
Functional/validation	Verify required business behavior against known expectations.
Performance	Measure response, throughput and resource behavior under specified conditions.
Load	Assess expected and peak workloads and transaction mixes.
Volume	Assess large data populations and database sizes.
Stress	Assess behavior beyond normal capacity or under constrained resources, including failure and recovery.
Recovery	Prove restoration, resumption and reconciliation against recovery objectives.
Security	Verify security requirements and identify relevant vulnerabilities.
Regression	Check that changes have not damaged previously working behavior.
Sociability/coexistence	Verify operation alongside other systems and shared workloads.
Compatibility	Check supported platforms, versions and connections.

3.5.2 Software Testing

A test strategy connects risks and **requirements** to cases, environments, data, **expected results** and release criteria. Include normal, boundary, invalid, unauthorized and failure cases. Record defect severity and ownership; retest fixes and assess related regression risk. Any permitted acceptance of remaining defects needs the appropriate authority and a documented risk decision.

White-box testing uses knowledge of internal code and paths. **Black-box testing** evaluates observable behavior against requirements without depending on internal implementation knowledge. Top-down integration may use stubs to simulate lower-level components; bottom-up integration may use drivers to simulate higher-level callers.

Alpha testing usually occurs in a controlled developer or **supplier** setting. Beta testing involves selected external or intended users before general release. A **pilot** limits initial real-world use; parallel testing compares old and new processing on comparable inputs. Agreement with the old system is useful evidence, but the old system may also contain defects.

A **sandbox** is an isolated or constrained environment used to observe or test execution while limiting access to surrounding resources. Verify its isolation and escape paths. Sandbox describes an execution environment, not whether testers have source-code knowledge.

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Quality and acceptance responsibilities

Quality assurance (QA) concerns the processes used to achieve quality, including standards and reviews. Quality control (QC) examines outputs for defects through activities such as inspection and testing. Quality assurance testing (QAT), in technical acceptance terminology, checks technical compliance and quality against agreed specifications. **User acceptance** testing (UAT) establishes suitability for business needs.

Authorized business representatives own business acceptance. Technical and operational owners accept their respective responsibilities. Define **acceptance criteria** during **requirements** work and refine detailed test cases as design develops. A supplier demonstration or developer sign-off does not establish UAT.

3.5.3 Data Integrity Testing

Test **completeness**, **accuracy**, consistency and validity through processing, storage and transfer. Examine keys, relationships, required fields, duplicate prevention, calculations and reconciliations. Compare detail as well as totals because aggregate agreement can hide wrong allocations.

Data Integrity in Online Transaction Processing Systems

ACID property	Meaning and example
Atomicity	Related updates commit together or not at all: both sides of an account transfer.
Consistency	Transactions preserve defined integrity rules and constraints.
Isolation	Concurrent transactions receive the protection specified by the isolation model.
Durability	Committed changes persist through failures within the design's guarantees.

Database ACID support does not prove that business rules are correct. Isolation levels permit different concurrency behavior, and an external action may not participate in the same atomic transaction. Test partial failures, retries and compensating actions across system boundaries.

3.5.4 Application Systems Testing

Exercise end-to-end business flows and embedded controls using **representative data** and known **expected results**. Include period-end processing, invalid master data, unauthorized actions, rejection handling and restart. A small normal-case sample is insufficient for a critical application.

Test data techniques submit inputs with known outcomes. An **integrated test facility (ITF**) uses controlled fictitious entities in live processing and needs safeguards against contaminating business results. Parallel simulation independently processes actual data using auditor-controlled logic. Code/**configuration** review and reperformance can supplement these methods. Confirm that the tested or reviewed artifacts correspond to the relevant live version.

Automated Application Testing

Automated tests improve repeatability, but their value depends on coverage, credible **expected results** and action on failures. Protect test code, data, credentials and pipeline **permissions**. A green pipeline does not prove that every important requirement has been tested.

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IS Auditor's Role in Information Systems Testing

Evaluate risk coverage, **traceability**, environment suitability, **representative data**, defect decisions, retesting and acceptance. Verify that accepted and proposed release versions match. Audit may independently test controls while management remains responsible for the delivery testing and acceptance.

Use synthetic or effectively masked data where suitable. Any sensitive production data used for testing needs **authorization** and appropriate protection, including access, transfer and disposal. Masking must preserve the relationships needed for meaningful tests. Prevent **test data** and test transactions from affecting genuine business outputs.

3.5.5 System Implementation

Implementation brings the solution into operational use. **Readiness** includes accepted functionality, working controls, reconciled data, infrastructure, trained users and support. Installation success establishes only that a technical **activity** completed.

Implementation Planning

Define sequence, owners, **dependencies**, schedule, communications, go/no-go **criteria** and authority. Rehearse critical steps in a representative environment. Agree fallback triggers, the latest safe decision point and the time needed for restoration and reconciliation. Verify monitoring, runbooks, service-desk support, backups, recovery and handover before authorization.

Auditor Perspective

Acceptance needs the right owner

Users or designated **business owners** accept business suitability against agreed criteria. Developers provide technical evidence; successful unit tests do not establish UAT.

Ask whether critical **requirements** were tested and whether the accepted version matches the release, rather than relying only on a pass percentage.

Tracing requirements through positive and negative tests

Readiness concerns the system, controls, information, people, support and **dependencies**. Unit tests examine components; integration tests examine interactions; system tests examine the complete system against specified requirements. UAT establishes whether business users can accept the solution for intended use. Security, performance, recovery and operational acceptance require appropriate coverage and competent participants.

A traceability matrix links an important requirement to design, implementation, test cases, results, defects and acceptance. Include negative cases: rejected unauthorized activity, invalid input, failed dependencies, duplicate requests and emergency paths. **Test data** must be controlled and protected; production **personal information** is not automatically permitted in a test environment.

Example: Every normal payroll change passes, but no test attempts an unauthorized pay increase through the bulk-import interface. A high pass percentage does not address that **authorization** requirement. Add the relevant misuse cases and test actual enforcement. Acceptance decisions should consider unresolved defects and their consequences, not just total counts. Business acceptance does not replace technical assurance, and developer sign-off is not evidence that users accepted the

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business process. Define measurable **acceptance criteria** early and revise them through controlled change when **requirements** change.

CI CD pipeline controls and test evidence

Continuous integration (CI) merges changes frequently and uses automated builds and tests to detect integration problems. Continuous delivery keeps a tested **release** ready for **deployment**; a **production release** decision may remain manual. Continuous deployment automatically deploys changes that pass the defined pipeline. The abbreviation CI/CD can refer to either delivery or deployment, so establish which arrangement is actually used.

CI/**CD pipeline controls** protect source branches, build and deployment identities, secrets, **dependencies**, test logic, artifacts and release decisions. Enforce appropriate review, **segregation of duties** and risk-based gates, and record approved exceptions. Automation can enforce an approval rule, but a developer who can alter the rule and release an unreviewed change may bypass the intended control. [R10, R13]

Example: Trace a **production release** from its **commit** and approved merge through build, security and regression results to the exact deployed artifact. Check who could override a failed test, change a pipeline **policy** or use an emergency path. Retain the decision and evidence. A successful pipeline run proves only that the configured checks ran and passed under the recorded conditions.

Use a Negative Test to Challenge Readiness

Example. User acceptance testing confirms that approved orders reach billing. The project has no evidence that an unapproved order is rejected when submitted through an alternate interface. Add a safe negative case with an expected result, verify that the relevant route and version were tested, and record the resulting state and logs. Assess the importance of any gap before recommending readiness. Business owners approve acceptance, and audit evaluates whether the evidence supports that decision.

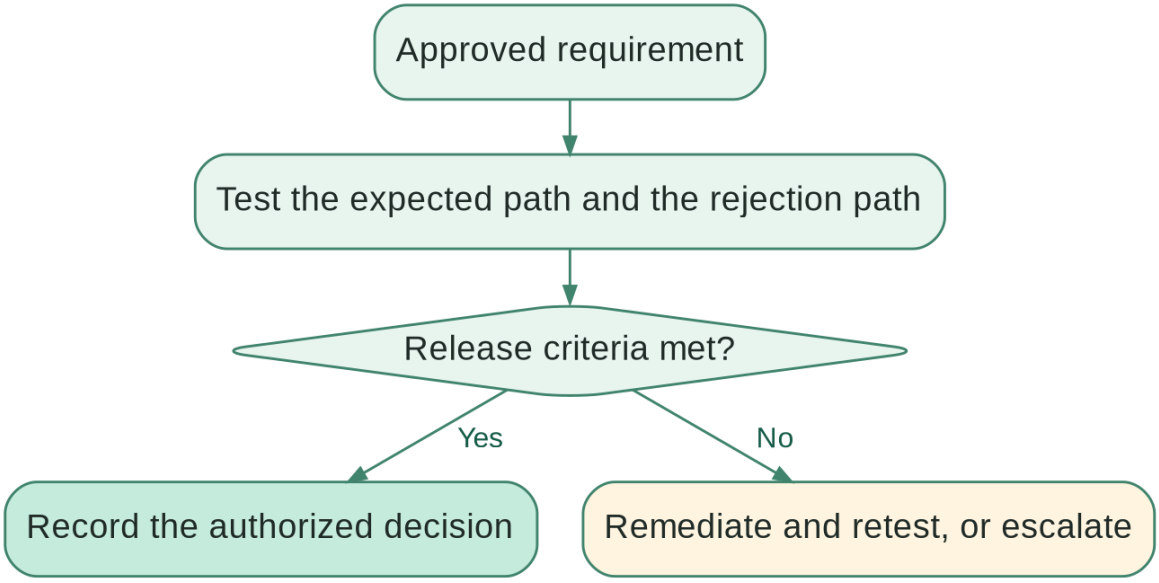


Diagram 3.1 Release readiness depends on traceable evidence and an authorized decision.

3.6 Implementation Configuration and Release Management

Maintain a trustworthy connection between what was specified, built, tested, approved and deployed. Without it, acceptance **evidence** may describe a different **configuration** from the one operating in production.

3.6.1 Configuration Management Systems

Configuration management identifies configuration items (CIs), their owners and relationships; establishes approved baselines; controls changes; records status; and verifies actual configurations. Items can include **source code**, libraries, parameters, infrastructure definitions, schemas and documentation. A configuration management database (CMDB) stores information and relationships; outdated records do not establish effective control.

A **baseline** is an agreed reference under change control. Status accounting tracks versions and states such as proposed, tested, approved and deployed. **Configuration** verification compares actual items with authorized records. Release management packages compatible approved changes; deployment management moves and verifies them in the target environment.

Example: Version 4.2 passes **UAT**, but a developer rebuilds it with an unreviewed library using the same filename. The name is unchanged; the accepted artifact is no longer established. Protect build inputs, artifact identity and promotion records, then verify the actual deployed version.

Restrict repository and production-library access, protect secrets, retain change approvals and logs, and verify **dependencies**. Emergency changes require controls suited to urgency, including **authorized access**, activity evidence and timely independent retrospective review.

Testing effective deployment authority and configuration

Configuration management identifies approved components and their relationships and maintains reliable baselines. Release management coordinates what is delivered and when. **Deployment** puts the change into its target environment. Assess these activities together while preserving their different objectives and evidence.

Segregation can be enforced through roles, independent **approval**, pipeline identities and protected gates rather than requiring a manual handoff for every release. Examine effective **permissions**, including emergency roles and the ability to edit the gate itself. A developer who cannot click Deploy may still control a service account or workflow that bypasses the restriction.

Example: Production accepts any image with an approved repository tag, but the tag can be moved after testing. Link **approval** to the actual immutable artifact or equivalent protected identity and verify deployed content. Review configuration drift, secrets handling, rollback criteria and evidence retention. Emergency deployment needs **defined authority**, proportionate testing, recorded actions and prompt **independent review**. Audit should determine whether a release was authorized and supported by appropriate evidence, without becoming the person who approves it for business use.

Match the Approval to the Deployment Configuration

Example. The approved application package passes testing. Production uses the same package with an environment override that disables a required authorization check. Review the **effective configuration** as well as the artifact identifier. Compare the approved deployment manifest, applicable environment settings, evidence of authorized changes and post-deployment tests. Trace who can

modify overrides or deploy outside the controlled route. Verify that fallback restores both the approved application and compatible settings.

3.7 System Migration, Infrastructure Deployment and Data Conversion

Technology deployment, **data migration**, **data conversion** and business changeover are related but distinct. Each needs appropriate ownership and evidence. The transition should preserve service, information integrity and control accountability.

3.7.1 Data Migration

Migration moves data from a source to a target. Conversion changes its representation, structure, coding or meaning to fit the target. Identify data owners and approve mappings, transformations, **exclusions** and validation before execution.

Refining the Migration Scenario

Inventory data, **interfaces** and historical **requirements**. Cleanse data through authorized rules; extract, securely transfer and load it in the correct sequence. Rehearse timing and repeatability, control script and mapping versions, and reconcile exceptions. Temporary adapters or data-redirection services during coexistence need clear ownership and removal plans.

Verify **completeness** using source populations, unique identifiers, counts, relevant totals and explained **exclusions**. Verify accuracy through significant field comparisons and approved transformations. Check keys and relationships for integrity. Account for transactions since the extract and protect sensitive staging files.

Worked example: Source balances are A = $100 and B = $200. Target balances are A = $150 and B = $150. Both populations contain two records totaling $300, yet both balances are wrong. Matching counts and totals are useful checks, not sufficient proof. Reconcile identities and values, including documented transformations.

Fallback (Rollback) Scenario

Fallback restores an acceptable operating state when transition **criteria** fail. Define triggers, authority, steps, dependencies, timing and reconciliation. Test the ability to restore, not only the existence of backups. Determine the point after which rollback is no longer feasible and forward recovery is required.

Transactions entered after the new system starts need particular attention. Restoring an earlier **backup** may discard them. Depending on design, use controlled journals, reverse conversion, **replay** or another tested method, and reconcile for omissions and duplicates.

3.7.2 Changeover (Go-Live or Cutover) Techniques

Choose according to **business impact**, architecture, **interfaces**, cost and recovery feasibility. Parallel operation can reduce transition exposure, but it is costly and may be impractical. No approach removes the need for testing, acceptance or fallback planning.

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Technique	Operation and main trade-off
Parallel changeover	Old and new operate together for a defined period. Comparison and continuity options improve, but workload, cost and reconciliation increase.
Phased changeover	Introduce functions or groups in stages. Each change is smaller, while temporary interfaces and dependencies become more complex.
Pilot changeover	Start with a selected site or user group, then expand. Limits initial exposure, but the pilot must represent significant wider conditions.
Abrupt/direct changeover	Stop the old and start the new at an agreed point. Concentrates transition risk; **fallback** can still exist if it is explicitly designed and tested.

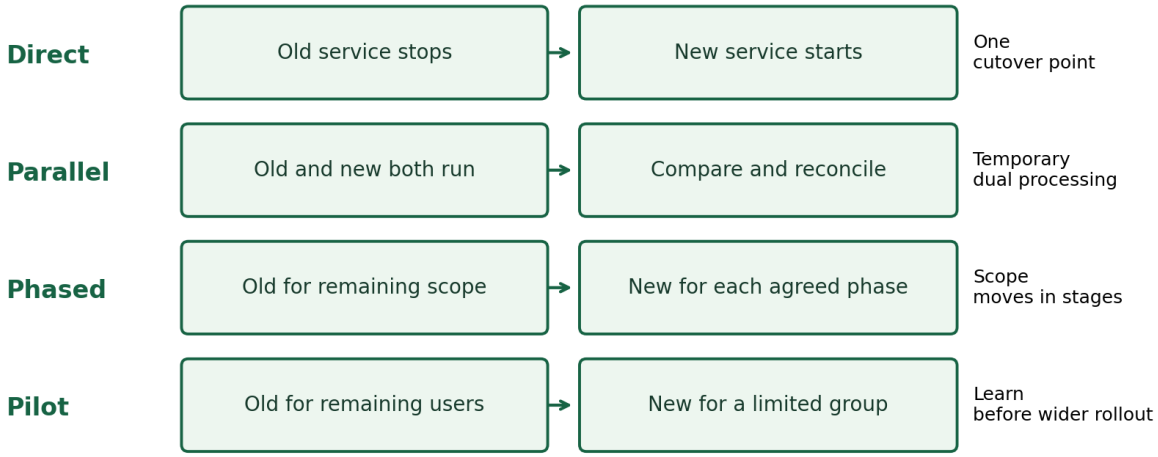


Figure 3.5 Cutover choices differ in timing, scope and duplicate processing; each needs explicit acceptance and fallback decisions.

Direct **cutover** concentrates transition risk at one point and avoids a long period of dual operation. A **fallback** may still be feasible, but only if its triggers, technical steps and treatment of transactions are designed and tested. Parallel changeover permits comparison while both systems operate; it also needs duplicate-processing controls and resources. Phased changeover moves defined functions or scope in stages and must control interfaces between old and new. A pilot limits **initial deployment** to a group or location; its evidence may not represent full-scale load or every business variation.

Auditor perspective: ask which business processes and **dependencies** actually changed, who authorized the transition, and whether the evidence supports the planned **scope**. A pilot can precede a phased rollout; these labels are not always mutually exclusive.

3.7.3 System Change Procedures and the Program Migration Process

Promote approved artifacts through controlled environments. Separate incompatible development, test and **deployment** duties as appropriate, restrict production changes, retain **approval** and deployment evidence, and reconcile release records to actual versions. Do not replace a controlled promotion with an unreviewed rebuild.

Critical Success Factors

Maintain a release manifest, **dependencies**, instructions, schedule, acceptance evidence, backup, **fallback** and verification criteria. Log who deployed what and when. Monitor critical controls and service

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performance after deployment. Exceptional developer access must be authorized, limited, logged, independently reviewed and removed promptly.

End-User Training

Train users in the changed process, responsibilities, controls and error handling, not just screen navigation. Provide role-based practice and support. Attendance proves participation; task demonstrations provide stronger evidence of competence. Investigate workarounds before assigning **cause**: they may reflect **requirements**, usability, training or operational problems.

3.7.4 System Software Implementation

Implement operating systems, databases and supporting software through an authorized plan. Verify **compatibility**, **capacity**, patches, security settings, privileges, licensing and dependencies. Preserve the prior state where needed, validate installation and recovery, update records and hand over procedures. Infrastructure readiness includes facilities, networking, hardening, logging, monitoring and failover.

3.7.5 Certification/Accreditation

In assurance terminology, certification assesses whether system controls meet specified **requirements**. Accreditation is the formal management authorization to operate and acceptance of associated **residual risk**. Some organizations use terms such as security assessment and authorization. The assessor provides evidence; the designated authority makes the risk decision. Record conditions, limitations and reassessment triggers.

Auditor Perspective

Rollback must account for new work

Before reverting, determine what happened after **cutover** and how those transactions will be preserved and reconciled. An old **backup** alone can discard valid business activity.

A direct **cutover** can have fallback if it is designed and tested. Evaluate the actual recovery window and dependencies.

Validating migration completeness accuracy and fallback

Migration moves applications, infrastructure or information; conversion changes data structure or meaning. Establish approved mappings, required populations, identifiers, balances, relationships, encodings, dates and exception treatment. Rehearse with representative volumes and realistic **dependencies**. Protect temporary extracts, credentials and retained copies.

Record counts and control totals support reconciliation but cannot by themselves prove **completeness** or **accuracy**. Equal counts can conceal missing and duplicated records; equal balances can conceal offsetting errors. Validate source and target populations, identities, significant fields, transformations and rejected records. Use sampling within its limits where full comparison is not feasible.

Example: Two customer balances are exchanged during conversion, leaving counts and total value unchanged. Identity-to-value comparison exposes the problem. Plan **fallback** or forward recovery explicitly, including transaction cutoffs, synchronization, restored state, **interfaces**, elapsed time and **reconciliation** of new activity. Direct cutover can have a tested fallback; parallel operation can be costly and can introduce duplicate processing or comparison errors. Choose an approach using business

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criticality, feasibility and demonstrated controls rather than declaring one method universally safest. Verify infrastructure capacity, security and support readiness before authorizing the transition.

Reconcile Records Through Cutover

Example. During a migration, customers continue submitting orders until the cutover window closes. The first reconciliation covers the initial load, while a final set of transactions is replayed later. Define the cutoff, identify the initial and incremental populations, and trace selected late changes through mapping, load and reconciliation. Check duplicates, rejects and unresolved differences before business sign-off. Confirm the trigger and evidence for fallback if results exceed the agreed limits. The audit conclusion covers the full cutover boundary.

3.8 Postimplementation Review

A postimplementation review (PIR) evaluates the live result against **business objectives**, **requirements** and delivery commitments. Consider benefits, lifecycle costs, performance, adoption, control effectiveness, support and remaining risks. Meeting the launch date does not establish that the investment succeeded.

3.8.1 IS Auditor's Role in Postimplementation Review

Assess outcomes using evidence from users, operations, financial records and live controls. Check access, **reconciliations**, incidents, **capacity**, recovery, documentation and unresolved issues. Compare actual results with approved baselines and distinguish cash savings from capacity or qualitative benefits. Preserve independence from the delivery decisions under review.

Schedule review after enough activity has occurred to assess the relevant outcomes. Some **benefits** require later follow-up; there is no universal fixed interval. Immediate stabilization and longer-term benefits evaluation serve different purposes. Assign actions, owners and dates, and retain **benefit ownership** after the project closes.

Example: A procurement platform promised a 20 percent reduction in handling time. It is stable, but manual **exceptions** have increased. Compare actual processing time with the **baseline**, investigate exception causes and assess the control effects. Successful installation is not evidence of the promised benefit.

Reviewing benefits controls and unresolved work after go live

A postimplementation review compares the delivered solution and outcomes with approved objectives, **requirements** and the **business case**. Allow enough operational experience for the matters being assessed, while checking urgent control failures promptly. The review is not simply a repeat of the final project status meeting.

Assess actual benefit realization, user capability, incidents, **performance**, data quality, controls, support and outstanding defects. Confirm accountable ownership of deferred items. A project can deliver the specified features within budget while failing its **business objective**; conversely, a justified approved change can make comparison with the obsolete baseline misleading.

Example: A new purchasing system was intended to reduce duplicate payments, but staff maintain an uncontrolled spreadsheet to bypass blocked transactions. Investigate **requirements**, usability, training, **interfaces** and workarounds before assigning a **cause**. Verify that corrective actions and benefit measurements use reliable evidence. Feed lessons into architecture, procurement and future project

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governance. Closure should transfer residual work and risks to accountable owners with appropriate authority, rather than letting them disappear when the project team disbands.

Reviewing a pilot before wider rollout

A **pilot implementation** introduces a system to a limited, selected operational group or location before wider rollout. It differs from phasing **deployment** by successive functions or stages, running old and new systems in parallel, or switching directly. A pilot can itself use one of those transition methods within its limited **scope**; see section 3.7.

In postimplementation review, compare **pilot** results with approved success criteria and record defects, control exceptions, reconciliation, **user acceptance** and support readiness. Assess whether the pilot population and load represent the wider rollout. Success in one small branch does not establish capacity, interfaces or recovery for the whole enterprise. The authorized **business owner** decides whether to expand, remediate or stop.

Measure the Live Result Against the Business Objective

Example. A new claims platform was intended to shorten approval time. The online stage is faster, but claims wait in a manual queue before review. Measure the end-to-end elapsed time using the approved baseline and comparable cases. Examine adoption, service stability and control exceptions, and check the benefit owner's action on the queue. The postimplementation review should explain the outcome and any corrective decision. Completion of the deployment milestone establishes the release event.

3.9 The Last Mile - Memory Aids and Exam Traps

Use these prompts after understanding the concepts. They support recall; the facts, **objectives and qualifiers** in a question determine the best answer.

Memory aids

Recall prompt	Meaning
WHY → WHAT → HOW → PROVE → MOVE → REVIEW	Justify the investment; define requirements; design/build/acquire; test and accept; implement; review outcomes. Work may overlap or iterate.
Owner accepts; auditor evaluates	Keep business acceptance and risk decisions with the authorized owners.
Trace both ways	Every important requirement needs evidence; every delivered change needs an approved basis.
Count + value + identity + meaning	Migration checks need populations, totals, identifiers and correctly transformed detail.
ACID	All-or-none updates; defined rules; protected concurrency; persistent committed results.
Range has two rails; limit has one boundary	Distinguish an allowed interval from a single threshold.
Accepted = deployed	Check artifacts, dependencies and parameters, not only the release filename.
Benefits outlive the project	Assign ownership and measurement after administrative closure.

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Calculation check: ROI needs a stated period; payback concerns recovery timing; NPV discounts cash flows. Total float uses early/late dates. A favorable EVM ratio does not prove accepted quality or realized benefit.

Exam traps

Tempting wrong answer	Exam answer
All tests passed, so the system is ready.	Check requirements coverage, critical scenarios, defects and operational readiness.
Using record counts to prove migrated data is accurate.	Counts and totals support **reconciliation** but can conceal substitutions and offsetting errors. Validate populations, identities, significant values, transformations and exceptions; interpret sampling within its limits.
A hash total is a cryptographic digest.	An arithmetic batch total and a cryptographic hash are different.
Choosing direct (big-bang) cutover for a critical system to save cost.	Direct **cutover** concentrates transition risk but can have a tested fallback. Parallel operation supports comparison when feasible but adds cost, workload and reconciliation risks. Compare the actual design and constraints.
Writing UAT test cases only after system testing ends.	Acceptance criteria are defined during requirements definition so that tests trace to requirements.
Agile removes documentation and controls.	Retain appropriate current evidence throughout delivery.
Developer sign-off establishes UAT.	Business users perform and sign off UAT; developer sign-off is not acceptance.
A backup is a rollback plan.	Include restoration, dependencies, new transactions, timing and reconciliation.
Treating persistent user workarounds as a coding defect.	A mismatch between specifications and business needs can indicate deficient requirements. Investigate training, usability, process and operating issues before assigning a cause.
QA, QAT and UAT mean the same thing.	Distinguish quality processes, technical acceptance and business acceptance.
Continuing after major cost or scope changes without review.	Re-evaluate the business case first; it must remain valid throughout the project.
PIR is only a budget check.	The PIR checks whether objectives and benefits were achieved. Hold it after adequate operation, led by someone independent of the project team.
Adding developers to a late project to recover the schedule.	This usually delays it further; re-baseline or reduce scope instead.
Letting developers move their own code into production.	This breaks segregation of duties; an independent function migrates approved changes.

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Decision owners at a glance

Use the assigned authority and evidence to distinguish delivery responsibility from oversight and assurance.

Role	Responsibility	Boundary
Sponsor	Owns business justification and business support.	Delivery outputs are not automatically realized benefits.
Project manager	Coordinates scope, schedule, cost, dependencies and issues.	Escalates decisions beyond delegated authority.
Business owner/users	Define needs and accept business suitability.	Developer sign-off is not UAT.
Operations/technical owners	Accept technical and support readiness within their duties.	Business and residual-risk acceptance remain authorized decisions.

Threats and control purpose

The examples show complementary control purposes. Their classification depends on how the control is configured and used.

Risk	Preventive direction	Detective evidence	Corrective direction
Incorrect business processing	Approved requirements and embedded authorization/validation.	Tests, reconciliations and controlled acceptance.	Fix, retest and deploy the accepted version.
Conversion loss or corruption	Approved mappings, controlled conversion and rehearsals.	Population, identity, value and relationship reconciliation.	Correct exceptions or execute a tested fallback.

3.10 Glossary

Look up terms alphabetically. Return to the lesson for the related example and audit decision.

Term	Definition
Acceptance criteria	Agreed conditions used to decide whether a deliverable is acceptable.
Accreditation	Formal management authorization to operate and accept associated residual risk.
Agile	Adaptive delivery through short feedback cycles and collaboration.
API	Application programming interface: defined interactions between programs.
Application controls	Controls over authorization, completeness, accuracy and validity within an application.
Atomicity	Related transaction changes commit together or not at all.
Baseline	Approved reference subject to controlled change.

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Term	Definition
Batch	Group of transactions processed together.
Benefits realization	Achieving and measuring the intended business value.
Black-box testing	Testing observable behavior without depending on internal code knowledge.
Brooks's law	Observation that adding people to a late software project can make it later.
Build provenance	Information about how an artifact was produced and the inputs and process supporting it.
Business case	Supported investment justification covering options, value, costs and risk.
Capability Maturity Model (CMM) versus CMMI	Capability Maturity Model (CMM) is the older model terminology; Capability Maturity Model Integration (CMMI) is its broader successor family. See 3.4.
CASE	Computer-aided software engineering: tools supporting lifecycle artifacts and activities.
Certification	Assessment of controls against specified requirements.
Changeover	Transition from the old operating arrangement to the new.
Check digit	Calculated digit used to detect certain identifier entry errors.
Checksum/CRC	Calculated error-detection values; CRC uses polynomial techniques.
CI	Configuration item: a managed component or artifact.
CI/CD pipeline controls	Controls over continuous integration and continuous delivery or deployment, including review, test gates, protected identities/artifacts and controlled exceptions. Delivery may retain manual production authorization. See 3.3 and 3.5.
CMDB	Configuration management database containing items and relationships.
CMMI maturity levels	Process maturity progression: 1 Initial, 2 Managed, 3 Defined, 4 Quantitatively Managed, 5 Optimizing; current CMMI also describes 0 Incomplete. Distinguish maturity from practice-area capability levels. See 2.10.4 and 3.4.
COCOMO	Constructive Cost Model: a software effort estimation model.
Common Gateway Interface (CGI)	The Common Gateway Interface (CGI) is a legacy interface through which a web server invokes a program or script to generate a response. See 3.3.5.
Completeness check	Check that required information is present; population completeness needs broader reconciliation.
Concurrency control	Mechanisms protecting simultaneous access and updates.
Configuration	Tailoring supported settings or parameters.
Configuration management	Identification, baselining, change control, status tracking and verification of configurations.
Consistency	Transaction property preserving defined integrity rules.

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Term	Definition
Control total	Aggregate used for comparison across processing stages.
Cost-benefit analysis	Cost-benefit analysis compares relevant costs and benefits of options over an appropriate period, using consistent assumptions. See 3.2.
Critical path method (CPM)	Dependency-network scheduling method. The longest-duration dependent path determines earliest completion under the model's assumptions; several critical paths can exist. See 3.1.10 and 3.3.
Customization	Changing or extending program logic beyond supported parameter settings.
DAST	Dynamic application security testing of a running application through exercised behavior.
Data conversion	Changing data representation, structure or coding.
Data migration	Moving data from a source to a target.
Data warehouse/data mart	Integrated analytical store; a mart serves a subject or group.
Decision table/tree	Tools representing condition-to-action rules or branching decisions.
Denormalization	Deliberate redundancy, often for performance, requiring consistency controls.
DevOps/DevSecOps	Integrated delivery and operations practices, with security embedded in DevSecOps.
DSS	Decision support system using data and models to assist decisions.
Durability	Persistence of committed changes within system guarantees.
Earned value analysis / management (EVA / EVM)	Integrates budgeted scope, schedule and cost. PV is planned value, EV earned value and AC actual cost. CV = EV - AC; SV = EV - PV; CPI = EV/AC; SPI = EV/PV. Value variance is not calendar delay. See 3.1.12 and 3.3.
EDI	Electronic data interchange of structured business documents.
EIS	Executive information system providing summarized management indicators.
Enterprise resource planning (ERP)	ERP integrates core business processes through coordinated application modules and shared or consistently integrated data. See 3.3.1.
ERD	Entity-relationship diagram modeling entities, keys and relationships.
ETL	Extract, transform and load: controlled movement and preparation of data.
Expert system	Knowledge base and inference engine applying expert rules.
Fallback/rollback	Controlled restoration to an acceptable prior operating state.
Feasibility	Practical viability within relevant constraints.
Float	Permissible activity delay within the schedule model without delaying the relevant milestone.

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Term	Definition
FPA	Function Point Analysis: functional size measurement, not directly an effort estimate.
Functional requirement	Required behavior or capability.
Gantt chart	Activities displayed against time.
GIS	Geographic information system for spatial data and analysis.
Hash total	Arithmetic total of a normally nonfinancial field used for control.
Human resource management system (HRMS)	An HRMS supports human-resources records and processes. See 3.3.1.
Idempotency	Repeated execution of a request does not repeat its intended business effect.
Integration testing	Testing interactions between components or systems.
IRR	Discount rate at which an investment's NPV is zero.
Isolation	Protection of concurrent transactions under the chosen isolation model.
ITF	Integrated test facility using controlled fictitious entities in live processing.
IT infrastructure	Supporting technology, facilities and operational resources used to deliver information services. See 3.3.7.
Key verification	Independent re-entry or comparison of entered data.
Library control	Protection and version control of source, object and related program artifacts.
Limit/range check	Single-boundary check versus lower-and-upper-boundary check.
Load/stress/volume testing	Expected workload, beyond-normal conditions and large data populations, respectively.
Middleware	Software connecting otherwise separate applications or components.
MIS	Management information system supplying routine structured reports.
Nonfunctional requirement	Required quality or constraint, such as performance, security or recovery.
Normalization	Relational data organization reducing redundancy and update anomalies.
NPV	Net present value: discounted cash flows including investment outlays.
Object-oriented encapsulation	Encapsulation groups state and behavior behind a defined interface and controls direct access to implementation details. See 3.3.5.
Object-oriented inheritance	Object-oriented inheritance allows a derived class to reuse or specialize members and behavior of another class. See 3.3.5.
Object-oriented polymorphism	Polymorphism allows a common interface or operation to have different implementations or behavior for different object types. See 3.3.5.
Object-relational mapping (ORM)	ORM maps application objects to relational database structures and operations. See 3.3.3.

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Term	Definition
Parallel simulation	Independent processing of actual data to compare results.
Parity	Bit-based error detection that can miss an even number of changed bits.
PERT	Program Evaluation and Review Technique, commonly using three-point duration estimates.
Pilot implementation	Limited operational deployment to a selected group or location before wider rollout. Broader readiness still requires representative evidence and an authorized decision. See 3.7 and 3.8.
PIR	Postimplementation review of live outcomes, controls, costs and benefits.
PMO	Project management office supporting or directing delivery under its mandate.
POC	Proof of concept testing selected feasibility assumptions.
Portfolio/program/project	Strategic investment collection; coordinated related work; temporary delivery endeavor.
Prototyping	Early models or versions used to clarify needs and obtain feedback.
QA/QC/QAT	Quality assurance processes, quality control of outputs and technical quality assurance testing.
RAD	Rapid Application Development using fast feedback, tools and often timeboxes.
Reasonableness check	Assessment of plausibility in the relevant business context.
Record count	Number of records; agreement is a useful but insufficient completeness check.
Referential integrity	Enforcement of valid relationships between related records.
Regression testing	Retesting to detect unintended effects of change.
Release/deployment	Packaging authorized changes versus moving and verifying them in an environment.
Requirements traceability	Connections among needs, design, implementation, tests and acceptance.
RFI/RFP/RFQ	Requests for information, proposals and quotations.
Run-to-run totals	Control information compared between processing stages.
Sandbox testing	A sandbox is an isolated or constrained environment used to observe or test execution while limiting access to surrounding resources. See 3.5.2.
SAST	Static application security testing of code or related representations without executing the application.
SBOM	Software bill of materials identifying components and versions; supports visibility rather than certifying security.
SDLC	Systems development life cycle from need and feasibility through implementation and review.

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Term	Definition
Slack time / total float	Total float, often called total slack, is the delay an activity can absorb without delaying the modeled project finish. See 3.1.10.
SOA/microservices	Related service-based architectural approaches with differing typical scope and deployment practices.
Sociability testing	Testing coexistence with other systems and workloads.
Software composition analysis	Analysis of third-party components, dependencies and relevant known issues.
Software testing	Evaluation against requirements or expected behavior using defined criteria and evidence. See 3.5.
Source-code escrow	Independent holding of agreed materials for release under specified conditions.
Stub/driver	Simulation of a called component versus simulation of a caller during testing.
System software	Software operating or supporting the computing platform, such as operating systems and utilities. See 3.3.9.
System testing	Testing the complete integrated solution against requirements.
Timeboxing	A fixed period for work, with scope prioritized within it; See 3.1.10.
UAT	User acceptance testing of business suitability by authorized representatives.
Unit testing	Testing an individual program unit or component.
Validity/existence check	Allowed-value check or confirmation of a reference/master record.
WBS/work package	Decomposition of work and its manageable assigned units.
White-box testing	Testing informed by internal logic, code and paths.
Work breakdown structure (WBS)	Decomposition of project work into manageable elements. See 3.1.4 and 3.1.9.

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3.11 Practice questions

These 12 original questions follow the CISA format: choose the one BEST answer from four options. Domain 3 carries 12 percent of the examination, so it has 12 of the book's 100 domain questions. Allow about 1.6 minutes per question, then check section 3.12. Before you look, name the **qualifier** and say why your **second choice** is weaker.

Question 3.01 Midway through an ERP rollout, costs are forecast to exceed budget by 40% and a regulatory reporting module has been removed from scope. What should the steering committee do **FIRST**?

A. Approve additional contract developers to protect the go-live date
B. Commission an independent quality review of completed deliverables
C. Re-evaluate the business case against the revised cost and scope
D. Rebaseline the project plan to reflect the updated cost and timeline

Question 3.02 An agile team working in fixed two-week timeboxes realises midway through a sprint that it cannot finish every committed story to the agreed definition of done. Which response is **MOST** appropriate?

A. Borrow developers from another team for the rest of the current sprint
B. Reprioritize and move lower-value stories to a later timebox
C. Defer regression testing of completed stories to the following sprint
D. Extend the current timebox by a few days to finish the committed stories

Question 3.03 An IS auditor reviews the business case for a new customer portal. Projected savings rely on cutting 30 call-centre posts, but HR confirms no headcount reduction is planned. The case also lacks a sensitivity analysis. Which is the **GREATEST** concern?

A. The main financial benefit depends on savings that will not be realized
B. The forecast has not been tested against changes in key assumptions
C. No option analysis compares the portal with continuing current operations
D. The business case does not name an owner for measuring realized benefits

Question 3.04 During UAT of a claims system, business testers and developers disputed whether several outputs were correct because expected results had never been agreed. In which phase should measurable acceptance criteria **FIRST** be established?

A. Requirements definition, alongside each business requirement
B. Feasibility study, as part of the approved business case
C. Detailed design, when screens and interfaces are specified
D. System testing, once the full solution has been integrated

Question 3.05 A bank is licensing a core lending application from a small, recently founded vendor and will not receive the source code. Which contract provision **BEST** protects the bank if the vendor ceases trading?

A. A right-to-audit clause covering the vendor's development practices
B. Service credits for missed support response times in the SLA
C. A source code escrow agreement triggered by vendor insolvency
D. A performance bond sized to cover the licence and implementation fees

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Question 3.06 A payroll clerk transposed two digits when keying an employee number, and the payment was posted to a different, valid employee. Which input control would **BEST** have prevented this error?

A. A hash total of employee numbers reconciled after batch posting
B. A validity check of the number against the employee master file
C. A reasonableness check on the payment amount for each employee
D. A check digit verified on the employee identification number

Question 3.07 During migration of customer accounts to a new core banking system, which reconciliation provides the **LEAST** evidence that individual account balances were converted accurately?

A. Tracing a sample of migrated balances back to the source account records
B. Matching each balance by account number between source and target
C. Reviewing accounts where approved mapping rules altered the balance
D. Comparing the count of records loaded with the count extracted from source

Question 3.08 Each module of a new billing system has passed unit testing, and the defects found have been fixed and retested. Which testing should be performed **NEXT**?

A. Regression testing of the existing legacy billing functions
B. User acceptance testing with business process owners
C. Integration testing of the interfaces between the modules
D. System testing of the complete solution against requirements

Question 3.09 A new treasury application has passed functional and load testing in a dedicated test environment. It will be deployed to a production server that also runs the general ledger and a payment gateway. Which additional testing addresses the **GREATEST** risk before go-live?

A. Sociability testing with the other applications on the server
B. Recovery testing of restart after a server or job failure
C. Stress testing beyond the expected peak transaction rates
D. Regression testing of the application's core treasury functions

Question 3.10 An auditor finds developers can promote their own code to production without independent authorization or enforced test gates. Developers can also bypass the deployment controls. Every promotion is logged for one year. What is the **PRIMARY** risk?

A. Failed deployments may lack a documented and tested rollback procedure
B. Logs may be insufficient to reconstruct events during investigations
C. Unauthorized or untested changes could be placed into production
D. Release volume may exceed the change advisory board's review capacity

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Question 3.11 A hospital is replacing its patient billing system. Billing errors would breach regulatory submission deadlines, and the vendor will support both systems for three months. Which changeover approach is **MOST** appropriate?

A. Phased changeover introducing billing functions in stages
B. Direct changeover backed by a tested fallback to the old system
C. Parallel changeover with reconciliation of both systems' outputs
D. Pilot changeover starting with the smallest outpatient clinic

Question 3.12 A procurement system went live two weeks ago, on time and under budget. The project manager proposes that her team perform the postimplementation review now so the project can be formally closed. Which response is **BEST**?

A. Schedule it after a period of operation, led by an independent party
B. Proceed now, as the team's knowledge of the system is still current
C. Proceed now, with internal audit co-leading the review with the team
D. Close the project now and include the review in next year's audit plan

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3.12 Answers and explanations

Each explanation states the **deciding principle**, then shows why the key is best and why each other option, although often a sound practice, does not answer the question asked.

Question 3.01 Answer: C

Project cost overrun and scope change · Qualifier: FIRST · Review section 3.1

Principle: When cost or scope changes materially, the **business case** must be re-validated before any recovery or rebaselining decision is made.

A. Adding staff is a recovery tactic; however, it presumes the **project** should continue and late additions can increase coordination effort.

B. Quality review can reveal delivery problems; however, it does not answer whether the changed investment still delivers **acceptable value**.

C. Correct. A 40% overrun plus lost regulatory **scope** may invalidate the justification; deciding whether to continue, change or stop must precede any recovery action.

D. Rebaselining is needed if the **project** continues; however, approving a new **baseline** before confirming the investment is still justified puts the decision out of order.

Question 3.02 Answer: B

Timebox management · Qualifier: MOST · Review section 3.1

Principle: Timeboxing holds the time fixed and flexes **scope** through authorized reprioritization, without weakening testing or quality.

A. Extra **capacity** can help some projects; however, adding unfamiliar staff mid-sprint raises onboarding and coordination effort and rarely recovers the timebox.

B. Correct. Scope is the variable in a timebox; deferring lower-value stories through the product owner keeps time and quality fixed as intended.

C. Deferring tests keeps velocity up; however, timeboxing must flex **scope**, not quality, so stories would be released without required evidence.

D. Finishing the committed work sounds responsible; however, stretching the **timebox** defeats its purpose and undermines schedule predictability.

Question 3.03 Answer: A

Business case benefits · Qualifier: GREATEST · Review section 3.2

Principle: A contradicted principal cash-benefit assumption materially undermines the **business case** and requires reassessment. This is more fundamental than uncertainty analysis of an otherwise credible forecast.

A. Correct. The principal projected cash saving conflicts with the stated staffing plan. Reassess the basis and decision, including other credible **benefits**, rather than assuming automatic cancellation.

B. Missing sensitivity analysis is a real weakness and the close runner-up; however, it tests uncertain **assumptions**, whereas the staffing saving is already known to be **false**.

C. Comparing with a do-nothing option is good practice; however, the stated facts show a more fundamental flaw in the **benefits** themselves.

D. **Benefit ownership** supports later measurement; however, assigning an owner cannot rescue a benefit that will never materialize.

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Question 3.04 Answer: A

Acceptance criteria timing · Qualifier: FIRST · Review section 3.3

Principle: Acceptance criteria should be defined with the requirements so each requirement is testable and later acceptance has an agreed basis.

A. Correct. Defining measurable **criteria** with each requirement makes it testable, supports traceability and prevents the disputes described in the scenario.

B. Feasibility sets high-level objectives and benefits; however, it lacks the detail needed for measurable **acceptance criteria** per requirement.

C. Design can refine expected behaviour; however, **criteria** first set here derive from the solution rather than the business need.

D. Integrated testing is when results are compared; however, setting **criteria** this late lets the build define what counts as correct.

Question 3.05 Answer: C

Software acquisition and escrow · Qualifier: BEST · Review section 3.3

Principle: Escrow gives the licensee access to **source code** and documentation if the vendor fails, allowing the application to be maintained.

A. Audit rights give insight into vendor practices; however, they provide no means to **maintain the software** after the vendor fails.

B. **Service credits** encourage support quality; however, they are worthless once the vendor has stopped trading.

C. Correct. A suitable escrow can **release** relevant code and supporting materials under defined vendor-failure triggers. Verify **completeness**, build capability and rights rather than assuming possession alone ensures maintainability.

D. A bond is the close runner-up as it recovers money; however, it does not let the bank keep **operating and maintaining** the system.

Question 3.06 Answer: D

Input controls: check digit · Qualifier: BEST · Review section 3.4

Principle: A suitably designed **check digit** detects specified transposition or transcription errors at entry. Coverage depends on the **algorithm**; not every valid-but-wrong identifier is detected.

A. A **hash total** could reveal the discrepancy; however, it is a detective batch control applied after posting, not a preventive entry check.

B. **Validity** checks are common; however, the transposed number belongs to another valid employee, so the check would pass.

C. Reasonableness checks catch implausible amounts; however, the amount was correct and only the **recipient identifier** was wrong.

D. Correct. A **check digit** algorithm is designed to reject transposed digits when the identifier is keyed, preventing the misposting.

Question 3.07 Answer: D

Data conversion reconciliation · Qualifier: LEAST · Review section 3.7

Principle: Counts provide limited **reconciliation** evidence and can conceal substituted, missing and duplicated records. Compare validated populations, identities and significant values.

A. Sampling migrated balances back to source directly tests value **accuracy** and is a standard conversion procedure.

B. Full account-level matching tests both identity and value, giving strong **accuracy** evidence.

C. Reviewing transformed balances confirms that approved conversion rules were applied correctly, which is **accuracy** evidence.

D. Correct. Matching counts alone cannot establish individual balance **accuracy** and can also conceal **population** errors. Identity and value reconciliation directly address the relevant conversion risks.

Question 3.08 Answer: C

Testing sequence · Qualifier: NEXT · Review section 3.5

Principle: Testing proceeds unit, integration, system, then acceptance, so verified modules are next tested together through their **interfaces**.

A. **Regression testing** protects existing functions after change; however, it does not replace the next level of the standard test sequence.

B. UAT is essential before go-live; however, it comes after integration and **system testing** have shown the solution works technically.

C. Correct. After **unit testing**, integration testing verifies that modules exchange data and control correctly through their interfaces.

D. **System testing** is the close runner-up; however, interface errors between modules should be found before testing the whole solution.

Question 3.09 Answer: A

Sociability testing · Qualifier: GREATEST · Review section 3.5

Principle: Sociability testing confirms a new system can coexist with other applications in its target environment without adverse effects.

A. Correct. Isolated testing cannot show resource contention or conflicts with the ledger and payment gateway, which could disrupt critical shared services.

B. Recovery testing is valuable; however, it addresses failure handling rather than the new **risk** of disrupting the ledger and payment gateway.

C. Stress testing explores behaviour beyond **capacity**; however, load is already tested and the untested exposure is interaction with critical co-resident systems.

D. **Regression testing** protects existing behaviour after changes; however, the treasury application is new and its functions have just passed testing.

Question 3.10 Answer: C

Developer access to production · Qualifier: PRIMARY · Review section 3.6

Principle: The stem explicitly gives developers unilateral authority to bypass **authorization** and testing. Logging alone does not prevent unauthorized or untested code reaching production.

A. **Rollback** planning is a sound release control; however, the finding concerns unauthorized promotion rather than recovery from failed deployments.

B. Log adequacy matters for investigation; however, logs are retained and are detective, so they do not address the core **access weakness**.

C. Correct. Self-promotion bypasses independent **approval** and testing gates, so unauthorized or defective code can reach production; logging only detects this afterwards.

D. CAB workload is an operational concern; however, the **finding** is about who can deploy, not how many changes are reviewed.

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Question 3.11 Answer: C

Changeover technique · Qualifier: MOST · Review section 3.7

Principle: Supported parallel operation permits controlled output comparison and continued processing in this scenario. It still requires resources, **reconciliation** and duplicate-processing safeguards.

A. Phasing reduces each step's size; however, it introduces temporary **interfaces** between old and new billing that add their own risk.

B. A tested **fallback** is the close runner-up; however, direct **cutover** concentrates risk at one point, and errors may surface only after deadlines are affected.

C. Correct. Running both systems and reconciling outputs detects errors before reliance shifts, and the vendor's dual support makes this feasible.

D. A **pilot** limits initial exposure; however, the smallest clinic may not represent full billing volume or complexity, and other sites remain unprotected.

Question 3.12 Answer: A

Postimplementation review · Qualifier: BEST · Review section 3.8

Principle: A PIR assesses whether objectives and **benefits** were realized, so it needs enough live operation and reviewers independent of the build team.

A. Correct. Allowing adequate operation produces evidence of realized **benefits**, and independent reviewers avoid the build team assessing its own delivery.

B. Current knowledge helps; however, two weeks is too soon to judge benefits, and the team would be **reviewing its own work**.

C. Audit involvement adds objectivity; however, co-leading compromises **audit independence** and the review is still premature.

D. Deferring the review has merit; however, folding it into an **audit plan** removes management ownership of the PIR and benefit follow-up.

3.13 Continue your preparation

Use your error log to identify the reason behind each mistake. Revisit the relevant explanation and **Last Mile** table, then test the concept with a fresh scenario. Register at certprepx.com for adaptive practice, timed full-length mocks and domain analytics. Available plans and features may change.

Scan to visit https://certprepx.com

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Domain 4: Information Systems Operations and Business Resilience

4.0 Domain overview

Reliable operations make information services dependable in normal conditions. Business **resilience** prepares the enterprise to continue or recover when conditions change. Both require more than working equipment: clear ownership, controlled processes, trustworthy data, capable people and evidence that **recovery arrangements** work.

Domain 4 represents **26 percent** of the current CISA exam outline. Sections 4.1–4.11 address information systems operations; Sections 4.12–4.16 address business **resilience**. The main headings and their numbers follow the supplied reference structure. Supporting numbered headings organize the explanations, examples and revision material.

Throughout this domain, ask what the business needs, what could prevent it, which control addresses that **risk** and what evidence demonstrates **performance**. A successful backup job is not proof of a successful restore. A server responding to a ping is not proof that users can complete their work.

Topics and reading route

Section	Subject
4.1	IT Components
4.2	IT Asset Management
4.3	Job Scheduling and Production Process Automation
4.4	System Interfaces
4.5	End-User Computing and Shadow IT
4.6	Systems Availability and Capacity Management
4.7	Problem and Incident Management
4.8	IT Change, Configuration and Patch Management
4.9	Operational Log Management
4.10	IT Service Level Management
4.11	Database Management
4.12	Business Impact Analysis
4.13	System and Operational Resilience
4.14	Data Backup, Storage and Restoration
4.15	Business Continuity Plan

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Section	Subject
4.16	Disaster Recovery Plans

Follow one service through the domain

Consider an online retailer. Orders depend on customer **authentication**, stock records, payment interfaces, databases, networks and support staff. Operations must identify those dependencies, control changes, investigate failures and meet **service commitments**. During disruption, the business needs an acceptable ordering capability, sufficiently **current data** and a tested route back to normal. Recovering the web page while payments and stock remain unavailable does not restore the service.

4.1 IT Components

IT components are the physical and logical building blocks that deliver services. An auditor needs enough technical understanding to identify trust boundaries, **dependencies**, control points and failure modes. Start with the service architecture and data flows rather than an isolated equipment list.

Information processing facility (IPF) means the computer room and its supporting areas. Apply access, environmental, location and recovery controls to the complete facility, including its support dependencies.

4.1.1 Networking

A network connects devices and services so information can be exchanged. A local area network (**LAN**) typically serves a limited location; a wide area network (**WAN**) connects dispersed locations. Wireless describes an access or transmission method, not a security boundary. Network diagrams should show significant connections, segments, remote access, external providers and management paths.

OSI and TCP/IP models

The Open Systems Interconnection (**OSI**) model separates communication into seven conceptual layers. The common four-layer **TCP**/IP model groups some functions differently. Use the models to locate responsibilities and faults; real devices can perform functions across multiple layers.

OSI layer	Purpose and common example
7 Application	Services used by applications, such as HTTP or DNS protocols.
6 Presentation	Representation, encoding and related data transformation functions.
5 Session	Coordination of communication sessions.
4 Transport	End-to-end transport, including TCP and UDP.
3 Network	Logical addressing and routing, including IP and routers.
2 Data link	Local link framing and addressing, commonly Ethernet and switches.
1 Physical	Transmission of signals over cables, fiber or radio.

TCP/IP application functions broadly cover **OSI** layers 5–7; transport maps to layer 4, internet to layer 3, and network access/link to layers 1–2. TCP provides an ordered reliable byte stream; UDP supplies

datagrams without equivalent delivery guarantees. IP routes packets; it does not itself guarantee delivery.

LANs, WANs and network administration

Ethernet switching commonly forwards frames within a **LAN**, while routing connects IP networks. VLANs create logical segmentation, but traffic between them still needs suitable routing and access controls. Common physical layouts include star, bus, ring and mesh; a star's central device or a ring's unprotected link can be a failure dependency. Redundant designs must be checked in their actual implementation.

Copper and fiber differ in distance, interference **exposure**, capacity and physical requirements. Fiber resists electromagnetic interference but remains vulnerable to cuts, misconnection and equipment failure. WAN services may use leased connectivity, packet-based provider networks, internet VPNs, cellular or other services. Assess **service commitments**, encryption where needed, monitoring and route dependencies.

Administration covers approved configurations, secure management access, accurate diagrams, device inventory, access control, fault handling and **performance monitoring**. Protect management credentials and protocols, separate management access where appropriate, and verify backups of device configurations. A network operations center (**NOC**) monitors service health; a security operations center (SOC) focuses on security monitoring and response. Their work must coordinate.

Network term	Basic meaning and control implication
Intranet	Uses internet-style technologies for internal organizational information and services. Internal location alone does not establish safe access.
Extranet	Provides controlled access to selected organizational resources for outside parties. Check identity, permitted information and boundary controls.
Internet service provider (ISP)	Supplies internet connectivity or related network services. Check the contract, actual availability and shared dependencies.
Media access control (MAC) address	Identifies a network interface at the relevant link layer. It can often be changed or spoofed, so it is not strong user authentication. Distinguish it from mandatory access control and a cryptographic message authentication code.

Converged protocols, IP networking and NAT

Convergence carries different services over shared infrastructure, such as voice and data or storage traffic. **Fibre Channel** supports storage networking; iSCSI transports SCSI commands over TCP/IP; Fibre Channel over Ethernet carries Fibre Channel frames over suitable Ethernet infrastructure. Shared infrastructure can reduce cost while concentrating availability and **capacity** risk. Check segmentation, **compatibility**, latency and quality of service.

IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses. Dual-stack and tunneling arrangements need explicit security and **monitoring** coverage. IPv6 does not automatically make a network secure. Network address translation (NAT) maps addresses, and port address translation (PAT) can multiplex connections using port mappings. Static mappings are fixed; dynamic mappings use an available pool. NAT is not a substitute for firewall **policy**, **authentication** or encryption.

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Voice over Internet Protocol (VoIP) carries voice communications over IP data networks. Assess call signaling and media paths, authentication, appropriate protection, segmentation, latency, packet loss, availability and power dependencies.

4.1.2 Computer Hardware Components and Architectures

The CPU executes instructions; **RAM** holds active working data and normally loses it without power. Persistent storage retains data, while firmware provides low-level device functions. Input/output components connect users, storage and networks. Multicore processors provide multiple processing cores; multiprocessor designs use multiple processors. Neither guarantees that a particular workload will run faster.

Enterprise environments can include mainframes, servers, workstations, mobile devices, thin clients and embedded systems. Virtualization allows multiple virtual machines on shared hosts; failures or overcommitment at that shared layer can affect many services. GPUs accelerate suitable parallel workloads. **Capacity**, **compatibility**, supportability and recovery requirements should determine architecture.

Platform	Basic meaning
Personal computer (PC)	A general-purpose computer intended principally for individual use, such as a desktop or laptop.
Mainframe	An enterprise computing platform designed for substantial centralized workloads and operational reliability; it still requires access, change, recovery and network controls.

4.1.3 Common Enterprise Back-End Devices

Switches connect local devices; routers direct traffic between networks; firewalls enforce traffic rules; intrusion **detection** systems report suspicious activity, while intrusion prevention systems can block it. Load balancers distribute requests. Each device has **configuration**, administrative access, logging, patching and availability requirements.

Proxy Servers

A **forward proxy** acts on behalf of clients, commonly for filtering, controlled internet access or caching. A **reverse proxy** sits in front of servers and can support access control, load distribution or TLS termination. Proxying changes the traffic path and visibility; inspect trust, certificate handling, logging and the protection of any decrypted data.

4.1.4 USB Mass Storage Devices

Universal Serial Bus (USB) is an interface used by peripherals, including removable storage. Not every USB device is storage: a device can expose other functions. Assess permitted functionality as well as the risk of files being copied.

Removable storage creates risks of data disclosure, malware introduction, loss and uncontrolled transfer. Define permitted use, restrict devices and ports where appropriate, protect approved media with suitable **encryption** and access controls, and monitor transfers. Encryption protects **confidentiality** if keys are protected; it does not make an infected file safe.

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4.1.5 Wireless Communication Technologies

A wireless local area network (WLAN) connects devices within a limited area using radio communications. Wi-Fi WLANs commonly use IEEE 802.11. Review access points, authentication, encryption, segmentation and rogue-device detection.

Wi-Fi supports local wireless networks, Bluetooth shorter-range connections, and cellular services wider-area mobile connectivity. Assess **authentication**, **encryption**, device management, interference and rogue access points. Separate guest and business access as needed. A backup cellular link may still share power, building access or provider infrastructure with the primary service.

Global System for Mobile Communications (GSM) is a family of cellular telecommunications standards associated with second-generation mobile services. Recognize the legacy term, but assess actual network, endpoint and VPN protections rather than assuming the technology name guarantees current security.

4.1.6 Hardware Maintenance Program

Preventive maintenance reduces avoidable failures through planned servicing and replacement. **Corrective maintenance** repairs a fault. Set authorized maintenance windows, retain service records and protect data and access during **supplier** work. Working on a live UPS or shared power path can immediately interrupt service; verify isolation, bypass procedures and safeguards rather than assuming redundant labels establish protection.

Review hardware errors, utilization trends, fault alerts, recurring incidents, maintenance history and end-of-support status. Match corrective actions to business **criticality**. Maintenance records show **activity**; declining failures and successful controlled tests provide evidence about results.

Understanding the service path and its technical dependencies

A business service depends on interacting components. The CPU executes instructions; RAM supports active work; storage retains data; operating systems manage resources; applications implement business functions. Firmware, device drivers, virtualization, middleware and network services introduce their own **dependencies** and **privileges**. More cores or memory help only when they address the actual workload constraint.

Use **OSI** and **TCP**/IP models to locate functions and failure boundaries. TCP provides an ordered reliable byte stream, while **UDP** provides datagrams without equivalent delivery guarantees. IP routes packets; DNS resolves names; routing, switching, address translation and time services can affect availability and auditability. The model supports understanding rather than proving that a particular device operates at only one layer.

Example: A payment application is healthy, but its name resolution and identity provider fail. Server uptime does not establish usable payment service. Trace representative transactions through endpoints, network paths, identity, application, **database** and external services. Review privileged administration, **monitoring**, capacity and failure handling at important **dependencies**. An inventory of hardware without these relationships provides limited support for impact analysis and recovery planning.

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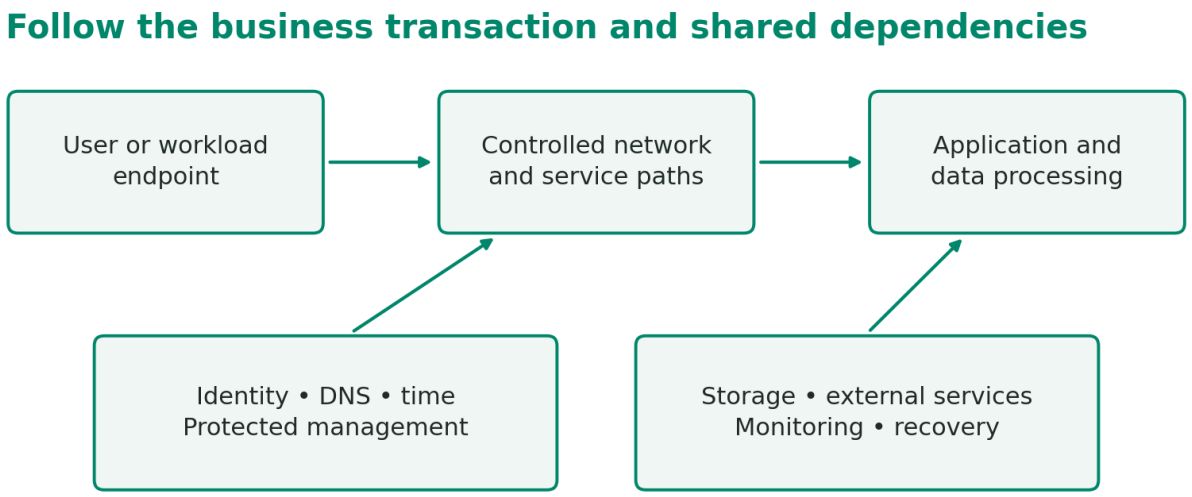


Figure 4.1 A business service depends on controlled paths and shared services.

Trace the Service Through Its Components

Example. An application processes transactions on several virtual servers. Each server uses the same storage service and network path. Start with a business transaction, then map the application, compute, network, storage and identity components needed to complete it. Verify the diagram with configuration and service records. Distinguish a component fault from the resulting service interruption. Redundant servers improve availability only to the extent that the shared dependencies and capacity support continued processing.

4.2 IT Asset Management

IT asset management establishes what the enterprise owns or uses, who is accountable and where each asset is in its lifecycle. Include physical devices, virtual and cloud resources, software, licenses and relevant information assets. Unrecorded resources may escape patching, recovery and access controls.

An **IT asset** can be owned, leased or supplied as a service. A cloud subscription belongs in the managed inventory even without a purchased server. Link the technical asset to its business service and information owner; section 2.7.1 introduces that distinction.

4.2.1 Inventory and lifecycle accountability

Record unique identifiers, type, owner, custodian, location, service relationships, classification, version, support status and lifecycle state. Reconcile procurement and financial records with discovery results, cloud inventories and disposal records. Discovery alone can miss disconnected devices, subscriptions or assets outside its **scope**.

An **asset inventory** emphasizes ownership, lifecycle and entitlement. A CMDB emphasizes **configuration** items and relationships supporting services. A data dictionary describes data definitions and structure. These records can be linked but serve different purposes.

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Manage acquisition, **receipt**, installation, assignment, maintenance, transfer, retirement and disposal. Remove access and account for retained information before reuse or return. Select sanitization appropriate to the media and sensitivity, and verify destruction or sanitization **evidence**. A disposal certificate is useful only when it can be linked to the correct assets and method.

4.2.2 Software entitlement and support

Compare actual use with license rights using the relevant metric: users, devices, processors, cores, instances or another contractual basis. Free or open-source software can still carry **obligations**. Overdeployment creates compliance and financial **exposure**; unused licenses create unnecessary cost. Track end-of-support dates and approved exceptions.

Following assets from acquisition to secure disposal

Asset management should identify what exists, who is accountable, where it is used, its **dependencies**, authorized configuration, support status and lifecycle obligations. Include physical equipment, virtual machines, applications, cloud resources, subscriptions and relevant information assets. Reconcile discovery sources with purchasing, deployment and financial records rather than trusting one inventory blindly.

Track licensing and permitted use, maintenance, renewal, **end of support** and disposal. Financial ownership and operational custody can differ, so establish both. Unsupported equipment can require replacement or a documented, **authorized and monitored** treatment plan; an inventory flag alone does not reduce exposure.

Example: An old application server is decommissioned, but its cloud disks, backup copies and privileged service principal remain. Verify removal or justified **retention** of all dependent assets and rights. Sanitization must fit the medium, sensitivity, intended reuse and applicable **requirements**. A disposal certificate is useful evidence but should be reconciled to the correct asset **population** and reliable execution records. Physical removal from the office does not establish secure information disposal.

Reconcile Short-Lived Assets to Ownership

Example. A development team creates cloud instances for short tests. The monthly inventory misses several that stored restricted data and were removed between reporting dates. Reconcile creation and deletion events with the approved asset inventory and owner records. Select assets from the provider's activity log and verify classification, access, retention and secure disposal. Use an inventory source and review frequency suited to the asset lifecycle. Determine who owns exceptions when a resource can be created outside the standard request route.

4.3 Job Scheduling and Production Process Automation

Scheduling tools automate the timing, **dependencies** and execution of production work. Their purpose is reliable business processing, not simply fewer operator keystrokes. A job that starts successfully may still produce incomplete or incorrect results.

Print spooling stores a print job temporarily in a queue so a printer can process it later. Protect spool files and queues against unauthorized reading, extra copies and excessive retention.

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4.3.1 Job Scheduling Software

Maintain authorized schedules, prerequisites, **service accounts**, parameters, calendars and completion **criteria**. Control access to create, alter, stop and rerun jobs. Use alerts and escalation for missed starts, abnormal duration, failure and unmet downstream dependencies. Protect scripts and secrets as production artifacts.

Example: Billing must wait until all approved sales feeds have arrived. A fixed 1 a.m. start without an input-**completeness check** can create an incomplete bill run. Configure **dependencies**, reconcile received inputs and make any authorized exception visible.

4.3.2 Scheduling Reviews

Review the schedule, actual execution history, failed jobs, manual overrides and reruns. Confirm that **exceptions** are investigated and outputs reconciled. Restart logic must avoid omitted or duplicated transactions. Separate **authorization** of schedule changes from routine operation where appropriate, and retain meaningful evidence for automated decisions as well as human intervention.

Controlling job dependencies reruns and exceptions

Production scheduling coordinates authorized jobs, **dependencies**, timing, input **availability** and output handling. Controls should detect missing, failed, late, duplicated and partially completed processing. Use reliable status evidence rather than treating a scheduler's green indicator as proof of the business outcome.

A rerun can create duplicate effects if the job is not designed or operated safely. Define restart points, checkpoints, **reconciliations** and **approval** for manual intervention. Protect scripts, service identities and schedule changes, including the ability to alter a dependency or suppress an alert. Test monitoring outside normal working hours and during provider outages.

Example: An overnight payment file is submitted successfully, but the acknowledgment is lost. A **blind** rerun sends duplicate payments. Unique transaction identity, receiver-side duplicate rejection and **reconciliation** can make retry behavior safe under the defined design. This property is often called idempotency when repeated handling does not repeat the intended business effect. Audit should trace an actual exception through **detection**, ownership, recovery and reconciled completion, including evidence of what was and was not processed.

Verify the Input Before Starting a Scheduled Job

Example. A billing job completes on time but uses yesterday's rate file because the current file had not arrived. The scheduler reports success because processing finished. Check the expected input date, dependency rules, rejection alerts and the reconciliation to the intended business period. A control that checks file presence addresses availability of input. A freshness check and period reconciliation address whether the job used the correct data. Examine what happens when the source is late, including authorization and effects of any rerun.

4.4 System Interfaces

Interfaces transfer information between components, systems, people or organizations. They can use APIs, message queues, files, shared databases or managed transfer services. Define the sender, recipient, format, timing, ownership and **processing acknowledgment**.

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A **message queue** holds messages for asynchronous processing between producers and consumers. Delivery, ordering and duplication guarantees depend on the design; review acknowledgments, retries, dead-letter handling and reconciliation.

A **file header record** contains control or identifying information such as file type, version, batch identifier or creation date. Validate it against expected processing requirements; a header does not authenticate content by itself.

4.4.1 Risk Associated With System Interfaces

Data can be lost, duplicated, delayed, misrouted, altered or interpreted differently. **Authentication** failures, expired certificates and incompatible schemas can stop processing. **Confidentiality**, authorization and availability matter alongside integrity; the leading concern depends on the information and scenario.

4.4.2 Controls Associated With System Interfaces

Control	Purpose and evidence
Agreed data contract	Define fields, formats, meaning, sequence, versions and error behavior.
Authentication and authorization	Establish approved endpoints and permitted exchanges.
Protected transfer	Use suitable encryption and integrity/authenticity protections with managed keys.
Completeness and accuracy reconciliation	Compare identifiers, counts, relevant totals and significant values; investigate differences.
Acknowledgment and exception handling	Distinguish receipt from successful business processing; assign rejected and delayed items.
Duplicate prevention	Use stable transaction identifiers and safe retry behavior.
Monitoring and change control	Alert on failures and delays; assess both endpoints before changes.

Example: A payment gateway acknowledges **receipt**, but three transactions fail downstream validation. The sender's "file delivered" message does not prove that payments posted. Reconcile **business outcomes**, including pending and rejected items, before closing the interface cycle.

Auditor Perspective

Receipt is not business completion

A transmission acknowledgment may prove delivery of a message without proving that each **transaction posted**. Reconcile accepted, rejected and pending outcomes.

Start with the exchange objective and investigate exceptions before treating the interface cycle as complete.

Reconciling interfaces across system boundaries

Interfaces can be batch files, messages, APIs or shared data services. Establish authorized endpoints, agreed schemas, identity, permitted actions, **validation**, sequence and ownership. Protected transport

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addresses only part of the risk; authenticated delivery can still carry incomplete, incorrect or duplicate business information.

Compare sent, received, accepted, rejected and pending populations. Define how invalid items are investigated and how late or out-of-order messages are handled. Monitor queues and dead-letter paths, and protect error records containing sensitive information. **Authentication** tokens, keys and certificates require controlled lifecycle management.

Example: The sender reports 10,000 records and the receiver reports 10,000, but the receiver accepted a repeated file while missing the current one. Sequence, file identity, business dates and record **reconciliation** matter alongside counts. Test unavailable endpoints, expired credentials, duplicates and malformed messages. An API gateway can enforce configured controls at its path, but cannot compensate for every downstream business **authorization** error. Trace one business event end to end and evaluate the **evidence** on both sides of the boundary.

Account for Acknowledgments and Rejections

Example. An order interface sends a receipt acknowledgment as soon as it accepts a message into a queue. Several messages then fail business validation. Trace a sample from source identifier through receipt, validation, posting and exception resolution. Reconcile the complete sent population to accepted, rejected and pending outcomes. Check which failures are retried and how duplicates are prevented. A receipt acknowledgment establishes receipt at a boundary. Completion requires evidence from the receiving business process.

4.5 End-User Computing and Shadow IT

End-user computing (EUC) includes user-created spreadsheets, databases, scripts and low-code tools. **Shadow IT** is technology used without the organization's expected IT visibility or **authorization**, including unsanctioned cloud services. EUC can be approved and controlled; it is not automatically shadow IT.

4.5.1 End-User Computing

The risk grows when a critical business process relies on logic that lacks review, testing, access control, documentation or backup. Inventory and classify important tools by **business impact**. Assign owners, validate inputs and formulas, protect key logic, control changes and retain **independent review**.

Example: A spreadsheet calculates loan provisions. Locking cells helps prevent accidental editing but does not prove formulas are correct. Reperform key calculations, test boundary cases, inspect links and macros, reconcile **source data** and review controlled changes.

4.5.2 Shadow IT

Understand why people bypass approved services, then provide workable alternatives and clear rules. Assess data sensitivity, **supplier** terms, identities, integrations, **retention** and exit. Discovery, monitoring and user education support governance. A blanket ban with no usable service may increase concealed use rather than reduce the risk.

Making end user solutions visible and proportionately controlled

End-user computing includes spreadsheets, databases, scripts and low-code applications managed outside traditional development arrangements. **Shadow IT** includes services or technology used without

appropriate organizational visibility or **authorization**. These capabilities may solve real needs, but critical processing can grow beyond the controls originally intended.

Discover significant use through business inquiry, purchasing, network or cloud visibility and data flows, subject to appropriate authority and privacy. Assess **criticality**, **sensitive data**, **dependencies**, ownership and change practices. Apply proportionate access, versioning, validation, review, backup, documentation and continuity controls. A universal ban may drive use further out of view without addressing the business need.

Example: A spreadsheet calculates lending decisions and exports results to the core platform. Inspect protected formulas, **input validation**, independent checks, change history, output **reconciliation** and trained backup staff. A password on the workbook does not establish reliable calculations or proper **authorization**. For low-code automation, also assess connectors, service identities, environments, promotion and the consequences of an owner leaving. Management should decide whether the solution remains appropriate or needs migration into a more controlled service.

Identify the Business Decision Behind an End-User Tool

Example. A department uses a low-code application to approve refunds. The inventory records the tool as a convenience application, but its output can initiate payments. Establish the owner, business purpose and data sensitivity. Trace changes to approval rules, access rights, input validation and payment exports. Assess version control, independent review and recovery proportionate to the exposure. The same tooling can support a minor worksheet or a material decision. Its business use determines the controls and audit coverage.

4.6 Systems Availability and Capacity Management

Availability means a service is usable when required. **Capacity management** ensures sufficient resources for current and expected demand at an acceptable cost. High utilization is not necessarily good **performance**: a saturated queue, storage device or network can make a business service unusable.

4.6.1 IS Architecture and Software

Architecture connects applications, operating systems, databases, communications, utilities and hardware. Identify shared **dependencies** and distinguish logical separation from physical **independence**. Ten virtual servers on one host do not provide ten independent failure domains.

4.6.2 Operating Systems

An **operating system (OS)** manages processes, memory, storage, devices and other resources. It enforces isolation and access boundaries. Protect privileged code and configuration, use supported versions and secure baselines, and control administrative accounts, remote access and service identities.

Software Control Features or Parameters

Parameters affect resource allocation, priorities, job behavior, **authentication**, auditing and data handling. Document approved settings and assess their impact before changes. Defaults are not evidence of suitability. Hardening removes unnecessary functions and restricts **exposure** while retaining required business capabilities.

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Software Integrity Issues and Operating System Reviews

Verify provenance and approved versions, protect program libraries and privileged processes, and detect unauthorized modification. Review patch status, isolation, file **permissions**, startup services, administrative utilities, logs and recovery settings. A supported operating system with a weak **configuration** can remain vulnerable.

4.6.3 Access Control Software, Communications and Utilities

Access control software enforces identity and **permissions**. Data communications software manages exchanges, sessions and protocols. Utility programs perform maintenance, backup, inspection or repair and may bypass ordinary **application controls**. Restrict powerful utilities, authorize their use and independently review material activity.

Simplex communication flows one way, half-duplex alternates direction and full-duplex supports both directions concurrently. These describe transmission behavior, not whether a business exchange is complete or secure. Reconcile **transaction outcomes** independently of transport success.

4.6.4 Software Licensing Issues and Source Code Management

Reconcile rights with installation and actual use, including virtualized and cloud deployments. Maintain controlled source repositories, access and change history, reviews, backups and recovery. Verify the relationship between source, builds and deployed objects. The right to **commit** code should not automatically confer unrestricted production **deployment** authority.

4.6.5 Capacity Management

Forecast demand using business growth, seasonality, planned changes and **measured bottlenecks**. Monitor CPU, memory, storage, I/O, network, database connections, response time, throughput and queue depth. Define thresholds and owners for action. **Average utilization** can hide short peaks or a single saturated dependency.

Example: A payment service has spare CPU but slow storage writes. Adding CPU may not improve response. Correlate business response times with I/O and queue metrics, validate the bottleneck and test an appropriate remedy.

Distinguishing averages from service capability

Availability measures require a defined **service boundary**, **observation** window and downtime treatment. If a service window contains 1,000 minutes and included downtime is 10 minutes, measured availability is 99 percent. A contract may exclude maintenance, but the **business impact** still exists. Compare reported measures with agreed definitions and actual user experience. Mean time between failures (MTBF) concerns reliability under the adopted definition; mean time to repair (MTTR) is an average, not a guaranteed recovery deadline.

Observed availability (%) = (agreed service-window time − included downtime) / agreed service-window time × 100. The preceding 1,000-minute window with 10 minutes' downtime gives 990/1,000 × 100 = **99%**. A 99.9% target over a 10,000-minute window allows 0.001 × 10,000 = **10 minutes** of included downtime.

MTBF = total eligible operating time / number of failures under the adopted measurement method. **MTTR = total repair time / number of completed repairs**. For 900 operating hours, 3 failures and 6 total repair hours over 3 completed repairs, MTBF = **300 hours** and MTTR = **2 hours**. Specify whether

the measure covers repair, wider restoration or another interval. Zero observed failures do not establish infinite reliability.

In a simplified steady-state repairable-system model, **availability** is approximately MTBF/(MTBF + MTTR). With 990 hours between failures and 10 hours to repair, the estimate is 99 percent. MTTR is an average repair measure, not a guaranteed deadline for an individual outage. Establish the **definitions** used before comparing different providers.

Capacity includes throughput, latency, concurrency, storage, network and **dependencies** under realistic peak demand. Example: CPU remains below 40 percent while transaction latency rises because **database** locks serialize the workload. More CPU is not the demonstrated solution. Examine bottlenecks, forecasts, headroom, demand changes and performance tests. Autoscaling requires appropriate limits, available resources and tested behavior; it can also amplify cost or fail to address a downstream constraint.

Test Capacity During Failover

Example. Both processing sites are available under normal demand. When one site fails, the remaining site can serve only part of peak demand and queues grow. Compare actual throughput, latency and capacity under a representative failover with the service objectives. Inspect the assumptions about reserved capacity, dependencies and priority work. Use availability measures for the agreed service window, and separately assess whether the surviving capacity keeps the critical service usable at the required load.

4.7 Problem and Incident Management

Incident management restores normal service after an interruption or reduction in quality. **Problem management** investigates underlying causes and reduces recurrence or impact. A workaround may restore service without eliminating the cause. The processes should connect rather than compete.

4.7.1 Problem Management

Analyze recurring or significant incidents, investigate causes, document known errors and workarounds, and manage permanent fixes through change control. A **known error** is an analyzed problem with documented information that supports resolution or workaround; do not assume every ticket has a confirmed **root cause**.

Example: A nightly job fails and is restarted successfully each morning. Restarts can restore service, but repeated failures and manual intervention warrant problem investigation. Confirm whether the **cause** is data, scheduling, **capacity** or code before choosing a fix.

4.7.2 Process of Incident Handling

Detect and record the event; assess **impact** and urgency; prioritize and assign it; diagnose, escalate and restore service; validate recovery with relevant users; and close with accurate **evidence**. Major incidents need coordination, communications and timely decisions. Preserve the timeline, actions, ownership, service effects and links to changes or related problems.

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4.7.3 Detection, Documentation, Control, Resolution and Reporting of Abnormal Conditions

Define thresholds for abnormal behavior and meaningful **alert** routing. Record timestamps, affected service, symptoms, severity, actions and outcome. Functional escalation seeks specialist skills; hierarchical escalation obtains management attention or decisions. Avoid closing incidents solely because an alert disappeared when the business outcome remains unknown.

4.7.4 Support/Help Desk and Network Management Tools

The **service desk** is a central contact for incidents and requests. **Identity verification** is important for password resets and access-related requests. Use authorization, ticket history, suitable knowledge records and escalation. Network tools provide status, traffic, faults and performance information; assess their coverage, secure management and response to alerts.

4.7.5 Problem Management Reporting Reviews

Review aging, repeated failures, time to restore, unresolved major causes and completion of corrective actions. A reduction in ticket counts can reflect underreporting rather than improved service. Trace selected serious incidents through **restoration**, **cause** analysis, authorized fixes and evidence that recurrence or impact declined.

Security incidents and major disruptions

A **security event** is an observable occurrence involving computing assets. An **alert** draws attention to activity that may need investigation; it can be erroneous or benign. A **security incident** meets the adopted criteria for actual or imminent compromise, unauthorized harm or a relevant security-policy violation. Establish criteria and authority before relying on a tool label. [R12]

Example: Repeated failed logins can generate an alert. Correlation may show a harmless expired service credential or an attempted account takeover. Evidence that an unauthorized actor is exporting customer records supports urgent incident handling; do not wait for a finished report before taking authorized action to limit ongoing harm.

A security incident also requires coordinated investigation and response. Prepare roles and authority in advance; assess **scope** and severity; contain harmful activity; remove causes; recover safely; and learn from the event. These activities can overlap and repeat. Once an incident is confirmed, **containment** is the first priority, with life safety always overriding.

Containment is urgent when ongoing compromise threatens harm, but the choice-isolating a host, blocking access, disabling credentials or another action-depends on operational **consequences** and authority. Preserve relevant evidence throughout where feasible. Escalation and legally required communications can occur in parallel with response; they should not automatically wait until containment is complete.

The computer security **incident response** team (CSIRT) coordinates relevant specialists. **Chain of custody** documents evidence identification, collection, handling, transfers and storage. Protect evidence integrity and record actions; avoid unnecessary wiping or unrecorded changes. Detailed forensics is developed further in Domain 5.

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Auditor Perspective

Restore service and investigate recurrence

An incident workaround can be appropriate for immediate **restoration**. Repeated failure also needs problem investigation and an authorized lasting remedy.

Do not confuse quick ticket closure with removal of the underlying cause.

Separating service restoration from root cause resolution

An incident is an interruption or degradation requiring response and **restoration**. **Problem management** investigates underlying causes and reduces recurrence. A workaround can restore service while the problem remains unresolved. A known error records a diagnosed condition and available treatment under the organization's process.

Prioritize using **business impact** and urgency, not simply arrival order or the most vocal user. Define escalation, communications, ownership and linkage among incidents, problems, changes and knowledge records. Security incidents need coordinated specialist handling without losing sight of service and **business consequences**.

Example: Restarting a service restores it every morning, so tickets meet the closure target. Repeated incidents show an unresolved **capacity** or software problem. Inspect **recurrence**, linked analysis, approved corrective change and measured improvement. A closed ticket is not **sufficient evidence** that the user's service was restored or that the cause was removed. Test major-incident coordination, handovers and provider dependencies. Audit should assess the process and outcomes without taking over the operational command or selecting unapproved emergency actions.

Keep Restoration and Cause Resolution Visible

Example. A service desk routes requests to an alternate queue to restore service. The routing rule that caused the failure remains active for another group. Review the incident record for restoration time and user impact. Trace the underlying issue to a problem record, approved correction and validation across the affected groups. Evidence that a workaround restored service supports incident closure under the applicable criteria. Closure of the underlying problem requires evidence that the cause or remaining exposure was addressed.

4.8 IT Change, Configuration and Patch Management

Change management evaluates and authorizes modifications so the enterprise receives the intended benefit without unacceptable disruption or control loss. Include code, infrastructure, configurations, schedules, databases, interfaces and security rules. Automation changes how controls operate, not whether they are needed.

4.8.1 Controlled changes and configuration baselines

Record the request, purpose, affected items, risk, impact, tests, approvals, schedule, communication and **fallback**. Verify the deployed result and close the record with evidence. Assess dependencies and segregation of incompatible responsibilities. **Approval authority** follows the enterprise's governance and risk; it is not universally a single job title.

Configuration management identifies items and relationships, maintains approved baselines, tracks status and verifies actual configurations. Detect drift through comparison and investigation. A CMDB containing an old **baseline** is not proof that production matches it.

Detecting unauthorized production changes

To test whether changes escaped **authorization**, start with **independent evidence** of actual changes-deployment records, **configuration** comparisons, system logs or artifact histories-and trace back to approved records. Starting only with the change register can miss changes omitted from that register. Also test approved changes forward to verify proper implementation. Assess the reliability and completeness of the actual-change population.

Emergency changes

An emergency process expedites necessary action under **defined authority**. Obtain emergency authorization before implementation where feasible, retain a minimum record and activity evidence, test proportionately, and complete formal documentation and **independent review** promptly afterward. A genuine emergency may require exceptional action under delegated authority; retrospective approval is not a standing license for uncontrolled changes.

4.8.2 Patch Management

A **patch** is a software update intended to correct defects, including security weaknesses. Patch management controls how relevant updates are assessed, tested, authorized, installed and verified. It is part of the wider change process; a patch being available does not prove that the correct assets have received it.

Inventory affected assets, monitor relevant vulnerabilities and patches, assess **exposure** and **compatibility**, prioritize risk, test appropriately, obtain required authorization, deploy and verify installation and service health. Track failures, exceptions and compensating controls. A patch tool's "deployed" label may not mean that installation or a required reboot succeeded.

Balance the **risk** of delay with the risk of implementation. A critical actively exploited weakness may justify expedited action under the approved process. Record the decision and verify the result; neither "always wait" nor "install everything immediately" is a reliable universal rule.

4.8.3 Release Management

A release packages compatible authorized changes. Plan content, **dependencies**, resources, testing, support **readiness**, communications, deployment and rollback. Major, minor and emergency releases can have different approval paths, but the deployed artifacts must still be identifiable and controlled. Preserve source/object relationships and protect production libraries.

4.8.4 IS Operations and IS Operations Reviews

Operations follows current runbooks for **monitoring**, jobs, backup, incident handling and escalation. Control operator **privileges**, manual interventions and handovers. Review whether procedures reflect actual services and whether exceptions lead to action. Trace a sample of failed jobs, privileged activities and operational alerts through approval, resolution and reconciliation.

Auditor Perspective

Trace changes in both directions

To find changes omitted from the register, start with reliable actual production **evidence** and trace back to **authorization**. Then test recorded approvals forward to implementation.

Emergency work needs defined authority and review, not a permanent bypass of accountability.

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Tracing actual changes back to authorization

Change control should assess need, impact, testing, **authorization**, deployment, verification and **recovery arrangements** in proportion to risk. Configuration management maintains reliable approved states and relationships. Patch management identifies relevant updates and evaluates, deploys and verifies them. These processes should exchange information without being treated as identical.

To find unauthorized changes, start from reliable actual-change evidence such as protected **configuration** histories or deployment events and trace to approvals. Sampling only registered tickets can exclude the unregistered changes the audit seeks. Validate the **completeness** and integrity of the selected source.

Example: A critical **database** patch has been assessed as relevant, but no active exploitation requires immediate action. Representative testing helps support a controlled rollout. An urgent active **threat** can justify an expedited authorized path with appropriate safeguards. Emergency change does not mean undocumented change: preserve the authority, actions, risk decision, proportionate validation and prompt **independent review**. A successful installation message is insufficient without verifying the intended version, service behavior, exceptions and updated configuration records.

Verify the State After a Patch and Rollback

Example. The change ticket shows a patch was installed. A later rollback returned one production instance to the vulnerable version while the inventory retained the patch status. Compare the approved change, deployment events and current versions of all relevant instances. Verify how rollback triggers update records and reopen exposure for remediation or authorized acceptance. Examine the temporary safeguards and deadline for reapplying the patch. The audit evidence should cover the state that is actually operating.

4.9 Operational Log Management

Logs record events such as access, changes, processing and failures. They support **accountability**, investigation, monitoring and evidence. Distinguish an audit trail from a database **transaction log**: transaction logs also support commit, rollback and recovery, so the statement “logs never support integrity” is too broad.

4.9.1 Types of Logs

Application logs describe business or application events; system logs describe operating behavior; security logs capture access and security events; **database** audit logs record selected actions; transaction logs support database recovery. Define what to capture according to **risk**. Collect sufficient **context**-time, source, actor, action, affected object and result-without unnecessarily recording secrets or sensitive payloads.

4.9.2 Log Management

The cycle covers generation, collection, transfer, storage, protection, analysis, alerting, reporting, **retention** and disposal. Synchronize time sources and document time zones. Detect collection failures, dropped events, parser problems and storage exhaustion. The absence of an **alert** means little if the relevant source stopped sending events.

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Storing, protecting and analyzing logs

Restrict access, separate privileged system administration from **independent review** where practical, and use appropriately protected centralized storage. Append-only or **immutable** mechanisms can strengthen protection, but configuration, retention and administrative access still matter. Describe controls as tamper-resistant and verifiable rather than assuming any product is absolutely tamper-proof.

Tune meaningful alerts, assign owners and review outcomes. Correlation can connect events across systems, while **false** positives can overwhelm responders and false negatives can hide incidents. Security information and event management (SIEM) centralizes and analyzes events; it does not replace ownership or response. Report unresolved concerns and trends to the appropriate **governance** level.

Write once, read many (WORM) storage permits writing under defined rules and subsequent reading while restricting alteration or deletion during the configured retention period. Verify actual retention and administrative bypass controls; distinguish WORM storage from a self-propagating malware worm.

Assessing whether logs can support a conclusion

Log management includes identifying required events, collecting them reliably, synchronizing time, preserving **integrity**, controlling access, retaining evidence and responding to relevant exceptions. The required coverage depends on the risk and purpose. A large volume of logs is not useful if the critical identity or event is missing.

Distinguish application audit trails, operating-system events, security alerts and **database** transaction logs. Transaction logs can support commit, **rollback** and recovery; selected audit logs can support accountability and investigation. Avoid a blanket claim that logs either guarantee integrity or have no integrity role. Check the particular control function.

Example: An administrator can delete local events before collection and the central system does not **alert** on missing data. A dashboard showing no suspicious activity gives weak assurance. Test sample events from **origin** through collection, parsing, storage, correlation and review, including failures. Confirm time sources, access segregation, tamper resistance and retrieval capability. **Retention** should meet justified business and applicable obligations while avoiding unnecessary sensitive content. Obtain evidence that someone acts on relevant events, not simply that the collector is running.

Establish a Trustworthy Event Timeline

Example. An access event appears to occur before the approval, but one source records local time and the other uses coordinated universal time. Confirm time zones, synchronization and any known clock drift before deciding the order. Preserve the original timestamps, document the normalization and trace records to a complete source. Check whether collectors detect missing events and protect retained copies. Consistent timestamps improve correlation when the sources and the interpretation of time are reliable.

4.10 IT Service Level Management

Service level management translates business needs into measurable commitments and monitors delivery. Define the service and customer perspective first. A server uptime target can be met while a broken **interface** prevents the business process from working.

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4.10.1 Service Level Agreements

Agreement	Relationship and purpose
SLA - service level agreement	Provider/customer agreement on defined services, targets, measures and responsibilities.
OLA - operational level agreement	Internal commitments between support groups that enable delivery of the service.
Underpinning contract	External supplier commitments supporting the service; assess consistency with business needs.

Specify service hours, availability, response and **restoration** targets, support, escalation, security, reporting, responsibilities and relevant remedies. Define **exclusions** and measurement methods. Supplier penalties may compensate financially but do not restore service; dependency commitments must support the end-to-end requirement.

4.10.2 Monitoring of Service Levels

Compare reliable actual availability and **performance** with the agreed **definitions**. Examine incidents, response and restoration times, missed commitments and improvement actions. Utilization reports answer a different question from availability compliance, though they can help explain causes. Review whether reporting excludes material outages or measures only a partial service.

4.10.3 Service Levels and Enterprise Architecture

Map each service to applications, infrastructure, suppliers, people and **interfaces**. Ensure supporting targets and **capacity** are compatible with the service commitment. A critical database cannot have a four-hour recovery design if the dependent service needs recovery in one hour without another viable route.

Connecting service agreements to business needs

An **SLA** specifies measurable commitments and conditions between service parties. Supporting operational-level agreements coordinate internal teams, while supplier arrangements support external dependencies. Define the **service boundary**, measurement source, response and resolution targets, exclusions, reporting, escalation and remedies clearly.

Targets should reflect the business service. A provider's infrastructure uptime can remain high while users cannot complete transactions because a shared identity or **interface** dependency is unavailable. Response time measures acknowledgment; resolution or **restoration** measures something different. Avoid interpreting one as the other.

Example: A payroll service meets its monthly availability percentage but repeatedly fails on payday. Review the service window, critical periods, **transaction outcomes** and breach handling. Averages can hide severe short interruptions or repeated impact on a subset of users. Assess whether breaches cause meaningful **corrective action** and whether supplier targets support the enterprise's own commitments. **Service credits** may compensate defined failures, but they do not establish recoverability. Reassess agreements when demand, architecture, providers or business requirements materially change.

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Connect Supplier Targets to the Service Commitment

Example. A service-level agreement requires the customer's application to be usable within two hours of an interruption. The supplier agreement measures two hours to acknowledge a support ticket. Map the customer commitment to internal team responsibilities and the supporting supplier terms. Check the measurement start and finish, exclusions and actual performance. If the supplier meets its acknowledgment target while the customer misses restoration, assess the coverage gap and who has authority to renegotiate the supporting arrangements.

4.11 Database Management

A **database** management system (DBMS) organizes and controls data storage, retrieval and change. It can provide access control, transaction processing, **integrity** constraints, concurrency and recovery. Its capabilities depend on correct design, configuration and use.

4.11.1 DBMS Architecture and Metadata

Schemas describe data organization and constraints. A **data dictionary** or catalog records metadata such as tables, fields, types, keys and relationships. Maintain authoritative **definitions** and restrict changes. Data independence reduces the need for applications to know physical storage details, but does not remove application dependencies on data meaning.

Java Database Connectivity (JDBC) is the Java API for database access through suitable drivers. An application programming interface is the general concept; JDBC is a specific database-access API. Review connection credentials, driver configuration, query construction, privileges and transaction handling. [R36]

4.11.2 Database Structure

A **relational database management system (RDBMS)** uses the relational model, commonly organizing data in tables related through keys and queried using Structured Query Language (SQL). It is one type of DBMS; the other models below organize information differently.

Model	Organization and consideration
Hierarchical	Tree-like parent/child structure; navigation follows defined relationships.
Network	Records can participate in multiple linked relationships; applications navigate those links.
Relational	Tables linked through keys, commonly queried with SQL; enforce appropriate constraints.
Object-oriented	Stores objects and their behavior-related structures in an object-based model.
NoSQL	Includes key-value, document, wide-column and graph approaches. Transaction and consistency guarantees vary by product and configuration.

4.11.3 Database Controls

Entity integrity requires a unique, non-null primary key in a relational design. **Referential integrity** enforces valid relationships, often through foreign keys. Domain integrity constrains allowed types,

ranges and values. Business rules may require additional application or database logic. A valid key does not prove that a transaction was genuine or authorized.

ACID comprises atomicity, consistency, isolation and durability. Commit makes a transaction's changes final under the DBMS model; **rollback** undoes uncommitted work. Concurrency controls prevent conflicting updates through suitable mechanisms. Test isolation and recovery behavior for the actual configuration rather than assuming every **database** provides identical guarantees.

Normalization reduces redundancy and update anomalies; denormalization intentionally adds redundancy, often for **performance**, and needs consistency controls. Restrict direct data changes, powerful utilities and administrator access. The database administrator (DBA) maintains the platform; **business owners** remain accountable for data meaning, classification and authorized use.

4.11.4 Database Reviews and Data Governance

Review roles, privileged activity, schema changes, constraints, patching, backups, **restoration**, performance and encryption/key dependencies. Independently monitor sensitive DBA actions where practical. Audit logs can detect unauthorized activity; constraints prevent particular invalid states; transaction logs support recovery. Select evidence according to the **control objective**.

Data governance assigns ownership, stewardship, standards and quality responsibilities. Define accuracy, completeness, consistency, timeliness and relevant lineage, then measure them. A technically valid database can still contain wrong business information. Trace important reports through their **source data**, transformations, exceptions and review.

Testing database integrity concurrency and recovery

A DBMS manages information structures, queries, access, transactions and recovery. Schemas and constraints define permitted relationships and values. Primary and foreign keys support identity and **referential integrity**; they do not establish that every permitted value is correct for the business. Views can restrict presentation, but underlying **privileges** and alternate access paths also matter.

ACID concerns atomicity, consistency, isolation and durability within the applicable transaction model. Concurrency anomalies include dirty reads, nonrepeatable reads, phantoms and lost updates. Locks, version checks and snapshot mechanisms offer different behavior. Verify the actual product, isolation level and **configuration** rather than assuming every **database** provides the same guarantees.

Example: A process commits a payment record, then fails before receiving an external bank acknowledgment. The local transaction's durability does not prevent duplicate payment on an unsafe retry. Evaluate end-to-end identity, reconciliation and exception handling. Assess **database** administration, **privileged access**, change, auditability, backup and tested recovery to a consistent point. NoSQL systems use varied data and consistency models; the label alone does not determine whether transactions, **integrity** checks or recovery controls are present.

Some database systems use **spool space** for temporary or intermediate query results. Quotas and workload controls can limit resource exhaustion. The term and implementation are product-specific and differ from a print spool.

A **deadlock** occurs when transactions wait on one another in a cycle and cannot progress without intervention such as aborting a transaction. Review detection, timeout/retry behavior and integrity controls.

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Check Integrity When a Bulk Load Uses an Alternate Route

Example. An application normally checks that each payment references a valid customer. A bulk loading utility bypasses that rule. Determine whether the database enforces the relationship or another safeguard covers the load. Inspect authorization for the utility, exceptions and post-load reconciliations. Test selected loaded records against the relevant reference data. Consider whether a failed batch can be recovered consistently, including partial processing and downstream effects. The control must cover the actual data entry routes.

4.12 Business Impact Analysis

A **business impact** analysis (BIA) determines how disruption affects business activities over time and what must be restored, when and with which resources. **Business process owners** provide essential impact and dependency information. IT translates approved needs into feasible technical capabilities.

The BIA informs recovery priorities and strategy. Establish continuity sponsorship, **policy**, **scope** and responsibilities, then use the BIA and accountable owner input to define criticality, dependencies and recovery requirements before choosing arrangements.

Exam reasoning: Once **policy**, sponsorship and scope are established, understand business impacts and approved recovery needs before selecting a site or drafting detailed recovery procedures. **Business process owners** supply essential impact and criticality information.

4.12.1 Classification of Operations and Criticality Analysis

Identify processes and evaluate financial, operational, legal, safety, customer and reputational effects over relevant time periods. **Criticality** depends on consequences and timing, not system size or the loudest department. Payroll may be time-sensitive near a payment deadline; a customer transaction service may be critical throughout its operating window.

Map people, facilities, applications, data, suppliers, networks, identity, keys and infrastructure. Sequence recovery according to business needs and technical **dependencies**. An identity service may need recovery before several higher-profile applications can be used. Validate **assumptions** with both business and technical owners.

Recovery objectives - distinguish the measures

Measure	Meaning and implication
RTO - recovery time objective	Target time to restore the required service capability after disruption. Drives recovery arrangements and must fit the business interruption tolerance.
RPO - recovery point objective	Target point to which data must be restored, commonly expressed as allowable data loss in time. Drives data-protection and replication design.
MTO / MTD / MAO	Maximum tolerable outage/downtime or maximum allowable/acceptable outage: **business interruption** limit before consequences become unacceptable. Treat these as equivalent only when the adopted boundaries and definitions match.
WRT - work recovery time	Time to reconcile backlogs and resume business work after technical restoration, where separately defined.

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Measure	Meaning and implication
SDO - service delivery objective	Required service capability during the alternate or recovery mode; it may be reduced but must be explicit.
MTBF/MTTR	Average interval between failures / average repair time under stated definitions. These are performance measures, not replacements for business recovery objectives.

Worked example: Order processing must resume within two hours, tolerate at **most** 15 minutes of data loss, and support 60 percent of normal transaction volume during recovery. The **RTO** is two hours, the RPO is 15 minutes and the alternate SDO is 60 percent. If technical recovery takes 90 minutes but **reconciliation** needs another hour before orders can be processed, the business capability is not restored within two hours.

Where technical **RTO** and **work recovery time** are measured separately, leave enough margin inside the maximum tolerable downtime for both. An RTO greater than the business tolerance is a clear design mismatch, but RTO merely equal to the tolerance may also leave no room for reconciliation, activation delays or other necessary work.

A shared resource must support the **earliest recovery need** of its dependent services unless there is a tested alternative. If applications require recovery in one, four and eight hours and all depend on the same database, a four-hour database strategy cannot satisfy the one-hour service.

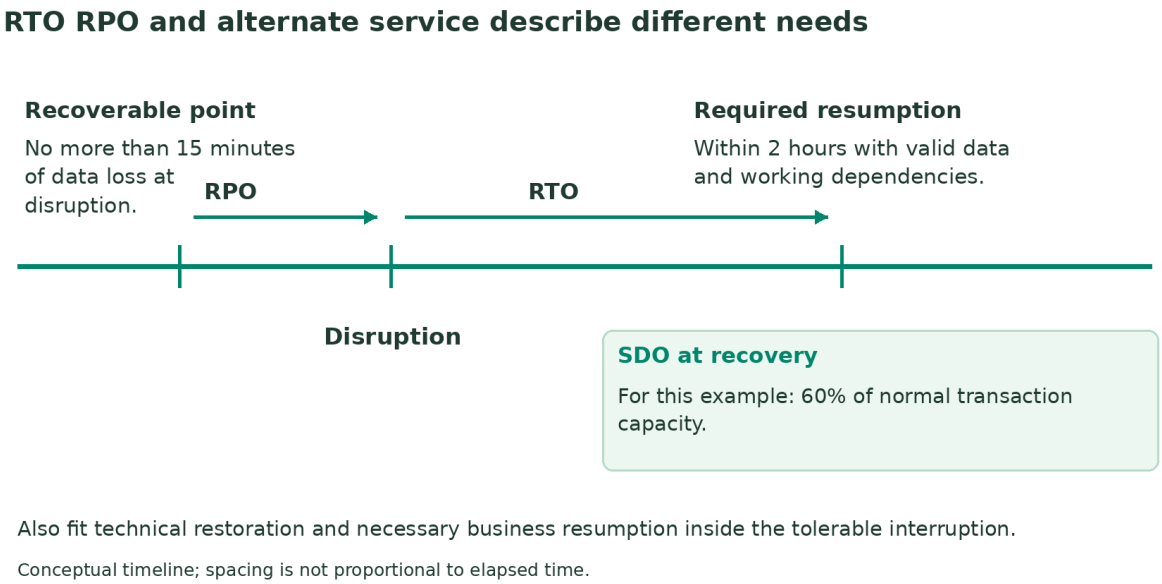


Figure 4.2 Recovery time, data currency and alternate capacity are separate objectives.

How RTO RPO WRT and MTD fit together

Maximum tolerable downtime (MTD) is the longest interruption the business can tolerate before consequences become unacceptable. **Work recovery time** (WRT) is the time needed after **technical restoration** to validate data, reconcile outstanding work and resume the agreed business activity. In a model that defines RTO as technical restoration time, plan RTO plus WRT within MTD, with an appropriate margin. If an organization defines RTO at the business-**service level**, necessary work recovery is already inside that target; do not add it a second time. Always establish the scope and start/end points.

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Example: disruption occurs at 10:00. **MTD** is four hours, so the business tolerance ends at 14:00. The technical **RTO** is two hours, targeting 12:00; planned WRT is one hour, targeting business resumption by 13:00. This leaves one hour of margin. The RPO is 15 minutes, so the recovery design must support a usable state no earlier than 09:45. A 09:50 **recovery point** meets the data-currency objective; a 09:30 recovery point does not, even if the servers start by noon.

The alternate **service delivery objective (SDO**) describes the capability required while operating in recovery, such as 60 percent of normal transaction volume. It is a **capacity** or service target, not another interval to add to RTO. After minimum service resumes, **restoration** to normal capacity can continue under the approved continuity arrangements. See NIST SP 800-34 Rev. 1 for recovery-objective and contingency-planning concepts.

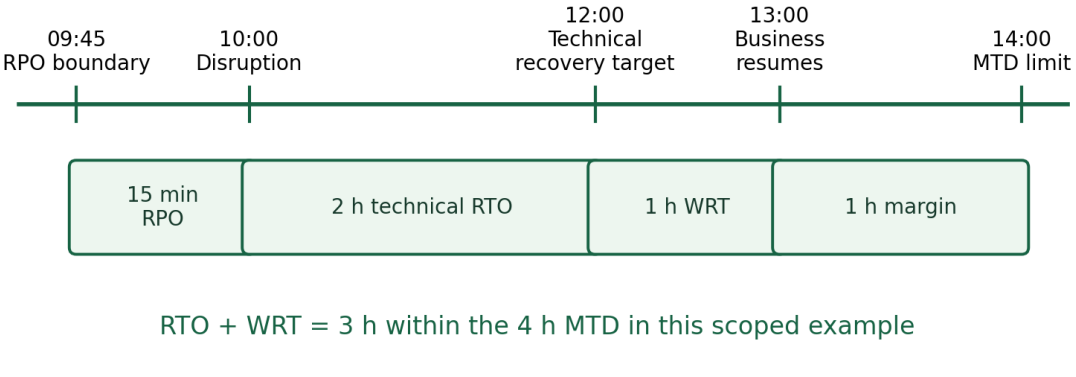


Figure 4.3 Recovery timeline with separate data, time and service-capacity objectives. RPO measures backward from disruption.

Auditor Perspective

Keep recovery measures separate

RTO concerns **restoration** time; RPO concerns the **recoverable data** point; SDO concerns the capability available during recovery. Read the measure the question actually asks about.

The technical recovery and business resumption arrangements must fit the maximum tolerable interruption, with appropriate time for necessary **reconciliation**.

Defining recovery targets at the correct boundary

The BIA identifies critical activities, interruption **consequences** over time, minimum service needs, dependencies and recovery priorities. Obtain accountable business input and approval. RTO concerns the targeted **recovery time**; RPO concerns the acceptable point to which information must be restored. A four-hour RTO and a one-hour RPO describe different requirements.

Maximum tolerable downtime is the interruption limit before consequences become unacceptable under the adopted definition. Where RTO means **technical restoration**, include subsequent **work recovery time** within that limit. Where RTO is defined at the usable business-**service level**, necessary validation is already within the target and should not be added twice. Establish start and end points before doing arithmetic.

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Worked example: A service must be usable within two hours and requires 30 minutes of **validation** after its shared **database** is ready. With no tested alternative, database readiness must be no later than 90 minutes. Averaging the RTOs of dependent applications would miss the critical service. Consider staff, identity, network, suppliers, facilities, interfaces and accumulated work as well as the server. Revisit the BIA after significant changes and validate that proposed **recovery arrangements** can actually meet approved needs.

Set Recovery Targets at the Business Boundary

Example. A critical application is restored within its technical target, but staff cannot complete business verification or process the required volume. Identify the activity and its critical period in the **business impact analysis**. Confirm the point at which the service counts as usable, the acceptable age of restored data and the capacity required. If the business RTO already includes verification and work recovery, count those steps within it. Trace the supporting dependencies to realistic recovery targets and test conditions.

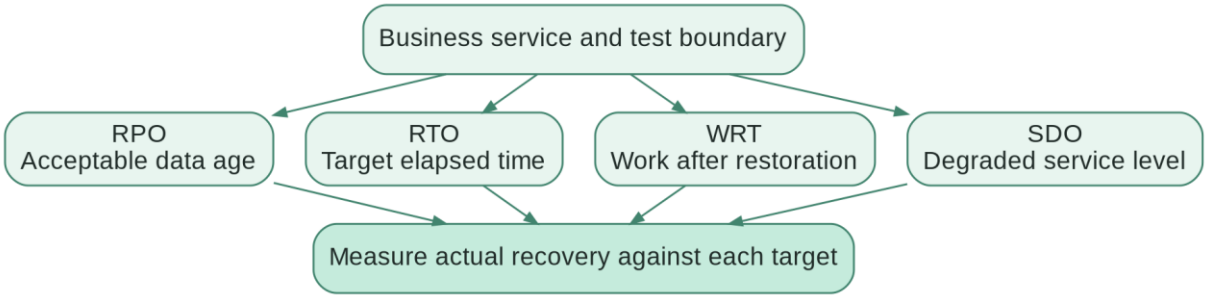


Diagram 4.1 Recovery targets measure different aspects of a usable business service.

4.13 System and Operational Resilience

Resilience combines the ability to withstand disruption, adapt and recover. **High availability** minimizes service interruption; **fault tolerance** maintains required operation through specified failures; disaster recovery restores capability after a disruptive event. All are bounded by their design assumptions. Redundancy against one disk failure does not establish resilience against fire or ransomware.

4.13.1 Application Resiliency and Disaster Recovery Methods

Use appropriate redundancy, clustering, failover, load distribution, replicated data and recoverable configurations. A cluster coordinates multiple nodes; designs can be active/passive or active/active. **Load balancing** distributes traffic but needs healthy back-end **capacity** and protection of its own failure dependencies. Failover switches service to an alternate component or location; failback returns it under control.

Examine state, sessions, transaction **consistency** and split-brain **risk**, where isolated nodes can both act as active owners. Quorum or fencing mechanisms can reduce that risk. Test behavior under network partition, storage failure and dependency outage, not only a clean planned switchover.

Synchronous replication waits for the required remote acknowledgment before completing a write under its **configuration**. It can reduce data-loss **exposure** but adds latency and depends on distance, capacity and connectivity. Asynchronous replication can reduce performance impact while permitting lag and potential data loss. Neither automatically protects against replicated deletion or corruption.

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4.13.2 Telecommunication Networks Resiliency and Disaster Recovery Methods

Identify single points of failure across routers, switches, firewalls, carriers, local loops, building entrances, power, DNS and management **dependencies**. Two suppliers or circuits may share a physical conduit, exchange, upstream provider or power source. Ask for evidence of **independence**.

Alternative routing provides another route or service; **diverse routing** emphasizes physical path separation. Provider terminology varies, so inspect the actual routes. Last-mile resilience needs particular attention to separate local access paths and building entry. Cellular backup can help, but shared regional failures or congestion can still affect it.

Example: Two internet circuits from different carriers enter through the same underground duct. A single excavation cut can remove both. The second carrier name does not establish diverse physical routing.

For converged voice and data, verify bandwidth, latency, jitter, packet loss and appropriate quality of service. Test that **backup** links can carry the required recovery workload. A link that permits a successful ping may be too small for the agreed service.

Continuity term	What independence to examine
Alternative routing	An alternate communications route available when the primary path is unavailable. Verify how failover is activated.
Diverse routing	Physical path or entry separation intended to reduce shared local failures. Check actual conduits, entrances and facilities.
Long-haul diversity	Independence of routes across the carrier/backbone portion beyond local access. Two brand names can still share infrastructure.
Last-mile protection	Protection of the local access connection between premises and the provider network. Examine local loops, building entrances, power and alternatives.

Terminology varies by provider. Trace the actual paths and failure domains; a contract label does not establish independent service.

Evaluating shared failure domains and degraded operation

Resilience is the ability to withstand, adapt to and recover from disruption while maintaining required outcomes. Redundancy can reduce a component failure but leave common causes untouched. Separate servers may share power, network, storage, identity, administrative control or a vulnerable software version.

Identify failure domains: groups exposed to one cause of failure. Assess concentration and correlated events across regions, facilities, providers and workforce. **High availability**, **fault tolerance** and disaster recovery address related but different mechanisms and timescales. Load balancing does not automatically provide independent failover, and failover itself can introduce data consistency or capacity risks.

Example: Two cloud regions share one compromised administrator and replication propagates encrypted data. Regional separation does not address that logical failure. Consider isolated **recovery**

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copies, protected credentials, clean rebuild, trusted **configuration** and controlled return to service. Test degraded modes and manual workarounds at realistic volumes, including how records are later reconciled. A workaround that functions for ten transactions may fail during a week-long outage. Validate resilience through scenarios and evidence, with appropriate operational **authorization** and safeguards.

Outage tolerance and alternate service capability

Maximum allowable outage (**MAO**), also called maximum acceptable outage in some sources, expresses the interruption limit before **business consequences** become unacceptable. **MTD** and MTO may express the same tolerance where the adopted definitions use the same boundary. Establish that boundary rather than assuming the labels are universally interchangeable. Unacceptable consequences need not mean irreparable damage. Section 4.12 explains how technical recovery, work recovery and business tolerance fit together.

The **service delivery objective (SDO**) is the required **service capability** during recovery, such as the approved transaction throughput. It is not another time interval. Assess resilience against time, data currency and usable service capacity together. A mirrored system that starts promptly but supports too little business work can still fail the approved recovery objectives.

Find a Dependency Shared by the Alternate Environment

Example. Primary and alternate systems are in separate locations but require the same certificate service and privileged cloud identity. Map failure domains across facilities, networks, data, credentials and control planes. Ask which independent authority can recover the service if the shared identity is unavailable. Examine alternate access arrangements and evidence from a representative exercise. Physical separation reduces some disruption scenarios. The service remains exposed to dependencies shared by both environments.

4.14 Data Backup, Storage and Restoration

Backups preserve **recoverable data** and supporting information. Storage resilience helps sustain access through certain failures. **Restoration** turns protected copies into usable business capability. Evaluate these together: backup success, media readability and functioning storage are each insufficient alone.

4.14.1 Data Storage Resiliency and Disaster Recovery Methods

RAID combines disks for **performance** and/or redundancy. RAID 0 stripes without redundancy; RAID 1 mirrors; RAID 5 uses distributed parity and commonly tolerates one disk failure; RAID 6 adds dual parity and commonly tolerates two; RAID 10 combines mirroring and striping, with failure tolerance depending on which disks fail. Rebuild performance and remaining **exposure** matter. RAID is not a backup.

Direct-attached storage connects directly to a host. Network-attached storage commonly provides file-level access; a storage area network commonly provides block-level storage connectivity. **Availability** depends on controllers, paths, fabrics, power and **configuration** as well as disks. Redundant paths must actually work.

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4.14.2 Backup and Restoration

Define **scope**, frequency, **retention**, recovery points, encryption, keys, access and test requirements from business needs. Include application-consistent data, configuration, source artifacts, certificates and recovery instructions where needed. A database backup without its necessary keys, logs or compatible software may be unusable.

A snapshot captures a point-in-time state. Some snapshots depend on the original storage or shared credentials, so they may not be independent recoverable copies. Replication provides another copy but may rapidly replicate deletion, corruption or **encryption** by malware. Assess **retention** and isolation, not only the number of copies.

4.14.3 Offsite Library Controls and Cloud Backup

Keep recoverable copies outside relevant shared failure domains. Assess environmental and physical protection, access, transport, inventory, location tracking and retrieval time. For irreplaceable information, protect against loss during transfer through an additional suitable copy or equivalent resilient arrangement. **Encryption** addresses disclosure; it does not recreate lost media.

Cloud backup requires clarity about service **scope**, regions, retention, immutable settings, deletion rights, keys, egress time, bandwidth and exit. Shared production credentials can make both live data and cloud copies vulnerable. An offsite copy is valuable, but independent access and recoverability also matter.

4.14.4 Media, Documentation and Periodic Backup Procedures

Select disk, tape, cloud or other media according to **capacity**, durability, access, **recovery time** and cost. Protect recovery documentation and credentials so they remain available during an outage while access stays controlled. Retain compatible readers, software and keys for the necessary retention period.

Monitor jobs, investigate failures, validate backup **completeness** and perform restore tests. Use transaction-consistent backups or the logs and procedures necessary for consistent recovery. Include new systems and data in the backup inventory as part of change and **asset management**.

4.14.5 Backup Schemes

The comparison below uses conventional file-**backup** terminology. Products can use synthetic fulls, incremental-forever or other mechanisms with different restore behavior. Verify the tool's actual **dependencies** rather than relying only on its labels.

Type	Copies	Typical restore dependency
Full	All selected data.	The chosen full backup, plus relevant later logs if point-in-time recovery is required.
Incremental	Changes since the previous relevant backup in the backup chain.	A suitable full and every required subsequent incremental, in order.
Differential	Changes since the last full backup.	The suitable full and latest required differential.

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Type	Copies	Typical restore dependency
Transaction-log backup/remote journaling	Transaction changes or journals for recovery to a chosen point.	Compatible base backup and an intact required log sequence.
Electronic vaulting	Electronic transfer of backup information to a remote location.	Depends on what is transferred, its completeness, consistency and restore method.

Worked example: Sunday is a **full backup**. Monday, Tuesday and Wednesday are conventional incrementals. To recover Wednesday's state, restore Sunday plus each required incremental in sequence. With differentials on those days, Sunday plus Wednesday's differential is typically sufficient. Missing an intermediate incremental can break the chain.

4.14.6 Rotation, Offsite Records and Backup Strategy

Grandfather-father-son is a rotation scheme using generations such as monthly, weekly and daily copies. Rotation is not the same as the backup method. **Retention** should reflect recovery scenarios and information **obligations**, including the need to find a clean point before delayed corruption is discovered.

Maintain records linking media or **backup** objects to contents, dates, protection, location, **retention** and disposal. Reconcile dispatched and received physical media. Protect access to catalogs and restore functions as well as the data.

RPO drives the protection design, but a scheduled 15-minute job does not guarantee a 15-minute achievable **recovery point**. Failures, transfer delays, replication lag and missing logs can extend loss. Verify the age and consistency of **recoverable data** at the location and time it is needed.

Testing recoverability and cyber recovery

Restore representative critical services into a controlled environment and verify **integrity**, functionality, timing and **data currency**. Include dependencies and the intended operating load. A successful restore of one small file cannot establish whole-service recovery.

Maintain appropriately isolated, offline or **immutable** copies according to the **threat** and design. Protect retention changes and deletion with separate authority. Cyber recovery may require a clean environment, validated identities and keys, malware assessment, clean restoration points and careful reconnection. Prevent reinfection while restoring business priorities.

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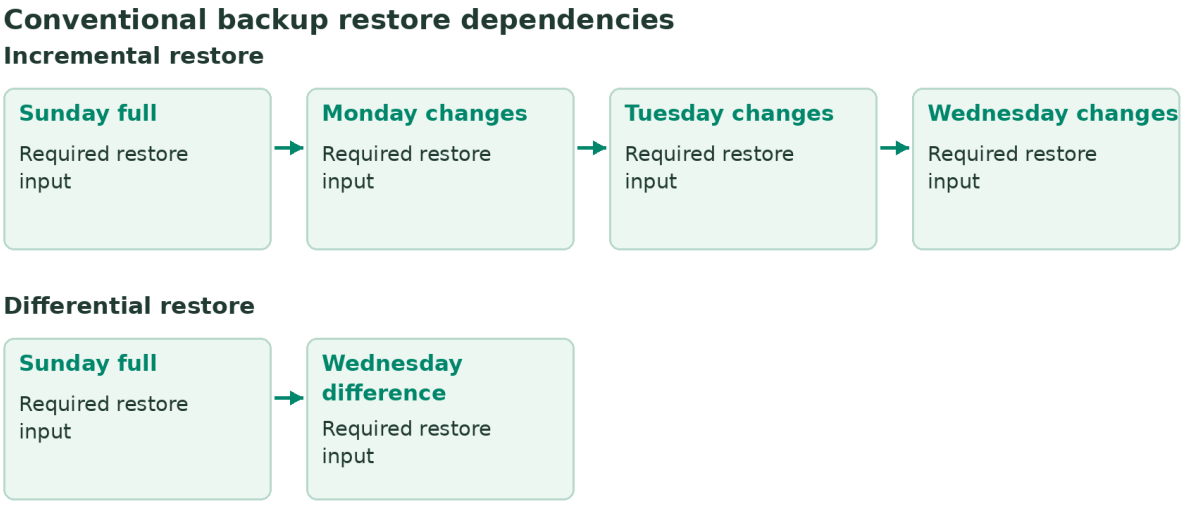


Figure 4.4 Restore dependencies for conventional incremental and differential backups.

Demonstrating usable recovery points

A **backup** is useful when the required information can be restored, trusted and used within the relevant objectives after the failure being considered. Protect copies against deletion, corruption, compromise and shared-site loss. **Replication** and RAID can support availability but can also propagate unwanted changes; they are not substitutes for appropriately independent recoverable copies.

Full, incremental and differential approaches differ in storage, backup effort and restore **dependencies**. Verify a complete consistent restore sequence, including logs, **encryption** keys, software, configuration and catalogs. Immutability reduces specified alteration or deletion paths under its implementation; assess administrator privileges, retention controls and recovery access as well.

Example: A 15-minute transfer schedule experiences two hours of failed copies. The schedule does not establish a 15-minute achievable **recovery point**. Examine the actual latest usable consistent data, transfer delays, missing logs and protected failure domains. **Remote journaling** requires a complete validated sequence and tested recovery, not merely a running agent. Restore representative business data in a suitable environment and reconcile it. Test that protected keys remain available to authorized recovery staff without exposing the copies to ordinary production compromise.

Disk mirroring and independent recovery copies

Disk mirroring maintains duplicate copies of data on separate disks; **RAID** 1 is a common implementation. It supports availability when a covered disk fails. The timing and failure protection depend on the implementation. Replication between systems or sites is a broader concept and should not automatically be treated as the same arrangement.

A mirror normally applies the same logical writes, so deletion, corruption or ransomware-driven changes may affect both copies. It is not an independent, retained point-in-time backup. Verify separate **recovery copies**, protected **retention**, required **data currency** and successful restoration. Example: Surviving a disk failure does not demonstrate recovery of yesterday's clean records after today's destructive write.

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Verify Restoration With the Required Keys and People

Example. The backup inventory shows protected copies with retention locks. A restore test fails because the recovery team cannot obtain the required decryption key from the affected environment. Inspect backup completeness, separate access, key recovery and the ability to validate application data after restore. Confirm that privileged accounts cannot silently shorten the protection needed for recovery. A retained copy addresses preservation. Successful recovery also depends on authorized access, usable keys, configuration and tested restoration procedures.

4.15 Business Continuity Plan

The **business continuity** plan (BCP) supports priority business activities through disruption. It covers people, premises, suppliers, processes, communications and technology. Disaster recovery plans support its technology-recovery needs; they do not replace the business-wide plan. Senior management sponsors continuity, and **business owners** remain accountable for their activities.

4.15.1 IT Business Continuity Planning

Align technology recovery with approved business priorities and **service levels**. Integrate IT plans with crisis management, emergency response, **supplier** arrangements and operating workarounds. Manual alternatives need capacity, controls, instructions and reconciliation; a statement that staff will “use spreadsheets” is not a tested strategy.

4.15.2 Disasters and Other Disruptive Events

Consider facility loss, utility interruption, telecommunications failure, cyberattack, key **supplier failure**, data corruption, severe weather and workforce unavailability. Assess correlated events that affect both primary and alternate arrangements. Focus on **consequences** and dependencies as well as named scenarios so the plan can adapt to unexpected causes.

Pandemic Planning, Reputation and Unanticipated Events

Workforce disruption can leave facilities and servers intact while key activities cannot be staffed. Plan **cross-training**, deputies, remote access, equipment, **supplier** resilience and sustainable workloads. Large-scale remote work can increase identity, connectivity and support demands.

A crisis can damage reputation through inaccurate information or poor handling even when systems recover. Define **authoritative communications**, approved channels and spokespersons. Scenario planning should allow flexible decisions rather than assume that every future event will match a predefined script.

4.15.3 Business Continuity Planning Process

Establish **policy**, **scope**, responsibilities and sponsorship; analyze business impacts and risk; evaluate and approve strategies; develop coordinated plans; train and exercise; and maintain and improve them. BIA identifies needs; strategy selects how to meet them; plans provide executable responsibilities and steps. Buying a site before understanding needs risks paying for the wrong capability.

4.15.4 Business Continuity Policy

State objectives, **scope**, **accountability**, risk expectations, planning and exercise requirements, maintenance responsibilities and reporting. Senior management should endorse the program and fund

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an appropriate capability. A policy document without owners or resources is not an operating continuity program.

4.15.5 Business Continuity Planning Incident Management

Define assessment, escalation, crisis coordination and **invocation authority**. Name primary decision-makers and deputies, and make thresholds and contact routes clear. Avoid relying on the person most likely to be unavailable. Protect **life and safety** and coordinate with emergency services as appropriate before treating technical recovery as the only priority.

4.15.6 Development and Components of Business Continuity Plans

Plans should be usable under disruption, with clear versions and accessible copies. Include **scope** and assumptions, activation, contacts, roles, dependencies, procedures, alternative service, communications, recovery priorities, resources and return-to-normal arrangements. Keep detailed sensitive credentials in appropriately protected **recovery arrangements** rather than ordinary uncontrolled plan copies.

Plan component	What it must enable
Decision authority	Who assesses, declares, invokes, escalates and authorizes return to normal, including deputies.
People and coordination	Teams, responsibilities, contacts, call trees, shifts and handovers.
Business procedures	Priority activities, alternate methods, acceptable controls and reconciliation.
Resources and dependencies	Facilities, technology, suppliers, equipment, data and essential supplies.
Communications	Staff, customers, suppliers, authorities and other stakeholders through approved channels.
Recovery and resumption	Sequence, validation, backlog handling, failback and accountable acceptance.
Maintenance and evidence	Training, exercises, issue actions, version control and updates after change.

4.15.7 Key Decision-Making Personnel and Required Supplies

Define crisis coordination, emergency response, damage **assessment**, business recovery, technology recovery, relocation and salvage responsibilities as appropriate to the organization. Ensure roles do not conflict and substitutes are trained. Supplies can include devices, connectivity, payment methods, forms and approved access to vital records. **Availability** matters during the event, not only on an inventory sheet.

4.15.8 Insurance

Insurance can transfer specified financial **consequences**, subject to policy terms, exclusions, limits and conditions. It does not perform operational recovery. Assess whether coverage fits exposures and whether claims require particular documentation or actions. **Business continuity** still needs people, resources, executable procedures and suitable technology.

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4.15.9 Plan Testing

Define objectives, **scope**, **assumptions**, scenario, participants, safety constraints and measurable success criteria. Exercise business processes and communications as well as technology. Select a method appropriate to the questions being tested; increasing disruption does not automatically make an exercise better designed.

Execute under control, document actual timelines, decisions and deviations, compare with objectives, analyze results and assign corrective actions. Retest significant **weaknesses**. A failed exercise can be valuable evidence for improvement; declaring success by excluding the difficult **dependencies** conceals risk.

4.15.10 Plan Maintenance and Good Practices

Update plans after material changes, incidents, exercises, organizational changes and **supplier** changes, as well as through scheduled reviews. Walkthroughs can reveal stale contacts or missing steps, but controlled change integration and accountable updates also keep content current. Distribution and **encryption** protect access; they do not ensure accuracy.

Keep continuity integrated with service ownership, **risk management**, procurement, change control and training. Monitor whether approved objectives are actually supported. Reassess costs and **benefits** of alternatives without weakening necessary resilience merely to make a plan look cheaper.

4.15.11 Auditing Business Continuity

Assess sponsorship, BIA, strategy alignment, plan ownership, **dependencies**, contracts, resources, exercises and **follow-up**. Interview business and technical personnel to confirm they understand their roles. Compare the plan with actual infrastructure, staffing, suppliers and recovery evidence.

Evaluate offsite storage access, protection and retrieval; inspect alternate-processing commitments for capacity, availability during shared disasters, test rights and technical fit. Review insurance in context. Audit provides an assessment; **management owns** invocation decisions, capability and acceptance of **residual risk**.

Planning continuity beyond technology restoration

Business continuity addresses prioritized activities, people, facilities, suppliers, information, communications and workarounds. Technology recovery supports these activities but does not by itself establish continuity. Sponsorship, **policy**, scope and an understanding of impacts should guide the strategy and executable plans.

Define activation authority, roles, deputies, contact methods, minimum resources, staff welfare, communications and decision records. Consider workforce unavailability, inaccessible facilities, regional events, **supplier** loss and prolonged disruption. Keep instructions accessible when ordinary systems and **authentication** are unavailable. Protect sensitive plan information appropriately.

Example: The recovery application works, but no authorized staff can release urgent payments from the alternate location. Exercise business roles, access, verification and communications as well as technical recovery. Manual processing needs realistic **capacity**, fraud controls, records and later **reconciliation**. Insurance covers defined **consequences** under its terms; it does not supply every missing capability. Maintain the plan after changes and follow exercise findings through assigned actions and retesting. A recent approval date does not establish that the resources and assumptions remain valid.

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Mobile sites and service bureau arrangements

A mobile site, or **mobile recovery site**, is transportable **recovery capability**, often in a vehicle, container or modular facility. Its usefulness depends on delivery time, site access, power, environmental support, connectivity, compatible equipment and **current data**. Confirm that staff can reach and use it during the modeled event. Equipment in transit does not satisfy an immediate service need.

A **service bureau** is a commercial provider of outsourced processing services, potentially serving several customers. It is not exclusively a **disaster recovery** facility. If selected for recovery, verify contracted compatible **capacity**, activation time, security, data transfer, customer priority and simultaneous-demand assumptions. A general subscription does not reserve adequate recovery capacity. Compare these dependencies with the alternatives in section 4.16.

Exercise the Manual Workaround Through Reconciliation

Example. During an outage, staff record urgent orders on paper. At service resumption the same orders can be entered twice if the paper sequence is not reconciled. Check the continuity plan for authority to invoke the workaround, numbering, approvals, custody and backlog reconciliation. In an exercise, trace selected orders from manual creation through entry and verification after recovery. Assess who owns the decision to return to normal processing and how exceptions are resolved.

4.16 Disaster Recovery Plans

A **disaster recovery** plan (DRP) describes how to restore technology and data to support the business's required **capability**. It must be executable with resources available during the relevant event. A fully equipped recovery room is of little value without usable data, identities, keys, connectivity and trained people.

4.16.1 Recovery Point Objective, Recovery Time Objective and Mean Time to Repair

Use the approved objectives from the BIA and validate actual **performance** against them. RTO concerns elapsed **recovery time**; RPO concerns the recoverable state of data; MTTR is an average repair measure. A faster average component repair does not prove that a whole service meets its RTO.

Recovery effectiveness includes timely restoration, acceptable **data currency**, functionality, integrity and the **required service** level. Meeting RTO alone is not sufficient if recovered information is wrong or the required process cannot run. Improving the recovery design can balance time, cost and risk; "optimized" does not mean every improvement reduces both cost and time.

4.16.2 Recovery Strategies

Select a feasible combination of alternative processing, resilient infrastructure, data protection, staffing, suppliers and business workarounds. Assess shared failure domains, geographic **exposure**, lead times, configuration compatibility, capacity and contractual priority. Compare demonstrated capability with targets rather than assuming that a site label proves a **recovery time**.

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4.16.3 Recovery Alternatives

Alternative	Capability and principal consideration
Mirrored/active-active or standby arrangement	Maintained alternate capability can support rapid recovery, depending on replication, consistency, failover and independent dependencies. It is not automatically zero downtime or zero loss.
Hot site	Equipped compatible capability intended for relatively rapid recovery. Current data, connectivity, configuration and activation still need verification.
Warm site	Partial readiness requiring additional equipment, configuration or data preparation. Measure the actual work and time.
Cold site	Facilities and utilities with substantial setup needed. Lower standing cost, but procurement and installation may be too slow.
Mobile site / mobile recovery site	Transportable recovery capability. Verify delivery time, site access, power, connectivity, compatible equipment, current data and staff logistics.
Reciprocal agreement	Mutual assistance with another organization or unit. Verify compatibility, spare capacity, rights, priorities and shared-event exposure.
Cloud recovery/DR as a service	Recoverable resources hosted through a provider. Test identity, network, quotas, regions, data, capacity, automation and exit dependencies.
Service bureau	Commercial outsourced processing provider. For recovery, verify reserved compatible capacity, activation time, security, customer priority and simultaneous-demand exposure; see 4.15.

Site labels have no universal recovery hours. A contract offering a **hot site** may exclude data loading, network setup or application work. Validate **scope**, total cost and demonstrated capability against the approved need. Geographic separation should address relevant threats while remaining technically workable.

Exam reasoning: Compare actual preparation, compatible **capacity**, data, **dependencies**, demonstrated recovery and cost. A cold site can suit a long recovery window only when setup and procurement are feasible. Reciprocal arrangements require evidence of capacity, rights, compatibility and availability during a shared event.

4.16.4 Contractual Provisions and Procuring Alternative Hardware

Define guaranteed **capacity**, compatible configurations, delivery and access times, shared-site priorities, test arrangements, support, security, **termination** and remedies. A reciprocal promise without reserved resources can fail when both parties need them. Procurement plans should account for shortages, logistics, compatible software and licensing; vendor assurances alone do not establish delivery capability.

4.16.5 Development of Disaster Recovery Plans

Document activation **criteria** and authority, **dependencies**, recovery sequence, resources, technical steps, backups and keys, contacts, communications, validation, alternate operation and return to normal. Provide accessible protected copies. Detail should allow trained substitutes to act, not require undocumented knowledge held by one engineer.

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Identify scenario differences such as site loss versus cyber compromise. Restoring into a contaminated environment can reintroduce the failure. Account for identity, **DNS**, time, certificates, networks, security tooling and management access before dependent applications. Define data **reconciliation** and business acceptance of recovered operations.

4.16.6 Organization and Assignment of Responsibilities

Assign coordination, **assessment**, infrastructure, application/data recovery, communications, business validation, relocation and salvage roles as needed. Specify handovers and **decision rights**. Protect damaged equipment and evidence; salvage decisions should not compromise safety or investigation. Audit may observe tests and evaluate evidence without directing recovery.

4.16.7 Disaster Recovery Testing Methods

Choose a recovery exercise for the evidence needed. A **paper test** reviews the plan; a tabletop or structured walkthrough discusses decisions. A **preparedness test** executes selected recovery activities, while simulation, parallel processing and **full interruption** have different scopes. Labels vary, so define exactly what will be executed and which service remains live. The comparison in section 4.16.8 links each method to its evidence and limits. [R5]

Exam reasoning: Document and discussion reviews identify gaps but do not demonstrate technical recovery. Execution exercises provide evidence within their **scope** and conditions. Select a safe authorized exercise capable of answering the objective; parallel testing is not universally most cost-effective. Preparedness exercise is not an exact synonym for **parallel test**.

4.16.8 Testing and Test Results

Test against explicit RTO, RPO, functionality, **data integrity** and SDO **criteria**. Measure activation delays and dependency work, not only a selected restore command. Verify business transactions and representative workload. Record what was excluded, pre-staged or simulated so conclusions stay within the evidence.

Worked example: A test recovers servers in 90 minutes against a two-hour RTO, but the database is four hours old against a 30-minute RPO. The recovery does not meet the approved objectives. A second test restores **current data** on time but cannot process payments; it also has not demonstrated recovery of the **required service**.

Retain findings, owners, deadlines and retest results. A test without **follow-up** can repeatedly expose the same failure. Include failback: synchronize changes, reconcile transactions and return under **authorization** without introducing loss or duplicates.

Choose the exercise for the evidence needed

The methods below answer different questions. They are not a mandatory sequence with a fixed disruption level: the agreed **scope**, execution and safeguards determine both **evidence** and operational risk. Component preparedness work can support a larger simulation or recovery exercise. [R5]

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Method	What it demonstrates	Risk and limits	Auditor focus
Paper test / checklist review	Plan contents, contacts, resources and obvious gaps are reviewed.	Usually low disruption; document review does not execute recovery.	Validate accuracy, ownership and unresolved gaps.
Tabletop / structured walkthrough	Participants discuss a scenario, roles, decisions and dependencies.	Usually low disruption; discussion does not prove restore speed or capacity.	Observe coordination, escalation and assumptions.
Preparedness test	Selected recovery procedures or components are actually exercised.	Impact depends on what is executed; not automatically nondisruptive or a parallel test.	Check scope, authorization, safeguards, timing and actual results.
Simulation	A scenario is rehearsed through agreed responses; execution can be partial or simulated.	Realism and operational impact depend on scope. Some steps may remain assumed.	Separate demonstrated actions from simulated outcomes.
Parallel test	Recovery systems process representative work while production continues.	Needs resources and isolation; actual production cutover remains untested.	Check representative load, reconciliation, timing and exclusions.
Full interruption	Production is intentionally interrupted and recovery arrangements support the service.	Potentially substantial business impact; requires authorization, safeguards and fallback.	Evaluate end-to-end function, integrity, RTO, RPO and SDO.

Management owns exercise **authorization**, safety, invocation decisions and remediation. The auditor evaluates objectives and safeguards, observes or tests independently, verifies the stated results and follows significant weaknesses to closure. Acting as the recovery operator or accepting business risk would impair that independent role. A **full interruption** is not automatically the BEST choice when a narrower exercise can safely answer the question.

4.16.9 Invoking Disaster Recovery Plans

Assess the event, protect safety, invoke through **defined authority**, communicate and execute the relevant sequence. Reassess conditions as facts change. Maintain an action and decision record, monitor restored services and **reconcile work** before declaring normal operations. Afterward, review lessons and update both technical and business plans.

Auditor Perspective

Recover the complete business service

A working server does not establish recovery if **authentication**, data, payments or another required dependency is unavailable. Measure the agreed end-to-end capability.

Meeting **RTO** alone is insufficient when the recovered data breaches RPO or the required functions do not work.

Selecting and exercising recovery arrangements using evidence

Recovery alternatives range from relatively unprepared facilities to equipped, replicated or highly synchronized arrangements. Hot, warm, cold and mirrored labels describe broad preparation concepts,

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not universal recovery hours. Assess actual **capacity**, compatible configuration, protected data, staff, licenses, dependencies and demonstrated restoration time against the approved **business objectives**.

Exercises provide different evidence. A document review identifies omissions; a walkthrough tests understanding; a simulation exercises selected decisions or actions. A **parallel recovery test** can use alternate processing while primary service continues. **Full interruption** exercises a live transition under controlled conditions and can create significant operational risk. A **preparedness exercise** is a practical exercise category and should not be treated as an exact synonym for parallel testing.

Example: A proposed **hot site** demonstrates the **required service** in four hours, while the available warm arrangement needs more than a day. For a six-hour target, the tested capability is the meaningful basis for selection. Exercise conclusions remain limited to tested data, workloads, dependencies and conditions. Include activation, data consistency, business validation, communications, failback and outstanding work. Choose **test scope** using risk, objectives and authorization rather than assuming one test is universally most cost-effective.

Read a Recovery Test at the Level It Demonstrates

Example. An isolated exercise restores the application and database at the alternate site. The production identity service and external payment interface were simulated. Compare the test scope and actual times with the business recovery requirements. Identify which dependencies and traffic changes still need evidence. Select an authorized follow-up test that exercises those paths at appropriate risk. A successful isolated restore supports the components and conditions tested. Record the remaining limitations before accepting end-to-end readiness.

4.17 The Last Mile - Memory Aids and Exam Traps

Use these aids to organize recall after understanding the explanations. Each exam-trap row contrasts a common error with a supported response; apply the **stated facts** and the chapter's reasoning.

Memory aids

Recall prompt	Meaning
Know it → own it → control it	Inventory the asset, assign accountability and apply lifecycle controls.
Incident restores; problem investigates the cause	A workaround can restore service while a permanent fix remains outstanding.
Trace from reality	To find unapproved changes, start with reliable actual-change evidence and trace to authorization.
Customer / colleagues / contractor	SLA with the customer; OLA between internal teams; underpinning contract with a supplier.
Time / point / capability	RTO = recovery time; RPO = recoverable data point; SDO = required alternate service.
Need → strategy → plan → prove → improve	BIA identifies need; strategy chooses a route; plans make it executable; exercises provide evidence; follow-up improves it.
Copies must survive together-failures	Separate relevant physical, administrative and cyber failure domains.

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Recall prompt	Meaning
Full + chain; full + latest difference	Conventional incremental restore needs the required chain; differential restore needs the suitable full and latest differential.
RAID keeps disks going; backup keeps recovery options	Disk redundancy is not protection from every deletion, corruption or disaster.
Recover the service, not just the server	Include identities, network, data, interfaces, controls and business validation.
Route names do not prove route independence	Check conduits, entry points, exchanges, providers and power.
Test all the objectives	Time, data currency, integrity, functionality and alternate capacity all matter.

Calculation check: Define uptime, downtime, failures and repair intervals before calculating availability. A scheduler's success, a backup job or RAID protection does not alone establish a complete business result.

Exam traps

Tempting wrong answer	Exam answer
Choosing audit logging as the BEST control for data integrity.	Selected audit logs support accountability and detection; database transaction logs also support commit, rollback and recovery. Assess the specific log and control purpose.
Starting continuity planning by selecting a recovery site or writing the plan.	Once sponsorship, policy and scope exist, use the BIA and owner input to establish criticality, dependencies and approved recovery requirements before selecting arrangements.
Meeting RTO proves recovery.	Recovery is effective when the RTO is met, data is within the RPO and required functions are validated.
Accepting an RTO longer than the maximum tolerable downtime.	Recovery targets must fit the approved interruption tolerance with time for necessary validation. Compare the actual scenario's exposures rather than applying a universal ranking.
A 15-minute backup schedule guarantees a 15-minute RPO.	RPO drives backup frequency, but only successful, recoverable copies meet it. Data currency at the recovery site follows the RPO.
Replication is a backup.	Replication and RAID are not backups; they copy deletion and corruption. Keep independent offsite backups.
Choosing a hot site for every system regardless of its RTO.	Match demonstrated end-to-end recovery capability, preparation, capacity, dependencies and total cost to the approved business need. Site labels do not establish fixed hours.
Two carriers eliminate telecommunications risk.	Verify actual alternate and physically diverse paths, including shared ducts, entrances, exchanges, providers and power. Carrier count and terminology alone do not establish independence.

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Tempting wrong answer	Exam answer
Treating an emergency change deployed before approval as a violation.	Use defined emergency authority and obtain authorization before implementation where feasible. Preserve actions and decisions and complete prompt independent review and documentation afterward.
A complete change register proves no unauthorized changes.	Start from actual production changes and trace them back to approved records.
Naming the recovery site as the most critical DR element.	Protected **recoverable data** is essential where restoration depends on it. Also assess activation authority, resources, dependencies and timing rather than naming one universal most-critical element.
Ranking a stale contact list above a structural plan failure.	Assess objective feasibility and activation authority, then compare consequences and likelihood using the **stated facts**. A stale contact can also be critical where no working alternative exists.
Setting a shared server's RTO at the average of its applications.	It must be ready early enough for the earliest dependent service, including subsequent validation, unless a tested alternative removes that dependency.
Restarting a failing job repeatedly to close incidents.	Incident management restores service; recurring failures need problem management to remove the root cause.
Relying on a document review to prove the DR plan works.	Execution tests demonstrate capability within their **scope**, conditions and time. Discussion and document reviews identify gaps but do not establish technical recovery; assess actual evidence and exercise risk.

Decision owners at a glance

Use the assigned authority and evidence to distinguish delivery responsibility from oversight and assurance.

Role	Responsibility	Boundary
Business process owner	Defines criticality, impacts and acceptable recovery capability.	Technical convenience should not set business tolerance.
IT/service owner	Delivers operation and recoverability against approved needs.	Restore commands alone do not prove a recovered service.
Crisis/invocation authority	Makes plan-activation and major response decisions.	Named deputies must also be able to act.
Recovery teams	Execute, validate, reconcile and hand back operation.	Failback needs the same control of data and authority.

Threats and control purpose

The examples show complementary control purposes. Their classification depends on how the control is configured and used.

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Risk	Preventive direction	Detective evidence	Corrective direction
Recurring service failure	Supported design, capacity and controlled changes.	Operational alerts and incident/problem trends.	Restore service and remove the investigated cause.
Loss of primary processing	Independent dependencies and suitable recovery strategy.	Realistic recovery exercises and backup monitoring.	Recover, reconcile and validate the required service.

4.18 Glossary

Look up terms alphabetically. Return to the lesson for the related example and audit decision.

Term	Definition
ACID	Atomicity, consistency, isolation and durability: transaction properties under the DBMS model.
Alternative routing	Another communications route or service; shared physical dependencies may remain.
API	Application programming interface defining program interactions.
Asset inventory	Record of assets, identifiers, ownership, lifecycle and relevant attributes.
Audit log	Record of selected events supporting accountability, detection and investigation.
Availability	Usability of a defined service when required.
Backup	Recoverable copy maintained under a defined protection and retention arrangement.
BCP	Business continuity plan supporting priority business activities through disruption.
BIA	Business impact analysis of disruption consequences over time and required recovery capabilities.
Capacity management	Planning and maintaining resources for present and expected demand.
Chain of custody	Documented identification, handling, transfer and storage of evidence.
Checkpoint	Recorded processing state supporting controlled restart.
Cloud backup	Backup service using cloud resources under defined protection and recovery responsibilities.
Cluster	Coordinated group of nodes providing a service; designs and failure behavior vary.
CMDB	Configuration management database of items and service relationships.
Cold site	Alternate facilities requiring substantial equipment, configuration and recovery work.
Commit	Finalization of a transaction’s changes under the database model.

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Term	Definition
Concurrency control	Protection against conflicting simultaneous access or updates.
Configuration baseline	Approved reference configuration under change control.
Corrective maintenance	Repair or restoration following a fault.
CSIRT	Computer security incident response team coordinating security response.
Database deadlock	A deadlock occurs when transactions wait on one another in a cycle and cannot progress without intervention such as aborting a transaction. See 4.11.
Database query spool space	Some database systems use spool space for temporary or intermediate query results. See 4.11.
Database schema	Defined organization and constraints of database structures.
Data dictionary	Catalog of data definitions, structures and related metadata.
Data governance	Ownership, standards and accountability for data management and quality.
DBA	Database administrator responsible for technical database administration.
DBMS	Database management system controlling data storage, access and processing.
Denormalization	Deliberate redundancy, often for performance, requiring consistency controls.
Differential backup	Conventional backup of changes since the last full backup.
Disk mirroring	Maintaining duplicate disk data, commonly through RAID 1. Supports availability for covered failures but may copy destructive logical changes; not an independent retained backup. See 4.14.
Diverse routing	Communications paths with verified physical separation.
DNS	Domain Name System translating names and other resource information.
DRP	Disaster recovery plan for restoring technology and data supporting business needs.
Electronic vaulting	Electronic transfer of backup information to a remote location.
Emergency change	Expedited change under defined authority, evidence and review arrangements.
Entity integrity	Unique, non-null primary key requirement in a relational design.
EUC	End-user computing, such as user-developed spreadsheets, scripts and applications.
Extranet	An extranet provides controlled access to selected organizational resources for external parties such as suppliers or customers. See 4.1.1.
Failback	Controlled return from an alternate arrangement to the intended normal environment.

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Term	Definition
Failover	Transfer of service to an alternate component or location.
Failure domain	Resources exposed to a shared failure cause, such as power, region, identity or administrative authority.
Fault tolerance	Continued required operation through specified failures within design limits.
Fibre Channel	Technology commonly used for storage networking.
File header record	A file header record contains control or identifying information such as file type, version, batch identifier or creation date. See 4.4.
Firewall	Control enforcing defined network traffic rules.
Foreign key	Field or fields used to reference a related key under the database design.
Forward proxy	Intermediary acting for clients when accessing other services.
Full backup	Copy of all selected data within the backup scope.
Full-interruption test	Controlled switch from production to recovery operation.
GFS	Grandfather-father-son backup generation/rotation scheme.
Global System for Mobile Communications (GSM)	GSM is a family of cellular telecommunications standards associated with second-generation mobile services. See 4.1.5.
Hardening	Reduction of unnecessary exposure through secure configuration.
High availability	Design and operating arrangements intended to minimize interruption.
Hot site	Equipped alternate capability requiring verified data, compatibility and activation readiness.
Idempotency	Safe repetition of a request without repeating the intended business effect.
IDS/IPS	Intrusion detection system reports suspicious activity; intrusion prevention system can also block it.
Immutable backup	Backup protected from alteration or deletion under defined retention and access mechanisms.
Incident management	Coordinated restoration of service and management of incident effects.
Incremental backup	Conventional backup of changes since the previous relevant backup in its chain.
Information processing facility (IPF)	Information processing facility (IPF) means the computer room and its supporting areas. See 4.1.
Interface	Connection through which systems or parties exchange information.
Internet service provider (ISP)	An ISP supplies internet connectivity or related network services. See 4.1.1.
Intranet	An intranet uses internet-style network or application technologies for an organization’s internal information and services. See 4.1.1.

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Term	Definition
IP	Internet Protocol providing logical addressing and packet routing.
iSCSI	SCSI storage commands transported over TCP/IP.
Java Database Connectivity (JDBC)	Java API for database access through suitable drivers; a specific example of an application programming interface. See 4.11.1.
Job scheduler	Tool controlling timing, dependencies and execution of workloads.
Known error	Analyzed problem with documented information supporting handling or resolution.
LAN	Local area network serving a limited location.
Last mile/local loop	Access connection between premises and the provider network.
Load balancing	Distribution of requests or workload among resources.
Log management	Collection, protection, analysis, alerting, retention and disposal of event records.
MAC address / media access control	A media access control address identifies a network interface at the relevant link layer. See 4.1.1.
Mainframe	A mainframe is an enterprise computing platform designed for substantial centralized workloads and operational reliability. See 4.1.2.
MAO	Maximum allowable or acceptable outage: interruption limit before business consequences become unacceptable. May align with MTD/MTO when definitions and boundaries match. See 4.12 and 4.13.
Message queue interface	A message queue holds messages for asynchronous processing between producers and consumers. See 4.4.
Metadata	Information describing data, structures or their management.
Mobile site / mobile recovery site	Transportable recovery capability requiring timely delivery, suitable location, connectivity, power, compatible resources and usable data. See 4.15 and 4.16.
MTBF	Mean time between failures under stated measurement assumptions.
MTD/MTO	Maximum tolerable downtime/outage before consequences become unacceptable.
MTTR	Mean time to repair; an average measure, not a guaranteed service recovery time.
NAS	Network-attached storage commonly offering file-level access.
NAT/PAT	Address translation / use of port mappings to multiplex translated connections.
NOC	Network operations center monitoring network and service operation.
Normalization	Organization of relational data to reduce redundancy and update anomalies.

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Term	Definition
NoSQL	Family of nontraditional database models with varying transaction and consistency guarantees.
OLA	Operational level agreement among internal support groups.
Operating system (OS)	An operating system (OS) manages computing resources and provides services to applications. See 4.6.2.
OSI	Seven-layer Open Systems Interconnection reference model.
Paper test	Document or checklist review of a recovery plan for completeness, contacts and obvious gaps. Discussion or review alone does not demonstrate technical execution. See 4.16.
Parallel recovery test	Recovery execution in an alternate environment while production continues.
Patch	Corrective software update, including an update addressing a security weakness. See 4.8.2.
Patch management	Risk-based assessment, testing, authorization, deployment and verification of patches.
Personal computer (PC)	A personal computer (PC) is a general-purpose computer intended principally for individual use. See 4.1.2.
Preparedness exercise	A practical exercise using actual resources for selected recovery capabilities; not an exact synonym for parallel testing.
Preparedness test	Exercise that demonstrates selected recovery activities or components. Scope and safeguards determine operational impact; it is not automatically nondisruptive or equivalent to a parallel test. See 4.16.
Preventive maintenance	Planned servicing intended to reduce avoidable failures.
Print spooling	Print spooling stores a print job temporarily in a queue so a printer can process it later. See 4.3.
Problem management	Investigation and management of underlying causes and recurrence.
QoS	Quality of service mechanisms prioritizing or managing traffic performance.
RAID	Redundant Array of Independent Disks; some levels provide redundancy, while RAID 0 does not.
RAM	Working memory that normally loses contents without power.
Reciprocal agreement	Mutual recovery assistance requiring validated capacity, compatibility and rights.
Referential integrity	Valid relationships between related records, commonly enforced with foreign keys.
Relational database management system (RDBMS)	An RDBMS is a database management system using the relational model, commonly organizing data in tables linked through keys and accessed using SQL. See 4.11.2.

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Term	Definition
Release	Identifiable package of authorized compatible changes.
Remote journaling	Transfer of transaction journals to a remote location for recovery.
Replication	Copying information to another system, synchronously or asynchronously.
Resilience	Ability to withstand, adapt to and recover from disruption.
Retention	Defined period and rules for keeping information or recovery copies.
Reverse proxy	Intermediary presenting services on behalf of back-end servers.
Rollback	Undoing uncommitted work, or restoring a prior configuration under a defined change plan.
RPO	Recovery point objective: target recoverable data point, often expressed as allowable loss in time.
RTO	Recovery time objective: target time to restore the required capability.
Runbook	Operational instructions for performing and verifying defined tasks.
SAN	Storage area network commonly providing block-level connectivity.
Security event / alert / incident	An event is observable activity; an alert requests attention; an incident meets adopted compromise or violation criteria. See 4.7.
Service bureau	Commercial provider of outsourced processing services. When used for recovery, compatible reserved capacity, priority, security and activation arrangements require verification. See 4.15 and 4.16.
Service delivery objective (SDO)	Required service capability during recovery, such as agreed transaction throughput; not an additional time interval. See 4.12 and 4.13.
Service desk	Central contact for service incidents and requests.
Shadow IT	Technology used outside expected organizational IT visibility or authorization.
SIEM	Security information and event management for centralized event analysis and monitoring.
SLA	Service level agreement defining provider/customer commitments and measures.
Snapshot	Point-in-time state; independence and recovery capability depend on implementation.
SOC	Security operations center coordinating security monitoring and response in this context.
Split brain	Conflicting active ownership among nodes after loss of coordination.
SPOF	Single point of failure whose failure alone can stop the required service.
Synchronous/asynchronous replication	Required remote acknowledgment before completion versus later remote propagation.

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Term	Definition
Tabletop / structured walkthrough	Scenario discussion in which participants work through roles, decisions and dependencies. Tests understanding and coordination, not actual restoration speed. See 4.16.
TCP/UDP	Reliable ordered stream transport versus datagram transport without equivalent delivery guarantees.
Telecommunication continuity methods	Alternative routing supplies a backup path; diverse routing separates relevant physical paths; long-haul diversity examines carrier/backbone paths; last-mile protection addresses the local access connection. See 4.13.2.
Transaction log	Database change record supporting transaction handling and recovery.
Underpinning contract	External supplier commitment supporting service delivery.
Universal Serial Bus (USB)	Universal Serial Bus (USB) is an interface used by peripherals, including removable storage. See 4.1.4.
UPS	Uninterruptible power supply providing limited power continuity under its design.
VLAN	Virtual LAN providing logical link-layer segmentation.
Voice over Internet Protocol (VoIP)	VoIP carries voice communications over IP data networks. See 4.1.1.
WAN	Wide area network connecting geographically dispersed locations.
Warm site	Partially prepared alternate capability requiring additional recovery work.
Wireless local area network (WLAN)	A wireless local area network (WLAN) connects devices within a limited area using radio communications. See 4.1.5.
Workaround	Temporary method reducing incident impact without necessarily eliminating the cause.
Write once, read many (WORM)	WORM storage permits writing under defined rules and subsequent reading while restricting alteration or deletion during the configured retention period. See 4.9.2.
WRT	Work recovery time for backlog reconciliation and business resumption where separately defined.

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4.19 Practice questions

These 26 original questions follow the CISA format: choose the one BEST answer from four options. Domain 4 carries 26 percent of the examination, so it has 26 of the book's 100 domain questions. Allow about 1.6 minutes per question, then check section 4.20. Before you look, name the **qualifier** and say why your **second choice** is weaker.

Question 4.01 An IS auditor wants to determine whether any program changes reached the production environment of a payroll system without approval. Which of the following audit procedures is **BEST**?

A. Trace a sample of production library changes back to approved requests
B. Interview developers about how code is promoted into production
C. Select a sample of approved change tickets and verify each was tested
D. Review change advisory board minutes for evidence of approval decisions

Question 4.02 The service desk has closed the same database connection failure eleven times in two months, each time by restarting the service. Which of the following actions would be **MOST** appropriate?

A. Add the restart procedure to the service desk knowledge base
B. Automate the restart so the service recovers without operators
C. Raise the incident priority so restarts are performed sooner
D. Log a problem record to investigate the underlying cause

Question 4.03 A retailer's order system sends a nightly file of shipped orders to the billing system. The interface relies on a transmission acknowledgment and has no other checks. Which of the following is the **GREATEST** risk?

A. Delivered records may fail to post, leaving orders unbilled
B. The billing system may be unavailable when the file arrives
C. The file may be intercepted in transit and read by outsiders
D. Transmission delays may push billing past the daily cut-off

Question 4.04 Sales invoices in a relational database are found to reference customer numbers that no longer exist in the customer table. Which of the following controls would **BEST** have prevented this?

A. Requiring a unique, non-null primary key in the invoice table
B. Logging deletions from the customer table for weekly review
C. Restricting the customer number field to a defined value range
D. Enforcing a foreign key between invoices and the customer table

Question 4.05 Finance calculates monthly loan-loss provisions in a large spreadsheet maintained by one analyst, and the results post directly to the general ledger. The **PRIMARY** risk an IS auditor should evaluate is that:

A. the file may be lost if the analyst's laptop fails
B. the analyst may share the file with unauthorized colleagues
C. the spreadsheet tool lacks vendor support and patching
D. formula changes are not independently tested or reviewed

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Question 4.06 During a review of a server hosting customer data, an IS auditor notes that system administrators can reconfigure and delete the local audit logs. Which of the following recommendations would **MOST** effectively address this?

A. Enable file-integrity alerts on the server's local log directory
B. Forward logs in real time to a store outside admin control
C. Have the administrators review the local audit logs each week
D. Extend local log retention to meet the regulatory requirement

Question 4.07 Users report that a payment application slows sharply at month-end, although monthly reports show average CPU utilization of 45 percent. What should be done **FIRST**?

A. Correlate peak response times with I/O and queue data
B. Move the month-end batch jobs to an overnight window
C. Lower the CPU alert threshold to flag the peaks sooner
D. Add processor capacity ahead of the next month-end peak

Question 4.08 An IS auditor compares the IT asset inventory with network discovery results and finds 40 active servers that are absent from the inventory. Which of the following should the auditor do **NEXT**?

A. Determine the owners and purpose of the unrecorded servers
B. Rerun discovery to confirm the scan covered each subnet
C. Recommend the inventory be updated with the discovered servers
D. Report the unrecorded servers as a finding to senior management

Question 4.09 Business units complain that IT is not meeting its agreed 99.5 percent availability commitment for the order system. Which of the following would **BEST** help an IS auditor evaluate the complaint?

A. The underpinning contract with the hosting supplier
B. Server utilization reports for the order system's hosts
C. Availability reports measured against the SLA definitions
D. Operational level agreements between support teams

Question 4.10 At a mid-sized insurer, a single database administrator holds full privileges on the claims database. Which of the following would be of **GREATEST** concern to an IS auditor?

A. DBA activity is logged, but the logs are reviewed by the DBA
B. Schema changes are documented by the DBA after implementation
C. Some tables were denormalized to speed up claims reporting
D. The DBA account password is changed once a quarter

Question 4.11 A critical database patch affects an online banking service. Applicability, business impact and urgency have been assessed; no active exploitation requires immediate deployment. Which action should be completed **FIRST** before authorizing production rollout?

A. Schedule an outage window and notify banking customers
B. Test the patch in an environment that mirrors production
C. Obtain the system owner's approval to deploy the patch
D. Deploy the patch to production with a rollback plan ready

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Question 4.12 An IS auditor is reviewing controls over the automated job scheduler for a bank's end-of-day processing. All of the following are appropriate controls **EXCEPT**:

A. making each job wait until its prerequisite input files arrive
B. reconciling job outputs and investigating failed or rerun jobs
C. relying on job completion codes as proof outputs are complete
D. restricting who can create, alter or rerun scheduled jobs

Question 4.13 An auditor reviews preventive maintenance using a stable equipment population, comparable workload and consistent incident reporting. Which provides the **MOST** persuasive outcome evidence when assessed alongside servicing records?

A. An inventory showing which devices remain under warranty
B. A maintenance contract specifying vendor response times
C. A declining trend in unplanned hardware failures and outages
D. Vendor service records confirming scheduled visits took place

Question 4.14 A manufacturer's board has approved a business continuity policy and named an executive sponsor. The IT manager now wants to sign a hot-site contract before the budget year closes. What should be done **FIRST**?

A. Assess threats and vulnerabilities facing the data center
B. Document recovery procedures for the core production systems
C. Compare hot-site and warm-site proposals on cost and capacity
D. Determine critical processes and their recovery objectives

Question 4.15 An IS auditor is evaluating how criticality rankings were assigned in a retailer's business impact analysis. Which source would provide the **MOST** reliable basis for ranking application criticality?

A. The enterprise risk register maintained by the risk function
B. IT operations staff who support and maintain the applications
C. Incident records showing historical outage frequency per system
D. Business process owners who depend on the applications

Question 4.16 While reviewing a bank's continuity documentation, an IS auditor notes several issues with the funds-transfer service. Which presents the **GREATEST** concern?

A. The SDO during recovery is set at half of normal volume
B. The approved RTO is eight hours, against an MTD of six hours
C. The BIA for the service was last refreshed fourteen months ago
D. Recovery of the service has not been exercised at the warm site

Question 4.17 Three applications with service RTOs of 2, 12 and 48 hours depend on one shared database. No alternative database is available, and the two-hour service requires 30 minutes of validation after the database is restored. Which target is **BEST** for database readiness?

A. Database readiness within the average of the three service RTOs
B. Database readiness within 90 minutes, leaving the required validation time
C. Database readiness within 48 hours to minimize recovery cost
D. Database readiness within 12 hours to satisfy most applications

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Question 4.18 A logistics firm states that its WAN is resilient because the data center is served by circuits from two different telecommunications carriers. What should the IS auditor do **FIRST**?

A. Test automatic failover from the primary to the secondary circuit
B. Recommend a cellular backup link for the critical branch sites
C. Obtain evidence of the physical cable paths and building entries
D. Review each carrier's SLA for availability and repair commitments

Question 4.19 To meet a near-zero RPO, an insurer synchronously replicates its claims database to a second data center and has discontinued periodic backups. Which is the **GREATEST** risk of this arrangement?

A. Failover to the second site may depend on manual DNS changes
B. Both data centers may sit within the same regional threat zone
C. Write latency on production transactions may increase with distance
D. Corrupted or maliciously encrypted data would reach the replica

Question 4.20 A BIA has set a one-hour RPO for an order-entry database that is currently backed up in full each night to tape stored offsite. Which change **BEST** addresses the gap?

A. Validated remote transaction journaling with a complete sequence and tested recovery points less than one hour old
B. Rebuild the database volumes on a dual-parity RAID 6 array
C. Contract a hot site so systems can be running within one hour
D. Switch nightly backups to disk to shorten restore duration

Question 4.21 A regional insurer has confirmed activation authority, compatible recovery equipment, staff availability and tested restore procedures. Its core systems require business data after total loss of the primary site. Which remaining prerequisite is **MOST** important to verify?

A. Restore procedures are documented for each critical application
B. Current backup copies of critical data are held offsite
C. The hot-site contract provides compatible hardware and capacity
D. Recovery team call trees contain current contact details

Question 4.22 The CIO of a hospital group led the drafting of a new business continuity plan with input from department heads. Which party should hold **PRIMARY** responsibility for approving the plan?

A. The continuity coordinator responsible for maintaining the plan
B. The CIO, who directed the plan's development and IT recovery
C. Senior management acting as sponsor of the continuity program
D. Internal audit, after independently reviewing plan completeness

Question 4.23 During a tabletop exercise of a retailer's business continuity plan, the IS auditor records the following observations. Which represents the **GREATEST** concern?

A. Authority to declare a disaster is not assigned to any role
B. Call-tree entries for several warehouse supervisors were outdated
C. Two recovery team members had not attended this year's training
D. Contact details for a secondary courier supplier were missing

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Question 4.24 In a parallel DR test observed by the IS auditor, systems were restored at the alternate site within the two-hour RTO, but the restored data was six hours old against a one-hour RPO. The auditor has confirmed these results against test logs. What should the auditor do **NEXT**?

A. Recommend moving recovery to a hot site with faster restoration
B. Report the RPO shortfall and its business impact to management
C. Redesign the backup schedule with IT to align it with the RPO
D. Request an immediate full interruption test to confirm results

Question 4.25 A distributor needs its core order service within six hours of primary-site loss. A mirrored site is unaffordable. Independent testing shows that a proposed hot site restores required data, interfaces and workload within four hours; the proposed warm and cold arrangements need more than a day. Which alternative **BEST** meets the need?

A. A warm site with network links and selected peripherals installed
B. A reciprocal agreement with a similar firm in another region
C. A hot site with compatible equipment configured and ready
D. A cold site with power and cooling, plus quick-ship hardware

Question 4.26 Which test would cause the **LEAST** disruption to live operations while still demonstrating that critical systems can run at the alternate site using current data?

A. Full interruption test with production switched to the alternate site
B. Parallel test processing recent transactions at the alternate site
C. Structured walkthrough of each recovery step with the team leaders
D. Simulation in which teams respond to a scripted disaster scenario

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4.20 Answers and explanations

Each explanation states the **deciding principle**, then shows why the key is best and why each other option, although often a sound practice, does not answer the question asked.

Question 4.01 Answer: A

Detecting unauthorized changes · Qualifier: BEST · Review section 4.8

Principle: To find unauthorized changes, **sample** from actual production changes and trace back to approvals, because an unrecorded change never appears in the change register.

A. Correct. Starting from evidence of what actually changed in production is the only direction that can surface changes with no **approval** record.

B. Useful for understanding the process; however, inquiry is weak **evidence** and does not test whether unauthorized changes occurred.

C. Valid for testing that approved changes were tested; however, sampling from tickets cannot reveal a change that **never had a ticket**.

D. Valid **evidence** that the CAB functions; however, minutes only cover changes that were submitted, not those that bypassed the process.

Question 4.02 Answer: D

Problem management · Qualifier: MOST · Review section 4.7

Principle: Recurring incidents are handed to problem management, which identifies and removes the **root cause** rather than repeatedly restoring service.

A. Documenting a workaround supports faster **restoration**; however, it institutionalises the symptom fix and leaves the cause in place.

B. Automation shortens outages; however, it hides the **recurrence** and still does not identify why the connection fails.

C. Faster **restoration** reduces impact per incident; however, it treats each recurrence as new rather than preventing the failure.

D. Correct. Repeated identical incidents signal an underlying problem; **problem management** investigates the cause and drives a permanent fix through change control.

Question 4.03 Answer: A

System interfaces · Qualifier: GREATEST · Review section 4.4

Principle: An acknowledgment proves **receipt**, not business processing, so the dominant interface risk here is incomplete posting, addressed by control totals and reconciliation.

A. Correct. Without control totals or **reconciliation**, rejected or dropped records go unnoticed, causing direct revenue loss; receipt is not completion.

B. **Availability** is a genuine interface risk; however, an unavailable recipient would fail the acknowledgment and be noticed, unlike silent posting gaps.

C. **Confidentiality** matters for interfaces; however, shipped-order data is moderately sensitive and the scenario's gap is completeness, not transport security.

D. Timeliness matters; however, a delay is visible and recoverable, whereas unposted records silently **cause** lost revenue.

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Question 4.04 Answer: D

Database integrity · Qualifier: BEST · Review section 4.11

Principle: Orphaned references are a **referential integrity** failure, prevented by foreign key constraints; logs detect after the fact and do not enforce integrity.

A. **Entity integrity** prevents duplicate or null invoice keys; however, it does not validate that a referenced customer exists.

B. **Logging** supports accountability and detection; however, it is detective and cannot stop a deletion that orphans invoices.

C. Domain **integrity** limits values to a valid format or range; however, a well-formed number can still point to a deleted customer.

D. Correct. A **foreign key** constraint enforces referential integrity, blocking deletion of customers with invoices or insertion of invalid references.

Question 4.05 Answer: D

End-user computing · Qualifier: PRIMARY · Review section 4.5

Principle: The main EUC **risk** is the absence of development, testing and change controls over logic that drives material financial figures.

A. **Availability** and backup are real EUC concerns; however, a lost file is noticed and rebuilt, while a wrong formula silently misstates the ledger.

B. **Confidentiality** matters for financial data; however, sharing affects disclosure, not the accuracy of figures posted to the ledger.

C. Supportability is a consideration; however, mainstream spreadsheet tools are patched, and the **risk** lies in the user-built logic.

D. Correct. Untested, unreviewed logic changes can misstate provisions without **detection**, which is the defining EUC risk for a critical calculation.

Question 4.06 Answer: B

Log protection · Qualifier: MOST · Review section 4.9

Principle: Administrators must not be able to alter logs of their own **activity**, so logs should be moved promptly to protected storage outside their control.

A. **Integrity** alerts add detection; however, administrators with local control can disable the alerting before altering logs.

B. Correct. Real-time forwarding to a store outside administrator control preserves evidence of their actions and enables **independent review**.

C. Regular review is useful; however, administrators reviewing logs they can delete is **self-review** and does not protect the record.

D. **Retention** addresses how long logs are kept; however, logs that admins can delete are not reliable for any period.

Question 4.07 Answer: A

Capacity management · Qualifier: FIRST · Review section 4.6

Principle: Averages hide short peaks and saturated **dependencies**, so the bottleneck must be identified from peak-period metrics before any remedy is chosen.

A. Correct. Correlating peak response with I/O and queue data validates the actual bottleneck, which is the prerequisite for choosing an effective remedy.

B. Rescheduling can relieve contention; however, it is a remedy chosen before the **cause** is known and may not address the bottleneck.

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C. Better alerting improves visibility; however, CPU may not be the **constraint**, and tuning thresholds does not diagnose the slowdown.

D. Adding **capacity** may help; however, the constraint may be storage or queues, so CPU spending could fail to improve response.

Question 4.08 Answer: A

IT asset management · Qualifier: NEXT · Review section 4.2

Principle: Before reporting or recommending, the auditor establishes the significance of the exception, which means **finding** who owns the servers and what they do.

A. Correct. Identifying owners and purpose shows whether these are **shadow IT**, unpatched or sensitive systems, which determines the finding's risk.

B. Validating **scope** is useful when discovery may miss assets; however, these servers were already found, so rescanning adds little.

C. Updating the inventory is a likely remediation; however, recording servers without knowing owners or purpose leaves the underlying **risk** unassessed.

D. Reporting will follow; however, the auditor needs to understand ownership and use first to rate the **finding**'s significance.

Question 4.09 Answer: C

Service level monitoring · Qualifier: BEST · Review section 4.10

Principle: The **SLA** is the provider-customer commitment, and availability reports measured on its agreed definitions show whether uptime was actually met.

A. The **supplier** contract shows external commitments supporting the service; however, it does not show what availability the business actually received.

B. Utilization can help explain **performance** causes; however, it answers a different question from whether agreed availability was delivered.

C. Correct. Availability measured on the **SLA**'s agreed service window and exclusions directly tests the commitment made to the business units.

D. OLAs define internal support commitments; however, they describe enablers of the service, not the **uptime experienced** by the customer.

Question 4.10 Answer: A

Database administration · Qualifier: GREATEST · Review section 4.11

Principle: Privileged DBA activity must be logged and independently reviewed; **self-review** leaves the most powerful user without effective oversight.

A. Correct. A privileged user reviewing their own activity defeats **segregation of duties**, so unauthorized data changes could go undetected.

B. Late **documentation** weakens change control; however, it concerns structure rather than unmonitored access to claims data.

C. Denormalization adds redundancy and **integrity** risk; however, it is a design trade-off that consistency controls can manage.

D. Password age matters; however, quarterly rotation is a minor weakness compared with unmonitored privileged **activity**.

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Question 4.11 Answer: B

Patch management · Qualifier: FIRST · Review section 4.8

Principle: After applicability and **impact assessment**, representative testing supports a controlled decision in the stated non-immediate emergency. Urgency can justify an expedited authorized process in other circumstances.

A. Scheduling and communication are needed; however, the window cannot be planned sensibly until testing shows effort and **impact**.

B. Correct. Testing in a production-like environment reveals **compatibility** and impact, giving the owner a sound basis to approve deployment.

C. Owner **approval** is required; however, it should be informed by test results, so it follows testing rather than preceding it.

D. Urgency matters for critical patches; however, deploying untested to a banking service risks an outage the **rollback** may not contain.

Question 4.12 Answer: C

Job scheduling · Qualifier: EXCEPT · Review section 4.3

Principle: A job that ends successfully may still produce incomplete or incorrect results, so completion codes need to be supported by output **reconciliation**.

A. Appropriate: **dependency checks** stop jobs from running on incomplete inputs, such as a bill run starting before all feeds arrive.

B. Appropriate: **reconciliation** and exception follow-up confirm processing was complete and that reruns did not omit or duplicate transactions.

C. Correct. Not appropriate: a completion code shows the job ended normally, not that every record was processed correctly, so it cannot replace **reconciliation**.

D. Appropriate: **restricting schedule changes** and reruns prevents unauthorized or erroneous processing in production.

Question 4.13 Answer: C

Hardware maintenance · Qualifier: MOST · Review section 4.1

Principle: Comparable failure and outage outcomes, assessed with servicing records and other causes, provide stronger result **evidence** than **activity** alone. A trend is not automatic proof of causation.

A. Warranty status supports lifecycle and cost decisions; however, it says nothing about whether **maintenance reduced failures**.

B. Contract terms show what the vendor committed to; however, they describe **obligations**, not whether failures were actually prevented.

C. Correct. Under the comparable conditions specified, fewer unplanned failures is useful **evidence** of outcomes. Interpret the trend alongside servicing, workload and other relevant changes.

D. Service records **evidence** that maintenance was performed; however, activity is not proof that it prevented failures.

Question 4.14 Answer: D

Business impact analysis · Qualifier: FIRST · Review section 4.12

Principle: With **policy** and sponsorship in place, the BIA comes next because criticality, RTO and RPO determine every later strategy and site decision.

A. Valid part of the **risk analysis**; however, threats are evaluated against processes already ranked as critical, so criticality must be established first.

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B. Valid plan development; however, procedures are written after the **BIA** and strategy decide what must be recovered, in what order and how fast.

C. Valid strategy work; however, comparing site options before RTOs are known risks buying the wrong **capability** at the wrong cost.

D. Correct. The BIA establishes which processes are critical and their **RTO** and RPO; without it, no site type or **procedure** can be shown to fit business needs.

Question 4.15 Answer: D

Criticality analysis · Qualifier: MOST · Review section 4.12

Principle: Criticality reflects **business consequences** of disruption over time, which business process owners are best placed to judge.

A. Valid enterprise-level **context**; however, the register is too aggregated to rank individual applications and is itself built from owner input.

B. Valid input on technical **dependencies** and recovery effort; however, IT staff cannot judge the financial and operational impact of losing a business process.

C. Valid evidence of likelihood; however, how often a system failed says nothing about the **business impact** when it does fail.

D. Correct. Process owners understand the financial, operational, legal and customer consequences of an outage over time, which is what criticality measures.

Question 4.16 Answer: B

Recovery objectives · Qualifier: GREATEST · Review section 4.12

Principle: An **RTO** longer than the maximum tolerable downtime means the plan fails by design even if recovery goes perfectly.

A. Valid design choice; however, a reduced service delivery objective is acceptable when the **business approves** it, and it is not a defect in itself.

B. Correct. The recovery target exceeds the stated maximum tolerable interruption, so the design fails the approved business tolerance. Testing and document updates alone do not correct that mismatch.

C. Valid maintenance gap; however, a slightly stale BIA may still be accurate, whereas the **RTO**-MTD mismatch is a certain failure.

D. Valid testing gap; however, an untested plan might still work, whereas an eight-hour target cannot fit within a **six-hour tolerance**.

Question 4.17 Answer: B

Recovery objectives · Qualifier: BEST · Review section 4.12

Principle: The shared dependency must be ready early enough for the tightest service target, including subsequent **validation**. Targets cannot simply be averaged.

A. An average would miss the earliest **required service** and has no business basis in this scenario.

B. Correct. Database readiness within 90 minutes leaves the specified 30 minutes for validation within the two-hour service target.

C. A long low-cost recovery window does not meet the **critical-service requirement**.

D. Twelve hours would meet a less urgent application's **limit** but fail the two-hour service.

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Question 4.18 Answer: C

Telecommunication resilience · Qualifier: FIRST · Review section 4.13

Principle: Assess actual physical and logical **dependencies** before relying on carrier diversity. Shared ducts, entry points, exchanges, providers or power can defeat apparent separation.

A. Valid operational test; however, **failover** can succeed while both circuits share a conduit, so it does not reveal a common physical failure point.

B. Valid supplementary measure; however, recommending a fix before confirming the design is weak puts the remedy ahead of the **evidence**.

C. Correct. Different carriers can share important failure domains. Establish actual path **independence** through suitable evidence and **corroboration** rather than inferring it from carrier count.

D. Valid contractual check; however, **SLA** penalties compensate after an outage and do not show whether one cable cut can sever both circuits.

Question 4.19 Answer: D

Replication versus backup · Qualifier: GREATEST · Review section 4.14

Principle: Replication can propagate unwanted changes. Assess protected independent recovery points and the actual **retention** and failure-domain design rather than relying on ordinary replication alone.

A. Valid gap that may lengthen **recovery time**; however, a slower failover still recovers data, whereas a corrupted replica cannot.

B. Geographic concentration also needs **assessment** against the disruption scenarios. In this case, relocating replicas alone would not prevent the same **logical corruption** from propagating to all copies.

C. Valid **performance** consideration of synchronous replication; however, latency degrades speed rather than destroying the ability to recover.

D. Correct. Without independent point-in-time backups, corruption or ransomware is replicated at once and no clean **recovery point** remains, which defeats recovery entirely.

Question 4.20 Answer: A

Backup frequency and RPO · Qualifier: BEST · Review section 4.14

Principle: A one-hour **RPO** requires demonstrated usable recovery points within that tolerance and a protection design surviving the relevant failure. A job schedule alone does not establish this.

A. Correct. The option includes remote capture, log continuity and tested usable recovery points within one hour. Scheduled transfer alone would not establish that **capability**.

B. Valid protection against disk failure; however, **RAID** is not a backup and does nothing to limit data loss in a site disaster.

C. Valid way to shorten **recovery time**; however, a hot site addresses **RTO** and still restores data that may be up to a day old.

D. Valid improvement to restore speed; however, faster restores address **RTO**, and data written since the last nightly backup is still lost.

Question 4.21 Answer: B

Offsite backup · Qualifier: MOST · Review section 4.14

Principle: The stem establishes other prerequisites and asks about data-dependent **restoration** after site loss. Protected **recoverable data** outside that failure domain is essential in this specific case.

A. Valid documentation check; however, procedures are worthless if there is no **current data** to restore, and skilled staff can adapt them.

B. Correct. Recoverable **current data** protected from primary-site loss supplies the stated missing prerequisite. This is not a universal ranking of all disaster-recovery elements.

C. Hardware and space are essential, but the scenario already establishes the recovery site's **readiness**. They do not replace a usable, sufficiently current copy of the required data.

D. Current contacts support activation and must be maintained. Checking them does not establish that the required **data can be recovered** at this otherwise prepared site.

Question 4.22 Answer: C

BCP ownership · Qualifier: PRIMARY · Review section 4.15

Principle: The **BCP** is a business-wide plan that senior management sponsors, approves and is accountable for; IT performs recovery but does not own the plan.

A. Valid custodian of plan content; however, a coordinator lacks the authority to **commit** resources and accept risk on behalf of the enterprise.

B. Valid contributor; however, IT performs technology recovery, and the **BCP** covers people, premises and processes beyond the CIO's authority.

C. Correct. Senior management is accountable for continuity across the business, funds the capability and must accept the **residual risk** the plan leaves.

D. Valid assurance role; however, approving the plan would impair **audit independence**, as the auditor evaluates rather than owns the plan.

Question 4.23 Answer: A

BCP incident management · Qualifier: GREATEST · Review section 4.15

Principle: Without designated authority to declare a disaster, the plan may never be invoked, so every other arrangement can sit unused.

A. Correct. If no one is empowered to declare a disaster, invocation is delayed or never happens, and recovery cannot start within its objectives.

B. Stale contacts require correction and can seriously delay recovery. The more fundamental gap in these stated facts is the absence of anyone with defined **authority to invoke** the plan.

C. Valid **readiness** gap; however, other trained members and documented steps can compensate for two people missing training.

D. Valid **supplier** dependency gap; however, a secondary courier is a fallback resource whose absence affects one activity, not plan activation.

Question 4.24 Answer: B

DR test results · Qualifier: NEXT · Review section 4.16

Principle: The auditor observes DR tests and reports established findings to management; designing fixes or directing tests belongs to management.

A. Valid if **recovery time** were the problem; however, the **RTO** was met, and a hot site does not make the restored data any more current.

B. Correct. The shortfall is already evidenced, so the auditor reports it and its **impact** to management, which owns the remediation and retest.

C. Valid remediation; however, designing the backup schedule is management's job, and doing it would impair the auditor's **independence**.

D. Valid way to gather more evidence; however, results are already confirmed, and a **full interruption** test adds disruption without new information.

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Question 4.25 Answer: C

Recovery alternatives · Qualifier: BEST · Review section 4.16

Principle: Choose the alternative whose demonstrated end-to-end **capability** meets the recovery window at acceptable cost, rather than infer fixed hours from a site label.

A. Valid lower-cost choice; however, a **warm site** lacks the main computer and needs equipment and configuration that typically take days.

B. Valid low-cost arrangement; however, spare **capacity**, compatibility and enforceability are uncertain, so a six-hour recovery cannot be relied on.

C. Correct. The stated test demonstrates required data, **interfaces** and workload within four hours, inside the six-hour target; the other available arrangements fail the constraints.

D. Valid cheapest standby; however, delivering, installing and configuring hardware at a **cold site** takes far longer than six hours.

Question 4.26 Answer: B

DR testing methods · Qualifier: LEAST · Review section 4.16

Principle: A **parallel test** can exercise alternate processing while production continues. Conclusions remain limited to tested data, workload, **dependencies** and conditions; live transition may remain untested.

A. Valid and the **most** rigorous test; however, deliberately interrupting production carries the greatest operational risk of all test types.

B. Correct. Systems are actually brought up and process real data at the alternate site while production keeps running, so recovery is demonstrated with minimal disruption.

C. Valid low-disruption exercise; however, discussing steps tests **understanding and coordination** but does not show that systems actually run.

D. Valid for practicing responses; however, a simulation exercises selected actions without necessarily recovering systems on **current data**.

4.21 Continue your preparation

Use your error log to identify the reason behind each mistake. Revisit the relevant explanation and **Last Mile** table, then test the concept with a fresh scenario. Register at certprepx.com for adaptive practice, timed full-length mocks and domain analytics. Available plans and features may change.

Scan to visit https://certprepx.com

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Domain 5: Protection of Information Assets

5.0 Domain overview

Protecting information assets means preserving the business's ability to use trustworthy information without unacceptable disclosure, alteration or interruption. Security combines **governance**, people, physical protection, technology and response. A strong product cannot compensate for unclear ownership, excessive access or an untested recovery process.

Domain 5 represents **26 percent** of the current CISA exam outline. Sections 5.1–5.9 address information asset security and control; Sections 5.10–5.15 address security event management. The main headings and their numbers follow the supplied reference structure. Supporting numbered subdivisions organize the explanations and examples.

For each scenario, identify the asset, the **threat**, the relevant security objective and the control's actual capability. Distinguish prevention from **detection**, confidentiality from integrity, and identification from proof of identity. Several controls may contribute; choose according to the facts rather than a memorized ranking.

Topics and reading route

Section	Subject
5.1	Information Asset Security Policies, Frameworks, Standards and Guidelines
5.2	Physical and Environmental Controls
5.3	Identity and Access Management
5.4	Network and Endpoint Security
5.5	Data Loss Prevention
5.6	Data Encryption
5.7	Public Key Infrastructure
5.8	Cloud and Virtualized Environments
5.9	Mobile, Wireless and Internet of Things Devices
5.10	Security Awareness Training and Programs
5.11	Information System Attack Methods and Techniques
5.12	Security Testing Tools and Techniques
5.13	Security Monitoring Logs, Tools and Techniques
5.14	Security Incident Response Management

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Section	Subject
5.15	Evidence Collection and Forensics

Follow one information flow

A customer submits an online payment. The business must authenticate the right user, authorize the transaction, protect transport and stored information, restrict administrative access, detect unusual **activity** and respond if something goes wrong. A valid certificate helps establish the service's identity; it does not prove that every payment is legitimate. **Encryption** protects data within its **scope**; it does not correct an unauthorized transaction.

5.1 Information Asset Security Policies, Frameworks, Standards and Guidelines

An **information asset** is valuable information identified for ownership, use and protection within the enterprise's classification scheme. Its value can depend on confidentiality, accuracy, availability and business use. Protecting an application and protecting the information it processes are connected responsibilities, with distinct owners and evidence.

Security **governance** sets direction, accountability and acceptable risk. Management translates that direction into a funded program and controls. Begin with **business objectives**, information needs and risk, then select protection proportionate to the consequences of compromise.

5.1.1 Information Asset Security Policies, Procedures and Guidelines

A **policy** states management intent and expectations. It needs appropriate approval, scope, ownership, communication, exception handling and review. **Approval authority** follows the organization's governance; not every detailed policy must be approved personally by the board. Policies should be enforceable and usable, not merely aspirational.

Document	Purpose	Example
Policy	High-level mandatory direction.	Sensitive information must be protected against unauthorized disclosure.
Standard	Specific mandatory rule implementing direction.	Approved encryption and authentication requirements for a defined use.
Baseline	Minimum approved configuration for a class of assets.	Required logging and hardening settings for production servers.
Procedure	Steps and responsibilities for performing work.	How to request, approve, provision and revoke access.
Guideline	Recommended practice allowing judgment.	Suggested approaches to organizing secure team collaboration.

A new protection **program** needs agreed ownership and rules before a tool can enforce them consistently. During an active incident, however, urgent protective action may precede a broad **policy** revision. "Policy first" is a planning principle, not a reason to leave immediate harm unaddressed.

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5.1.2 Security Objectives and Accountability

Confidentiality prevents unauthorized disclosure. Integrity protects against improper modification or destruction and supports trustworthy information. Availability ensures **authorized access** when needed. Authenticity concerns whether an identity or information is genuine. Non-repudiation provides evidence supporting attribution and resistance to denial; its strength depends on the technical, operational and legal context.

The board and senior management oversee risk and direction. A CISO or equivalent coordinates the security program. **Business owners** define information value and authorized use; custodians operate protection under assigned responsibilities. Users follow handling rules and report concerns. Outsourcing changes delivery responsibilities without eliminating the enterprise's own **accountability**.

Defense in depth places complementary safeguards across people, process and technology. Preventive controls restrict harmful activity; detective controls identify events or conditions; corrective controls restore or remediate; deterrent controls discourage. A control can serve more than one purpose, and a layer's effectiveness depends on its **configuration** and operation.

5.1.3 Information Security Frameworks and Standards

An information security management system (ISMS) organizes risk-based security management. **ISO/IEC 27001** specifies ISMS **requirements**; ISO/IEC 27002 provides control guidance. ISO/IEC 27701 addresses privacy information management systems. Use the relevant edition and scope rather than assume every reference describes the same version.

NIST Cybersecurity Framework 2.0 groups outcomes into Govern, Identify, Protect, Detect, Respond and Recover. **COBIT** addresses **governance** and management of enterprise information and technology. Frameworks help organize outcomes and assurance; adopting a name or obtaining certification does not prove that every asset and risk is covered.

5.1.4 Information Security Baselines and Access Standards

Establish minimum configurations and access expectations by asset class, **exposure** and **business impact**. Document exceptions with owners, compensating controls and review dates. Verify actual settings against the baseline. A spreadsheet showing the intended configuration is not evidence that production matches it.

5.1.5 Data Classification, Privacy and the Information Lifecycle

Classification assigns handling **requirements** according to the **consequences** of unauthorized disclosure, alteration or unavailability and relevant obligations. Define the scheme and criteria; inventory significant data; assign owners; classify and label where useful; then apply access, sharing, storage, retention and disposal rules. Aggregated information can be more sensitive than individual records.

Example: A customer list may become highly sensitive when combined with identity numbers, account balances and **authentication** details. The classification should reflect the combined **impact**, not the low sensitivity of an isolated customer name.

Privacy concerns **appropriate processing** of **personal information**, not only keeping it secret. Identify purposes, applicable permissions or other lawful grounds, minimization, accuracy, retention, transparency and relevant individual rights. Consent is not the only possible basis for processing. Controller and processor roles, duties and cross-border requirements depend on the applicable framework and actual arrangement.

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A **privacy** impact **assessment** evaluates how proposed processing can affect individuals and what safeguards are needed. Build privacy into design and defaults. Pseudonymization separates identifying information but can leave data personal and linkable; effective anonymization must address reasonably possible re-identification. Removing names alone does not establish anonymity.

Protect the lifecycle from creation and collection through use, sharing, archive and disposal. Select sanitization appropriate to the medium and sensitivity: degaussing applies to suitable magnetic media and does not sanitize solid-state storage. Verify the method and outcome, including copies and cloud arrangements, rather than relying on a disposal **policy** alone.

Applying security frameworks to an actual risk

An information security program connects **business objectives** and **obligations** to accountable protection, monitoring and improvement. Frameworks organize outcomes and responsibilities; standards can provide requirements; baselines specify configurations. A mapping between frameworks can support coverage but does not establish that the mapped controls are implemented or effective.

NIST CSF 2.0 includes Govern alongside Identify, Protect, Detect, Respond and Recover. These connected functions operate concurrently across the lifecycle rather than prescribing one fixed sequence. COBIT addresses **governance** and management more broadly. An information security management system establishes a managed approach within its defined **scope**; certification scope and exceptions still require examination. [R14]

Example: Management presents an approved security framework and a high self-**assessment** score. Trace a material risk such as privileged account compromise into responsibilities, implemented safeguards, alerts, response and recovery evidence. Test actual paths and unresolved **exceptions**. Security decisions should consider confidentiality, integrity and availability together; a restrictive safeguard that makes a critical service unusable can introduce another significant risk. Audit assesses the program and **control evidence** without taking over the security owner's decisions.

Relate a Security Requirement to Its Operating Scope

Example. An assurance certificate covers the central service, while a shared support team outside that scope has privileged access to it. Read the certificate and scope statement, identify the relevant obligations and trace the support access to the applicable controls. Inspect approval, actual use, monitoring and revocation. Consider whether the exclusion affects reliance on the service. A policy or assurance claim establishes a stated boundary. Evidence of relevant activity determines how far the audit can rely on it.

5.2 Physical and Environmental Controls

Physical controls restrict access to assets; environmental controls protect them against conditions such as heat, water, fire and power interruption. Evaluate both through the **business impact** and the actual facility design. Protect people and emergency egress as well as equipment.

TEMPEST concerns investigation and control of unintended emanations from information-processing equipment that could reveal information. Distinguish information leakage from ordinary interference or loss of power; use specialist requirements where the risk warrants them. [R46]

5.2.1 Environmental Exposures and Controls

Assess power quality and continuity, cooling, humidity, water ingress, fire, contamination, cabling and equipment placement. Use appropriate **detection**, alarms, maintenance and escalation. A UPS

provides power continuity within its **capacity** and duration; a generator supports longer outages only if startup, fuel, switching and maintenance arrangements work.

Fire **detection** and suppression require a professionally engineered system suited to occupancy, equipment and applicable safety **requirements**. Clean agents, inert gases and water-based systems have different constraints; no agent is universally harmless or suitable. Review testing, maintenance, alarms, release procedures and coordination with facility specialists.

Monitor temperature and humidity against appropriate equipment and facility **requirements**. Protect water-sensitive areas with **detection** and suitable routing. Review common dependencies: two power supplies connected to the same failed source do not provide independent supply. Environmental alarms need a timely response, not only a dashboard.

Emergency power off (EPO) is an emergency control for removing electrical power from specified equipment or areas. Evaluate its scope, accessibility, protection against accidental activation and approved life-safety design with facility specialists.

Term	Meaning and review point
CO2	Carbon dioxide, used as a gaseous suppression agent. Suitability and life-safety design require qualified assessment.
Halon	A family of fire-extinguishing agents found in legacy references or systems. Consider applicable environmental restrictions and approved maintenance/replacement arrangements.
Wet-pipe sprinkler	A sprinkler system whose piping normally contains water. Distinguish its operation from dry-pipe or preaction arrangements rather than declaring one system best for every facility.

5.2.2 Physical Access Exposures and Controls

Use layers such as perimeter controls, staffed reception, verified credentials, restricted areas, controlled doors, cabinets and **monitoring**. A mantrap or interlocked entrance can reduce tailgating when properly designed and operated; it is not infallible. Visitors need **approval**, identity checks, appropriate escort and access expiration.

Badges can be lost, shared or cloned. Biometrics can be spoofed or reject valid users. CCTV can deter and provide evidence but needs suitable placement, **retention**, review and privacy controls. The “strongest” control depends on the **threat**; visitor escort does not replace reliable control of unattended entry points.

Piggybacking and tailgating

Piggybacking (or **tailgating**) occurs when an unauthorized person follows an authorized person through a controlled door without presenting a credential. It defeats badge and biometric readers because the control authenticates only the first person. Courtesy, social pressure and propped doors make it common.

A **mantrap** (also called a deadman door or interlocking entrance) uses two doors so that only one person can pass after authenticating; the second door will not open until the first has closed. Anti-passback rules, turnstiles, guards at high-**risk** entries, CCTV review and staff awareness support it.

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Exam reasoning: An interlocking vestibule directly addresses following-through when occupancy sensing and operation enforce single-person passage. Examine bypasses, **exceptions** and life-safety behavior. Stronger credentials alone do not stop an unauthenticated follower.

5.2.3 Auditing Physical Access

Observe actual entry and exit practices, reconcile **access rights** to authorized personnel, inspect visitor handling and test removal of obsolete access. Review alarms, **exceptions**, forced-door events and restricted-area access. Include contractors, cleaners, delivery routes and emergency procedures. Compare records with actual conditions; a signed policy does not prove consistent enforcement.

5.2.4 Industrial Control Systems Security

Industrial control systems (**ICS**), including supervisory control and data acquisition (**SCADA**) and programmable logic controllers (PLCs), monitor or control physical processes. Safety, availability, timing and equipment constraints can dominate decisions. A control appropriate for an office laptop may disrupt a production process.

Inventory assets and **dependencies**, segment operational technology from other networks, restrict remote access, use controlled gateways or industrial DMZs, monitor appropriately and maintain recoverable configurations. Coordinate changes and patching with engineering, safety and operations. Use tested compensating controls when immediate patching is impractical. Active scanning or testing needs **authorization** and assessment of process impact.

Testing physical boundaries and environmental dependencies

Physical protection includes site and room boundaries, **identity verification**, visitors, escorts, restricted zones, **monitoring** and timely removal of access. Evaluate actual paths, including deliveries, maintenance, emergency exits and shared spaces. A badge policy cannot compensate for an uncontrolled door through which people routinely follow authorized staff.

An interlocking vestibule can reduce following-through when occupancy **detection** and operation enforce single-person passage. Two interlocked doors alone do not guarantee one person at a time. Examine **exceptions**, accessible entry, emergency behavior and applicable life-safety requirements. Test physical and logical access records together where both are relevant to sensitive administration.

Environmental protection includes suitable power, cooling, fire **detection** and suppression, water **exposure** and equipment conditions. Example: Redundant power feeds share the same distribution unit and maintenance **procedure**. Verify the actual common cause and tested transfer behavior rather than counting feeds. Review inspection, maintenance, alarms, response ownership and provider evidence. Select safeguards for the site’s risks and obligations; audit should not prescribe one suppression technology as universally correct for every facility.

Verify the Response at a Physical Boundary

Example. A monitored door opens during a maintenance visit. The system logs the event, but the monitoring team receives no notification because that alarm class was disabled for a prior job. Compare the access and alarm arrangements with the approved site requirements. Review authorizations, temporary overrides, restoration to normal settings and response records. Use authorized exercises that respect the site's safety procedures. An event in the log supports detection only when the appropriate people or system receive and act on it.

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5.3 Identity and Access Management

Identity and access management (**IAM**) establishes who or what receives access and maintains that access throughout its lifecycle. Include employees, contractors, customers, services, applications and devices. A correct login is only one part of **effective access** control.

5.3.1 IAM Lifecycle and Best Practices

Identity proofing establishes an identity during **enrollment** to the required assurance. Identification claims an identity; authentication verifies control of its authenticator; authorization determines permitted actions; accountability links actions to identifiable actors. Shared accounts weaken attribution unless strong individual identification and session evidence compensate.

Joiner, mover and leaver processes should connect authoritative personnel or business records to approved provisioning, timely changes and **revocation**. Remove unnecessary access after role changes and **termination**, including remote sessions, tokens, service credentials and supplier access. Periodic recertification detects accumulated or orphaned rights; it does not replace prompt event-driven removal.

Example: A finance employee transfers to procurement but retains both **supplier** creation and payment **approval**. The employee's account is active and authenticated correctly, yet the authorization and segregation controls are weak. Reconcile actual entitlements with the new role and incompatible duties.

5.3.2 Authentication, Authorization and Accountability

Concept	Meaning and audit question
Authentication	Does the claimant control the required authenticator? Assess enrollment, strength, recovery and bypass paths.
Authorization	Can the identity perform only permitted actions on permitted resources? Test server-side enforcement.
Least privilege	Only access needed for the assigned task, at the needed time and scope.
Need-to-know	Access to information is limited to justified business need.
Segregation of duties	Separate incompatible capabilities or provide suitable compensating review.
Accountability	Actions are attributable through unique identities, protected records and review.

Authentication Factors and Passwords

Factors commonly involve something known, possessed or inherent, such as a password, cryptographic device or biometric characteristic. Multi-factor **authentication** combines different factor types; two passwords remain one factor type. Strength also depends on **enrollment**, recovery, phishing resistance and session handling. MFA can be defeated by weak recovery or stolen sessions.

Prefer suitable phishing-resistant cryptographic **authentication** for higher-risk uses. Password controls should support sufficient length, checking against compromised choices, throttling and secure recovery. Store passwords with an appropriate salted password-**hashing** or derivation function and cost settings, not reversible encryption or a fast general-purpose hash alone. Do not expose passwords in logs.

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A **personal identification number (PIN)** is a memorized secret used in an authentication process. A PIN and another memorized secret do not automatically create two different factor categories. Evaluate how the PIN activates or supplements an authenticator.

In **challenge-response authentication**, a verifier issues a challenge and checks a response produced using the required secret or private-key capability. Freshness and protocol binding matter; replay resistance alone does not establish protection against phishing, interception or session hijacking.

5.3.3 Zero-Trust Architecture

Zero trust removes implicit trust based only on network location or ownership. Evaluate identity, **device posture**, resource sensitivity and context to grant limited access, with reassessment as appropriate. Policy decision and enforcement components need trustworthy signals. A VPN connection or an internal IP address should not automatically grant broad access.

5.3.4 Privileged Access Management

Privileged access management (PAM) controls powerful human and service access through appropriate **approval**, vaulting, time-limited elevation, credential rotation, session controls and independent monitoring. Break-glass accounts need protection, availability and prompt review of use. A vault does not solve unrestricted authorization or a service account with excessive rights.

5.3.5 Directory Services, IGA and IDaaS

Directory services maintain identities, groups and related information. Identity **governance** and administration (IGA) adds access requests, role governance, certifications and **policy** oversight. Identity as a service (IDaaS) provides identity capabilities through a supplier. Assess directory administration, synchronization, availability, customer configuration and recovery dependencies.

Protect groups and roles with accountable owners and controlled membership. Verify authoritative sources and synchronize **revocation** reliably. A disabled account in one directory can remain usable in a disconnected application if its provisioning or session controls do not reflect the change.

5.3.6 System Access Permission and Access Control Models

Model	How access is decided	Main consideration
DAC	Discretionary access control permits authorized owners to grant rights.	Flexible; prevent inappropriate onward sharing.
MAC	Mandatory access control applies centrally enforced labels and clearances.	Owners cannot simply override the mandatory policy.
RBAC	Role-based access control assigns permissions through roles.	Control role design, membership, inheritance and incompatible combinations.
ABAC	Attribute-based access control evaluates subject, resource, action and context attributes.	Protect attribute quality and test policy combinations.
Rule-based control	Rules permit or deny actions under specified conditions.	Resolve precedence, exceptions and conflicting rules.

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5.3.7 External Parties, Logical Access and Remote Access

Identify all access paths: user **interfaces**, APIs, management channels, databases, utilities, remote support and emergency routes. Enforce approval, **least privilege**, expiry, monitoring and appropriate authentication for third parties. Contracts should address security duties and incident cooperation, but actual access controls still need verification.

Digital rights management (**DRM**) can restrict copying, printing or use of protected content. It depends on identities, keys, compatible applications and enforcement. It cannot universally prevent photography or all onward disclosure. Assess usability, legitimate access and recovery.

Remote access needs protected transport, **device posture**, restricted **permissions**, secure credentials and session controls. Split tunneling changes traffic and monitoring paths; assess its actual risk rather than assuming every implementation has the same exposure. Unmanaged endpoints can undermine an otherwise strong encrypted channel.

5.3.8 Biometrics

Biometrics uses physical or behavioral characteristics, such as fingerprints, iris patterns, facial features or typing behavior. Protect templates and **enrollment**, assess spoofing and presentation-attack **detection**, and provide an appropriately controlled fallback. Biometric characteristics are difficult to replace after compromise.

Measure	Meaning	Trade-off
FAR - false acceptance rate	Unauthorized claimant is incorrectly accepted.	Lowering FAR can reduce unauthorized entry but may increase rejection of valid users.
FRR - false rejection rate	Legitimate claimant is incorrectly rejected.	Can disrupt operations and drive unsafe workarounds.
CER/EER	Threshold where false acceptance and false rejection rates are equal.	A comparison measure under specified test conditions; it does not describe all real-world risks.

In CISA terms, **false** rejection is a Type I error and false acceptance is a Type II error. **Accuracy** depends on threshold, population, sensors, environment and attack resistance. FAR is important for unauthorized acceptance; a claim about overall accuracy needs more than one rate.

Exam reasoning: False acceptance directly admits an impostor, while false rejection can affect availability and trigger risky **fallback**. Compare FAR, FRR and CER/EER under comparable conditions and at relevant operating thresholds. The preferred balance depends on the consequences and actual use.

Failure-to-enroll rate (FER or FTE) measures the proportion of eligible individuals who cannot complete the defined enrollment procedure. It concerns establishing a usable biometric reference, rather than rejecting an already enrolled genuine user during verification. State the population, permitted retries and final enrollment outcome before calculating it.

System-level rate examples: define the test population, attempt rules and threshold. Do not treat comparison-level false-match rates, system acceptance rates and enrollment failures as interchangeable; retries and decision processes can change their relationship. [R43]

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Rate	Calculation under the stated test	Example
FAR	False acceptances / impostor attempts × 100%.	2 / 10,000 × 100 = 0.02%.
FRR	False rejections / genuine attempts × 100%.	3 / 500 × 100 = 0.6%.
FER/FTE	Individuals unable to enroll / eligible individuals undergoing the defined enrollment procedure × 100%.	4 / 200 × 100 = 2%.
CER/EER	Operating point where the measured FAR and FRR curves cross under comparable testing.	Not the average of FAR and FRR at an arbitrary threshold.

5.3.9 Federated Identity Management and Single Sign-On

Single sign-on (SSO) lets an **authentication** serve multiple applications. Federation lets a relying service accept identity assertions from an identity provider across a trust arrangement. Security depends on the provider, assertion **validation**, trust configuration, sessions and recovery. Concentrated identity services need strong protection and resilience.

SAML commonly carries **authentication** and attribute assertions. OAuth 2.0 supports delegated **authorization**; it is not by itself an authentication protocol. OpenID Connect adds an identity layer. Validate issuer, audience, signature, lifetime and relevant request binding; protect tokens and redirect configurations.

Kerberos uses a key distribution center, ticket-granting tickets and service tickets, relying heavily on symmetric cryptography. Time synchronization, service keys and ticket protection are important. The **KDC** is a critical dependency, but replicated domain/KDC arrangements can provide redundancy; it is not inevitably a single physical point of failure.

5.3.10 Auditing Logical Access

Trace approved access to actual entitlements and actual entitlements back to **approval**. Include privileged, dormant, shared, service and external identities, groups and inherited rights. Test **revocation** and restricted functions, not only login success. Review logs and exceptions independently, and confirm that support or recovery paths do not bypass the intended assurance.

Auditor Perspective

Test the bypass as well as the login

Strong MFA can be undermined by weak **enrollment**, recovery, sessions or service-account paths. Examine how access is actually obtained and removed.

Authentication establishes confidence that a claimant controls the required authenticator. **Authorization** decides what that identity may do. Match the control to the question: successful login alone does not establish that **access rights** are appropriate.

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Assessing effective access and zero trust decisions

Access assurance follows identity establishment, **authentication**, **authorization**, activity and removal. Test joiners, movers, leavers, privileged users, emergency access and workload identities. Effective privilege can arise through group nesting, inherited cloud roles, service identities and the ability to change policies. A list of directly assigned roles may omit important access.

Practical extension: NIST **zero trust** architecture focuses on protecting resources and avoids granting implicit trust solely because of network location or ownership. The policy engine makes the access decision using policy and available signals; the policy administrator establishes or terminates the relevant communication path; the **policy enforcement point** enforces the decision. The engine and administrator form the **policy decision point**'s logical functions. These functions need not be separate physical appliances. [R9]

Evaluate trustworthy device and identity **context**, resource policy, session monitoring, re-evaluation, enforcement and revocation. Continuous evaluation does not mean asking the user to re-enter a password for every packet. Example: Device compliance becomes unacceptable after login, but a privileged session continues indefinitely. Test whether the signals reach the **decision process** and whether access is restricted as designed. Microsegmentation can limit lateral movement but is one architectural approach, not proof that every **zero trust** objective is met. Strong authentication, sound authorization and protected recovery must work together. [R15]

Logical function	Audit question	Useful evidence
Policy engine	Are decisions based on approved policy and reliable context?	Policy, signals and decision records
Policy administrator	Are permitted paths and credentials established and revoked correctly?	Session and credential lifecycle evidence
Enforcement point	Can alternate paths bypass the decision?	Positive, negative and revocation tests

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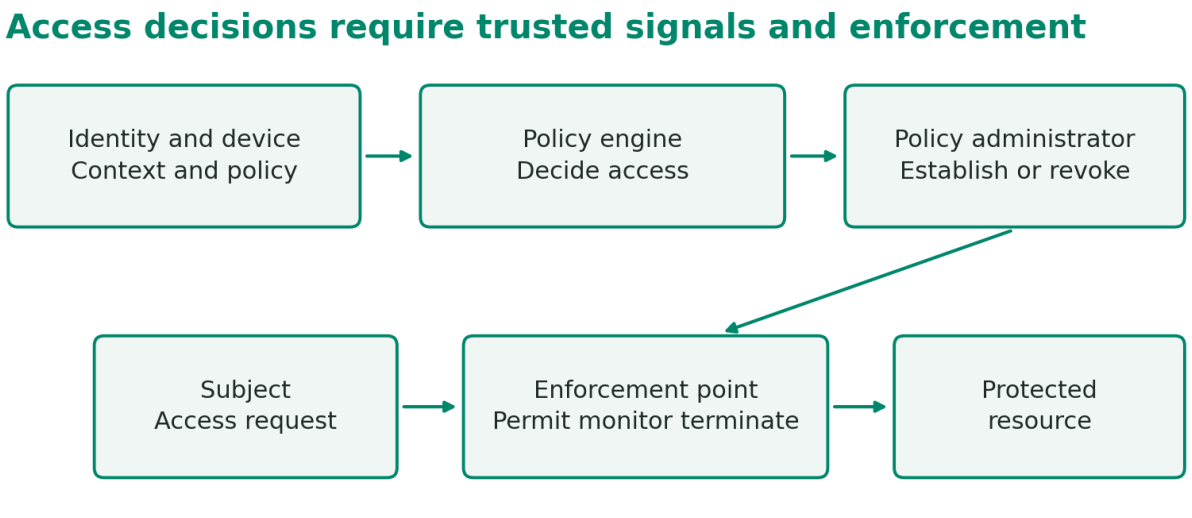


Figure 5.1 Logical access decisions require trustworthy signals and effective enforcement.

Need to know and least privilege

The **need-to-know principle** limits access to information required for an authorized business duty. **Least privilege** limits the **permissions** and capabilities needed to perform that duty. They reinforce each other, but a high clearance, senior role or successful login does not by itself authorize access to all information at that level. Verify both the legitimate information need and the allowed actions.

Example: A payroll analyst may need assigned employee records without needing other departments' files or **permission** to approve their own pay changes. Review role membership, record-level restrictions, exceptional grants and periodic recertification against actual duties. A broad role name is insufficient **evidence** of justified access.

Test the Decision and Its Enforcement in Zero Trust

Example. A device is approved to use a sensitive service. Its compliance status later changes, but one direct administrative route keeps accepting an existing session. Trace the **policy decision point**, including the engine and administrator where applicable, to the **policy enforcement point** on each relevant path. Verify the approved re-evaluation and revocation rules with an authorized test. Include workload identities, exceptions and service-to-service paths. A successful decision at the gateway supplies evidence for the path it controls. Test alternate routes and the effect of a change in device or identity context. [R9]

5.4 Network and Endpoint Security

Network and **endpoint** controls protect communication paths and the devices using them. Map data flows, trust boundaries and management paths before evaluating individual products. A permitted encrypted connection can still carry an unauthorized action from a compromised endpoint.

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5.4.1 IS Network Infrastructure

Assess local and wide-area networks, internet, partner connections, remote access and cloud connectivity. Maintain diagrams, ownership, secure **configuration** and monitoring. Apply suitable segmentation and restricted management access. Network access control (NAC) can evaluate identity and **device posture**, but its decisions depend on reliable attributes and enforcement.

Network address translation is not a **firewall policy**. A VLAN is a logical boundary, not proof that traffic is filtered appropriately. Check inter-segment routes, access lists and bypass paths. Management, identity and backup networks can be as sensitive as business traffic.

5.4.2 Virtual Private Networks

A VPN protects traffic through a tunnel under its protocol and **configuration**. IPsec commonly protects IP traffic; TLS-based arrangements can provide remote access through different architectures. Review cryptography, peer identity, key/certificate lifecycle, **endpoint** controls, routing, access **scope** and logging. A VPN does not make the remote device trustworthy or encrypt information after it leaves the tunnel.

5.4.3 Network Services, NAS, CDNs and NTP

Secure DNS and directory services, file sharing, network-attached storage (NAS) and administrative protocols. Disable unnecessary services, restrict management **exposure** and protect credentials. A file share requires appropriate **permissions** and recovery even when the storage device itself is hardened.

A content delivery network (CDN) distributes or caches content near users. Assess **origin** protection, cache rules, sensitive content, third-party access, TLS **termination** and location requirements. Avoid caching private responses under shared keys. Network Time Protocol (NTP) supports consistent time for logs and time-sensitive authentication; restrict and monitor time sources and their configuration.

Protocol	Basic purpose	Protection boundary
HTTP / HTTPS	Hypertext Transfer Protocol exchanges web requests and responses; HTTPS uses HTTP through a TLS-protected connection.	Connection protection does not prove truthful content or correct application authorization.
FTP / FTPS / SFTP	File Transfer Protocol transfers files; FTPS adds TLS to FTP, while SFTP is a separate SSH-based file-transfer protocol.	Basic FTP does not itself encrypt credentials/content. Validate the actual protocol and configuration.
Telnet	Supports remote terminal sessions.	Traditional Telnet does not itself encrypt the session; changing its port does not supply protection.

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5.4.4 Applications in a Networked Environment

Client-server systems divide processing between clients and services; middleware connects components and may hold powerful credentials. Treat every interface as an **authorization** and **validation** boundary. On-demand services and APIs need secure identity, tenant isolation, capacity limits and change control. A trusted front end should not allow a user to bypass server-side checks.

5.4.5 Firewalls

A firewall enforces traffic rules between defined boundaries. Review **business justification**, least-permitted connectivity, rule ordering, **logging**, management access and periodic recertification. Broad exceptions, stale rules and unrestricted outbound paths can defeat a well-chosen architecture.

Type	Capability	Limit
Packet filtering	Checks fields such as addresses, protocols and ports.	Limited application context; stateless rules assess packets without connection state.
Stateful inspection	Tracks connection state when assessing traffic.	A valid connection can still contain malicious permitted content.
Circuit-level gateway	Mediates sessions or connections.	Usually provides less application-content inspection than an application proxy.
Application proxy/gateway	Terminates and mediates selected application protocols.	Needs suitable protocol support, configuration and capacity.
NGFW	Combines capabilities such as application awareness and intrusion prevention.	Features are not assurance; inspect enabled policies, updates and encrypted-traffic visibility.
WAF	Web application firewall applies HTTP/application-focused rules.	Does not replace secure code or correctly enforced authorization.

Firewall Architectures and the DMZ

A screening router filters at a boundary. A dual-homed host has **interfaces** in separate networks; it must not inadvertently bridge the boundary. A screened host combines filtering with a hardened host. A screened subnet or **DMZ** isolates public-facing services from internal services using controlled paths. One or multiple devices can implement zones; two physical firewalls are not a universal requirement.

Example: An internet mail gateway in a DMZ scans and relays to an internal service through restricted rules. Review direct bypass paths, permitted relay behavior, administrative access and **monitoring**. The drawing is useful only if actual routes and rules enforce it. No architecture is automatically strongest against every **threat**.

5.4.6 Unified Threat Management and Network Segmentation

Unified **threat** management (UTM) combines multiple security functions. It can simplify operations while concentrating failure, performance and administration dependencies. Provide suitable resilience and management separation. Segment by business function, sensitivity and exposure; use **internal controls** and, where appropriate, microsegmentation to restrict lateral movement.

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5.4.7 Endpoint Security

An **endpoint** is a device or computing system that connects to and uses networked services, such as a workstation, laptop, server or mobile device. Endpoint security protects that system and its activity. A secure network path does not compensate for a compromised endpoint using valid credentials.

Use supported configurations, patching, **least privilege**, **application control**, encryption, host firewalls and malware protection according to risk. Endpoint protection platforms can combine several preventive and detective methods; modern antimalware is not limited to static signatures.

Endpoint detection and response (EDR) adds endpoint telemetry, investigation and response. Extended detection and response (XDR) correlates across sources such as endpoints, identity, network and cloud. Capabilities overlap across products; verify coverage, agents, telemetry, response authority and actual operation. A product label alone does not establish protection.

Examining network paths and endpoint enforcement

Network security depends on defined trust boundaries, permitted traffic, managed devices, protected administration and **monitoring**. A public-service zone or DMZ limits **exposure** through controlled paths; one or several devices can enforce it. Merely naming a subnet DMZ does not demonstrate isolation.

Inspect actual firewall rules, security groups, routing, remote access, east-west paths and management **interfaces**. IDS detects specified activity; IPS can block within its configured path. Evaluate coverage, encrypted traffic visibility, tuning, **exceptions** and bypasses. A security appliance outside the attack path provides little direct protection against that path.

Example: Web-to-**database** traffic is restricted, but an unrestricted administration network permits the same server to reach all databases. Trace effective connectivity and **privileges** rather than reviewing only the intended diagram. **Endpoint** controls should address secure configuration, patching, execution, detection, response and sensitive local data. Remote access requires appropriate identity, device and session controls. Assess both preventive and detective layers and verify the response to loss of telemetry or a disabled agent. Technical products do not remove the need for accountable configuration and tested operation.

Network protocols and boundary controls

A **proxy server** acts as an intermediary. An application proxy or **application gateway** mediates a supported application protocol; a **forward proxy** acts for clients, while a reverse proxy fronts services. Protocol termination, payload inspection, TLS visibility and address exposure depend on the architecture and configuration. The word proxy alone does not guarantee complete Layer 7 inspection or hide every internal address.

A **demilitarized zone** (DMZ) is a network zone that separates exposed services from more trusted internal services through controlled paths. A **bastion host** is a deliberately hardened host exposed at a security boundary, such as a controlled gateway or administrative jump host. Limit its functions, **privileges** and management access, maintain patches and inspect its logs. Neither concept requires exactly two firewalls or one universal interface design.

IP source routing allows a sender to specify route information in an IP packet. Assess whether such traffic is needed and how it is filtered. Disabling source routing does not prevent every form of IP spoofing. [R45]

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IPsec protocols and operating modes

Authentication Header (AH) and **Encapsulating Security Payload** (ESP) are IPsec protocols, not the names of its transport and tunnel modes. The **security association** specifies the negotiated protection, algorithms, peers and keys. Transport mode protects the upper-layer payload; tunnel mode encapsulates an inner IP packet within an outer packet. Inspect the actual negotiated policy rather than assuming the VPN label proves confidentiality. [R20, R21, R22]

Protocol	Protection and audit distinction
AH	Supports integrity and **data-origin authentication** for covered packet fields, with anti-replay protection when enabled and checked. It provides no confidentiality. Mutable IP-header fields are not all authenticated.
ESP	Can provide confidentiality and integrity/authentication according to the selected **security association**. Verify the required combination, algorithms and replay checks; the protocol name alone does not prove all services are enabled.

Infrastructure services and local network risks

Term	Purpose and risk	What to verify
DHCP	**Dynamic Host Configuration Protocol** leases IP addresses and supplies options such as subnet mask, gateway and DNS servers. Rogue servers can redirect clients; resource exhaustion can deny leases.	Authorized servers, trusted relay paths, appropriate DHCP snooping, capacity and useful **lease history**. DHCP supplies host configuration; it does not simply allocate entire subnets. [R23]
SNMP	Simple Network Management Protocol monitors and manages devices. SNMPv1/v2c community strings lack cryptographic confidentiality. SNMPv3 levels are noAuthNoPriv, authNoPriv and authPriv.	Verify **authentication** and privacy separately: authNoPriv authenticates without encrypting, while authPriv provides both. Use approved algorithms, restrict management sources and read/write permissions, and remove default or shared secrets. [R24]
RPC	Remote procedure call lets a program invoke a procedure on another system. Implementations can support authentication and protected communication; RPC is not inherently unauthenticated.	Identify the implementation, exposed services and required ports; restrict reachability, authenticate and authorize calls, patch and log. Do not infer an exploitable service solely from a port number. [R25]
ARP poisoning	**False** Address Resolution Protocol mappings associate an IPv4 address with an attacker's MAC address on a local network. This can enable interception or denial of service.	Validate trusted bindings, configure dynamic ARP inspection where supported, protect its trust **assumptions** and segment access. DHCP snooping can supply bindings; static devices need appropriate handling. IPv6 uses neighbor discovery rather than ARP. [R26]

Example: A rogue **DHCP** server supplies an attacker-controlled gateway, while an exposed management service accepts weak credentials. Review the complete path from infrastructure

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configuration to administrative access. A firewall purchase does not resolve every local-network trust issue. For discovery and testing techniques, see sections 5.11 and 5.12.

Verify Effective Separation Between Network Paths

Example. An application segment has approved firewall rules. A shared management network still lets an administrator reach the same service outside the expected application route. Map effective paths, including management connections and traffic between workloads. Compare allowed and denied flows with the approved business need and verify them using authorized tests. Review the rights to alter routing, endpoint rules and exceptions. Network segmentation is effective for a stated purpose when the relevant paths enforce the required boundary.

5.5 Data Loss Prevention

Data loss prevention (**DLP**) identifies sensitive information and monitors or restricts prohibited use, transfer or disclosure. It does not prevent every form of loss, including all corruption or unavailability. Define information ownership, sensitivity, approved use and response before relying on tool rules.

5.5.1 Types of DLP and Data States

State/control point	Example	Limitation
At rest	Discover sensitive records in file stores or cloud repositories.	Coverage and permissions can leave stores unexamined.
In motion	Inspect email, web uploads or other transfers.	Encryption, unsupervised channels and unsupported formats can limit inspection.
In use/endpoint	Control copying, printing or removable-media transfer.	Requires suitable endpoint coverage and cannot eliminate every physical disclosure.

5.5.2 Content Analysis Methods and Deployment

Methods include patterns, keywords, lexicons, fingerprints, exact-data matching, labels and contextual analysis. A lexicon uses a curated set of terms, potentially with proximity or weighting, to improve **context**. Pattern matches can resemble sensitive information without being sensitive, while unsupported formats can hide genuine content.

Inventory data and channels, agree policies, **pilot** in **monitoring** mode where appropriate, tune rules and assign exception owners before expanding enforcement. Accurate classification helps, but useful discovery can also identify previously unclassified information. DLP does not require every file to have a manual label before it can provide value.

Example: Blocking every number resembling an account identifier disrupts ordinary documents. Combine appropriate **validation**, document context, destination and approved use; test known positive and negative cases. Protect DLP findings because captured evidence may itself contain **sensitive data**.

5.5.3 DLP Risk and Audit Considerations

Assess blind spots, authorized overrides, **performance**, **false** positives, missed events, privacy and insider misuse of the tool. Verify that blocked transfers cannot simply use an unmonitored path. Measure investigations and resolved exposure, not only alert volume. DLP complements identity, encryption, acceptable-use rules and incident handling.

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Testing DLP coverage through real disclosure paths

DLP seeks to identify and control specified sensitive-data handling across relevant paths. Establish owners, protection objectives and permitted use, then map email, web, cloud sharing, endpoints, removable media, printing and other relevant channels. Network inspection alone cannot see every local device action or encrypted application exchange.

Classification helps decide handling rules, while discovery can identify sensitive content that has not been labeled. Do not delay all discovery until every file has a manual label. Assess content and **context** rules, precision, **false** positives, encryption, exceptions, user override and the possibility that an alert contains sensitive information itself.

Example: A user copies a customer extract to USB without sending it through the network DLP gateway. **Endpoint** or device controls are needed for that path, with appropriate **monitoring** and investigation. Test representative permitted and prohibited cases, including cloud links, compressed files and alternate formats where relevant. A blocking demonstration establishes only the tested route and rule. Verify response ownership, authorized **exceptions** and whether discovered gaps are corrected. DLP supports a wider protection program and does not by itself establish a lawful processing purpose or prevent all disclosure.

Test a Disclosure Route Outside the Main Gateway

Example. A data loss prevention policy blocks unapproved email attachments, while a managed browser can upload the same report to a personal sharing service. Identify the information classification, intended destination and routes the rule covers. Use synthetic data in authorized tests of email, browser and endpoint paths, including approved exceptions. Inspect alerts and whether the response meets policy. A blocked attachment supports one route. Coverage depends on how the rule applies across the channels that handle the information.

5.6 Data Encryption

Encryption transforms plaintext into ciphertext using an algorithm and key so authorized decryption can recover it. **Hashing** creates a digest for comparison. Digital signatures support integrity and attribution. Choose the service needed and verify the trust and key arrangements that make it meaningful.

5.6.1 Elements of Encryption Systems

Technique	Key arrangement	Typical purpose
Symmetric encryption	A shared secret key for the relevant encryption/decryption operations.	Efficient bulk confidentiality; authenticated modes can also protect integrity.
Asymmetric cryptography	Related public/private key pair.	Depending on the algorithm: signatures, key establishment or public-key encryption.
Cryptographic hash	No secret key in an ordinary hash operation.	Digest comparison against a trusted reference; not confidentiality or origin authentication by itself.
MAC/HMAC	Shared secret authentication key.	Integrity and source authentication among key holders; not public-verifiable attribution.

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Technique	Key arrangement	Typical purpose
Digital signature	Private signing key and public verification key.	Integrity, signatory authentication and evidence supporting non-repudiation.

AES is a symmetric algorithm used for suitable bulk **encryption**. RSA supports specified public-key operations; elliptic-curve techniques support uses including signatures and key agreement. Signing and verification follow the selected algorithm. Describe a signature as created with the sender's **private signing key** and verified with its trusted corresponding **public key**, rather than as generic encryption of a hash.

5.6.2 Link Encryption and End-to-End Encryption

Link **encryption** protects traffic on individual links, potentially exposing plaintext at intermediate processing points. End-to-end encryption protects content between designated endpoints. Define those endpoints precisely: TLS terminating at a **reverse proxy** protects the client-to-proxy leg, while the onward leg and proxy processing need their own protection.

Encryption at rest protects selected stored representations, but authorized applications may access **plaintext**. Full-disk encryption helps protect a powered-off lost device when keys are protected; it does not prevent an authenticated compromised application from reading accessible data.

5.6.3 Key Establishment, Digital Envelopes and Signatures

For public-key encryption, use the intended recipient's **public key** so the corresponding **private key** can decrypt under the scheme. In a digital envelope, a random symmetric content key encrypts the message and is protected for the recipient using a suitable public-key mechanism. This combines efficient bulk encryption with manageable key distribution.

A sender signs using the sender's **private signing key**; the recipient verifies using the associated public verification key. Verification depends on trusting that key and checking the relevant certificate or other binding. A signature does not hide the content. Signature evidence can be weakened by compromised keys, poor **identity proofing** or unreliable signing processes.

Which key does what. Match the security objective to the cryptographic operation, key role and trust assumptions.

Objective	Key used	Applied to
Confidentiality	Recipient's public key encrypts; recipient's private key decrypts.	The message, or the symmetric session key in a digital envelope.
Authentication and non-repudiation	The sender's private signing key creates a signature; its trusted corresponding public key verifies it.	The message under the selected signature algorithm, including required hashing or encoding.
Integrity	The sender's private signing key creates a signature; its trusted corresponding public key verifies it.	The message under the selected signature algorithm, including required hashing or encoding.
Confidentiality plus signature	An appropriate signed-and-encrypted scheme using sender signing and recipient decryption keys.	Protected message and relevant content-key material under the scheme.

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Objective	Key used	Applied to
Digital envelope	A symmetric key encrypts the data; the recipient's public key protects that key.	Bulk data and the symmetric key.

Worked example: A **supplier** distributes a software update with a signature. Verify it against a trusted supplier key and the exact downloaded bytes. A matching digest posted on an attacker-controlled page does not establish authenticity. A valid supplier signature also does not prove that the software contains no defects.

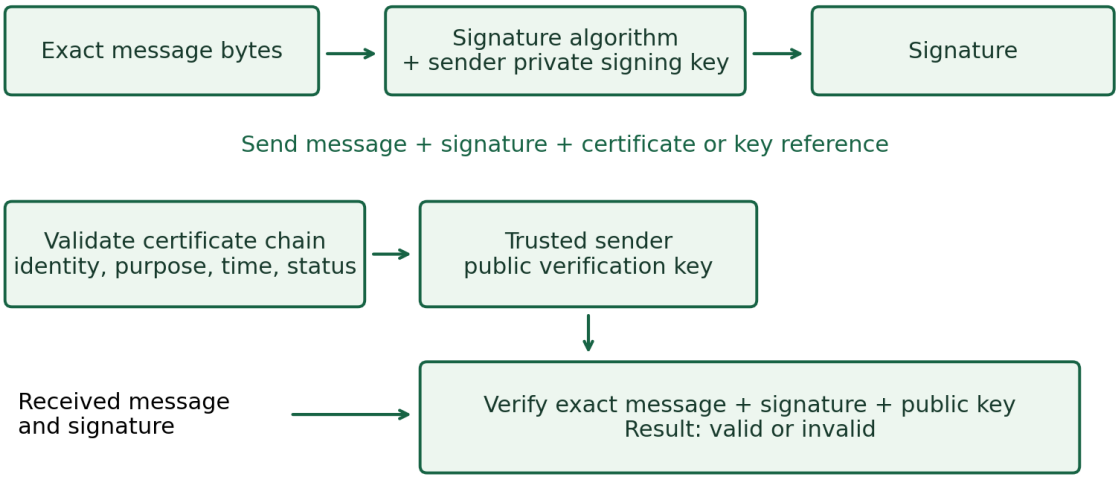


Figure 5.2 Signature creation and verification. Trust in the public key and protection of the private key remain essential.

A signature scheme processes the message using its specified algorithm, including **hashing** or encoding as required. RSA-based signatures, ECDSA and EdDSA do not share a generic private-key **encryption** operation. A valid result supports integrity and **signatory authentication** under key and identity assumptions; it does not establish confidentiality, truth or malware-free content. See NIST FIPS 186-5 for digital-signature concepts.

5.6.4 Hashes, Password Protection and Cryptographic Limits

A **message digest** is the output of a hash function applied to a message. A cryptographic hash can reveal alteration when compared with a trusted expected value. If an attacker can replace both file and digest, an unkeyed comparison does not authenticate the file. Checksums, CRCs and parity are primarily error-detection mechanisms and are not substitutes for adversary-resistant authentication.

A **salt** makes identical passwords produce different stored results and defeats reuse of precomputed tables across accounts. It is normally stored with the derived value and need not be secret. Password protection also requires an appropriate costly password-**hashing** function, access restrictions and online guessing controls. Salt alone does not prevent guessing a weak password.

Use approved algorithms, modes, key sizes, random generation and libraries for the purpose. Protect nonces and initialization values according to the scheme; some modes fail badly when values are

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reused. **Authenticated encryption** protects **confidentiality** and integrity within its scope. Key length alone cannot fix a broken implementation.

5.6.5 Elliptic Curve, Quantum and Homomorphic Cryptography

Elliptic-curve techniques can provide comparable classical security with smaller keys than some older public-key systems, but key lengths across **algorithm** families are not directly comparable. They commonly support signatures and key agreement rather than bulk data **encryption**.

Sufficiently capable quantum computing threatens widely used RSA and elliptic-curve public-key systems. NIST has finalized initial post-quantum standards, including key-encapsulation and signature schemes. Plan inventory, **cryptographic agility** and **migration** according to **exposure** and information lifetime. Quantum key distribution is a different technology with specialized channel and operational requirements; it does not solve all endpoint security.

Homomorphic encryption supports specified computations on ciphertext so results can later be decrypted. Partially, somewhat and fully homomorphic approaches support different operation sets or depth. Assess **performance**, supported operations, keys and leakage outside the cryptographic computation; it is not a universal replacement for ordinary processing.

5.6.6 Applications of Cryptographic Systems

TLS establishes protected communication using authenticated key establishment or applicable pre-shared arrangements and symmetric record protection. Modern certificate-based TLS commonly uses ephemeral key agreement and signatures; it does not universally encrypt a session key with the server's **public key**. **Forward secrecy** concerns protection of past sessions against later long-term key compromise under the chosen mode.

IPsec protects IP traffic through security associations defining keys, algorithms and related parameters. SSH protects remote sessions and associated transfers; verify host keys, user keys, **authorization** and rotation. Kerberos supplies tickets for authenticated service access and needs protected keys and reliable time.

DNSSEC authenticates relevant DNS data through signatures and a **trust chain**; it does not encrypt DNS queries or make a website trustworthy. S/MIME can sign and encrypt email under certificate and key arrangements. SPF, DKIM and DMARC address aspects of domain-level email **authentication** and **policy**; they do not prevent all phishing or authenticate every display name.

5.6.7 Encryption Audit Procedures

Identify what is protected, against whom and at which points plaintext exists. Review algorithm/mode suitability, key generation, access, distribution, rotation, backup, compromise response and disposal. Validate **configuration**, certificate handling, recovery and vendor **dependencies**. Test that backups of encrypted data remain recoverable with appropriately protected keys.

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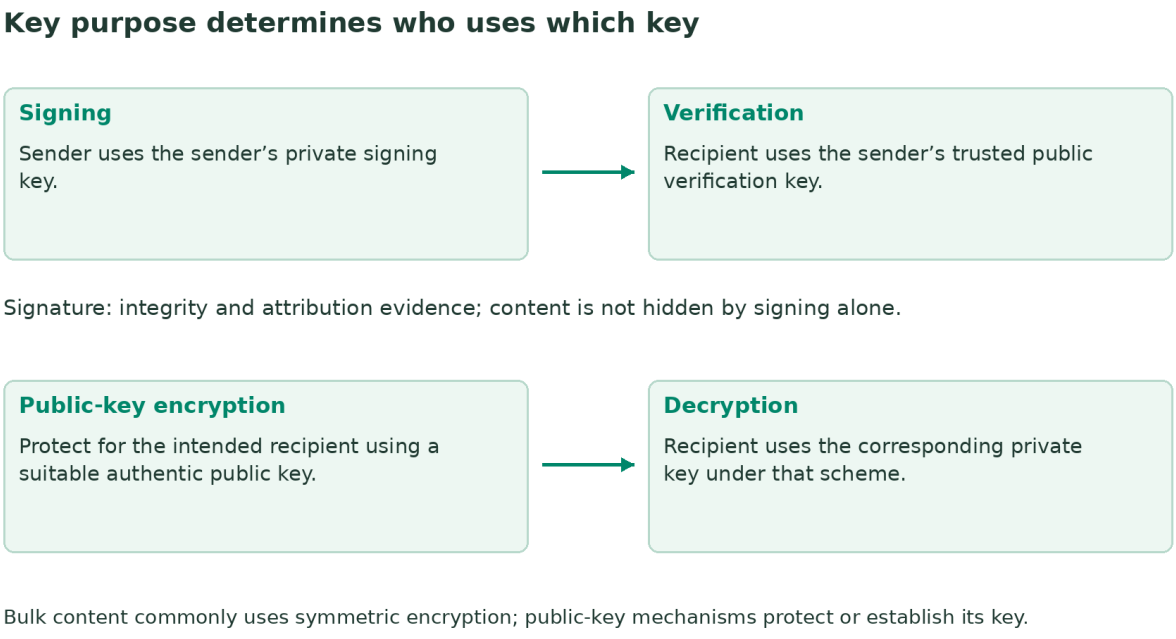


Figure 5.3 Signing, verification, encryption and decryption use keys for different purposes.

Auditor Perspective

Identify the cryptographic service

For a signature, the sender signs with a **private signing key** and recipients verify with its trusted public key. For suitable public-key **encryption**, protect content for the recipient's key.

A matching unkeyed digest from an untrusted source does not establish **origin**. A signature does not provide confidentiality by itself.

Using cryptographic terms accurately

Symmetric encryption uses shared secret key material for suitable encryption and decryption operations. Public-key techniques use related public and **private** keys for specified operations such as signing, verification or key establishment. A **digital envelope** commonly protects bulk data with a symmetric content key and protects or establishes that key for the recipient. These patterns do not mean every algorithm can perform every operation.

A **digital signature** is created with the **private signing key** and verified with the trusted corresponding **public key** using the selected algorithm. It supports integrity, signatory authentication and non-repudiation evidence under the key, identity and process assumptions. It does not conceal the message or establish that its content is true. Describing all signatures as encrypting a hash with a private key is technically incorrect.

Encryption alone does not necessarily detect alteration; **authenticated encryption** can provide confidentiality and integrity. An unkeyed hash supports comparison only when the expected digest and collection process are trustworthy. A MAC depends on a shared secret and does not provide the same third-party signatory evidence as a **digital signature**. Example: A malicious download and its digest are replaced together on an untrusted page. Matching them does not authenticate the publisher. Assess keys, trust, algorithms, modes, lifecycle and implementation rather than the presence of a cryptographic acronym.

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Legacy cryptography and nonrepudiation limits

Nonrepudiation, also written non-repudiation, concerns evidence that supports attribution and resists false denial of an action. A verified **digital signature** supports evidence about the **signed content** and **signing identity** under trustworthy identity, key and process assumptions. It does not prove that a recipient received or read the message. Evidence of receipt needs an appropriate receiving or acknowledgement process and reliable records. [R6]

Algorithm	Recognition point	Audit conclusion
Data Encryption Standard (DES)	Legacy symmetric block cipher with a 56-bit effective key and 64-bit block.	Inadequate for modern protection; identify remaining use and an approved replacement.
Triple DES / 3DES / TDEA	Three-key construction has a nominal 168-bit key bundle, but about 112-bit security strength and a 64-bit block. Nominal length is not effective strength.	NIST disallowed applying new **TDEA** protection after 2023 and withdrew SP 800-67 Rev. 2 on 1 January 2024. Processing previously protected data has separate transition allowances. [R27]
AES	Modern symmetric block cipher with a 128-bit block and 128-, 192- or 256-bit keys.	Use an approved mode and appropriate key management for the requirement. AES alone does not provide signatures or establish origin.

An obsolete algorithm is not made suitable by keeping its name off the network diagram or rotating a weak key more often. Inventory actual **dependencies**, data **retention** and interoperability, then plan migration with accountable owners. A modern algorithm still needs appropriate modes, keys, implementation and access controls.

Secure Sockets Layer (SSL) is obsolete: SSL 2.0 and SSL 3.0 should not be enabled. **TLS** 1.0 and TLS 1.1 are also deprecated; assess supported TLS 1.2/1.3 configurations and current **policy**. POODLE is associated with SSL 3.0; BEAST also affected TLS 1.0, so these attack names do not define a simple SSL-versus-TLS boundary. The informal phrase SSL certificate does not mean the obsolete SSL protocol is required. [R7, R28, R29]

Concealment watermarking and signature images

Steganography hides information within another carrier, such as an image or audio file, to conceal its existence. Encryption protects intelligibility; the two can be combined. A detector's clean result is limited by the formats and methods it can examine. [R39]

Technique	Meaning	Important distinction
Digital watermarking	Embeds identifying, ownership or tracking information in content; it can be visible or imperceptible.	Does not itself provide confidentiality or the same verification as a cryptographic signature.

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Technique	Meaning	Important distinction
Scanned signature image	An electronic image of a handwritten signature that can be copied.	Does not itself provide cryptographic integrity or signer-key verification.
Digital signature	Created and verified through the specified signing algorithm and trusted key arrangement.	Its evidence depends on identity, key protection, validation and process. See 5.6.3 and 5.6.

Check Where Decryption Keys and Plaintext Become Available

Example. Stored records are encrypted, and a service account can retrieve the key and export cleartext. Define the threat and the boundary of protection. Inspect the encryption configuration, key authorization, use logs and controls on export. Verify that rotation and recovery arrangements preserve authorized access to needed data. A digest helps compare content, while a verified digital signature can support origin and integrity under its trust assumptions. Select evidence according to the specific protection objective.

5.7 Public Key Infrastructure

Public key infrastructure (PKI) supports trusted use of public keys through certificates, **validation** and lifecycle processes. It answers a central question: why should a relying party believe that this key belongs to the claimed subject for the intended purpose?

5.7.1 Digital Certificates and PKI Components

An X.509 certificate binds a **public key** to a subject under the issuer's signed assertions and applicable policy. It commonly contains issuer, subject, serial number, validity and permitted uses. A certificate is public information; the associated **private key** must remain protected.

Component	Role	Important distinction
CA - certification authority	Issues and signs certificates under its policy; manages certificate lifecycle responsibilities.	A trusted root is a trust anchor. Intermediate CAs can also issue certificates.
RA - registration authority	Verifies applicants and approves registration requests under delegated arrangements.	Acts in the registration role; certificate issuance/signing belongs to the CA role.
X.509 certificate	Signed binding of public key, subject and relevant constraints.	It is not the private key and does not encrypt data by itself.
CRL - certificate revocation list	Signed published information identifying revoked certificates within its scope.	The list reports revocation; it does not destroy or disable a private key.

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Component	Role	Important distinction
OCSP - Online Certificate Status Protocol	Signed status response for a certificate under the responder's arrangements.	Check response authenticity and freshness; it is not an absolute real-time guarantee.
Repository/directory	Publishes or distributes certificates and related information.	Storage/distribution does not confer CA issuance authority.
Key pair	Related public and private cryptographic keys.	Protect the private key and use each key only for approved purposes.
Relying party	Uses certificates and validation evidence to make a trust decision.	Must enforce validation and application-specific identity requirements.

5.7.2 Certificate Validation and Trust Chains

Build and validate the chain to an accepted **trust anchor**; check signatures, validity periods, constraints, key usage and the intended identity or service name. Apply appropriate **revocation**/status policy. A certificate from an untrusted issuer, for a different hostname or for the wrong purpose should not become acceptable merely because its date has not expired.

Example: A payroll site presents a validly signed certificate for an unrelated domain. **Encryption** may be possible, but the intended server identity has not been established. Ignoring the name mismatch defeats a key part of the trust decision.

Separate certificate trust from the protected connection

In a typical certificate-based TLS 1.3 connection, the client validates the server certificate for the intended service and the server proves possession of the associated **private signing key** over the handshake. Ephemeral key agreement and the key schedule derive traffic keys; symmetric **authenticated encryption** protects application records. A certificate is not itself a session key, and it does not encrypt the whole conversation. Client certificates are used only where the chosen **authentication** arrangement requires them.

5.7.3 Key Management

Protect generation, **enrollment**, distribution, storage, use, rotation, backup where appropriate, **revocation** and destruction. Hardware security modules (HSMs) can protect sensitive keys and operations, subject to correct configuration, roles and lifecycle controls. Separate duties and apply dual control or split knowledge where justified.

Encryption-key recovery may be essential to restore business information. Signing-key recovery or escrow has different attribution risks and needs deliberate **policy**. Do not treat all private keys as interchangeable. Inventory certificates and renew them through controlled automation to avoid both outages and unmanaged issuance.

5.7.4 Certificate Revocation, CRLs and OCSP

Revocation indicates that a certificate should no longer be relied upon under the applicable **policy**, for reasons such as key compromise or incorrect issuance. Publish usable status information and ensure relying applications enforce the intended behavior. CRLs can be periodic and OCSP responses can be cached; verify freshness, scope and outage handling.

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Revoking a certificate does not erase a key or stop every use by parties that do not check status. Following compromise, contain exposure, revoke affected certificates as appropriate, replace keys and certificates, and investigate affected use. Consider trust-anchor compromise and the extent of required reissuance.

5.7.5 PKI Infrastructure Risk and Audit Procedures

Review CA and RA roles, **identity proofing**, issuance approvals, **key protection**, certificate policies, status publication, renewal, logging and recovery. Protect trust stores and privileged changes. Test validation failures and inventory completeness. A legitimate domain-validated certificate does not prove the operator is honest or that the site's content is safe.

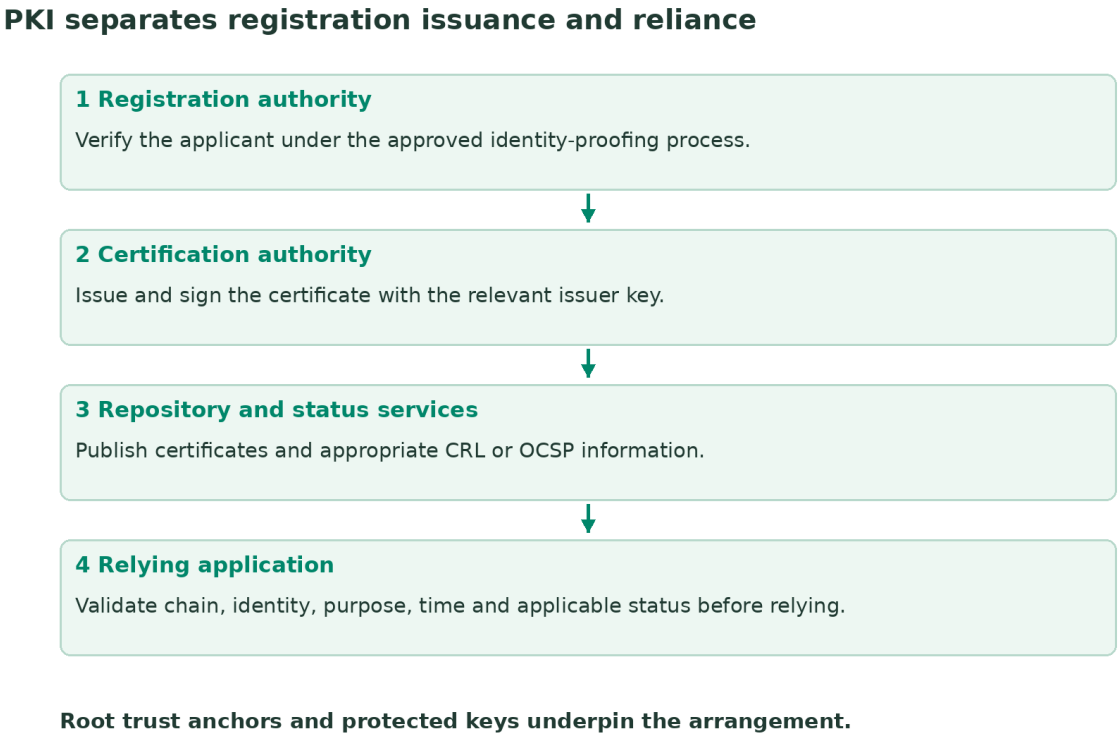


Figure 5.4 Registration, issuance, publication and reliance have distinct responsibilities.

Auditor Perspective

Validate the binding and its status

The CA issues, the RA verifies registration and the repository publishes. A certificate must also be suitable for the intended identity, purpose, time and **trust chain**.

Revocation reports a trust decision. It does not erase the **private key** or force applications with weak validation to stop trusting it.

Following certificate trust and revocation to the relying party

A PKI includes certificate policies, **certification** authorities, registration functions, repositories, status services, subscribers and relying parties. The registration authority verifies applicants under its delegated role; the CA issues and signs certificates under the applicable process. Roles may be combined operationally, but their functions and **accountability** remain distinct.

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Certificate validation considers a suitable **trust chain**, signatures, validity period, names, key usage, relevant **policy** and status. A trusted root anchors the chain; intermediate CAs can issue within their authority. CRLs and OCSP communicate status. Verify authenticity, freshness, responder authority and how a client behaves if status is unavailable. A response is not an absolute real-time guarantee of all aspects of certificate validity.

Example: A server key is compromised and a replacement certificate is installed, but the old certificate remains valid and trusted. Request **revocation** through the authorized process and assess relying-party checking, key **containment** and necessary communications. Revocation does not erase the **private key** or alter the mathematics of an old signature. Protect key generation, storage, access, backup where appropriate, recovery, rotation and destruction. A valid certificate identifies an asserted binding under its trust process; it does not certify that the application is free of vulnerabilities.

Certificate policy and certification practice statements

A **certificate policy** (CP) states the rules and applicability for certificates in a particular community or class of use. A **certification practice statement** (CPS) describes the practices a certification authority follows in issuing and managing certificates. In study terms, the CP describes **requirements** and permitted use; the CPS explains how the CA implements its practices. Their scope and obligations must be read in the actual documents. [R30]

Review **identity proofing**, issuance **approval**, private-**key protection**, renewal, revocation, publication and audit arrangements against the applicable CP and CPS. A published CPS is a statement of practice, not proof that every practice operated. Obtain relevant operating evidence. Certificate validity and a trusted chain also do not justify obsolete SSL or weak TLS settings; see section 5.6.

Test Certificate Validation at the Point of Use

Example. A client trusts the issuing certificate authority and accepts a valid chain. Its configuration also accepts a certificate for the wrong service name. Verify the intended identity, chain, time, usage and applicable status checks with a safe negative test. Confirm how the relying application behaves on validation failure and how approved exceptions are recorded. Examine who can change trust settings. A chain can establish a trusted issuer while the application still needs to verify the intended service identity.

5.8 Cloud and Virtualized Environments

Cloud computing provides network access to shared computing services that can be provisioned and released as needed. Common characteristics include self-service, shared resources, elasticity and measured use. A hosted server is not automatically the whole cloud model. Section 2.9.3 explains the service models and retained responsibilities. [R33]

Virtualization abstracts computing resources so multiple logical environments can use shared physical resources. A virtual machine has its own guest environment managed through a hypervisor. Cloud services may use virtualization, but the two terms describe different ideas. Assess the actual architecture and responsibility boundary.

Cloud and virtualization change where services run and how responsibilities are divided. They can improve scale and resilience while introducing shared infrastructure, powerful management planes and **supplier dependencies**. Identify the actual service and contract rather than infer control from the word “cloud.”

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5.8.1 Virtualization and Typical Controls

A **virtual machine (VM)** is a logical computer environment running on shared host resources. The hypervisor provides the layer that creates or manages that separation. Guest isolation, management privileges and shared-host dependencies all require control.

A **hypervisor** manages virtual machines and resources. Type 1 runs directly on the platform; type 2 runs above a host operating system. Security depends on design, patching, **configuration** and administration, not only the type. Hypervisor compromise can affect hosted workloads, but the actual impact depends on the compromise and isolation.

Control administrator access, management APIs, images, snapshots, virtual networks and resource allocation. Prevent VM sprawl through inventory, ownership and retirement. Inspect **VM escape**, insecure images, shared storage, host concentration and backup **dependencies**. Protect snapshots because they can contain secrets and sensitive memory or disk state.

5.8.2 Virtual Circuits, VLANs and Virtual Storage

A virtual circuit is a logical communication path over shared infrastructure. VLANs provide logical network separation; virtual storage networks or storage zoning organize access to storage resources. These mechanisms have different purposes. Logical separation needs verified **configuration**, access restrictions and **monitoring**; it does not necessarily provide encryption.

5.8.3 Software-Defined Networking

Software-defined networking (SDN) separates network control from forwarding functions and uses programmable control **interfaces**. Centralized **policy** can improve consistency while concentrating controller and API risk. Protect administrative identity, controller availability, authorized applications, configuration changes and communication channels. Validate actual forwarding behavior against policy.

5.8.4 Containerization

Containers package applications and **dependencies** while commonly sharing a host kernel. They differ from full virtual machines. Protect image provenance, registries, build pipelines, host/kernel security, secrets, orchestration access and runtime **permissions**. Avoid unnecessary privileged containers, restrict network paths and keep images maintained. A scanned image does not prove that its runtime configuration is safe.

5.8.5 Secure Cloud Migration and Shared Responsibility

Inventory applications, data and **dependencies**; assess suitability and risk; assign responsibilities; design access, logging, encryption, resilience and exit; then test **migration** and operation. Validate data movement and configuration rather than assume the provider supplies every customer control.

Service model	Typical customer scope	Typical provider scope
IaaS - infrastructure as a service	Guest operating systems, applications, identities, data and customer configuration.	Underlying facilities, hardware and virtualization within service scope.
PaaS - platform as a service	Customer code, data, identities and relevant application configuration.	Underlying infrastructure and managed platform/runtime.

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Service model	Typical customer scope	Typical provider scope
SaaS - software as a service	Appropriate use, identities, access, data handling and customer settings/integrations.	Application and supporting platform/infrastructure within its contractual scope.

The exact division varies by service. Public, private, community and hybrid deployment models describe arrangements, not guaranteed cost or security rankings. The customer retains its own duties and **business risk** decisions, while providers also have contractual and potentially legal duties. A contract is important but does not replace correct customer access and **configuration**.

5.8.6 Cloud Risk, Assurance and DevSecOps

Assess tenant isolation, management-plane access, exposed storage, residency, **supplier** and subprocessor use, incident notification, **encryption**/key access, retention, deletion, service continuity and exit. A cloud access security broker (CASB) can add visibility and policy enforcement for supported services; inspect coverage and bypass paths.

Evaluate relevant independent assurance reports for **scope**, period, exceptions, subservice organizations and customer control obligations. A **SOC 2 report** is not a universal certificate of security. Contracts should support appropriate assurance, incident cooperation, data return and deletion, while technical evidence demonstrates operation.

DevSecOps integrates security into design, build, testing and operation. Protect source, **dependencies**, build identities, secrets and release artifacts; run suitable checks and act on failures. Infrastructure as code improves repeatability only when **definitions** and deployment authority are controlled.

Auditing cloud posture entitlements and shared responsibility

Responsibilities vary across **IaaS**, PaaS, SaaS and the particular contract and **configuration**. Map provider and customer duties for identity, network, application, data, **monitoring**, resilience and exit. Public, private, community and hybrid describe arrangements; they do not establish a universal cost or security ranking. Virtualization also requires controlled hosts, hypervisors, management planes, images, snapshots and guest separation.

Practical extension: **Cloud security posture management (CSPM)** capabilities assess cloud configurations against selected policies and **exposure** conditions. cloud infrastructure entitlement management (CIEM) capabilities examine effective identity entitlements, including inherited access, service principals, workloads, unused privileges and escalation paths. These are capability categories with varying implementation; a product score is not assurance that the cloud estate is complete or secure.

Example: A storage bucket is private, but a neglected automation identity can grant public access and disable **logging**. Review **configuration** and effective authority together. Reconcile discovered accounts, tenants and resources; inspect **policy** exceptions; test correction; and validate that monitoring cannot be silently disabled. Evaluate protected keys, data location, provider assurance, isolated recovery and control-plane dependencies. For containers and serverless functions, include image or package provenance, runtime identity, secrets and short-lived resources that a periodic inventory might miss. Cloud outsourcing changes responsibilities without removing the customer's accountable risk decisions. [R18]

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Focus	Evidence to examine	Unsupported inference
CSPM	Inventory, selected checks, findings and verified correction	A high score proves every resource secure
CIEM	Effective permissions, identities, activity and privilege changes	Unused access is harmless
Shared responsibility	Service-specific contract and implemented responsibilities	The provider owns every customer control

Combine Cloud Posture and Entitlement Evidence

Example. An environment has storage encryption enabled, but an inherited workload role can copy data to a second account. Identify which service and tenant own each safeguard before selecting tests.

Cloud question	Evidence to examine
Is the resource configured within the approved baseline?	**CSPM** coverage, configuration history, detected deviations and timely remediation.
Who or what can perform the sensitive action?	**CIEM** or other effective-permission analysis, including inherited roles, workload identities and relevant policy boundaries.
Can an approved exception or alternate path bypass a safeguard?	Effective access tests under authorized conditions, exception scope, review and monitoring records.

Check that the inventory includes new and short-lived resources. Compare the observed behavior with the provider and customer responsibilities. A tool's coverage and permissions must support the conclusion.

5.9 Mobile, Wireless and Internet of Things Devices

Mobile and connected devices move information beyond conventional facility boundaries. Their security depends on **enrollment**, ownership, connectivity, updates, access and information handling across the whole lifecycle.

5.9.1 Mobile Computing and Device Management

Assess loss, theft, malicious applications, unsupported versions, insecure **interfaces** and compromised accounts. Mobile device management (MDM) or unified **endpoint** management can enforce relevant settings, inventory, application restrictions, encryption and recovery actions. Verify actual enrollment and compliance rather than assume every device is managed.

Encryption and strong device authentication work together. Encryption depends on **key protection** and device state; it does not make every stolen device unreadable in every condition. Remote wipe can reduce exposure if the device receives and executes the command, so it is not a substitute for protection already in place.

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5.9.2 Bring Your Own Device and Mobile Payments

Bring your own device (BYOD) means using personally owned devices for organizational work under agreed arrangements. Ownership differs from permission to access business information. For example, an employee's phone may be approved for a managed email application while local export of customer records remains prohibited.

BYOD needs agreed responsibilities, permitted data, **privacy** boundaries, support, incident handling and exit. Managed application containers and selective wipe can separate corporate information, subject to implementation limits. Define what the organization may inspect or remove and maintain usable business alternatives.

Mobile payments need trustworthy **enrollment**, transaction **authorization**, protected tokens/keys, application integrity and secure interfaces. Consider account takeover, lost devices, malicious overlays, interception and fraudulent transactions. Tokenization can reduce exposure of payment credentials but shifts risk to the token service, mappings and authorized use.

5.9.3 Wireless Networks

Wi-Fi is a local wireless technology; cellular services provide wider-area connectivity; personal-area technologies such as Bluetooth support short-range links. Ad hoc connections can bypass expected infrastructure controls. Review discovery, pairing, **configuration** and unauthorized access paths.

WEP and legacy WPA/TKIP are unsuitable as modern protection. **WPA2** with appropriate AES-based protection and WPA3 provide stronger arrangements when correctly configured. WPA3-Personal uses SAE; enterprise modes use different **authentication** arrangements, commonly 802.1X with an authentication service. They are not interchangeable labels.

Enterprise **authentication** can support individual **accountability** and revocation. Validate authentication-server certificates and client configuration to resist impersonation. A hidden SSID or MAC-address allowlist is not strong authentication because identifiers can be observed or imitated. Assess coverage and transmit power against operational needs; lower power alone does not secure the network.

A rogue access point is unauthorized; an **evil twin** impersonates a legitimate network. Detect unauthorized infrastructure, restrict bridging, protect administration and isolate guests. Wireless testing should evaluate actual **authentication**, encryption, segmentation and monitoring, not only visible SSID names.

Extensible Authentication Protocol (EAP) is an authentication framework supporting multiple methods, commonly used in enterprise network access. Security depends on the selected method, certificate validation and configuration; the acronym alone does not guarantee strong authentication. [R44]

5.9.4 Internet of Things

The **Internet of Things (IoT)** connects physical devices that sense, act on or exchange information about their environment. Connected cameras, environmental sensors and building controllers are examples. A device can have little user interface while still holding credentials, collecting sensitive data or affecting a physical process. [R34]

IoT devices can contain weak credentials, limited patch support, exposed services and sensitive sensors. Inventory and assign owners, change defaults, restrict **interfaces**, segment according to risk,

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secure updates and monitor behavior. Evaluate **supplier** support and retirement before purchase. Physical consequences may make safety and availability critical alongside confidentiality.

Example: A connected camera shares a network with payroll servers. Even if camera recordings are not highly sensitive, a compromised device can become a route toward other assets. Restrict its permitted communications and management path rather than treating it as a harmless appliance.

Following device identity updates and data through the lifecycle

Mobile, wireless and IoT controls should fit device capability, sensitivity, connectivity and the **consequences** of failure. Identify ownership and inventory, authorized **enrollment**, secure configuration, updates, authentication, data storage, network access and retirement. Unmanaged embedded devices can remain in service long after their update support ends.

Device management can enforce **encryption**, screen locking, configuration and permitted applications, but its coverage and exceptions require testing. Remote wipe depends on execution and device reachability. Lost-device exposure also depends on **key protection**, device state and what was already accessible. BYOD arrangements must address business information, user privacy, support and removal of access.

Example: A building sensor has a shared default password, unrestricted outbound access and an unsupported firmware image. Segment its necessary communications, address **authentication** and support, and determine whether replacement is needed under authorized **risk** decisions. For wireless networks, assess appropriate authentication and protected protocols, rogue access points, guest separation and management **interfaces**. A hidden network name or MAC filtering alone is weak identity assurance. Test removal of certificates, tokens, local data and provider access when a device or user leaves.

Wireless discovery and war driving

War driving is searching an area for wireless networks, often using a moving vehicle and **detection** equipment; location recording may support mapping. Discovery can reveal authorized, rogue or misconfigured networks, but a detected network is not necessarily vulnerable. An authorized wireless assessment needs defined locations, permitted collection and handling of incidental information. Discovery does not authorize association, interception or exploitation. **War dialing** is the separate legacy telephone-line discovery technique described in section 5.11.

Confirm Revocation and Data Handling on a Personal Device

Example. A departing employee used an approved personal device. The device management system reports selective removal of managed data, while a persistent cloud session is still active. Compare the approved BYOD arrangement with enrollment, device compliance, session revocation and selective removal evidence. Verify which data and applications are covered, including local exports or synchronized copies. Evaluate user privacy and the scope of any management action. Removing a managed container and revoking service access address different paths.

5.10 Security Awareness Training and Programs

Awareness helps people recognize **risk**; training develops skills for assigned work; education develops broader understanding. Programs should make secure behavior practical and supported by processes and technology. People should be able to report concerns without unnecessary friction or blame.

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5.10.1 Needs Assessment and Program Design

Identify audiences, responsibilities, incident trends, **threat exposure**, policy changes and observed gaps. Include executives, technical staff, business users and relevant third parties. Tailor content for administrators, developers, finance approvers and support staff rather than repeat the same generic module for everyone.

Cover phishing, **identity verification**, data handling, access requests, physical security, safe use of tools and reporting. Combine onboarding, refreshers, timely reminders and role-specific practice. Training is one layer against social engineering; phishing-resistant **authentication**, filtering, payment verification and usable workflows can also be important.

5.10.2 Measuring Effectiveness

Attendance measures participation; quizzes measure some knowledge. Behavioral evidence includes appropriate reporting, secure handling and **performance** in realistic exercises. Interpret phishing-simulation click rates alongside scenario difficulty, reporting quality and **exposure**; a single low rate does not prove the workforce is protected.

Example: After training, staff report suspicious payment changes more quickly and independently verify requests through approved contacts. Those outcomes provide stronger **evidence** than completion percentages alone. Use findings to improve both the training and the business process.

Measuring awareness through behavior and useful response

An awareness **program** should address the risks and decisions people actually face. Provide role-specific learning for payment staff, developers, privileged administrators, support teams and executives. Training completion establishes attendance or completion, not necessarily understanding or changed behavior.

Use comparable exercises, reporting behavior, appropriate knowledge checks and **incident** experience to assess results. Interpret **phishing** simulations using difficulty, **population**, reporting opportunities and measurement consistency. A lower click rate may reflect easier messages or fear of reporting, while a healthy increase in reports can improve response. Avoid turning the measure into a target people can satisfy without learning.

Example: Payment staff receive a convincing voice message requesting an urgent bank-account change. Awareness helps, but a trusted independent callback **procedure** and enforced **approval** controls directly support verification. Combine people, process and technical safeguards. Test whether staff know how to report and whether reports reach timely triage. Protect exercise information and maintain a learning environment so staff do not hide mistakes. The auditor evaluates **program** design, coverage and evidence of useful outcomes without assuming training is the single best defense for every social-engineering path.

Physical observation and discarded information

Dumpster diving searches discarded material for useful information, such as customer records, credentials or device details. Reduce **exposure** through secure collection, protected disposal areas, suitable destruction and accountable handling of paper and media. A disposal contract should be supported by evidence that the required process actually occurred.

Shoulder surfing observes a person entering credentials or viewing sensitive information on a screen or keypad. Position screens, use appropriate **privacy filters**, shield entry and control visitor proximity.

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Short session timeouts help with unattended sessions, but do not prevent **observation** while an authorized person is actively working. Include these situations in role-specific awareness and physical-control reviews.

Measure the Behavior the Program Is Intended to Change

Example. Help desk staff complete awareness training. In a controlled exercise, several still reset accounts after a persuasive call without checking the approved identity evidence. Define the behavior the program is meant to improve, and compare training completion with exercise results, reported events and actual verification records. Examine whether the procedure is usable under pressure. Follow up after coaching or process changes. Completion is a participation measure. Evidence of correct verification supports a conclusion about the control behavior.

5.11 Information System Attack Methods and Techniques

A **threat** is a circumstance or event with the potential to cause harm. An **attack** is an attempt to exploit a weakness or bypass a safeguard. Identify what the method targets, how it reaches the target and which controls can interrupt it. **Passive attacks** observe. **Active attacks** change, impersonate or disrupt.

5.11.1 Fraud Risk Factors and Computer Crime Exposures

Pressure, opportunity and rationalization can contribute to fraud. Computers can be targets, tools or repositories of evidence. Assess collusion and management override as well as individual access. Technical controls need **reconciliation**, **independent review** and meaningful exception handling.

5.11.2 Network, Identity and Social-Engineering Attacks

Attack	Mechanism	Relevant control direction
Eavesdropping/sniffing	Observing traffic or signals.	Appropriate encryption, endpoint protection and physical safeguards.
Traffic analysis	Inferring information from timing, size and communication patterns.	Limit exposed metadata; specialized padding or cover techniques where justified.
Man-in-the-middle	Interposing between parties to observe or alter communication.	Authenticated protected protocols and correct identity/certificate validation.
Replay	Reusing captured valid messages or authentication material.	Fresh challenges/nonces, bounded lifetimes and correct replay validation.
Session hijacking	Taking over an authenticated session.	Protect tokens, sessions, endpoints and reauthentication for sensitive actions.
IP spoofing / masquerading	Forging source information or identity.	Appropriate anti-spoof filtering and strong identity checks; IP alone is weak identity.
Phishing/spear phishing/whaling	Deceptive messages, targeted messages or executive-focused deception.	Layered messaging controls, reporting, robust authentication and independent verification.
Smishing/vishing	Deception through text messages or voice.	Trusted verification channels and role-based processes.

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Attack	Mechanism	Relevant control direction
Pharming/DNS spoofing	Redirecting name resolution or hosts information to a false destination.	Protect resolution and endpoints; validate service identity. DNSSEC has a defined scope.
Watering hole	Compromising a site the target population visits.	Secure browsing, endpoint protection, updates and supplier/site vigilance.
DoS/DDoS	Exhausting or disrupting a service, from one or multiple sources.	Capacity, filtering, rate controls and upstream mitigation suited to the traffic.
Reflection/amplification	Using third-party responses to overwhelm a spoofed victim.	Anti-spoofing, secure exposed services and upstream protection.
Tailgating/pretexting	Following an authorized entrant or using a false story.	Controlled entry, verification, escort where appropriate and reporting.

5.11.3 Application, Password and Transaction Attacks

Attack	Mechanism	Relevant control direction
SQL injection	Untrusted input changes the meaning of database commands.	Parameterized queries, appropriate validation and least-privileged database access.
XSS	Untrusted content executes as script in a user's browser.	Context-appropriate output encoding, safe templating and suitable sanitization; defense-in-depth browser policies.
CSRF	Inducing an unwanted authenticated request from a user's browser.	Anti-CSRF protections, appropriate cookie settings and sensitive-action verification.
Buffer overflow	Writing beyond intended memory boundaries.	Memory-safe design, bounds controls, updates and platform exploit mitigations.
Race condition/TOCTOU	Exploiting a gap between checking a condition and using it.	Atomic operations, synchronization and correct authorization at use.
Brute force / dictionary / password spraying	Many guesses; likely words; limited guesses across many accounts.	Rate controls, compromised-password checks, strong authentication and monitoring.
Credential stuffing	Trying breached credentials against other services.	Unique credentials, MFA and detection of automated abuse.
Rainbow-table use	Using precomputed hashes to accelerate password recovery.	Unique salts and appropriate costly password derivation.
Privilege escalation	Obtaining rights beyond those authorized.	Patching, least privilege, isolation and controlled administration.
Salami / data diddling	Small repeated thefts or improper input alteration.	Reconciliation, independent review, controlled code and transaction authorization.
Resource enumeration / URL manipulation	Discovering or changing identifiers to reach unauthorized resources.	Server-side authorization on each relevant resource and action.

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Attack	Mechanism	Relevant control direction
DNS tunneling	Carrying covert traffic through DNS.	Resolver controls, traffic analysis and appropriate egress restrictions.

5.11.4 Malware and Targeted Intrusion

A virus replicates by infecting host content; a worm can self-propagate across systems; a Trojan presents as legitimate while delivering malicious behavior. Spyware and keyloggers collect information. Rootkits conceal **activity** or privileged presence. A logic bomb triggers under a condition; a backdoor bypasses expected access. A **botnet** is a set of controlled compromised devices, and **cryptojacking** misuses resources for mining.

Ransomware can encrypt, steal or threaten disclosure of information. An advanced persistent **threat** describes sustained, targeted activity, not one specific malware family. A zero-day exploit targets a weakness before effective patching or public knowledge is available in the relevant **context**; an immediate patch may not exist.

Use secure **configuration**, reduced **exposure**, supported updates, application control, behavior-based detection, segmentation and recoverable isolated backups. Patching reduces exposure to known flaws; it does not prevent every worm or unknown exploit. Modern endpoint tools may detect some unknown attacks, so “antivirus is always useless against zero-days” is also too broad.

5.11.5 Ransomware and Control Selection

Coordinate **containment**, evidence, credential protection, clean recovery and impact assessment. Test **recovery copies** and consider theft as well as encryption. Payment does not guarantee recovery, deletion of stolen data or freedom from further attack; decisions require appropriate management and specialist input within applicable obligations.

Do not treat every suspicious redirect as the same attack or every input weakness as the same fix. **SQL injection** and XSS differ in execution **context** and primary design defenses. Security assessment should examine the actual mechanism and the control’s enforceable scope.

Auditor Perspective

Choose the defense for the mechanism

SQL injection changes database-command meaning; parameterization separates commands from data. XSS executes in the browser; safe rendering and context-appropriate encoding address that path.

A generic “**input validation**” answer may be less direct than the control that targets the stated execution context.

Connecting modern attack paths to controls

An attack technique matters because of the path it uses and the business consequence. **SQL injection** changes query structure when untrusted input is interpreted as commands; **parameterized queries** help preserve the distinction between code and data. Replay reuses a previously valid message; protected uniqueness or freshness information must be checked and enforced by the receiver. A **timestamp** or nonce that nobody validates does not prevent repeated business action.

Practical extension: **Prompt injection** introduces adversarial instructions through user input, documents, websites or retrieved material. Separate trusted instructions from untrusted content,

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constrain tool authority, validate outputs and require suitable **authorization** for consequential actions. An LLM deciding whether its own input is trustworthy is not a complete boundary.

Data poisoning manipulates training, fine-tuning or retrieval sources to influence behavior. Assess provenance, permitted changes, access, validation and monitoring. Sensitive-**data leakage** can occur through prompts, retrieved records, outputs, logs or provider retention, and does not require every provider to use every submission for model training. Example: A retrieved invoice tells an assistant to email the entire customer table. Enforce data and action permissions outside the model and test indirect-injection cases. Also evaluate bias, unsupported generated statements, drift and **human oversight** in the actual use context. [R11, R17]

Reconnaissance methods and their control implications

Reconnaissance gathers information that may support a later intrusion. The same technical discovery method can be used in an authorized **assessment**. Distinguish what was discovered from a confirmed **vulnerability** or successful compromise. The following comparison links terms to controls; authorization and agreed scope remain necessary for testing. [R31]

Method	What it identifies or exposes	Audit and control focus
Port scanning	Reachable network listeners and apparent port states; results can suggest services.	Validate service identity, permitted exposure and scanner limits. An open port alone is not proof of a vulnerability; see section 5.12.
War driving	Wireless networks within detection range, including potentially unauthorized access points.	Compare discovery with inventory and approved settings; assess coverage and location assumptions. See section 5.9.
War dialing	Telephone numbers that respond as modems or other remote-access devices.	Identify and remove or secure unapproved legacy dial-up paths. A modem response does not by itself prove successful access.
Dumpster diving	Information discarded in paper, equipment or media.	Review secure disposal, physical access and destruction evidence. See section 5.10.
Shoulder surfing	Information observed during entry or display.	Inspect visibility, shielding, proximity and user behavior. A timeout alone does not prevent live observation. See section 5.10.

ARP poisoning is an active manipulation of local IPv4 address resolution, rather than merely discovering services. It can redirect or disrupt traffic. Section 5.4 explains the mechanism and the role of trusted bindings and dynamic ARP inspection. Choose controls for the actual attack path instead of treating every reconnaissance or network attack as a password problem.

Trace a Generative AI Risk From Input to Action

Example. A support assistant retrieves documents and can prepare outbound messages. An untrusted document contains instructions intended to redirect the assistant. Audit uses synthetic data in an approved test and follows the path from retrieval to the proposed action.

Risk	Evidence to obtain
Prompt injection	Check whether untrusted instructions can trigger a restricted action. Inspect tool permissions, approval gates and logged outcomes.

Risk	Evidence to obtain
Data poisoning	Verify provenance, review and integrity checks for knowledge sources or training data before they are used.
Information leakage	Check access before retrieval, what the model can send, where data are retained and whether outputs disclose restricted content.
Bias and misleading results	Inspect representative validation cases, independent checks of outputs and evidence that exceptions receive human review.

Distinguish the **model's generated text** from the **system's ability to act**. Test the application and its controls together, with explicit scope and safe data. [R11]

5.12 Security Testing Tools and Techniques

Security testing supplies evidence about selected controls and **weaknesses** under defined conditions. It needs **authorization**, scope, rules and competent interpretation. A clean result applies to the assessed population, methods and time; it is not proof of universal security.

5.12.1 Security Assessments, Audits and Vulnerability Assessments

A **risk assessment** evaluates threats, **weaknesses** and consequences. An audit evaluates evidence against objectives and criteria and can include technical testing; it is not limited to paperwork. A vulnerability assessment identifies and analyzes weaknesses using automated and manual techniques. Validation may occur, while controlled exploitation is a more characteristic focus of penetration testing.

5.12.2 Penetration Tests and Rules of Engagement

A **penetration test** uses authorized attack techniques to demonstrate feasible paths and impact within scope. Agree targets, timing, permitted methods, exclusions, safeguards, data handling, emergency contacts, **stop conditions** and reporting before execution. The authorization must cover the relevant systems and parties, including supplier or cloud conditions.

Black-box, gray-box and white-box describe the tester’s supplied knowledge. Blind, double-blind and targeted labels can also describe how much the defenders know. **Definitions** vary, so state the arrangement explicitly. Even a double-blind exercise requires informed authorization and a **safe coordination** path.

Plan, discover, validate or exploit within limits, analyze **impact**, report and retest. Keep sensitive results restricted and preserve agreed evidence. Neither manual nor automated testing covers the entire **attack surface**; combine methods according to objectives. Red teams emulate adversaries, blue teams defend, and purple-team work shares learning.

5.12.3 Testing and Audit Considerations

Assess exposed and internal paths, identities, remote access, application and network controls, **configuration**, production resources and exception handling. Include **authorization** failures, not only authentication. Evaluate compensating controls and the risk of bypasses. Prioritize findings by business exposure and credible attack paths, assign remediation owners and verify closure.

Example: A scan reports an outdated component. Confirm whether it is present, reachable and vulnerable in the actual **configuration**, then assess impact and treatment. A **false positive** wastes effort; dismissing a real weakness because no exploitation was attempted also lacks support.

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Defining security testing scope and limits

Vulnerability assessment identifies and analyzes weaknesses; penetration testing can demonstrate exploitation and impact within an authorized scope. Neither establishes that every weakness was found, and a failed exploit attempt does not prove safety. Code review, configuration review, scanning, fuzzing and adversarial exercises provide different evidence.

Establish **written authorization**, objectives, boundaries, permitted methods, timing, sensitive-data handling, **stop conditions**, communications and recovery arrangements. A technically useful test can disrupt a fragile production service if constraints are ignored. Ensure competent interpretation and suitable follow-up rather than treating the tool's severity rating as a final business conclusion.

Example: A scanner reports no critical issues on a payment application but does not test business **authorization** or the internal API. Examine coverage and perform suitable additional procedures. Validate material findings and distinguish confirmed effects from potential consequences. Prioritize treatment using exposure, exploitability, asset criticality, controls and **business impact**. Retest the corrected condition and relevant bypass paths. A clean report supports a bounded conclusion about its **scope**, criteria, method and time, not a universal declaration of security.

Blind testing and the limits of scanning

Port scanning discovers reachable ports and listeners under the scan's conditions. Filtering, timing, protocol and vantage point affect the result. Service identification, **vulnerability** validation and controlled exploitation are further procedures; one does not automatically establish the others. Specify authorization and production safeguards before testing. [R31]

Arrangement	Knowledge and notice	Assurance boundary
Blind test	The tester has limited prior internal information. Defenders may be informed under the agreed arrangement.	Specify what knowledge is withheld and whether detection or only technical resistance is being assessed; the label is used inconsistently.
Double-blind test	The tester has limited prior information and the operational defenders have no advance notice of the exercise.	An authorized sponsor or **control group** must still know, approve scope, coordinate safety and provide stop authority. It does not mean nobody in management is informed.

State these conditions in the **rules of engagement** rather than relying on labels alone. A defensible conclusion identifies the permitted targets, supplied knowledge, **detection** objectives, excluded actions and time window. A covert exercise without appropriate authorization is not an independent assurance method.

Match the Security Test to the Assurance Question

Example. A scanner confirms that the service responds on an approved port. Management wants assurance that users cannot approve orders above their delegated limit. Define the authorized application test, expected denial and safe accounts or data needed for it. Review how the result maps to the relevant routes and versions. Use a vulnerability scan for weaknesses within its coverage, and test the business authorization rule directly when that is the objective. Document conditions and limitations.

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5.13 Security Monitoring Logs, Tools and Techniques

Monitoring combines telemetry, analysis, decisions and response. It supports detection and accountability and can also trigger preventive or **corrective action** when integrated with enforcement. Recording more events without reliable collection or ownership can create cost without useful assurance.

5.13.1 Intrusion Detection and Prevention

IDS detects and reports suspicious **activity**; IPS can act inline to block it. Signature methods look for known patterns; anomaly methods compare activity with an expected model. **False** positives can overwhelm teams or block legitimate work; false negatives leave attacks undetected. Tune and test both effects.

Placement depends on objectives. External sensors can observe attempted traffic; internal sensors can observe allowed traffic and lateral activity. **Encryption**, blind spots, packet loss and missing context limit visibility. No sensor placement sees every attack, and IDS cannot repair weak application authorization or poor **identity proofing** by itself.

Honeypots or honeynets attract or observe suspicious **activity** through controlled decoy resources. Isolate them, define ownership and prevent them from becoming routes to production. Their telemetry complements normal **monitoring** but does not replace it.

Heuristic detection uses indicative rules or characteristics to identify potentially suspicious activity. It can produce false positives and false negatives. Compare it with signatures and statistical approaches under the actual detection design; no method universally has the lowest false-positive rate.

5.13.2 Audit Logging and Protection

Capture meaningful identity, time, source, action, object and outcome. Synchronize time, control access, monitor collection failures and retain information for justified periods. Logs may contain **personal information**, credentials or investigation details, so minimize unnecessary content and protect access.

Place relevant evidence beyond the sole control of the person whose actions are being logged. Central collection, restricted administration, append-only or **immutable** retention, integrity verification and **independent review** can reduce tampering risk. Signatures detect alteration only when their keys and verification are trustworthy; encryption alone protects confidentiality rather than all alteration.

Audit reduction filters or summarizes records for review; trend and **variance** analysis identifies deviations; signature analysis recognizes known patterns. Retain sufficient original evidence according to **policy**. A summary that discards context can frustrate an investigation.

5.13.3 SIEM and Security Monitoring Tools

A SIEM centralizes relevant events, normalizes them and correlates activity across sources for analysis and alerting. Its value depends on source coverage, parsing, time, **detection** logic, staffing and response. **Endpoint**, identity, network, DLP and cloud telemetry can contribute. Not every function is real-time, and collecting records alone is not full detection.

The security operations center coordinates **monitoring**, triage and escalation. Define **alert** owners, severity, playbooks and handovers. Test whether significant scenarios produce usable detections and appropriate action. Monitor gaps in telemetry and overdue cases rather than using alert volume as the sole success measure.

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Testing the full monitoring and response chain

Monitoring requires relevant telemetry, reliable collection, useful **detection** logic, appropriate triage and an authorized response. SIEM tools aggregate and correlate events; endpoint detection and response tools provide endpoint-specific visibility and actions. Automation can coordinate responses, but needs protected workflows, suitable permissions and safeguards against harmful actions.

Assess coverage, time synchronization, parsing, tuning, **alert** thresholds, **retention** and loss-of-visibility alerts. A detector that generates too many irrelevant alerts can leave important events unexamined. A low alert count can indicate effective prevention, weak coverage or disabled collection. Inspect evidence of actual review and escalation.

Example: A simulated privileged account misuse creates a local event but no **alert** because the parser drops the identity field. Test from event generation through collection, correlation, analyst action and recorded outcome. Include off-hours, **supplier** responsibilities and failed automation. Use reliable metrics such as coverage of relevant scenarios and measured handling under defined conditions. Detection speed alone does not establish correct containment or complete recovery. The auditor should evaluate the chain while **management owns** the operational rules and response authority.

Host and network detection with controlled decoys

Capability	Visibility	Limits to examine
HIDS	Host-based intrusion detection uses host evidence such as logs, file changes and process activity.	Installed coverage, permissions, tamper protection, collection health and alert handling. Host compromise can affect local evidence.
NIDS	Network-based intrusion detection examines traffic visible at a sensor, TAP, mirror port or equivalent cloud traffic source.	Sensor placement, missing paths, packet loss, encrypted payload visibility and detection rules. A sensor does not see traffic never supplied to it.
Honeypot / honeynet	A honeypot is a controlled decoy resource; a honeynet is a network or coordinated environment of decoys used to observe suspicious activity.	Isolate and monitor decoys, limit collected data and prevent pivoting to real services. They need not be production systems or deliberately vulnerable, and they complement normal **monitoring**.

Example: A network sensor sees an encrypted connection but not the malicious command inside it; host telemetry may provide additional **evidence**. Confirm collection and response across both sources instead of assuming one tool supplies complete coverage. A decoy **alert** can be useful, but silence at the decoy does not establish that the real application was safe.

Verify That Events Survive Collection and Normalization

Example. A central monitoring rule looks for changes by a privileged user. The source records the user, but normalization drops the identity field and the rule misses the event. Trace a safe test event from source through collection, parsing, correlation, alert and response. Inspect which sources and fields are required and how missing data are detected. Verify time alignment and role-based access to the logs. An alerting rule's effectiveness depends on the telemetry and workflow it actually receives.

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5.14 Security Incident Response Management

A **security incident** requires coordinated action under the organization's incident criteria and authority. Distinguish the underlying event, a detection alert and the incident decision; section 4.7 introduces these terms. Response must fit the evidence, urgency and potential business harm.

Incident response limits harm and supports trustworthy recovery. Prepare authority, people, tools, suppliers, communications and evidence arrangements before incidents occur. Response is iterative: new **facts** can change containment and recovery decisions.

An **incident response plan (IRP)** specifies authority, coordination and response arrangements for incidents. It should define escalation, communications and responsibilities, with procedures adaptable to the incident's facts and urgency.

5.14.1 Incident Response Process

Activity	Purpose and evidence
Preparation	Defined incident criteria, authority, roles, contacts, playbooks, exercises and tools.
Detection and analysis	Establish what happened, scope, severity, affected information and likely business consequences.
Containment	Limit ongoing harm through actions appropriate to the situation and authority.
Eradication	Remove causes and persistence, correct exposure and address compromised identities.
Recovery	Restore validated capability, verify integrity and monitor for recurrence.
Learning and improvement	Document lessons and implement accountable changes to controls, plans and training.

This sequence is a useful teaching model, not a rule that every action waits for the previous stage to finish. Current NIST guidance integrates **incident response** across cybersecurity **risk management**. Safety, evidence, containment, communications and recovery can require parallel work.

Example: A confirmed compromised server is actively sending sensitive files externally. Authorized isolation or blocking can urgently limit harm while responders preserve evidence and assess **scope**. The correct action for a safety-critical system may require a different **containment** method; "always disconnect power first" is unsafe reasoning.

5.14.2 CSIRT, Plans and Communications

The CSIRT coordinates technical responders with business, management, legal, privacy, communications and other relevant parties. Define who can make disruptive decisions and who speaks externally. Prioritize by **business impact** and credible **exposure**, not alert count alone.

Incident plans should cover escalation, evidence, internal and external communications, **supplier** cooperation, **restoration** and records. Required notification can run alongside containment and assessment; do not assume it always waits until the incident ends. Use applicable obligations and specialist advice for specific decisions.

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5.14.3 Security Orchestration, Automation and Response

SOAR coordinates tools and playbooks and can automate actions such as enrichment, case creation or approved **containment**. Protect its privileged connections, validate inputs, test playbooks and use human **approval** where impact warrants it. An erroneous automated block can interrupt a critical service. Record decisions and permit controlled reversal.

Auditor Perspective

Read the incident's current state

If harmful transmission is active and authorized isolation is appropriate, **containment** is urgent. If the event is already contained, assess what **eradication** or recovery now requires.

Safety, **evidence**, required communications and impact can require parallel action. For the examination: once an incident is confirmed, contain it first; if it is already contained, move to **eradication** and recovery.

Coordinating containment evidence communications and recovery

Incident management needs preparation, **detection**, analysis, authorized response, recovery and improvement. NIST SP 800-61 Rev. 3 integrates **incident response** with cybersecurity risk management; older phase models can still help organize study when their scope is clear. Treat the activities as coordinated work with feedback rather than a rigid rule that all communication waits until recovery. [R12]

When confirmed harm is ongoing, appropriate **containment** can be urgent. Preserve relevant evidence and consider safety, business effects, specialist authority and required communications in parallel. Powering off a system may destroy volatile evidence or create unsafe **consequences**; isolation, credential restriction or another action may better fit the facts.

Example: A confirmed compromised server is exfiltrating customer data. Authorized isolation or blocking can limit harm while responders preserve evidence, investigate **scope** and meet applicable duties. Validate **eradication**, trusted **restoration**, data consistency and permitted return to service. Ransomware can involve theft as well as encryption, so a successful restore does not close every exposure. Track lessons into accountable improvements and retest them. Audit reviews preparedness and response evidence without assuming operational incident command.

Coordinate Containment With Evidence Preservation

Example. Monitoring confirms that a compromised session is sending restricted data. The response team has an authorized way to revoke the session and protect the service. Assess the urgency, authority, possible safety or service effects, and evidence that can be preserved alongside containment. Record the decisions and later verify eradication, recovery and monitoring for recurrence. Evaluate communications and required escalation within the applicable process. Audit checks preparedness and response evidence while authorized responders direct the incident.

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5.15 Evidence Collection and Forensics

Forensics collects, preserves and analyzes information to establish **facts**. The purpose may be operational, administrative, regulatory, civil or criminal. **Scope**, authority, privacy and evidence requirements depend on the context. A chain-of-custody record supports trustworthiness but does not alone guarantee admissibility or automatically settle a case.

5.15.1 Types of Investigations and Computer Forensics

Specialists may examine disks, memory, networks, databases, email, cloud records, mobile devices or **malware**. Each has different collection constraints. Do not assume all cloud logs or mobile information can be acquired like a disk image. Obtain the appropriate authority and coordinate with relevant specialists.

5.15.2 Protection, Acquisition and Imaging

Preserve relevant sources and prevent unnecessary alteration. Document the scene, timing, system state and actions. Collect **volatile information** first where justified and safe because some data disappears on shutdown; live acquisition itself changes system state and needs competent handling. **Operational safety** and ongoing harm can change the collection order.

For suitable media, make a **forensic image** using appropriate write protection, verify it with trusted cryptographic hashes and preserve the original. Analyze verified working copies where feasible. A **matching hash** supports identity of bytes at comparison points; it does not establish who created the original information.

5.15.3 Extraction, Analysis and Reporting

Extract relevant artifacts, correlate timelines, test competing explanations and document tools, versions, methods and limitations. Distinguish observed **facts** from inference. Account for time zones, clock errors, overwritten data and missing logs. Reporting should be understandable to the intended audience and reproducible to the extent the **evidence** permits.

5.15.4 Chain of Custody and Audit Considerations

Record evidence identifiers, source, collection method, time, collector, hashes where applicable, transfers, handlers, purpose, storage and access. Seal and protect physical items and secure digital repositories. Document every material handling change. Unexplained gaps can reduce evidential weight; **consequences** depend on the relevant proceeding and **facts**.

An auditor who encounters suspected wrongdoing should preserve relevant information, avoid unnecessary alteration and escalate through authorized channels. Do not improvise intrusive collection outside competence or authority. Evaluate forensic **readiness**, **retention**, access and supplier cooperation before an incident makes evidence urgently necessary.

Preserving defensible evidence without universal collection rules

Forensic work collects, preserves, analyzes and explains information for an authorized purpose. Establish authority, relevance, privacy, **scope**, competence and specialist involvement before selecting methods. **Requirements** differ across internal investigation, regulatory work and legal proceedings. A

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chain-of-custody form supports trustworthiness but does not alone guarantee admissibility or evidential weight.

Record identification, collection method, time, handlers, transfers, storage, access and analysis. Use appropriate **integrity** checks, protected originals and controlled working copies. A **matching hash** can show that content is unchanged since the comparison point; it does not establish that the original content was truthful or lawfully collected. Tools and collection procedures themselves need validation.

Example: A powered-on workstation contains relevant **volatile memory** that will disappear on shutdown. After authority, scene **documentation** and safety are established, a competent specialist can capture the relevant **volatile information** before destructive action. This is not a universal command to collect memory first in every system. Cloud and SaaS evidence may require provider exports, control-plane logs and coordinated preservation rather than a physical disk image. Document unavailable evidence, time differences and collection limitations so conclusions do not exceed what the preserved material supports.

Establish What a Cloud Evidence Export Contains

Example. A provider supplies a signed log export. Its hash matches the value recorded at collection, but the exporting role did not have access to administrator events. Verify the tenant, accounts, event sources, filters and time window. Document collection authority, custody and integrity checks, and obtain authorized additional records for the gap. A matching hash supports unchanged content relative to the trusted reference. A scope check establishes which activity the evidence can support.

5.16 The Last Mile - Memory Aids and Exam Traps

Use these prompts after reading the explanations. They help recall distinctions; the asset, **threat**, timing and stated objective still determine the answer.

Memory aids

Recall prompt	What to remember
CIA: disclosure / alteration / interruption	Confidentiality, integrity and availability. Several can be affected together.
Claim → prove → permit → trace	Identification, authentication, authorization and accountability.
Know / have / are	Different factor categories; two passwords do not establish MFA.
FAR admits the wrong person; FRR rejects the right person	FAR admits an impostor; FRR rejects a legitimate user. Both have consequences. Compare CER/EER under comparable conditions and assess operating thresholds and fallback.
Owner directs; custodian implements	Assign business accountability and operational duties explicitly.
Sign with mine; verify with mine	Create the signature with the **private signing key** and verify with the trusted corresponding public key. It supports integrity, signatory authentication and non-repudiation evidence; it does not conceal the message.
Encrypt for them	In public-key encryption, recipient's public key protects content for the recipient's private key.

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Recall prompt	What to remember
CA issues; RA verifies; repository publishes	Separate certificate authority, registration and distribution roles.
Revocation changes trust, not the key's mathematics	A CRL/OCSP communicates status; it does not erase the private key.
A digest needs a trusted comparison	An unkeyed hash alone does not establish source authenticity.
DLP needs meaning and coverage	Know sensitive information, permitted use, relevant channels and response.
Find / demonstrate / monitor / respond	Vulnerability assessment, penetration testing, detection and incident response supply different evidence.
Preserve → verify → analyze → explain	Protect sources, validate copies, analyze competently and document conclusions and limits.

Term check: WORM retention differs from a malware worm; a scanned signature differs from a cryptographic signature. Separate enrollment failure, false rejection and false acceptance. Define an event, alert and incident before deciding how to respond.

Exam traps

Tempting wrong answer	Exam answer
Encryption always guarantees integrity.	Encryption alone need not detect alteration. **Authenticated encryption** can provide confidentiality and integrity; trusted hash comparisons, MACs and signatures provide integrity-related evidence under their respective assumptions.
Using a digital signature to keep a message confidential.	The sender signs with a **private signing key** and recipients verify with its trusted public key. This supports integrity, signatory authentication and non-repudiation evidence. Confidentiality requires suitable encryption.
Any matching hash authenticates the download.	Trust the expected digest or an authenticated signature/MAC and its key.
OCSP always proves current validity in real time.	Verify responder authority, signature, freshness and relevant certificate validation.
Every CA is the root trust anchor.	Intermediate CAs can issue; the accepted root anchors the chain.
OAuth alone is user authentication.	OAuth delegates authorization; OpenID Connect adds an identity layer.
Counting two passwords, or a password and a security question, as MFA.	MFA requires different factor types: something you know, have or are.
Letting file owners override mandatory access labels.	MAC is system-enforced and owners cannot override it; DAC is owner discretion; RBAC assigns access by role.

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Tempting wrong answer	Exam answer
Placing a public web server on the internal network.	Isolate public services with controlled paths. One or multiple devices can enforce a boundary. IDS detects; IPS can block within its configured enforcement path.
Relying on technical filters alone against social engineering.	Combine role-appropriate awareness with independent verification and technical safeguards suited to the attack. Measure behavior and useful outcomes under comparable conditions.
A zero-day can always be fixed by installing its patch immediately.	A patch may not yet exist; reduce exposure and use relevant compensating controls.
SQL injection and XSS have identical primary defenses.	Parameterization protects database commands; context-aware encoding/safe rendering addresses XSS.
A contract alone secures the cloud.	Verify provider assurance and actual customer identity, configuration and data controls.
Relying on device tracking to protect data on lost mobile devices.	Enforce suitable device encryption and protected keys. Remote wipe complements protection when it executes; assess coverage, exceptions, device state and actual exposure.
A passed audit or clean scan proves complete security.	The conclusion is bounded by scope, criteria, methods, evidence and time.
Notifying customers or restoring systems before containing a confirmed incident.	For confirmed ongoing harm, authorized **containment** can be urgent. Preserve evidence, protect safety and meet communications duties in parallel as appropriate, then validate eradication and recovery.
Chain of custody guarantees court acceptance.	Document custody and preserve **integrity**. Choose volatile collection and imaging methods according to authority, relevance, safety and system type; they support but do not guarantee admissibility.
Treating false rejection as the security-critical biometric error.	False acceptance directly admits impostors. Assess FAR, FRR, comparable CER/EER, thresholds, fallback and the consequences of each error.
Saying the directory server issues certificates.	The CA issues and signs; the RA verifies identity; the directory distributes certificates.
Deploying DLP or encryption before classifying data.	Establish owners, protection objectives and handling rules. Classification guides controls; discovery can also identify unclassified sensitive information.
Treating a vulnerability scan as proof of exploitability.	A vulnerability assessment identifies and analyzes weaknesses; penetration testing can demonstrate exploitation and impact within scope. A non-exploited issue is not automatically safe.

Decision owners at a glance

Use the assigned authority and evidence to distinguish delivery responsibility from oversight and assurance.

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Role	Responsibility	Boundary
Business/data owner	Defines information sensitivity and justified use.	Technology teams do not decide business value alone.
Security/IAM teams	Implement and monitor assigned security and access controls.	Protect privileged bypass and recovery paths.
CA/RA/relying party	Issue; verify registration; validate and decide reliance, respectively.	A repository distributes rather than issues certificates.
CSIRT and authorized management	Coordinate incident response and business decisions.	Audit can assess readiness and evidence without owning response.

Threats and control purpose

The examples show complementary control purposes. Their classification depends on how the control is configured and used.

Risk	Preventive direction	Detective evidence	Corrective direction
Credential misuse	Suitable authentication, authorization and controlled recovery.	Identity, endpoint and transaction monitoring.	Contain sessions, correct exposure and recover securely.
Data disclosure	Classification-based access, encryption and controlled sharing.	Appropriate DLP and security-event analysis.	Contain disclosure, assess impact and apply response duties.
Malicious alteration	Controlled changes and authenticated integrity mechanisms.	Trusted verification and protected audit evidence.	Investigate and restore a validated state.

5.17 Glossary

Look up terms alphabetically. Return to the lesson for the related example and audit decision.

Term	Definition
ABAC	Attribute-based access control using subject, resource, action and context attributes.
Accountability	Ability to attribute actions to identifiable actors using reliable evidence.
AES	Advanced Encryption Standard, a symmetric block cipher.
AH / ESP	IPsec protocols: Authentication Header provides integrity/authentication without confidentiality; Encapsulating Security Payload supplies selected confidentiality and/or integrity/authentication services. Transport and tunnel are separate modes. Verify negotiated settings. See 5.4.
Anonymization	Processing intended to prevent reasonably possible identification under the relevant context.
Application control	Restriction of which software or actions may execute under policy.

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Term	Definition
APT	Advanced persistent threat: sustained, targeted adversarial activity.
ARP poisoning / spoofing	Falsifying local IPv4 address-to-MAC mappings to redirect or disrupt traffic. Trusted bindings and correctly configured dynamic ARP inspection can reduce exposure. See 5.4 and 5.11.
Asymmetric cryptography	Public/private-key techniques for specified uses such as signing or key establishment.
Authentication	Verification of control of the required authenticator.
Authorization	Decision and enforcement of permitted resource actions.
Availability	Authorized access to usable information and services when needed.
Baseline	Approved minimum or reference configuration under controlled change.
Biometric template	Stored representation used for biometric comparison; protect its lifecycle.
Blind / double-blind testing	Knowledge and notice arrangements for an authorized test. Double-blind commonly withholds advance notice from operational defenders, while an informed sponsor or control group retains authorization and safeguards. Define the arrangement. See 5.12.
Botnet	Set of compromised systems controlled for malicious activity.
Bring your own device (BYOD)	Personally owned devices used for organizational work under agreed arrangements. See 5.9.2.
Brute force	Repeated guessing or exhaustive trial against a secret or access mechanism.
CA	Certification authority issuing and signing certificates under policy.
CASB	Cloud access security broker providing supported cloud visibility and policy controls.
CDN	Content delivery network distributing or caching content near users.
CER/EER	Crossover/equal error rate where specified biometric false acceptance and rejection rates coincide.
Certification practice statement (CPS)	Statement of the CA's certificate-management practices. Distinguish from the certificate policy (CP), which states rules and applicability. Operating evidence is still needed. See 5.7.
Chain of custody	Documented collection, handling, transfer, access and storage of evidence.
Challenge-response authentication	In challenge-response authentication, a verifier issues a challenge and checks a response produced using the required secret or private-key capability. See 5.3.2.
CIA	Confidentiality, integrity and availability.
CIEM	Cloud infrastructure entitlement management capabilities assessing effective cloud identity privileges and excess access.
Ciphertext/plaintext	Encrypted representation / readable input or recovered content.

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Term	Definition
Cloud computing	Network access to shared services that can be provisioned and released as needed. See 5.8.
Cloud shared responsibility	Division of security duties according to the actual service and agreement.
CO2, halon and wet-pipe suppression	CO2: carbon dioxide suppression agent; halon: legacy family of suppression agents; wet-pipe: sprinkler piping normally filled with water. See 5.2.1.
Confidentiality	Protection against unauthorized disclosure.
Container	Application packaging/runtime isolation commonly sharing a host kernel.
Credential stuffing	Trying previously breached credentials on other services.
CRL	Certificate revocation list: signed status publication for certificates within its scope.
Cryptographic agility	Ability to change cryptographic mechanisms and dependencies in a controlled way.
Cryptographic hash	Digest function used for adversary-resistant comparison under appropriate assumptions.
Cryptojacking	Unauthorized use of resources for cryptocurrency mining.
CSIRT	Computer security incident response team.
CSPM	Cloud security posture management capabilities assessing configurations and exposure against selected policies.
CSRF	Cross-site request forgery inducing an unwanted authenticated browser request.
DAC	Discretionary access control allowing authorized owners to grant rights.
Data classification	Assignment of handling requirements according to impact and obligations.
Data custodian	Role implementing assigned data-management and protection responsibilities.
Data Encryption Standard (DES)	DES stands for Data Encryption Standard. See 5.6.
Data owner	Accountable business role directing classification and authorized use.
Data poisoning	Manipulation of training, fine-tuning or retrieval material to influence behavior.
Defense in depth	Complementary protection across multiple layers and failure paths.
DHCP	Dynamic Host Configuration Protocol supplies leased IP addresses and host options. Review authorized servers, rogue responses, scope capacity and lease records. See 5.4.
Digital certificate	Signed binding of a public key to a subject and related constraints.
Digital envelope	Content encrypted symmetrically with its key protected for intended recipients.

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Term	Definition
Digital signature	Private-key signing with public-key verification supporting integrity and attribution.
Digital watermarking	Digital watermarking embeds identifying, ownership or tracking information into content. See 5.6.
DLP	Data loss prevention identifying and controlling prohibited sensitive-information use or disclosure.
DMARC	Domain-based Message Authentication, Reporting and Conformance for email-domain policy.
DMZ / bastion host	A demilitarized zone separates exposed services from internal services through controlled paths. A bastion host is a hardened boundary host. Neither specifies one mandatory physical firewall design. See 5.4.
DNSSEC	DNS Security Extensions authenticating relevant DNS data; not query encryption.
DoS/DDoS	Denial of service / distributed denial of service.
DRM	Digital rights management restricting supported uses of protected content.
Dumpster diving	Obtaining information from discarded paper, media or equipment. Control through protected collection, suitable destruction and accountable disposal. See 5.10 and 5.11.
ECC	Elliptic-curve cryptography, commonly supporting signatures and key agreement.
EDR	Endpoint detection and response.
Emergency power off (EPO)	EPO is an emergency control for removing electrical power from specified equipment or areas. See 5.2.1.
Endpoint	A device or computing system that connects to and uses networked services. See 5.4.7.
EPP	Endpoint protection platform combining supported protective capabilities.
Evil twin	Wireless access point impersonating a legitimate network.
Extensible Authentication Protocol (EAP)	EAP is an authentication framework supporting multiple methods, commonly used in enterprise network access. See 5.9.3.
Failure-to-enroll rate (FER/FTE)	Failure-to-enroll rate (FER or FTE) measures the proportion of eligible individuals who cannot complete the defined enrollment procedure. See 5.3.8.
FAR	False acceptance rate: unauthorized claimant is incorrectly accepted.
Federation	Trust arrangement allowing services to rely on an identity provider’s assertions.
File Transfer Protocol (FTP)	FTP is an application protocol for transferring files. See 5.4.3.
Firewall	Control enforcing defined traffic policy between boundaries.

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Term	Definition
Forward secrecy	Protection of past session secrets against later long-term key compromise under the chosen mode.
FRR	False rejection rate: legitimate claimant is incorrectly rejected.
Heuristic detection / filtering	Heuristic detection uses indicative rules or characteristics to identify potentially suspicious activity. See 5.13.1.
HIDS / NIDS	Host-based intrusion detection analyzes host evidence; network-based intrusion detection analyzes traffic supplied to its sensors. Placement, encryption and collection health affect visibility. See 5.13.
HMAC	Hash-based message authentication code using a shared secret key.
Homomorphic encryption	Encryption supporting specified computations over ciphertext.
Honeypot / honeynet	Controlled decoy resource / network of decoys used to observe suspicious activity. Isolate and monitor; decoys do not replace monitoring of actual services. See 5.13.
HSM	Hardware security module protecting keys and cryptographic operations under its controls.
HTTP / HTTPS	HTTP (Hypertext Transfer Protocol) defines web request/response communication. See 5.4.3.
Hypervisor	Virtualization layer managing virtual machines and resources.
IaaS/PaaS/SaaS	Infrastructure, platform and software as a service.
IAM	Identity and access management across human and nonhuman identities.
ICS/SCADA/PLC	Industrial control systems / supervisory control and data acquisition / programmable logic controller.
IDaaS	Identity as a service.
Identification	Claiming an identity.
Identity proofing	Establishing an identity at enrollment to the required assurance.
IDS/IPS	Intrusion detection / intrusion prevention systems.
IGA	Identity governance and administration.
Incident response plan (IRP)	An incident response plan (IRP) specifies authority, coordination and response arrangements for incidents. See 5.14.
Integrity	Protection against improper modification/destruction and support for trustworthy information.
Internet of Things (IoT)	Connected physical devices that sense, act on or exchange information about their environment. See 5.9.4.
IPsec	Protocol suite protecting IP communications through configured security associations.
IP source routing	IP source routing allows a sender to specify route information in an IP packet. See 5.4.
ISMS	Information security management system.

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Term	Definition
KDC/TGT	Key distribution center / ticket-granting ticket.
Kerberos	Ticket-based authentication using a key distribution center and protected service keys.
Key escrow	Controlled holding or recovery arrangement whose suitability depends on key purpose and risk.
Least privilege	Only the access necessary for the task, scope and time.
MAC - access model	Mandatory access control using centrally enforced policy and labels.
MAC - cryptography	Message authentication code using a shared secret.
Malware	Software with malicious purpose or behavior.
Man-in-the-middle	Adversarial interposition between communicating parties.
Mantrap	Controlled interlocked entry intended to reduce unauthorized following.
MDM	Mobile device management.
Message digest	A message digest is the output of a hash function applied to a message. See 5.6.4.
MFA	Multi-factor authentication using different factor categories.
Microsegmentation	Fine-grained restriction of communications among workloads or resources.
NAC	Network access control using relevant identity and posture policy.
Need-to-know principle	Information access requires a justified business duty even when clearance or role is broader. Complements least privilege over permitted actions. See 5.3.
NGFW	Next-generation firewall with integrated advanced traffic-control functions.
Nonce	Value used with freshness/uniqueness requirements according to the protocol.
Nonrepudiation / non-repudiation	Evidence supporting attribution and resistance to false denial under trustworthy key, identity and process arrangements. A sender's signature alone does not prove recipient receipt. See 5.6.
NTP	Network Time Protocol.
OAuth/OpenID Connect	Delegated authorization framework / identity layer built on OAuth.
OCSP	Online Certificate Status Protocol providing signed certificate-status responses.
PAM	Privileged access management.
Password spraying	Limited common-password guesses across many accounts.
Penetration test	Authorized attack-based testing demonstrating feasible paths and impact within scope.
Personal identification number (PIN)	A PIN is a memorized secret used in an authentication process. See 5.3.2.

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Term	Definition
Pharming	Redirection to a false destination through resolution or related endpoint manipulation.
Phishing	Deceptive communication seeking harmful action or information.
PIA	Privacy impact assessment.
PKI	Public key infrastructure supporting certificate-based key trust and lifecycle.
Policy enforcement point	Logical component permitting, monitoring or terminating access under a policy decision.
Port scanning	Discovery of reachable listeners and apparent port states under the scan's conditions. An open port is not by itself a confirmed vulnerability. See 5.11 and 5.12.
Post-quantum cryptography	Schemes designed to resist relevant classical and quantum attacks.
Private/public key	Secret key held by its owner / related key distributed for approved public operations.
Prompt injection	Adversarial instructions in input or retrieved material intended to redirect an AI application across its trust boundary.
Proxy server / application gateway	Intermediary that mediates communication; an application gateway handles a supported application protocol. Inspection, TLS termination and address exposure depend on configuration. See 5.4.
Pseudonymization	Separation or replacement of identifiers while linkability can remain.
RA	Registration authority verifying applicants under delegated registration arrangements.
Rainbow table	Precomputed values used to accelerate recovery of inadequately protected passwords.
Ransomware	Malicious extortion involving encryption, theft, disclosure threats or related disruption.
RBAC	Role-based access control.
Remote procedure call (RPC)	Mechanism for invoking a procedure on another system. Security depends on the implementation, authentication, authorization and permitted exposure. See 5.4.
Replay	Reuse of captured valid information to repeat or impersonate an action.
Revocation	Status decision that a certificate should no longer be relied upon under the relevant policy.
Rootkit	Mechanism concealing malicious presence or privileged activity.
SAE	Simultaneous Authentication of Equals, used in WPA3-Personal.
Salt	Per-password value preventing reuse of identical precomputed hash results across accounts.
SAML	Security Assertion Markup Language used for federated assertions.

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Term	Definition
Scanned signature image versus digital signature	A scanned handwritten-signature image can be copied and does not by itself supply public-key signature verification. See 5.6.
SDN	Software-defined networking separating control from forwarding functions.
Segregation of duties	Separation of incompatible capabilities or suitable compensating controls.
Shoulder surfing	Observing confidential entry or display. Positioning, shielding and privacy controls address active observation; session timeouts alone do not. See 5.10 and 5.11.
SIEM	Security information and event management for collection, correlation and analysis.
S/MIME	Secure/Multipurpose Internet Mail Extensions supporting signed and encrypted email.
Snapshot	Point-in-time representation whose independence and protection depend on design.
SNMP	Simple Network Management Protocol. v1/v2c community protection lacks cryptographic confidentiality; v3 security depends on configured authentication and privacy levels. See 5.4.
SOAR	Security orchestration, automation and response.
SOC	Security operations center in this context; elsewhere SOC can refer to assurance reporting.
Spear phishing/whaling	Targeted phishing / executive or high-value-target phishing.
SQL injection	Untrusted data altering database-command meaning.
SSH	Secure Shell protocol for protected remote sessions and related transfers.
SSL deprecation	SSL 2.0/3.0 are obsolete; TLS 1.0/1.1 are also deprecated. Review supported TLS configurations and current requirements, not the informal phrase SSL certificate. See 5.6 and 5.7.
SSO	Single sign-on serving multiple applications from an authentication arrangement.
Steganography	Steganography conceals information within another carrier, such as an image or audio file, to hide its existence. See 5.6.
Symmetric encryption	Encryption using a shared secret key arrangement.
Tailgating	Unauthorized following of an authorized entrant.
Telnet	Telnet supports remote terminal sessions. See 5.4.3.
TEMPEST / compromising emanations	TEMPEST concerns investigation and control of unintended emanations from information-processing equipment that could reveal information. See 5.2.
TLS	Transport Layer Security protecting communications under negotiated identity and cryptographic arrangements.
TOCTOU	Time-of-check/time-of-use gap exploited through a race condition.

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Term	Definition
Tokenization	Replacement of sensitive values with tokens under a controlled mapping/use arrangement.
Trust anchor	Accepted root of a certificate-validation trust decision.
UTM	Unified threat management combining security functions.
Virtualization	Abstraction of resources into logical environments using shared physical resources. See 5.8.
VLAN	Virtual local area network providing logical link-layer separation.
VM escape/sprawl	Breakout from VM isolation / uncontrolled proliferation of VMs.
VPN	Virtual private network protecting traffic within its defined tunnel.
Vulnerability assessment	Identification and analysis of weaknesses through selected methods.
WAF	Web application firewall.
War driving / war dialing	War driving discovers wireless networks in an area; war dialing searches telephone numbers for modems or remote-access devices. Discovery is not permission to connect or exploit. See 5.9 and 5.11.
Watering hole	Compromise of a site visited by a target population.
WEP/WPA/WPA2/WPA3	Wireless protection families; assess version, mode, algorithms and configuration.
Workload identity	An identity used by software or an automated service rather than an interactive human user.
Write blocker	Mechanism restricting writes to suitable media during forensic acquisition.
X.509	Certificate format and related public-key framework.
XDR	Extended detection and response across supported telemetry sources.
XSS	Cross-site scripting: untrusted content executing as script in a browser.
Zero day	Weakness or exploitation before effective patching/public knowledge in the relevant context.
Zero trust	Access approach removing implicit trust based only on network location or ownership.

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5.18 Practice questions

These 26 original questions follow the CISA format: choose the one BEST answer from four options. Domain 5 carries 26 percent of the examination, so it has 26 of the book's 100 domain questions. Allow about 1.6 minutes per question, then check section 5.19. Before you look, name the **qualifier** and say why your **second choice** is weaker.

Question 5.01 A retailer plans DLP enforcement for customer records but has no agreed classification scheme, accountable data owners or approved handling requirements. Which action should the IS auditor recommend **FIRST** to establish the policy basis for enforcement?

A. Encrypting removable media issued to customer service staff
B. Defining a data classification scheme with assigned data owners
C. Running the DLP tool in monitor mode to baseline outbound traffic
D. Tuning DLP content-matching rules against sample customer files

Question 5.02 After fingerprint-reader recalibration, an auditor compares measurements under the same population and conditions. PIN fallback remains independently authorized and monitored with no observed control deterioration. Which trend presents the **GREATEST** direct risk of unauthorized biometric admission?

A. A crossover error rate above that of a rival vendor's product
B. More frequent fallback PIN use at the dusty loading-bay reader
C. An increase in the false acceptance rate at the new threshold
D. An increase in the false rejection rate reported by shift staff

Question 5.03 During a walkthrough, the IS auditor sees employees at the badge-controlled data centre door routinely letting colleagues follow them in without badging. Which control would **BEST** address this exposure?

A. An interlocking vestibule that detects occupancy and enforces one-person passage
B. Awareness sessions reminding staff not to let others follow
C. CCTV recording the entrance with footage reviewed each week
D. Biometric readers replacing the existing badge readers

Question 5.04 A defence contractor requires that access to design files be decided by the system by comparing each file's sensitivity marking with the user's clearance, and that file owners cannot override these decisions. Which access control model is **MOST** appropriate?

A. Mandatory access control enforced by the operating system
B. Role-based access control mapped to project job functions
C. Access control lists maintained by each department's lead
D. Discretionary access control administered by file owners

Question 5.05 A purchasing department digitally signs every purchase order it sends to suppliers. What is the **PRIMARY** assurance a supplier gains by verifying the signature?

A. The order contents stayed confidential while in transit
B. The order came from the buyer and has not been altered
C. The buyer's internal approval workflow was completed
D. The supplier's receipt of the order can be proved later

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Question 5.06 An engineering firm must send very large design files to a partner so that only the partner can read them, while keeping encryption efficient. Which approach would **BEST** meet both requirements?

A. Signing a hash of the files with the firm's own private key
B. A random symmetric file key protected with the partner's public key
C. Encryption of the files with the firm's own private key
D. Asymmetric encryption of the files directly with the partner's public key

Question 5.07 An employee reports that his laptop, which stored his private signing key under a weak passphrase, was stolen yesterday. What should be done **FIRST**?

A. Generate a new key pair and issue him a replacement certificate
B. Notify trading partners who rely on his digitally signed orders
C. Request revocation of the certificate bound to the stolen key
D. Review documents signed with the key since the laptop was taken

Question 5.08 In a public key infrastructure, which component is **MOST** directly responsible for verifying a subscriber's identity before a certificate request is approved?

A. Certification authority
B. Certificate directory
C. Validation authority
D. Registration authority

Question 5.09 An organization will host a public web server that must be reachable from the internet while limiting the risk to its internal network if the server is compromised. Where should the server **BEST** be placed?

A. Outside the perimeter firewall, on the internet router segment
B. On the internal network in a dedicated VLAN filtered by ACLs
C. On the internal LAN, reached by port forwarding at the perimeter
D. In a screened subnet between the external and internal firewalls

Question 5.10 Management argues that because its SaaS payroll provider holds ISO/IEC 27001 certification, the company no longer needs to review which of its employees can access the payroll application. Which response by the IS auditor is **MOST** appropriate?

A. Advise that the company remains responsible for its users' access
B. Recommend a contract clause making the provider liable for breaches
C. Obtain the provider's SOC 2 Type 2 report to confirm access controls
D. Check that the certificate's scope includes access management

Question 5.11 Which of the following authentication combinations provides the **LEAST** assurance when claimed as multifactor authentication?

A. A one-time code from a hardware token plus a palm-vein scan
B. A smart card combined with a PIN entered on the card reader
C. A password combined with a fingerprint scan at the workstation
D. A password combined with a memorized security-question answer

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Question 5.12 A network DLP appliance inspects outbound traffic at the internet gateway, including traffic decrypted by the web proxy. It would typically be able to detect each of the following **EXCEPT**:

A. Source code uploaded to an external server over unencrypted FTP
B. Customer records pasted into an external website's input form
C. Card numbers sent in the body of a personal webmail message
D. Design drawings copied from a laptop onto a USB flash drive

Question 5.13 An auditor reviews stored-password protection. Hashing schemes and cost parameters have been validated as suitable, and online guessing is effectively rate-limited independently of account lockout. Which finding presents the **GREATEST** avoidable exposure of recoverable plaintext passwords?

A. Passwords encrypted with a key held on the application server
B. Salted password hashes readable by database administrators
C. Salts stored in the same table as the password hashes
D. Account lockout disabled on the customer login page

Question 5.14 Field sales staff carry company tablets holding customer contracts and pricing. Several tablets were lost last quarter. Which control would **BEST** reduce the exposure of data on lost tablets?

A. Enabling device location tracking to support recovery of lost units
B. Requiring staff to report lost tablets within one business day
C. Installing reputable anti-malware software on each tablet
D. Enrolling tablets in MDM that enforces encryption and remote wipe

Question 5.15 A warehouse wireless network uses WEP with a hidden SSID and a MAC-address allowlist. Which recommendation would **MOST** improve the security of this network?

A. Rotating the shared WEP key quarterly and after any staff departure
B. Migrating to WPA3, or WPA2 with AES and 802.1X authentication
C. Lowering access point transmit power to limit signal outside the site
D. Reviewing the MAC address allowlist monthly to remove stale entries

Question 5.16 During a facilities audit, an IS auditor finds that internet-connected HVAC controllers still use vendor default passwords and sit on the same flat network segment as the finance servers. Which of the following is the **GREATEST** concern?

A. HVAC telemetry is retained longer than the records retention policy allows
B. Facilities staff lack documented procedures for configuring the controllers
C. Controller firmware updates are applied by the vendor during business hours
D. A compromised controller could provide an attack path to finance servers

Question 5.17 Management wants evidence that security awareness training changed employee behavior. Phishing exercises use comparable difficulty and a stable reporting process. Which provides the **BEST** evidence?

A. An increase in average post-training quiz scores from 72% to 88%
B. Rising report rates and falling click rates in phishing simulations
C. Completion of the annual module by 98% of staff before the deadline
D. Annual review and approval of the training content by the CISO

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Question 5.18 Investigation confirms that poisoned DNS records in the bank's recursive resolvers redirected customers to a counterfeit login page. Customer hosts and the legitimate application are uncompromised. Which control **MOST** directly addresses this confirmed path?

A. Awareness training that teaches customers to spot deceptive emails
B. Multi-factor authentication for online banking customer logins
C. Hardening DNS servers and enabling DNSSEC validation on resolvers
D. Email gateway filtering with link rewriting and attachment scanning

Question 5.19 An attacker captured a valid, digitally signed funds-transfer message and resubmitted it hours later, causing a duplicate payment. Which control would **BEST** prevent this attack?

A. Signed unique transaction identifiers and timestamps with receiver-side freshness checks and rejection of reused identifiers
B. Verifying each sender's digital signature against a trusted certificate
C. Requiring dual authorization for transfers above a set value threshold
D. Encrypting the messaging channel between the two parties with TLS

Question 5.20 During an authorized test, a crafted search string returns other customers' records. Database traces confirm that the input changed the SQL command structure rather than remaining a data value. Which remediation **MOST** directly addresses this weakness?

A. Rewriting the database calls to use parameterized queries throughout
B. Encoding application output according to its HTML rendering context
C. Removing write and delete rights from the application's database account
D. Placing a web application firewall with injection signatures in front

Question 5.21 Management has asked for an external penetration test of the customer portal, which is hosted by a cloud service provider. Which of the following should occur **FIRST**?

A. Choosing a gray-box approach so testers receive partial credentials
B. Running an automated vulnerability scan to identify likely targets
C. Agreeing scope and rules of engagement with written authorization
D. Telling the SOC so that alerts generated by testers can be suppressed

Question 5.22 The **PRIMARY** advantage of a penetration test over a vulnerability assessment is that a penetration test:

A. causes less disruption to production systems during execution
B. identifies a broader range of weaknesses across the environment
C. can be performed with no prior knowledge of the target architecture
D. demonstrates whether weaknesses can be exploited and their impact

Question 5.23 An IS auditor plans to rely on database audit logs as evidence of privileged activity by the database administrators (DBAs). The auditor should **FIRST** verify that:

A. the log files are encrypted on the server to protect their confidentiality
B. the logs are held beyond the control of the DBAs whose actions they record
C. alerts are generated for repeated failed DBA login attempts each day
D. log retention meets the period defined in the organization's policy

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Question 5.24 An organization is reviewing how to get more value from its security information and event management (SIEM) system. Which of the following would contribute **LEAST** to that goal?

A. Adding identity and cloud telemetry to the sources correlated by the SIEM
B. Adopting monthly alert volume as the program's main performance measure
C. Synchronizing time sources across systems that feed events to the SIEM
D. Assigning alert owners and documented triage playbooks for key use cases

Question 5.25 The security operations center has confirmed that a compromised file server is actively transferring sensitive files to an external IP address. What should the incident response team do **FIRST**?

A. Isolate the server or block traffic to the external destination
B. Restore the server from the most recent known-clean backup image
C. Perform root-cause analysis to identify the initial entry point
D. Notify affected customers and the regulator of the data breach

Question 5.26 A powered-on workstation is relevant to a payment-fraud investigation. The specialist has authority, documented the scene and confirmed that live capture is safe. Volatile memory is relevant and will be lost on shutdown. What should be done **NEXT**?

A. Shut down the workstation to stop further changes to the disk
B. Review the browser history on the machine to confirm the suspicion
C. Capture the contents of volatile memory before power is removed
D. Create a forensic image of the hard drive using a write blocker

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5.19 Answers and explanations

Each explanation states the **deciding principle**, then shows why the key is best and why each other option, although often a sound practice, does not answer the question asked.

Question 5.01 Answer: B

Data classification before DLP · Qualifier: FIRST · Review section 5.1

Principle: Missing ownership and handling **requirements** leave no agreed **policy** basis for enforcement. Discovery can inform design and does not need to await manual classification of every item.

A. Media **encryption** is a sound protective control; however, it addresses one channel and presumes the data's sensitivity has already been determined.

B. Correct. Accountable classification and handling decisions supply the missing **policy** basis. This does not imply that discovery must wait for a manual label on every file.

C. Baselining reduces **false** positives at rollout; however, the baseline cannot be interpreted until the organization knows which data is sensitive.

D. Rule tuning is necessary for **accuracy**; however, without a classification scheme there is no agreed definition of what the rules should detect.

Question 5.02 Answer: C

Biometric error rates · Qualifier: GREATEST · Review section 5.3

Principle: Under the comparable conditions specified, increased **false** acceptance directly raises impostor-admission **exposure**. False rejection and fallback can also have security effects in other circumstances.

A. **CER** is a useful comparison measure when selecting products; however, it does not show that the deployed readers are admitting impostors.

B. **Fallback** can create serious **exposure**, but the stem establishes independent authorization, monitoring and no observed deterioration. Rising false acceptance is the more direct admission concern here.

C. Correct. A rising **FAR** means impostors are now more likely to be admitted, directly undermining the purpose of the control over a high-**risk** area.

D. Higher **FRR** disrupts staff and may encourage workarounds; however, it denies legitimate users rather than admitting unauthorized ones.

Question 5.03 Answer: A

Mantrap and tailgating · Qualifier: BEST · Review section 5.2

Principle: Enforced single-person passage directly addresses an unauthenticated follower. Effectiveness depends on occupancy enforcement and actual operation, not merely the presence of two doors.

A. Correct. Occupancy **detection** and enforced single-person passage address following-through. Emergency operation must still meet relevant safety requirements.

B. Awareness addresses the behavioural cause; however, it depends on **staff compliance** and social pressure often overrides the guidance.

C. CCTV deters and provides **evidence**; however, it is detective and weekly review identifies tailgating long after the unauthorized entry occurred.

D. Biometrics strengthen **authentication** of the person badging; however, a follower still enters behind them without presenting any credential.

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Question 5.04 Answer: A

Access control models · Qualifier: MOST · Review section 5.3

Principle: MAC enforces system-assigned labels and clearances that users and owners cannot change, matching the stated requirement.

A. Correct. MAC compares object labels with subject clearances under system **policy**, so owners cannot override the decision.

B. **RBAC** simplifies administration by job function; however, it grants rights by role rather than by comparing sensitivity labels with clearances.

C. ACLs give granular control; however, they are a **discretionary mechanism** whose entries can be changed by whoever maintains them.

D. **DAC** is common and flexible; however, it lets owners grant or change access at their discretion, which the requirement explicitly forbids.

Question 5.05 Answer: B

Digital signatures · Qualifier: PRIMARY · Review section 5.6

Principle: A correctly verified signature supports **integrity** and **signatory authentication** under the trusted key and identity assumptions, with non-repudiation evidence. It does not conceal content.

A. Confidentiality requires encryption, typically with the recipient's **public key**; a signature leaves the content readable.

B. Correct. Verifying with the buyer's **public key** confirms the order was signed with the buyer's private key and matches the exact bytes signed.

C. A signature shows the holder of the signing key approved the content; however, it cannot **evidence** that internal workflow steps were followed.

D. The buyer's signature supports evidence of the signed order's **origin** and integrity. Proof of supplier receipt needs appropriate **receiving-process evidence**; the buyer's signature alone does not establish it.

Question 5.06 Answer: B

Digital envelope · Qualifier: BEST · Review section 5.6

Principle: A digital envelope uses fast symmetric encryption for bulk data and the recipient's **public key** to protect the session key.

A. A signature provides **integrity** and origin assurance; however, the files remain readable by anyone who intercepts them.

B. Correct. The digital envelope combines efficient symmetric bulk encryption with public-**key protection** of the content key, so only the partner can recover it.

C. Anything encrypted with the firm's **private key** can be read by anyone holding the firm's public key, so it gives origin, not confidentiality.

D. This achieves **confidentiality** for the partner; however, asymmetric encryption of large files is slow and impractical compared with an envelope.

Question 5.07 Answer: C

Private key compromise · Qualifier: FIRST · Review section 5.7

Principle: Request **revocation** through the compromise process. Its effect depends on relying parties checking relevant status and **policy**; replacement alone does not invalidate the old certificate.

A. Re-keying is needed to restore his **capability**; however, issuing a new certificate does nothing to stop trust in the compromised one.

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B. Partner notification is useful; however, **revocation status** reaches relying parties systematically through the CRL or OCSP once published.

C. Correct. Revocation is the containment step: it publishes that the certificate must no longer be relied on, limiting further misuse of the key.

D. Investigating affected signatures is part of the response; however, it should follow **containment** so the exposure stops growing.

Question 5.08 Answer: D

PKI components · Qualifier: MOST · Review section 5.7

Principle: In the stated separation of roles, the RA verifies subscriber identity and the CA issues and signs under its **policy** and approval process.

A. The CA issues and signs certificates and is ultimately trusted for them; however, **identity verification** is typically delegated to the RA.

B. The directory or repository distributes certificates and **revocation** lists; it has no role in proving who the subscriber is.

C. A **validation** authority (for example an OCSP responder) reports certificate status to relying parties after issuance, not before it.

D. Correct. The RA verifies the applicant's identity and supporting information before forwarding an approved request to the CA for issuance.

Question 5.09 Answer: D

DMZ placement · Qualifier: BEST · Review section 5.4

Principle: A screened public-service zone with controlled paths limits internal **exposure**. Two firewalls are one implementation, not a universal requirement for every DMZ.

A. This isolates the server from the **LAN**; however, it leaves the server itself unprotected by any firewall filtering.

B. **VLAN** segmentation with ACLs adds some separation; however, the server still resides inside the internal network and VLAN controls are weaker than a **DMZ**.

C. Port forwarding makes the server reachable; however, a compromised server would sit directly on the internal **LAN**.

D. Correct. The screened subnet lets the external **firewall** admit only web traffic while the internal firewall blocks the server from reaching internal hosts.

Question 5.10 Answer: A

Cloud shared responsibility · Qualifier: MOST · Review section 5.8

Principle: Under **shared responsibility** the SaaS customer still owns its data and user access decisions, so it must continue reviewing access.

A. Correct. In **SaaS** the customer remains accountable for its data and for granting and recertifying its own users' access, whatever the provider's certifications.

B. Contractual liability can transfer some financial impact; however, **accountability** for the data and its access cannot be outsourced.

C. Provider **assurance** reports are valuable; however, they test the provider's controls, not the customer's own entitlement decisions.

D. Confirming **scope** tests whether the certification is relevant; however, the provider's ISMS cannot cover which customer employees the company itself has authorized.

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Question 5.11 Answer: D

Multifactor authentication · Qualifier: LEAST · Review section 5.3

Principle: MFA requires different factor types; a password and security answer are both something you know, so it is single-factor.

A. This combines possession (the token) with inherence (the vein pattern), two distinct factor types and strong **assurance**.

B. This combines possession (the card) with knowledge (the PIN), so it is genuine two-factor **authentication**.

C. This combines knowledge (the password) with inherence (the fingerprint), satisfying the **different-factor requirement**.

D. Correct. Both elements are knowledge factors, so one phishing or social-engineering attack can capture both; this is not true multifactor **authentication**.

Question 5.12 Answer: D

DLP coverage · Qualifier: EXCEPT · Review section 5.5

Principle: The described local USB copy does not pass through network inspection. Evaluate **endpoint** or device controls and relevant supporting telemetry for that path.

A. Unencrypted FTP traffic crosses the gateway in clear text, so network **DLP** can inspect its contents for sensitive patterns.

B. Form submissions travel as web traffic through the decrypting proxy, so the gateway **DLP** can inspect the posted data.

C. Webmail traffic passes through the proxy and gateway, so its decrypted content is visible to network **DLP** inspection.

D. Correct. The local copy bypasses the described network gateway. **Endpoint** or device controls can monitor or restrict this path; other telemetry can also supply supporting **evidence**.

Question 5.13 Answer: A

Password storage · Qualifier: GREATEST · Review section 5.6

Principle: Reversibly encrypted passwords can all be recovered if the server and key are compromised; salted **one-way hashing** is the expected control.

A. Correct. Reversible **encryption** with the key on the same server means one compromise exposes every password in clear text.

B. **DBA** access should be restricted. Properly salted password hashes still require candidate guessing rather than direct decryption with the server-held key described in the correct option.

C. This looks weak but is normal practice: **salts need not be secret**, since their purpose is to defeat precomputed tables.

D. Missing lockout deserves **assessment**, but the stem already states independent **rate limiting**. It does not create the same direct recovery of the password store as compromise of reversible encryption and its co-located key.

Question 5.14 Answer: D

Mobile device management · Qualifier: BEST · Review section 5.9

Principle: Enforced device **encryption** with protected keys reduces lost-device disclosure **exposure**. Remote wipe is complementary and depends on execution; device state and management coverage matter.

A. Valid as a recovery aid; however, tracking does not stop someone holding the device from **reading its data**, and many lost devices are never recovered.

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B. Valid as a procedural trigger for response; however, **prompt reporting** does nothing to protect the data unless technical protection is already enforced on the device.

C. Valid against malicious applications; however, anti-**malware** does not protect stored data when a device is physically lost or stolen.

D. Correct. Suitable enforced **encryption** and protected keys reduce **exposure**, while remote wipe can remove data when it executes. Validate actual coverage and exceptions rather than assuming wipe always succeeds.

Question 5.15 Answer: B

Wireless security · Qualifier: MOST · Review section 5.9

Principle: WEP is broken, and hidden SSIDs and MAC filtering are weak identifiers, so replacing the protocol with WPA2/WPA3 strong **encryption** and **authentication** addresses the root weakness.

A. Key rotation can be useful for suitable shared-key schemes, but it does not repair **WEP**'s protocol **weaknesses**. Replace the obsolete protection rather than relying on rotation alone.

B. Correct. WEP can be cracked regardless of key changes; WPA3 or WPA2-AES with 802.1X provides strong **encryption** and individual **authentication**, removing the underlying weakness.

C. Valid for reducing casual **exposure**; however, attackers with directional antennas can still reach the signal, and lower power does not fix WEP's broken **encryption**.

D. Valid housekeeping; however, MAC addresses are visible in traffic and easily spoofed, so a cleaner allowlist still provides weak **authentication**.

Question 5.16 Answer: D

Internet of Things · Qualifier: GREATEST · Review section 5.9

Principle: Weakly secured IoT devices on an unsegmented network become a pivot point toward high-value assets, which outweighs operational or **documentation** issues.

A. Valid compliance gap; however, temperature telemetry has low sensitivity, so excess **retention** carries little impact compared with exposure of finance servers.

B. Valid **control weakness** that may explain the default passwords; however, missing procedures are a contributing **cause**, while the attack path is the actual exposure.

C. Valid **availability** consideration for building systems; however, update timing is an operational scheduling issue **far** less material than an open path to financial systems.

D. Correct. Default credentials make the controllers easy to compromise, and the flat network lets an attacker move from them to the finance servers; this **exposure** of critical assets is the **most** material risk.

Question 5.17 Answer: B

Security awareness effectiveness · Qualifier: BEST · Review section 5.10

Principle: Awareness effectiveness is measured by changed behaviour in realistic conditions, not by attendance, knowledge scores or content **approval**.

A. Valid measure of knowledge gained; however, knowing the right answer in a quiz does not show that staff **apply it** when facing a real lure.

B. Correct. Simulation results show how staff actually behave when targeted; more reporting and fewer clicks are behavioural **evidence** that the **program** is changing outcomes.

C. Valid measure of participation and coverage; however, completing a module shows attendance, not whether staff **behave more securely** afterward.

D. Valid **governance** control over content quality; however, approved content says nothing about whether the workforce has absorbed and applies it.

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Question 5.18 Answer: C

Attack methods: pharming · Qualifier: MOST · Review section 5.11

Principle: The stem establishes DNS poisoning. Hardened resolvers and appropriate DNSSEC **validation** target forged answers within that **scope**; they do not prevent every redirection mechanism.

A. Valid against **phishing**; however, no lure message was involved here, and customers who typed the correct address had nothing suspicious to recognise.

B. Valid for reducing the value of stolen passwords; however, it does not stop the redirection, and **real-time proxies** on a fake site can still capture one-time codes.

C. Correct. Resolver hardening reduces compromise **exposure** and correctly configured DNSSEC validation rejects forged signed-zone answers under its **trust chain**. This does not certify every endpoint or website safe.

D. Valid against **phishing** delivered by email; however, no messages were used, so filtering mail does not address the redirected name resolution.

Question 5.19 Answer: A

Attack methods: replay · Qualifier: BEST · Review section 5.11

Principle: A valid signed message can be replayed unless the receiver enforces freshness and rejects reused transaction identity. A **nonce** or **timestamp** without checking is insufficient.

A. Correct. Protected transaction identity and time information is checked at the receiver, which rejects duplicates or stale requests before repeating the business effect.

B. Valid for **authentication** and integrity; however, a replayed message carries the original genuine signature, so verification succeeds.

C. Valid fraud control for large payments; however, the replayed message **reuses already-authorised content**, and transfers below the threshold remain exposed.

D. Valid for protecting **confidentiality** in transit; however, once a message is obtained, **encryption** does not tell the receiver that it has already been processed.

Question 5.20 Answer: A

Attack methods: SQL injection · Qualifier: MOST · Review section 5.11

Principle: Input that changes the meaning of database commands is SQL injection, and **parameterized queries** separate data from commands at the source.

A. Correct. Parameterized queries pass input strictly as data, so crafted strings cannot alter the SQL command; this corrects the defect that let records leak.

B. Valid defence against cross-site scripting in the browser; however, this attack executes in the **database**, not in rendered page output.

C. Valid least-privilege measure limiting modification; however, the attack **read data**, so an account without write rights would still return other customers' records.

D. Valid compensating layer; however, signatures can be bypassed by encoded payloads, and the **vulnerable code** remains in place.

Question 5.21 Answer: C

Penetration testing · Qualifier: FIRST · Review section 5.12

Principle: Establish scope, **rules of engagement** and appropriate written authorization before intrusive testing. Confirm provider terms and obtain any additional permission they require.

A. Valid design decision about tester knowledge; however, it is one element of the **rules of engagement**, decided as part of agreeing scope.

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B. Valid early reconnaissance step; however, scanning is itself testing activity and must not start until the **scope** and authorization are in place.

C. Correct. Scope, constraints, **stop conditions** and appropriate authorization must be established, including compliance with the provider's terms. Authorization does not eliminate operational risk.

D. Valid coordination concern; however, how defenders are informed is set within the **rules of engagement**, and suppressing alerts may defeat the test's purpose.

Question 5.22 Answer: D

Vulnerability assessment vs penetration testing · Qualifier: PRIMARY · Review section 5.12

Principle: Controlled penetration testing can demonstrate exploitation and impact within agreed limits; **vulnerability** assessment identifies and analyzes weaknesses. Neither provides universal coverage.

A. Valid scheduling consideration; however, exploitation usually carries more risk of disruption than scanning, which is why **rules of engagement** set safeguards.

B. Valid goal of security testing; however, breadth is the strength of a **vulnerability** assessment, while a penetration test goes deeper into fewer paths.

C. Valid description of **black-box testing**; however, vulnerability assessments can also run without prior knowledge, so this is not a distinguishing advantage.

D. Correct. Controlled exploitation can demonstrate a reachable attack path and **impact** under the test conditions. It does not establish that all **weaknesses** were found or exploited.

Question 5.23 Answer: B

Audit logging and protection · Qualifier: FIRST · Review section 5.13

Principle: Logs provide **accountability** only if the people whose actions they record cannot alter them, so **log integrity** must be established before the logs are relied on.

A. Valid for **confidentiality** of log content; however, DBAs with server access can often still delete or replace encrypted files, so **integrity** is not assured.

B. Correct. If DBAs can edit or delete the logs, entries recording their own actions cannot be trusted; **independence** of the log store is the precondition for relying on it.

C. Valid **detective control** over access attempts; however, it concerns failed logins, not whether the record of successful privileged activity can be trusted.

D. Valid compliance check that ensures **evidence** exists for the period; however, retained logs are worthless as evidence if the DBAs could have altered them.

Question 5.24 Answer: B

SIEM and security monitoring · Qualifier: LEAST · Review section 5.13

Principle: SIEM value comes from coverage, reliable correlation and response; **alert** volume says nothing about detection quality and can reward noise.

A. Valid and important: wider source coverage lets the **SIEM** correlate account misuse and cloud activity it cannot otherwise see.

B. Correct. Alert volume measures activity, not effectiveness; it can rise with **false** positives and fall when telemetry is missing, so it adds little value.

C. Valid and important: correlation across sources depends on accurate, consistent timestamps, so time synchronization directly improves **SIEM** analysis.

D. Valid and important: alerts deliver value only when someone **owns them and responds** consistently, which owners and playbooks provide.

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Question 5.25 Answer: A

Incident response · Qualifier: FIRST · Review section 5.14

Principle: For the confirmed active exfiltration, authorized **isolation** or blocking is the urgent protective action. Safety, **evidence** and required communications need coordinated attention and can proceed in parallel.

A. Correct. Containment stops the active exfiltration and limits further loss of **sensitive data**, while evidence is preserved for the steps that follow.

B. Valid recovery step; however, restoring before **containment** and eradication destroys evidence and may reintroduce the attacker's access.

C. Valid and needed to prevent recurrence; however, analysis while **data is still leaving** the network allows avoidable harm to continue.

D. Valid obligation that may run in parallel; however, notification follows **assessment** of **scope** under the plan and legal requirements and does not stop the ongoing loss.

Question 5.26 Answer: C

Evidence collection and forensics · Qualifier: NEXT · Review section 5.15

Principle: After authority, scene **documentation**, relevance and safety are established, capture the needed volatile evidence before an action destroys it. Collection order is context-dependent.

A. Valid concern about preserving disk state; however, powering off destroys memory evidence such as running processes, network connections and **encryption** keys.

B. Valid line of enquiry; however, browsing the live system alters **evidence**, and analysis belongs on verified copies after acquisition.

C. Correct. Memory contents disappear at shutdown and may hold processes, connections and keys; capturing them first preserves **evidence** before disk imaging.

D. An offline disk image can be necessary, but shutting down first would lose **volatile memory**. Capture justified volatile **evidence** before that offline acquisition; imaging does not universally require shutdown.

5.20 Continue your preparation

Use your error log to identify the reason behind each mistake. Revisit the relevant explanation and **Last Mile** table, then test the concept with a fresh scenario. Register at certprepx.com for adaptive practice, timed full-length mocks and domain analytics. Available plans and features may change.



Scan to visit https://certprepx.com

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Mixed Scenario Practice

Use these 20 additional **original scenarios** after the domain questions. They test supporting concepts and cross-domain reasoning, including selected Practical extensions. They are not a separate examination blueprint or a full-length mock. Identify the **qualifier** and justify why the closest alternative is weaker.

Mixed practice questions

Question M.01 A valid attribute sample has a 3 percent observed deviation rate and a 7 percent upper deviation limit at the planned confidence level. The tolerable rate is 5 percent. Which conclusion is **BEST** supported?

A. The sample does not support the planned reliance within tolerance
B. The sample supports reliance because the observed rate is below tolerance
C. Add five convenient items and ignore the statistical upper limit
D. No control failures occurred in the untested population

Question M.02 A provider emails an unsigned spreadsheet of privileged accounts. The customer has protected historical identity logs and a reconciled account inventory. What should **MOST** influence the auditor's reliance?

A. The lower number of accounts in one source
B. The spreadsheet's attractive summary charts
C. Relevance, authenticity, completeness and collection controls for each source
D. External source location alone

Question M.03 An analytics join triples several payments because the vendor reference table contains repeated vendor identifiers. What should be done **FIRST** before treating the output as duplicate-payment evidence?

A. Validate join cardinality and correct the procedure against known cases
B. Report all returned rows as duplicate payments
C. Increase the payment sample while keeping the same join
D. Remove every vendor with repeated identifiers from the audit

Question M.04 A business risk exceeds board-approved tolerance. The service manager says a senior title permits immediate acceptance, but the delegation requires governing approval for this exposure. What is **MOST** appropriate?

A. Have internal audit accept the exposure
B. Ignore the delegation because the service is urgent
C. Accept because the manager has a senior title
D. Use the required approval or escalation route and document the response

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Question M.05 A team wants to send support recordings to an external model for training. The recordings are encrypted in transit. Which unanswered question is **PRIMARY** before approving this new use?

A. Whether every support agent prefers the new tool
B. Whether the provider's name appears in the asset register
C. Whether the presentation uses the company logo
D. Whether this processing purpose and provider arrangement are permitted

Question M.06 Customer counts and total balances match after conversion, but two customers' balances have been exchanged. Which procedure **MOST** directly identifies this error?

A. Compare the same aggregate total again
B. Ask whether the migration job ended normally
C. Reconcile customer identities to their source and target balances
D. Compare storage utilization

Question M.07 Planned value is 100, earned value is 80 and actual cost is 90. Which interpretation is correct under the stated earned-value definitions?

A. Behind the planned earned value and over cost for the work earned
B. Ahead of plan and under budget
C. CPI is 1.25
D. Exactly 20 calendar days late

Question M.08 A supplier delivers an SBOM identifying the components in a release. Which conclusion is **BEST** supported by that document alone?

A. The declared component inventory can support further dependency analysis
B. The signing keys were never compromised
C. Every dependency was built securely
D. The release contains no exploitable weaknesses

Question M.09 A payroll test set passes all approved changes but never attempts an unapproved change through the import interface. What **BEST** addresses this gap?

A. Add negative authorization tests through the import path
B. Declare readiness from the high pass rate
C. Repeat the same positive tests with different employee names
D. Replace business acceptance with developer sign-off

Question M.10 Developers cannot click the production deployment button, but can edit the pipeline service identity to bypass all approval gates. Which is the **GREATEST** demonstrated concern?

A. The pipeline uses automation
B. Builds run outside office hours
C. Developers can view test results
D. Effective authority defeats the intended independent deployment control

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Question M.11 A payment is accepted by a receiver, but its acknowledgment is lost. Which control **MOST** directly prevents duplicate business payment during a retry?

A. A different file name for each retry without business checks
B. A longer application log retention period
C. Protected unique transaction identity with receiver-side duplicate rejection
D. A faster network connection

Question M.12 A service window is 1,000 minutes, with 10 minutes of downtime included under its agreed definition. What is the measured availability?

A. 100 percent
B. 90 percent
C. 99.9 percent
D. 99 percent

Question M.13 A service must be usable within two hours. Business validation takes 30 minutes after database readiness, and no alternative database exists. What is the LATEST database-readiness target?

A. Two hours after disruption
B. Thirty minutes after validation ends
C. The average RTO of all applications
D. 90 minutes after disruption

Question M.14 A backup runs every 15 minutes, but failures mean the latest usable copy is two hours old. The RPO is 30 minutes. Which conclusion is **BEST**?

A. The server RTO determines whether the data is current
B. The current usable recovery point does not meet the stated RPO
C. The schedule establishes compliance with RPO
D. No control concern exists if failure notifications were sent

Question M.15 A parallel recovery test processes representative transactions while primary production continues. Which conclusion would exceed the evidence unless separately tested?

A. The tested results can be compared with expected transactions
B. The test can identify missing recovery steps
C. The selected alternate processing operated during the test
D. A live production cutover and failback succeeded

Question M.16 A policy engine detects that an administrator's device is no longer compliant, but access continues through an alternate unmonitored path. Which gap is **MOST** direct?

A. The device has an inventory record
B. The user authenticated earlier
C. The policy uses context
D. The access decision is not enforced across the relevant paths

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Question M.17 A private cloud-storage resource can be made public by a forgotten service principal that also disables logs. Which review **MOST** directly adds to configuration-only posture checking?

A. The number of vendor certifications
B. The color used in the monitoring dashboard
C. The resource's display name
D. Effective workload permissions and privilege-escalation paths

Question M.18 A retrieved document instructs an AI assistant to send confidential records to an external mailbox. Which control **MOST** directly limits the harmful action even if the model follows the text?

A. Assume retrieved documents are trusted because search found them
B. Enforce data and action permissions outside the model with suitable authorization
C. Allow every tool action and review only monthly
D. Ask the model to promise obedience to policy

Question M.19 A captured signed payment message is submitted twice. Its signature validates both times. Which statement is correct?

A. Freshness and duplicate-transaction enforcement are needed at the receiver
B. A signature cannot validate if a message is replayed
C. Signature validation proves the payment must be executed twice
D. Encrypting the message removes the need for duplicate checks

Question M.20 A forensic copy has the same cryptographic hash as the collected original. Which conclusion is **BEST** supported by this comparison alone?

A. The evidence is automatically admissible in every court
B. The compared content is unchanged relative to that integrity reference
C. The original content was truthful
D. The collection was legally authorized

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Mixed answers and explanations

Question M.01 Answer: A

Sampling evaluation | Review sections 1.6

Principle: Evaluate the method-based upper **limit** against the planned tolerance; the observed rate alone does not establish acceptable reliance.

A. Correct. The 7 percent upper **limit** exceeds the 5 percent tolerance. Investigate and reconsider reliance or further procedures.

B. The observed rate does not account for the specified **sampling uncertainty**.

C. Convenient additions do not form a valid **statistical repair**.

D. The **sample** cannot establish absence of deviations in untested items.

Question M.02 Answer: C

Source reliability | Review sections 1.7

Principle: Reliability and relevance depend on the evidence and collection process; external **origin** is not an automatic superiority rule.

A. A smaller count can indicate omissions or a different **scope** rather than better evidence.

B. Presentation does not establish **evidence** quality.

C. Correct. These considerations determine what the sources can support for the **audit objective**.

D. External **origin** is one factor, not a sufficient reliability test.

Question M.03 Answer: A

Analytics validation | Review sections 1.8

Principle: Validate the **procedure** before interpreting its exceptions; faulty joins create nonsampling error.

A. Correct. Establish key relationships and test corrected logic before concluding.

B. The output contains a demonstrated **procedure** defect and does not yet support that finding.

C. More data does not correct the **join defect**.

D. Discarding the affected vendors can conceal relevant risk and **bias** coverage.

Question M.04 Answer: D

Governance and acceptance | Review sections 2.2, 2.5

Principle: Risk decisions must follow accountable authority and delegated limits.

A. Audit evaluates and challenges; it does not become the business **risk owner**.

B. Urgency calls for the applicable **decision process** rather than silent override.

C. A title does not override the **stated delegation**.

D. Correct. The stated limits require an authorized governing decision or further treatment.

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Question M.05 Answer: D

Privacy and AI processing | Review sections 2.6, 5.11

Principle: Transport protection does not establish **permission** for a new processing purpose or provider arrangement.

A. User preference can inform adoption but does not determine **permission**.

B. Inventory matters, but does not resolve the primary **processing question**.

C. Branding does not establish **appropriate processing**.

D. Correct. Establish applicable purpose, basis, **obligations** and terms; then assess safeguards and implementation.

Question M.06 Answer: C

Data lineage | Review sections 2.7

Principle: Identity-to-value **reconciliation** detects substitutions that aggregate counts and totals can conceal.

A. The **aggregate** already matches and hides this error.

B. Normal **termination** does not establish accurate conversion.

C. Correct. Linking each relevant identity to its value directly addresses the exchanged balances.

D. Storage utilization does not address business-value **accuracy**.

Question M.07 Answer: A

Earned value | Review sections 3.1

Principle: SPI is 0.80 and CPI is approximately 0.89; earned-value variance is not directly calendar delay.

A. Correct. EV is below PV and below AC, showing the specified schedule and cost variances.

B. Both ratios are **below one**, contradicting this description.

C. **CPI** equals EV/AC, or 80/90, approximately 0.89.

D. Calendar delay requires the actual schedule and **dependencies**.

Question M.08 Answer: A

Software supply chain | Review sections 3.3, 2.9

Principle: An **SBOM** supports component visibility; it is not a security certificate.

A. Correct. The listed components inform analysis, subject to inventory **completeness** and accuracy.

B. **Key protection** cannot be inferred from an inventory.

C. Build trust requires separate **evidence** about processes and inputs.

D. The inventory alone does not establish **vulnerability** or exploitability status.

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Question M.09 Answer: A

Authorization testing | Review sections 3.4, 3.5

Principle: Coverage must include the relevant misuse and alternate paths, not only successful permitted **activity**.

A. Correct. The missing cases directly test whether the **authorization** requirement is enforced at that interface.

B. The pass rate says nothing about the **omitted requirement**.

C. More positive cases do not test rejection of unauthorized **activity**.

D. Developer sign-off does not replace business acceptance or **authorization** testing.

Question M.10 Answer: D

Pipeline authority | Review sections 3.6

Principle: Evaluate effective **privileges** and control over the gate, not only visible user-interface permissions.

A. Automation can **enforce controls** when designed and protected appropriately.

B. Timing alone does not establish an **authorization** weakness.

C. Viewing results can support legitimate work and is not the **demonstrated bypass**.

D. Correct. The service identity provides a direct route around independent **authorization**.

Question M.11 Answer: C

Job retry integrity | Review sections 4.3, 4.4

Principle: Safe retries require control of the repeated business effect, not merely improved delivery or **observation**.

A. A new transport name can disguise the same **duplicate business request**.

B. **Retention** aids investigation but does not prevent duplicate execution.

C. Correct. Receiver-side enforcement can prevent the same business transaction from being applied again.

D. Bandwidth does not eliminate **lost acknowledgments**.

Question M.12 Answer: D

Availability metrics | Review sections 4.6, 4.10

Principle: Availability under the stated definition is (window minus included **downtime**) divided by window.

A. The specified **downtime** prevents a 100 percent result.

B. The stated downtime is **1 percent** rather than 10 percent of the window.

C. This would require only **one minute** of included downtime.

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D. Correct. (1,000 − 10)/1,000 equals 0.99.

Question M.13 Answer: D

Recovery dependencies | Review sections 4.12, 4.16

Principle: Necessary dependent work must fit inside the stated business-service target.

A. This leaves no time for required **validation**.

B. The sequence is reversed: **database** readiness precedes validation.

C. An average has no basis in this **service requirement**.

D. Correct. Ninety minutes plus 30 minutes equals the two-hour service target.

Question M.14 Answer: B

Recovery point evidence | Review sections 4.14

Principle: Achievable usable **data currency**, not the nominal schedule, determines whether the recovery-point objective is met.

A. Restoration speed and **data currency** are different objectives.

B. Correct. A two-hour-old usable point exceeds the 30-minute loss tolerance.

C. The demonstrated failures make the schedule insufficient **evidence**.

D. Notification helps response but does not restore the missing **capability**.

Question M.15 Answer: D

Exercise selection | Review sections 4.16

Principle: Parallel processing does not by itself demonstrate live transition and return.

A. Such comparisons can be part of the **parallel exercise**.

B. An exercise can reveal missing steps within its **scope**.

C. This is consistent with the described execution **evidence**.

D. Correct. Live **cutover** and failback were not established by the stated test.

Question M.16 Answer: D

Zero trust enforcement | Review sections 5.3

Principle: Signals and decisions must lead to effective enforcement; a bypass defeats the intended boundary.

A. Inventory does not enforce resource **authorization**.

B. Earlier **authentication** does not justify indefinite access despite changed conditions.

C. **Context** can improve decisions and is not the demonstrated defect.

D. Correct. The alternate route bypasses the intended decision and session restriction.

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Question M.17 Answer: D

Cloud entitlements | Review sections 5.8

Principle: Configuration posture and effective identity authority must be examined together.

A. Certifications do not establish the customer identity **configuration**.

B. **Dashboard** appearance does not address the control path.

C. A display name does not determine **permissions**.

D. Correct. The service principal's effective powers create the demonstrated **exposure** and monitoring bypass.

Question M.18 Answer: B

Indirect prompt injection | Review sections 5.11

Principle: Untrusted content must not acquire tool authority merely by appearing in the model **context**.

A. Retrieval relevance does not establish **trustworthiness**.

B. Correct. Independent **permission** and authorization enforcement can block the consequential action.

C. Delayed review can occur **after disclosure**.

D. A generated promise is not a dependable **enforcement boundary**.

Question M.19 Answer: A

Signature and replay | Review sections 5.6, 5.11

Principle: An unchanged **valid signature** does not establish a fresh transaction.

A. Correct. The receiver must enforce protected identity or freshness rules and reject reuse.

B. The unchanged bytes can retain a **valid signature**.

C. Signature **validity** does not authorize repeating the business effect.

D. **Confidentiality** alone does not prevent replay.

Question M.20 Answer: B

Forensic integrity | Review sections 5.15

Principle: A hash comparison supports content **integrity** relative to the reference, subject to the method; it does not establish all evidential properties.

A. Admissibility depends on the proceedings and other **requirements**.

B. Correct. The comparison supports this limited **integrity** conclusion.

C. Unchanged content can still be **false**.

D. Authority requires separate **evidence**.

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Cross-Domain Revision and Topic Finder

Apply the audit method across domains

Use the subject knowledge from the relevant domain with the **audit method** from Domain 1. The connections below are learning aids; they do not change the published exam weights.

Situation	Technical/governance knowledge	Audit method to apply
Supplier assurance	Domain 2 vendor oversight; Domains 4 and 5 operational/security scope.	Establish criteria, assess report relevance, test customer duties and report supported gaps.
System go live	Domain 3 acceptance, conversion and release; Domains 4 and 5 support/security.	Evaluate readiness evidence and limitations while preserving management's authorization responsibility.
Recovery exercise	Domain 4 priorities and recovery measures; Domain 5 clean recovery and access.	Trace evidence to criteria, identify exclusions, evaluate consequences and follow up correction.
Privileged access	Domains 2 and 5 ownership and identity controls; Domain 4 operating records.	Validate populations, test approval and removal, protect evidence and evaluate exceptions.

Personal revision record

Copy or reuse these fields in a notebook or spreadsheet. Record the principle and **changed reasoning**, not only the answer letter.

Field	What to enter
Topic and section	The concept and book reference.
My original reasoning	Why I preferred the wrong answer.
Error type	Knowledge, qualifier, stage, owner, evidence, reading or timing.
Correct distinction	Why the chosen answer fits and the strongest alternative does not.
Action and retest	A focused review task and a fresh scenario after a delay.

Coverage and topic finder

The following map locates the main subject areas of the five domains. Each domain also has The **Last Mile**, a glossary and practice questions.

Domain	Section	Main subject
1	1.1	IS Audit Standards, Guidelines, Functions and Codes of Ethics
1	1.2	Types of Audits, Assessments and Reviews

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Domain	Section	Main subject
1	1.3	Risk-Based Audit Planning
1	1.4	Types of Controls and Considerations
1	1.5	Audit Project Management
1	1.6	Audit Testing and Sampling Methodology
1	1.7	Audit Evidence Collection Techniques
1	1.8	Audit Data Analytics
1	1.9	Reporting and Communication Techniques
1	1.10	Quality Assurance and Improvement of the Audit Process
2	2.1	Laws, Regulations and Industry Standards
2	2.2	Organizational Structure, IT Governance and IT Strategy
2	2.3	IT Policies, Standards, Procedures and Guidelines
2	2.4	Enterprise Architecture and Considerations
2	2.5	Enterprise Risk Management
2	2.6	Data Privacy Program and Principles
2	2.7	Data Governance and Classification
2	2.8	IT Resource Management
2	2.9	IT Vendor Management
2	2.10	IT Performance Monitoring and Reporting
2	2.11	Quality Assurance and Quality Management of IT
3	3.1	Project Governance and Management
3	3.2	Business Case and Feasibility Analysis
3	3.3	System Development Methodologies
3	3.4	Control Identification and Design
3	3.5	System Readiness and Implementation Testing
3	3.6	Implementation Configuration and Release Management
3	3.7	System Migration, Infrastructure Deployment and Data Conversion
3	3.8	Postimplementation Review
4	4.1	IT Components
4	4.2	IT Asset Management
4	4.3	Job Scheduling and Production Process Automation

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Domain	Section	Main subject
4	4.4	System Interfaces
4	4.5	End-User Computing and Shadow IT
4	4.6	Systems Availability and Capacity Management
4	4.7	Problem and Incident Management
4	4.8	IT Change, Configuration and Patch Management
4	4.9	Operational Log Management
4	4.10	IT Service Level Management
4	4.11	Database Management
4	4.12	Business Impact Analysis
4	4.13	System and Operational Resilience
4	4.14	Data Backup, Storage and Restoration
4	4.15	Business Continuity Plan
4	4.16	Disaster Recovery Plans
5	5.1	Information Asset Security Policies, Frameworks, Standards and Guidelines
5	5.2	Physical and Environmental Controls
5	5.3	Identity and Access Management
5	5.4	Network and Endpoint Security
5	5.5	Data Loss Prevention
5	5.6	Data Encryption
5	5.7	Public Key Infrastructure
5	5.8	Cloud and Virtualized Environments
5	5.9	Mobile, Wireless and Internet of Things Devices
5	5.10	Security Awareness Training and Programs
5	5.11	Information System Attack Methods and Techniques
5	5.12	Security Testing Tools and Techniques
5	5.13	Security Monitoring Logs, Tools and Techniques
5	5.14	Security Incident Response Management
5	5.15	Evidence Collection and Forensics

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Formula and Decision Reference

Use this reference after studying the topic. Identify what is being measured, the units, the period and the model's assumptions before calculating. Equations support interpretation; they do not establish input reliability or a successful control. Supporting calculations are included for understanding without implying that every expression is separately tested on the examination.

Measure or model	Expression and boundary	Study section
Risk score	Likelihood score × impact score on defined scales; an ordinal result is not a monetary expected loss.	1.3.4
One-event expected loss	P × I for a defined event, consequence unit and period. P is a probability, not an ordinal score or ARO.	1.3.4
Risk shorthand	Asset × vulnerability × threat and inherent risk minus control require a defined model; use as conceptual shorthand only.	2.5.3
Audit risk	AR = IR × CR × DR in the stipulated conceptual audit-risk model. Keep residual business risk separate.	1.3.3
Sampling confidence	Sampling risk level = 1 − confidence level under the chosen method; 95% confidence corresponds to 5% risk.	1.6.2
Sample deviation	Observed deviations / tested items × 100%; a sample result is not automatically a population rate.	1.6.2
Mean and spread	$\bar{x} = \Sigma xi/n$. Sample $s = \sqrt{[\Sigma(xi - \bar{x})^2/(n - 1)]}$; population $\sigma = \sqrt{[\Sigma(xi - \mu)^2/N]}$.	1.6.2
Single loss	SLE = AV × EF; EF is a fraction of the asset/exposure value lost in the modeled event.	2.5.3
Annual loss	ALE = SLE × ARO; ARO is expected events per year and can exceed one.	2.5.3
Annual control benefit	ALE before − ALE after − annual control cost; compare consistent periods and included costs.	2.5.3
Simple ROI	(Total benefits − investment cost) / investment cost × 100%, using the stated cost/benefit convention and period.	3.2
Simple payback	Initial outlay / constant net inflow per period; uneven inflows need cumulative tracking and stated timing.	3.2
NPV and IRR	NPV = CF0 + Σ[CFt/(1+r)^t], t = 1 to n. IRR is a rate at which NPV = 0.	3.2
PERT duration	TE = (O + 4M + P)/6; O optimistic, M most likely, P pessimistic. P here is a duration.	3.1.10

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Measure or model	Expression and boundary	Study section
Total float/slack	TF = LS − ES = LF − EF; use consistent early/late dates and calendar. Free float has a different boundary.	3.1.10
Earned-value variances	SV = EV − PV; CV = EV − AC. PV planned value, EV earned value, AC actual cost. SV is not calendar days.	3.1.12
Earned-value indices	SPI = EV/PV; CPI = EV/AC. Interpret the ratios with scope, quality and reliable completion evidence.	3.1.12
Supporting EVM forecast	EAC = BAC/CPI if current cost efficiency continues; ETC = EAC − AC; VAC = BAC − EAC.	3.1.12
MTBF / MTTR	Eligible operating time / failures; total repair time / completed repairs. Define the observation and event conventions.	4.6
Observed availability	(Service-window time − included downtime) / service-window time × 100%.	4.6
Average-based availability	MTBF/(MTBF + MTTR) × 100% under the simplified repairable-system model.	4.6
Downtime allowance	Service-window time × (1 − availability target as a fraction), using the same boundary/exclusions.	4.6
Recovery constraint	Technical RTO + separately defined WRT ≤ MTD; retain a margin. Do not add WRT if already inside business-service RTO.	4.12.1
Recovery-point age	Disruption time − timestamp of latest usable recovered data; compare with RPO. A job schedule alone is insufficient.	4.12.1
FAR / FRR	False acceptances / impostor attempts; false rejections / genuine attempts; multiply each by 100 for percent.	5.3.8
Enrollment / crossover	FER uses the defined enrollment population. CER/EER is the crossover operating point, not an arbitrary average.	5.3.8
Check digit / parity	Use the stated digit algorithm; even parity makes the total count of 1 bits even. Both have detection limits.	1.4.2; 3.4.1

Selected technical reading

Use current versions of primary guidance to deepen unfamiliar subjects and confirm implementation details. The guide explains concepts for study; an organization must determine the **standards** and **requirements** applicable to its own work.

ISACA IT Audit Framework Read guidance

ISACA COBIT Read guidance

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NIST Secure Software Development Framework Read guidance

NIST Contingency Planning Guide Read guidance

NIST Digital Identity Guidelines Read guidance

NIST Incident Response Recommendations Read guidance

NIST Post Quantum Cryptography Read guidance

ISO Privacy Information Management Read guidance

X509 Certificate and CRL Profile Read guidance

Online Certificate Status Protocol Read guidance

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CISA COMPLETE STUDY GUIDE

Best of luck with your CISA exam!

Thank you for making this guide part of your preparation. On exam day, pay attention to words like **FIRST**, **BEST**, and **MOST**. They tell you what decision the question is asking you to make. Read the scenario carefully, consider the risk and the evidence, and answer from the auditor's perspective.

I hope this guide serves you well on exam day and in your work as an IS auditor.

Muhammad Aamir

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